BASE24 Remote Banking™

Transaction Processing Manual

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ACI Worldwide Inc.

Preface

The ***BASE24 Remote Banking Transaction Processing Manual*** describes how transactions are processed by the BASE24-telebanking and BASE24-billpay products. This manual includes basic transaction processing concepts and definitions, as well as specific information on how transactions are identified and processed. It also describes how to control and configure various aspects of BASE24-telebanking and BASE24-billpay transaction processing.

A BASE24 Remote Banking system can have both the BASE24-telebanking and BASE24-billpay products or it can have just the BASE24-telebanking product. This manual describes BASE24 Remote Banking systems with both the BASE24-telebanking and BASE24-billpay products and BASE24 Remote Banking systems with only the BASE24-telebanking product.

Audience

This manual is intended for BASE24 managers and operational staff who are responsible for configuring the BASE24 Remote Banking database for transaction processing and understanding how the transactions flow through a BASE24 Remote Banking system.

Prerequisites

Persons using this manual should be familiar with the BASE24 Remote Banking products, their basic functions, their terminology, and the BASE24 files and tables they use.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 and NET24 publications:

- The ***BASE24 Logical Network Configuration File Manual*** describes the assigns and params used in configuring BASE24-telebanking and BASE24-billpay processes.
- The ***BASE24 Core Files and Tables Maintenance Manual, BASE24-billpay Tables Maintenance Manual, BASE24 Base Files Maintenance Manual,*** and ***BASE24 ITS Kernel Files Maintenance Manual*** provide detailed descriptions and valid values for the configuration fields discussed in this manual.
- The ***BASE24-billpay Billing Application Manual*** provides screen descriptions, field descriptions, and operator instructions for tables associated with the BASE24-billpay Billing Application.
- The ***BASE24 Remote Banking Customer Service Support Manual*** provides detailed descriptions, valid values, and procedures for using the customer service representative (CSR) terminal.
- The ***BASE24 Remote Banking End-of-Period Processing and Reporting Manual*** provides information about end-of-period processing and reporting.
- The ***BASE24 ISO Host Interface Manual*** provides detailed processing information for the BASE24 ISO Host Interface.
- The ***BASE24 External Message Manual*** presents detailed field descriptions and configuration information for the ISO-based external message used between the BASE24 Remote Banking system and a host.
- The ***BASE24 Integrated Server Transaction Security Manual*** describes the support provided by BASE24 object-oriented products for PIN verification and encryption, message authentication, and dynamic key management, along with the database settings required to implement this support.
- The ***BASE24 Tokens Manual*** provides the procedures necessary for configuring the Token File (TKN).
- The ***BASE24 Log Message Reference Manual*** provides a complete explanation of log message documentation.
- The ***BASE24 ITS Kernel Configuration Manual*** provides information about operating the Extended Memory Table Build utility and making objects change to a revised extended memory table.

- The ***BASE24 ITS Kernel Objects Reference Manual*** provides detailed information about the function and operation of each kernel object.
- The ***BASE24 IEP Configuration Manual*** provides a detailed description of the Integrated Endpoint process.
- The ***BASE24 IEP Objects Reference Manual*** provides detailed information about the function and operation of each integrated endpoint object.
- The ***NET24-XPNET Network Control Operations Guide*** provides detailed information on service-based routing performed by the XPNET process.
- The ***NET24-XPNET Network Control Command Reference Manual*** provides detailed information on configuring process attributes.
- The ***NET24-XPNET Event Management Manual*** provides information about the XPNET implementation of the Tandem Event Management Service (EMS).

In addition to the BASE24 and NET24 publications, this manual contains references to the following International Organization for Standardization (ISO) publications:

- The ISO 3166-1-1997 standard, ***Codes for the Representation of Names of Countries,*** contains the country codes used in BASE24 Remote Banking Internal Transaction Data (ITD) fields.
- The ISO 4217-1995 standard, ***Codes for the Representation of Currencies and Funds,*** contains currency codes used in BASE24 Remote Banking Internal Transaction Data (ITD) fields.
- The ISO 8583-1993 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specification,*** contains descriptions of the fields used in standard ISO messages between the BASE24-telebanking product and an acquirer endpoint (e.g., an interactive voice response system.

Software

This manual documents standard processing for BASE24-telebanking release 1.3 and BASE24-billpay release 1.3.

Customer-specific code modifications (i.e., CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” introduces the reader to the basics of BASE24 Remote Banking transaction processing. It includes definitions and discussions of transaction originators and authorizers, issuers and acquirers, hosts and DPCs, files and tables, extended memory tables, supported transactions, and transaction messages.

Section 2, “BASE24 Remote Banking Transaction Path,” discusses the processes involved in transaction processing, how they are linked together to form various transaction paths, and the timers they use to control processing. It also introduces service-based routing.

Section 3, “BASE24 Remote Banking Transaction Processing Controls and Impacts,” describes the various transaction processing control options that enable a logical network to be tailored to the needs of its institutions. It also includes discussions of such topics as authorization levels and transaction routing.

Section 4, “BASE24-telebanking Message Flows,” presents a series of diagrams illustrating the paths taken by BASE24-telebanking transactions under various types of processing situations.

Section 5, “BASE24-billpay Message Flows,” presents a series of diagrams illustrating the paths taken by BASE24-billpay transactions under various types of processing situations.

Section 6, “BASE24 Remote Banking Transaction Impacts on the Database,” describes the impact of BASE24 Remote Banking transactions on the balance and usage accumulation totals in the Customer Table (CSTT), Interface Security File (ISF), and Positive Balance File (PBF). It also describes how BASE24 Remote Banking transactions impact data other than balances or counts in the BASE24 Remote Banking database.

Appendix A, “BASE24 Remote Banking Messages,” describes Internal Transaction Data (ITD) fields and the Telebanking Standard Internal Message (BSTM).

Appendix B, "BASE24 Remote Banking Action Codes," describes the action codes that can be carried in the ITD to identify the results of processing a transaction.

Appendix C, "BASE24 Remote Banking Default Processing Codes," lists the default processing codes provided with the BASE24 Remote Banking products.

Appendix D, "Object-Oriented Design," introduces the object-oriented form of programming and describes the objects used with BASE24-telebanking processes. This appendix also provides operator instructions for building extended memory tables, sending commands for server control, and warmbooting extended memory tables and objects.

Appendix E, "BASE24-billpay Configuration Guidelines," provides guidelines for configuring BASE24-billpay processes to the XPNET system and to the Pathway system.

Appendix F, "BASE24-billpay Processing Details," provides detailed processing information for several BASE24-billpay processes.

Readers can use the index by ITD field name to locate information about a particular Internal Transaction Data (ITD) field and the index by data name to locate information about a particular field from the ITD or ITS Transaction Log File (ITLF).

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (e.g., TB-AE000-07 for the ***BASE24 Remote Banking Transaction Processing Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (e.g., 07/2000 for July, 2000) identifies the last time material on a particular page has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions.

The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a *product* publication is modified in any way. Each print record entry includes the date of the change, a unique version number, and the type of change (new, update, reissue, or preliminary). Version 1 is assigned to a *new* manual. Later versions can be updates or reissues, depending on the changes being made. An *update* typically involves changes on a small number of pages throughout the manual. A *reissue* occurs when all pages of the manual are replaced because changes affect a large number of pages. A manual is identified as *preliminary* when it is released to customers on a limited basis, but it has not yet reached final form.

Conventions Used in this Manual

This section explains how command syntax is documented in this manual and a change in Nucleus process terminology.

Syntax Notation

Syntax descriptions in this manual use several symbols and conventions to denote how commands must be entered. These conventions are described in the table below.

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that you must enter exactly as shown.
lowercase italics	Identify variables that you must provide.
Brackets [ ]	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar (|).
Vertical bar |	Separates alternatives in a list of items.
Ellipsis …	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.

Notation	Meaning
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. You can insert additional spaces between any adjacent words or symbols, but not within a word or multiple character symbol.
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

Nucleus Process Terminology Change

There is one process that is the hub of an XPNET system. It manages message queues; controls all lines, stations, and processes in the system; and serves as an interprocess interface. Throughout this manual, this process is referred to as the XPNET process.

In other manuals, you may see references to the Nucleus process as performing these same functions. These two terms (Nucleus process and XPNET process) should be viewed as synonymous. The Nucleus process and the XPNET process perform the same functions and processing. As manuals are reissued, references to the Nucleus process will be replaced with references to the XPNET process.

Section 1

Introduction

This section provides an introduction to BASE24 Remote Banking transaction processing. It provides readers with some basic information on how a BASE24 Remote Banking system processes transactions.

Remote Banking Products

The BASE24 Remote Banking product family currently includes two products—BASE24-telebanking and BASE24-billpay.

The BASE24-telebanking product provides online banking services (e.g., balance inquiries, check clearance inquiries, statement downloads and statement closing downloads, check stop payments, and immediate transfers between two accounts) to customers using remote banking devices (e.g., a touch-tone telephone or personal computer) or calling a customer service representative (CSR) or branch employee.

The BASE24-billpay product provides automated payment and scheduled transfer services (e.g., immediate, future, and recurring payments, future and recurring transfers, and associated inquiries) to customers using remote devices or calling a CSR or branch employee. The BASE24-billpay product provides files that can be used as input to programs that generate checks and Automated Clearing House (ACH) output for payments; however, the BASE24-billpay product does not currently include these programs. The BASE24-billpay product also provides facilities for signing up new customers and accounts, accessing customer information, listing financial institutions and branches, and entering or inquiring on loan applications and miscellaneous requests.

Optional Enhanced Telebanking product modules provide financial institutions and service providers the ability to support all BASE24-telebanking transactions as well as the subset of banking transactions that are provided with the BASE24-billpay product. These product modules consist of a subset of the BASE24-billpay software.

Enhanced Telebanking provides for the scheduling and real-time history of immediate, one-time future transfers, and recurring transfers. The product modules monitor future and recurring transfers, and on the scheduled transaction date, automatically initiate the transfer transaction. When Enhanced Telebanking initiates a transfer, BASE24 routes the transaction to the appropriate authorizing entity for authorization. Various reports are available directly related to these transfer transactions.

Enhanced Telebanking also supports customer and account activation and maintenance transactions, financial institution and financial institution branch list inquiry transactions, loan application transactions, and miscellaneous request transactions.

The BASE24-telebanking software provides two functions used to support the services provided by the BASE24-telebanking, Enhanced Telebanking, and BASE24-billpay products—customer verification and funds authorization. Customer verification involves identifying the customer and verifying his or her authenticity. Funds authorization involves checking the customer's account to ensure that sufficient funds are available to perform a transfer or payment.

Because the BASE24-billpay and Enhanced Telebanking products rely on the BASE24-telebanking software to support its services, BASE24-billpay and Enhanced Telebanking cannot be installed without BASE24-telebanking. The BASE24-telebanking product can be installed without the BASE24-billpay or Enhanced Telebanking product; however, scheduled transfer capabilities and customer signup functions are available only if both the BASE24-telebanking product and either the Enhanced Telebanking or BASE24-billpay products are installed.

This manual documents transaction processing for the BASE24-telebanking, Enhanced Telebanking, and BASE24-billpay products. Throughout this manual, references are made to BASE24, BASE24 Remote Banking, BASE24-telebanking, Enhanced Telebanking, and BASE24-billpay, as follows:

- BASE24 includes BASE24-telebanking, Enhanced Telebanking, and BASE24-billpay, as well as other BASE24 products.
- BASE24 Remote Banking includes BASE24-telebanking, Enhanced Telebanking, and BASE24-billpay.
- BASE24-telebanking indicates the BASE24-telebanking product with or without the Enhanced Telebanking or BASE24-billpay product. For example, the BASE24-telebanking Integrated Authorization process withdraws funds from a customer's account for a BASE24-telebanking transaction, Enhanced Telebanking, or BASE24-billpay transaction.
- Enhanced Telebanking indicates the subset of BASE24-billpay transactions described above.
- BASE24-billpay indicates the BASE24-billpay product only.

Remote Banking Endpoint Devices

This manual uses the term *remote banking endpoint device* to describe any device, server, or system that acquires BASE24 Remote Banking transactions using the Remote Banking Standard Interface or the Customer Service Interface. The Customer Service Interface supports Pathway-based BASE24 Remote Banking CSR terminals only. The Remote Banking Standard Interface can receive transactions from any device, server, or system that uses the Remote Banking Standard Interface message formats. Examples of some devices that use the Remote Banking Standard Interface message formats include the following: interactive voice response system (IVR), call center, screen phone, personal computer, third-party processor, kiosk, personal digital assistant, and Web server.

When this manual refers to a device that communicates with BASE24 Remote Banking using message formats that conform to a particular interface standard, the term *<interface>-compliant device* is used, where *<interface>* is an acronym for the interface standard (e.g., RBSI for the Remote Banking Standard Interface). For example, dynamic key management is only supported for RBSI-compliant devices.

BASE24 Remote Banking Transaction Processing

A BASE24 Remote Banking system provides the means by which a transaction originator can request and receive authorization for an action on a customer account from a transaction authorizer. The role of a BASE24 Remote Banking system in transaction processing includes the following:

- Routing transactions from the transaction originator to the appropriate authorizer.
- Participating in or carrying out authorization on behalf of the institutions for which it processes.
- Logging all transactions for later processing and settlement.

A BASE24 Remote Banking system can acquire transactions from a variety of sources and can involve different authorizers in its processing. Several transaction originators and authorizers can exist in a BASE24 Remote Banking system, as described below.

Transaction Originators

A BASE24 Remote Banking system can acquire transactions for authorization from the following types of originators, which are also known as acquiring endpoints.

Customer-Operated Devices

Customer-operated devices such as telephones and personal computers can be used to originate BASE24 Remote Banking transactions. Telephones use interactive voice response systems (IVRs) to communicate with the BASE24 Remote Banking system. Personal computers can be connected directly to the BASE24 Remote Banking system or can be connected through a Web server.

The BASE24 Remote Banking Standard Interface controls communications with RBSI-compliant devices, which include interactive voice response systems, personal computers, and other devices or servers that use the ISO 8583-1993 message format for communication.

Customer Service Representative-Operated Devices

Customer service representatives (CSRs) or branch employees at a financial institution or service provider can also initiate BASE24 Remote Banking transactions on behalf of customers.

When transactions are initiated from the BASE24-provided CSR screens, the BASE24 Remote Banking Customer Service Interface controls communications with the CSR-operated device. Throughout this manual, this device is referred to as a CSR terminal.

When transactions are initiated from proprietary CSR screens, the BASE24 Remote Banking Standard Interface controls communications with the CSR-operated device. Throughout this manual, this device is referred to as an RBSI-compliant device.

Hosts

Host computers can be set up to control RBSI-compliant devices or CSR terminals and forward transactions to BASE24 Remote Banking products. In this case, communications between these acquirer hosts and BASE24 Remote Banking products are controlled by BASE24 ISO Host Interface processes.

Note: Some Remote Banking transactions cannot be acquired from a host because they require large messages, and large messages are not supported by the ISO Host Interface process. Refer to the ***BASE24 ISO Host Interface Manual*** for a list of the Remote Banking transactions that can be acquired from a host.

Transaction Authorizers

The BASE24 Remote Banking system can authorize transactions or route them to a host, as described below.

BASE24 Remote Banking System

The BASE24 Remote Banking system can be set up to authorize transactions in those situations where customer files are maintained on the BASE24 Remote Banking system.

Hosts

Hosts can be set up to authorize transactions in situations where customer files are maintained on the host computer. In this case, the BASE24 Remote Banking products can screen the transactions for customer or account verification before sending them to the host. BASE24 Remote Banking can also be set up to stand in and authorize certain transactions for a host in situations where communications between the BASE24 Remote Banking products and the host system are down. The BASE24-billpay product cannot stand in for a host to authorize a schedule payment or schedule transfer transaction.

Note: Some Remote Banking transactions cannot be sent to a host for authorization because they require large messages, and large messages are not supported by the ISO Host Interface process. Refer to the ***BASE24 ISO Host Interface Manual*** for a list of the Remote Banking transactions that can be sent to a host.

Object-Oriented Design

The following processes in the BASE24 Remote Banking transaction path have been developed using what is known as object-oriented design (OOD):

- Integrated Endpoint process
- Integrated Authorization Server process
- Interface Master process

The End-of-Period process also uses an object-oriented design. However, this process is not part of the BASE24 Remote Banking transaction path.

Object-oriented design divides the processing to be done according to the data involved and the action performed on that data. Each process designed this way contains one or more objects. Only one object has access to a source of data (e.g., only the Financial Institution Access object has access to the Institution Definition File). Only one object can perform a specific action (e.g., only the Authorization Combined Positive Customer object can approve a transaction). An object appears no more than once in each process.

The key to object-oriented design is that each object requests other objects for data and requests other objects to perform actions on data. Each object needs to know only which object can provide the needed data or perform the necessary actions. One object does not need to know how the other objects obtain the data or perform the action. Refer to the message flows in section 4 for examples of how objects interact when processing BASE24 Remote Banking transactions.

ACI uses the name integrated transaction services (ITS) to identify its products that use object-oriented design and uses the name procedural to identify its products that do not use object-oriented design.

The objects used in BASE24 Remote Banking processes are divided into three groups: kernel, integrated endpoint, and core. Refer to appendix D for descriptions of the objects that make up each of these groups.

Issuers and Acquirers

Transaction originators and authorizers are generally defined by the electronic funds transfer (EFT) industry terms acquirer and issuer. An acquirer, or transaction acquirer, is the financial institution or transaction processor that originates a transaction. An issuer is an institution that owns accounts that can be used in EFT transactions.

A BASE24 system is identified as an issuer or an acquirer, depending on where a transaction originates and who is to authorize the transaction. For example, when a BASE24 system sends a transaction to a back-end host for authorization, the BASE24 system represents the acquirer in the message exchange and the back-end host represents the issuer. On the other hand, when a transaction is sent to a BASE24 system for authorization, the RBSI-compliant device or customer service representative terminal represents the acquirer and the BASE24 system represents the issuer. A single BASE24 system can function as both an issuer and an acquirer.

Hosts and DPCs

Hosts can play a prominent role in the transaction processing performed by the BASE24 system. A host is an external mainframe computer system that can function as an issuer or an acquirer. An issuer host authorizes transactions received from the BASE24 Remote Banking system and updates customer records. An acquirer host sends transactions to the BASE24 Remote Banking system for authorization.

Although the term host generally refers to an actual computer or system of computers, the term also refers to the institution or organization responsible for the host computer system and its processing. For example, in the phrase, “the host can opt to receive advice messages,” the term host actually refers to the BASE24-defined institution in control of the host computer, rather than the computer itself.

The BASE24 system also refers to hosts as data processing centers (DPCs). Generally, the terms DPC and host are synonymous. However, the term host implies either a host computer or the organization responsible for a host computer; the term DPC always characterizes the host computer or computer network itself.

A DPC can be one mainframe computer communicating with a BASE24 system, or a DPC can represent several mainframe computers networked together at a host site. Regardless of the machine configuration in relation to the BASE24 system, a DPC represents a single host from the point-of-view of the BASE24 system. A host site can establish multiple DPCs; however, the BASE24 system considers each of these DPCs to be a separate host.

DPCs are assigned identification numbers and are defined to the BASE24 system using the Host Configuration File (HCF). Each DPC must have a record in the HCF to be recognized by the BASE24 system. The HCF contains options at the DPC-level that allow hosts to specify individual processing parameters for each DPC.

Accounts

A BASE24 system classifies all accounts as one of two types: credit or noncredit. This classification determines the account-level limits and totals that apply to a particular transaction.

Credit and Noncredit Accounts

Credit accounts involve funds advanced to an accountholder, by a financial institution or retailer, based on a credit agreement with the accountholder.

Noncredit accounts involve funds an accountholder has on deposit with a financial institution (e.g., checking or savings).

A BASE24 system allows processors to set up various transaction limits and totals. Throughout the BASE24 system, the account classification determines which limits and totals apply to a particular transaction.

Account Type Codes

BASE24 products use two-character codes to identify account types stored in BASE24 files, used by BASE24 processes, and appearing in internal and external messages. The account types available vary among BASE24 products. The codes used to identify a particular account type also vary between the BASE24 Remote Banking products and the other BASE24 products (i.e., BASE24-atm, BASE24-pos, and BASE24-teller). The BASE24 Remote Banking products use ISO standard account type codes while the other BASE24 products do not. Also, the BASE24 Remote Banking products use 1- to 6-character alphanumeric codes to display account types on files and tables maintenance screens and reports, while the other BASE24 products use the same two-character account type codes everywhere.

The Int Account Type column of the following table shows the account type codes used internally and in messages by the BASE24 Remote Banking products. For each code in the Int Account Type column, this table presents the equivalent account type used on BASE24 Remote Banking file and table maintenance screens and reports (Screen Account Type), the equivalent account type code used by other BASE24 products (BASE24 Account Type), a description of the account type, and whether the account type is a credit or noncredit account.

A check (✓) in the ATT column of this table identifies values in the Int Account Type and Screen Account Type columns that are defined in the Account Type Table File (ATT) ACI provides with BASE24 Remote Banking products. If you want to use additional account types in the Positive Balance File (PBF), you must define ATT records for the corresponding values shown in the Int Account Type and Screen Account Type columns. For example, if you want to use account type 14 in the PBF to identify a particular group of savings accounts, you must define a new ATT record with an internal account type of 1A and a screen account type of SAV1.

Account Type						
Int	**Screen**	**BASE24**	**ATT**	**Description**	**Credit**	**Non credit**
10	SAV	11	✓	Savings account		✓
1A	SAV1	14		Savings account		✓
1B	SAV2	15		Savings account		✓
1C	SAV3	16		Savings account		✓
1D	SAV4	17		Savings account		✓
1E	SAV5	18		Savings account		✓
1F	SAV6	19		Savings account		✓
20	DDA	01	✓	Checking account		✓
2A	DDA1	02		Checking account		✓
2B	DDA2	03		Checking account		✓
2C	DDA3	04		Checking account		✓
2D	DDA4	05		Checking account		✓
2E	DDA5	06		Checking account		✓
2F	DDA6	07		Checking account		✓
2G	DDA7	08		Checking account		✓
2H	DDA8	09		Checking account		✓
30	CR	31	✓	Credit account	✓	

Account Type			ATT	Description	Credit	Non credit
Int	**Screen**	**BASE24**				
3A	CR1	33		Credit account	✓	
3B	CR2	34		Credit account	✓	
3C	CR3	35		Credit account	✓	
3D	CR4	36		Credit account	✓	
3E	CR5	37		Credit account	✓	
3F	CR6	38		Credit account	✓	
3G	CR7	39		Credit account	✓	
38	LOCR	32	✓	Line of credit	✓	
58	CD	13	✓	Certificate		✓
59	IRA	12	✓	Retirement account		✓
90	NOW	21	✓	Interest-bearing checking account		✓
9A	CMRCL	43	✓	Commercial loan	✓	
9B	INSTL	41	✓	Installment loan	✓	
9C	MRTGL	42	✓	Mortgage loan	✓	

Duplicate Account Numbers

The BASE24 Remote Banking products provide account types that permit a financial institution to distinguish between different accounts for its own record keeping and reporting purposes. These account types include 1A through 1F for savings accounts, 2A through 2H for checking accounts, and 3A through 3G for credit accounts. For example, an institution that wants to distinguish between different types of checking accounts can assign type 20 to the first checking account type, 2A to the second checking account type, and 2B to the third checking account type.

If the only difference between two accounts is one of the extra account types, the BASE24 Remote Banking Integrated Authorization Server process considers the account numbers to be duplicates. Each checking account must have a unique account number. In the earlier example, the BASE24 Remote Banking Integrated Authorization Server process treats account types 20, 2A, and 2B the same during processing. If an account with account type 20 and an account with account type 2B (or any other combination of accounts with checking account types) have account number 12345, the Integrated Authorization Server process uses the first account it finds that is open.

An account qualifier is a user-defined code that can be used in place of the application account number and account type to identify and access an account. When the transaction message contains an account qualifier that matches a value in a Customer/Account Relation Table (CACT) row for the customer, the Integrated Authorization Server process uses the account qualifier to identify the correct account even if duplicate account numbers exist.

BASE24 Remote Banking Transactions

The BASE24 Remote Banking products support a complete set of transactions, identifying each of these transactions by a six-character processing code. The first two characters of the processing code contain the transaction code, which indicates the type of transaction to be performed (e.g., 30 for an available funds inquiry transaction). The next two characters of the processing code indicate the type of the first account on which the transaction is to be performed (e.g., 10 for a savings account), while the last two characters indicate the second account on which the transaction is to be performed (e.g., 20 for a checking account). If an account type is not applicable for a transaction, 00 is used for the account type in the processing code.

All transactions are considered to be standard or nonstandard. Standard transactions are defined by a default set of processing codes that is provided with the standard BASE24 Remote Banking products. The default transaction codes are presented on the following pages. Refer to appendix C for a complete list of default processing codes. All other transactions are considered to be nonstandard and use a value of 00 for each of the account types.

Nonstandard Transactions

Nonstandard transactions permit the customer or ACI to define transactions in addition to those defined in the set of default processing codes. Nonstandard transactions are categorized based on who is responsible for them, as follows:

- User-defined transactions can be defined by the customer. These transactions have a transaction code of U*x,* where *x* represents a value of 0 through 9 or A through Z. Refer to the topic "Configuring User-Defined Transactions" in section 3 for examples of user-defined transactions.
- Custom development transactions can be defined as part of customer-specific development work (i.e., RPQs) done by ACI Worldwide personnel. These transactions can have a transaction code of P*x,* Q*x,* or R*x,* where *x* represents a value of 0 through 9 or A through Z.
- Distributor transactions can be defined by the ACI regional marketing groups. These transactions can have a transaction code of X*x,* Y*x,* or Z*x,* where *x* represents a value of 0 through 9 or A through Z.
- ACI-use-only transactions are reserved for use with products other than BASE24 Remote Banking. These transactions can have a transaction code of W*x,* where *x* represents a value of 0 through 9 or A through Z.

Default Transaction Codes

The following table lists the default transaction codes provided with the BASE24 Remote Banking products. These transaction codes appear in the Customer Allowed Transaction Table (CATT) and the Processing Code Definition File (PCDF). Transaction codes available for each of the three BASE24 Remote Banking product offerings are identified with a check (✓) in the second through fourth columns of the table.

Code	TB	Enhanced TB	BP	Transaction Description
30	✓			Available funds inquiry.
3A	✓			Check clearance inquiry.
3B	✓			Last *count* of debit/credit transactions inquiry.*
3C	✓			Last *count* of account by source transactions inquiry. The transaction source is identified by a code set by the host and updated in the Transaction History File (THF) by the Enscribe Refresh process.*
3D	✓			Last *count* of check transactions inquiry.*
3E	✓			Last *count* of debit transactions inquiry.*
3F	✓			Last *count* of credit transactions inquiry.*
3G	✓			Last *count* of account transfer transactions inquiry.*
3H			✓	Customer vendor list inquiry.
3J			✓	Scheduled payments list inquiry. Allows customers to inquire on scheduled payments by customer ID or application account number.
3K		✓	✓	Scheduled transfers list inquiry. Allows customers to inquire on scheduled transfers by customer ID or application account number.
3L		✓	✓	History inquiry – payments and transfers. Allows customers to inquire on payments and transfers by customer ID or application account number.

Code	TB	Enhanced TB	BP	Transaction Description
3M		✓	✓	Scheduled transactions list inquiry. Allows customers to inquire on scheduled payments and transfers by customer ID or application account number.
3N	✓			Customer account list inquiry.
3P	✓			Multiple account balance inquiry/Account list inquiry with balances.†
3Q	✓			Statement closing download.
3R	✓			Statement download.
40	✓			Transfer – immediate.
4A		✓	✓	Transfer – future (internal transaction).
4B		✓	✓	Transfer – recurring (internal transaction).
50			✓	Payment – immediate (internal transaction).
5A			✓	Payment – future (internal transaction).
5B			✓	Payment – recurring (internal transaction).
90	✓			PIN change.
91	✓			PIN verify.
9A			✓	Schedule payment – immediate.
9B			✓	Schedule payment – future.
9C			✓	Schedule payment – recurring.
9D		✓	✓	Schedule transfer – future.
9E		✓	✓	Schedule transfer – recurring.
9F			✓	Scheduled payment delete.
9G		✓	✓	Scheduled transfer delete.

Code	TB	Enhanced TB	BP	Transaction Description
9H			✓	Scheduled payment update. Allows customers to change the date, amount, period type, number of periods, or to skip the next payment.
9J		✓	✓	Scheduled transfer update. Allows customers to change the date, amount, period type, number of periods, or to skip the next transfer.
9K		✓	✓	Customer add.
9L		✓	✓	Customer inquiry.
9M		✓	✓	Customer update. Allows customers to change the following: family name (last name), given name (first name), middle initial, title, street or mailing address, city, state, country, postal code, phone numbers, birth date, language, alternate contact, mother’s maiden name, government ID, customer information, customer type, customer profile, branch, default account number and type, and service fee account number and type.
9N			✓	Customer vendor add.
9P			✓	Customer vendor deactivate.

Code	TB	Enhanced TB	BP	Transaction Description
9Q			✓	Customer vendor update. Allows customers to change the following information for a customer vendor, regardless of the customer's verify status: account with vendor, budget category, customer vendor short name, and user information. This transaction also allows customers to update the actual vendor information in the Vendor Table (VNDR) and Vendor Branch Table (VNDB). The following information in these tables can be updated by any customer for most vendors. The only exceptions are public vendors with a vendor verify status of verified. Customers cannot update any information for these vendors because they are used by many customers. Therefore, the vendor and vendor branch information for these vendors is in effect *owned* by the system owner, not an individual customer. Vendor information that can be updated includes the following: vendor name, vendor branch name, address, city, state, postal code, phone number, account with vendor, customer vendor short name, budget category, customer vendor user information, vendor branch information, vendor branch remittance information, vendor information, and vendor remittance information.
9R			✓	Master vendor list inquiry.
9S			✓	Master vendor add (internal transaction).
9T			✓	Customer vendor request mailing. Initiates a request for a hard copy listing of vendors that can be sent to a customer.
9V		✓	✓	Loan application.
9W		✓	✓	Miscellaneous request.
9X		✓	✓	History inquiry – all transactions. Allows customers to inquire on all BASE24-billpay transactions by customer ID or application account number.
A0	✓			Check stop payment add.
B0		✓	✓	Customer account list inquiry (internal transaction).
B1		✓	✓	Bank list inquiry.

Code	TB	Enhanced TB	BP	Transaction Description
B2		✓	✓	Bank branch list inquiry.
B3			✓	Cancel immediate payment (internal transaction). BASE24-billpay processes this transaction as a reversal.
B4		✓	✓	Miscellaneous requests inquiry.
B5		✓	✓	Customer account add.
B6		✓	✓	Customer account activate.
B7		✓	✓	Customer account deactivate.
B8		✓	✓	Customer account delete.
B9		✓	✓	Customer account update. Allows customers to change the following account information, as long as the customer verify status is not closed: account qualifier, debit allowed flag, credit allowed flag, inquiry allowed flag, and account description.
BB		✓	✓	Customer activate.
BC		✓	✓	Customer deactivate.
BD			✓	Customer vendor activate.
BE			✓	Customer vendor mask verify inquiry.
BF	✓			Customer verify.
BG	✓			Customer select.

[*] The term *count* refers to the number of transactions, customer vendors, accounts, or account balances that can be included in each message sent to the RBSI-compliant device or customer service representative terminal.

[†] The multiple account balance inquiry transaction is available from the Remote Banking Standard Interface only. From the Customer Service Interface, this transaction is referred to as the account list inquiry with balances transaction because the transaction response is a combination of the account list inquiry (3N) transaction and the multiple account balance inquiry (3P) transaction.

Values in the following fields on Host Configuration File (HCF) screen 23 control the number of transactions, customer vendors, or accounts that can be returned in a single inquiry response message to a host.

- CUSTOMER VENDOR MAX COUNT*
- HIST INQ PMNT/XFER MAX COUNT
- LAST TRANSACTION MAX COUNT
- SCHEDULED TRANSFER MAX COUNT
- SCHEDULED PAYMENT MAX COUNT*
- ACCOUNT LIST MAX COUNT

* These fields are not currently used.

Values in the following fields on VRU Configuration Data (VCD) screen 18 control the number of transactions, customer vendors, or accounts that can be returned in a single inquiry response message to an RBSI-compliant device or a customer service representative (CSR) terminal.

- CUSTOMER VENDOR INQ
- HIST INQ PMNT/XFER
- LAST TRANSACTION
- MULTIPLE ACCT BAL*
- SCHEDULED TRANSFERS
- SCHEDULED PAYMENTS
- ACCOUNT LIST

* A maximum count field does not appear on HCF screen 23 for the multiple account balance inquiry transaction, because the number of accounts is controlled in the transaction structure itself. A maximum count is included on VCD screen 18 to ensure that the count in the transaction does not exceed the maximum number of accounts (13) that can be returned in a transaction authorized on a host.

The following fields on VCD screen 18 control the number of items that can be returned in a single inquiry response an RBSI-compliant device or a CSR terminal. Since these inquiry transactions cannot be acquired from a host, there are no corresponding maximum count fields on HCF screen 23.

- SCHED TRANSACTIONS
- HIST INQ ALL TRANS
- MASTER VENDOR INQ
- BANK LIST INQ
- BANK BRANCH LIST INQ
- MISCELLANEOUS REQ
- CUSTOMER VENDOR MASK
- STATEMENT DOWNLOAD
- STMT CLOS DOWNLOAD

The value in the MAX HISTORY RECORDS field on Customer Table (CSTT) screen 1 controls the total number of Transaction History File (THF) records that can be returned in a *last count of* transaction history inquiry transaction to an RBSI-compliant device for this customer. No limit is imposed on the total number of items that can be returned for any of the other inquiry transactions listed above or for *last count of* transaction history inquiry transactions entered from a customer service representative (CSR) terminal.

Financial Transactions

Transactions are classified as financial or nonfinancial. This classification impacts whether a BASE24-billpay transaction is authorized by the BASE24-telebanking Integrated Authorization Server process. Only transaction codes 40, 4A, 4B, 50, 5A, and 5B are financial transactions.

BASE24 Core Files and Tables

BASE24 core files and tables are used by one or more BASE24 Remote Banking products.

The BASE24-telebanking product uses all of the core files and tables for configuration and transaction processing, with or without the BASE24-billpay product installed.

The BASE24-billpay product uses all but three core files and tables—File Control File (FCF), Transaction History Configuration File (THCF), and Transaction History File (THF). However, the majority of these files and tables actually are used by BASE24-telebanking processes to perform BASE24-billpay transactions. BASE24-billpay processes themselves use only the following core files and tables:

- Account Type Table File (ATT)
- Customer/Account Relation Table (CACT)
- Customer Allowed Transaction Table (CATT)
- Customer/Personal ID Relation Table (CPIT)
- Customer Table (CSTT)
- Personal Information Table (PIT)
- Processing Code Definition File (PCDF)
- Transaction Type by Source Code Table (TTSC)

Each file or table has its own maintenance screens unless noted otherwise. Refer to the ***BASE24 Core Files and Tables Maintenance Manual*** for file and table maintenance screen illustrations and field descriptions.

The following table describes each of the core files and tables used by BASE24 Remote Banking products.

Account Type Table File (ATT)	This file contains a record for each valid account type in the BASE24 integrated transaction services (ITS) network. Each ATT record contains an ISO code and a corresponding description. Screens for BASE24 integrated server (IS) core files and tables, and BASE24 Remote Banking reports use the account type names defined in the ATT instead of account type codes.

Address Table File (ADRT)	This file contains geographic address information associated with the RBSI-compliant devices and customer service representative (CSR) terminals. This information is displayed on VRU Configuration Data (VCD) screens.
Customer/Account Relation Table (CACT)	This table contains one row for each account owned by a BASE24 Remote Banking customer. Each row includes information other than financial amounts for one account. This table is used with the Positive Customer with Accounts Authorization method (PCAA) or the Positive Customer with Balances/History Authorization method (PCBA). This information is displayed on CSTT screen 2.
Customer Allowed Transaction Table (CATT)	This table contains one row for each supported combination of FIID, customer profile, and processing code. CATT rows can be established for individual customers or groups of customers. The CATT is used to configure transactions allowed at the customer level. The Processing Code Definition File (PCDF) is used to configure transactions allowed at the issuer institution level. A transaction must be defined in the CATT and PCDF before it can be performed by a customer.
Customer/Personal ID Relation Table (CPIT)	This table is used to link the Personal Information Table (PIT) and the Customer Table (CSTT). With the CPIT, a single CSTT entry can be associated with multiple PIT entries and multiple CSTT entries can be associated with a single PIT entry. Use of this table is optional for all BASE24 Remote Banking product options.
Customer Table (CSTT)	This table contains one row for each customer allowed to perform BASE24 Remote Banking services. This row contains authorization parameters and usage accumulation information for the specified customer.
Exception Log File (ELF)	This file contains one record for each exception condition that prevents the internal transaction data (ITD) or external message from being logged to the ITS Transaction Log File (ITLF). The ELF cannot be accessed from a file maintenance screen.

File Control File (FCF)	This file contains version indicators that are used to identify the current version of each file or table that is audited by the Transaction Monitoring Facility (TMF) and is refreshed. For example, the Institution Routing Configuration File (IRCF) is audited by the TMF but is not refreshed, so it does not have an FCF record. The version indicator is used as part of the file name. Each FCF record contains a version indicator to identify the current file or table because audited files and tables cannot be renamed. The version indicator is appended to the file name specified in an LCONF assign. This file contains version records for the Transaction History File (THF).
Institution Periodic Reporting File (IPRF)	This file contains the historical information necessary for generating monthly institution transaction activity reports. The IPRF cannot be accessed from a file maintenance screen.
Institution Routing Configuration File (IRCF)	This file contains information used by institutions to define the routing and authorization parameters for their transactions. The IRCF can contain one record for each possible combination of product ID, route profile, route code, and account type in the BASE24 Remote Banking system. Using wildcard values reduces the number of IRCF records needed.
Interface Security File (ISF)	This file contains transaction security parameters for RBSI-compliant devices and customer service representative (CSR) terminals. The ISF contains one record for each record in the ITS Interface Configuration File (IFCF). The IFCF contains one record for all CSR terminals and one record for each group of up to 90 stations assigned to the same RBSI-compliant device. Having one ISF record for each IFCF record enables an institution to configure security parameters for the CSR terminals and each group of RBSI-compliant device stations. **Note:** The maximum number of CSR terminals is controlled by the Pathway configuration.

Interface Station Table File (IST)	This file associates the stations defined for remote banking endpoint devices with records in the ITS Interface Configuration File (IFCF). This information is displayed on VRU Configuration Data (VCD) screens.
Internal Message Translation File (IMTF)	This file contains information used by the Internal Message Translation Interface object to map authorization destinations to symbolic process names and DPC numbers. The IMTF is used when a BASE24 ISO Host Interface process is used to communicate with an institution's data processing center (DPC) or when routing transactions to an XPNET Billpay Server process. The IMTF is also used to validate a station or process when a message is received from certain non-object-oriented BASE24-billpay sources.
ITS Interface Configuration File (IFCF)	This file contains processing parameters for remote banking standard interfaces. The IFCF also contains the dynamic key management configuration information used when communicating with RBSI-compliant devices. The IFCF information is displayed on VRU Configuration Data (VCD) screens.
ITS Transaction Log File (ITLF)	This file contains a record for each BASE24 Remote Banking transaction posted by a BASE24-telebanking Integrated Authorization Server process during a given business day. The ITLF can also contain transactions normally posted only to the History Table (HIST).
Log Configuration File (LCF)	This file controls which ITS Transaction Log File (ITLF) an object uses. The LCF contains one record for each group of ITS objects that logs information to the ITLF. The LCF has limited use at the present time because all ITS objects must use a common logging destination. As a result, the LCF must contain only one record that applies to all log groups.
Personal Information Table (PIT)	This table contains information such as customer name, address, date of birth, and social security number. This table is optional and is used for customer verification with transactions performed by customer service representatives. It is also used for identifying customers on BASE24-billpay reports.

Processing Code Definition File (PCDF)	This file contains one record for each combination of product ID, route profile, and processing code that is supported by an institution. Subsets of the transactions defined in the PCDF can be established at the customer level in the Customer Allowed Transaction Table (CATT). A transaction must be defined in the CATT and PCDF before it can be performed by a customer.
Transaction History Configuration File (THCF)	This file contains information used with the Positive Customer with Balances/History Authorization method (PCBA). With this authorization method, account history information can be obtained from the Transaction History File (THF). The BASE24-telebanking product permits a processor to maintain multiple THFs instead of having to use a single file. The refresh group, FIID, and account number in the THCF determine which THF contains a particular transaction.
Transaction History File (THF)	This file contains account history information that can be obtained using check clearance and transaction history inquiry transactions. The THF must be updated using the BASE24 Enscribe Refresh process and is not accessible through file maintenance.
Transaction Type by Source Code Table (TTSC)	This table identifies transactions allowed from a specific source. Currently, only transactions performed at customer service representative (CSR) terminals can be configured in this table. This table can be used to specify the order in which transactions are displayed on CSR menus.

BASE24-billpay Files and Tables

The BASE24-billpay product uses several files and tables unique to the BASE24-billpay product for configuration and transaction processing. These files and tables are in addition to the BASE24 core files and tables listed earlier.

Each file or table has its own maintenance screens unless noted otherwise. Some files and tables are used for internal processing and cannot be accessed from a screen. Refer to the ***BASE24-billpay Tables Maintenance Manual*** for BASE24-billpay table maintenance screen illustrations and field descriptions.

Several tables can also be accessed from the customer service representative (CSR) terminal. Refer to the ***BASE24 Remote Banking Customer Service Support Manual*** for screens, field descriptions, and operator instructions associated with the CSR terminal.

Refer to the ***BASE24-billpay Billing Application Manual*** for screens, field descriptions, and operator instructions for tables associated with the BASE24-billpay Billing Application.

The following table describes each of the files and tables used exclusively by the BASE24-billpay product.

Bank Table (BANK)	This table contains institution level parameters and information. The screen used to maintain this table is accessed from Institution Definition File (IDF) screens 40 through 42. Refer to the ***BASE24 Base Files Maintenance Manual*** for the screen illustration and field descriptions.
Bank Advice Control Table (BKAC)	This table contains processing information used by the Bank Advice program, which permits additional BASE24-billpay transactions to be logged to the ITS Transaction Log File (ITLF). The BKAC cannot be accessed from a table maintenance screen.
Bank Billing Summary Table (BBLS)	This table contains the total number of transactions and fee amount for each FIID and statement cycle combination. The BBLS is used for the BASE24-billpay Billing Application and cannot be accessed from a table maintenance screen.

Bank Branch Table (BKBR)	This table contains information pertinent to a specific financial institution branch such as address, phone, and contact information.
Bank Notice Table (BKNT)	This table is used to identify error conditions that require customer notices to be generated.
Bank Transaction Advice Table (BKAV)	This table contains information used by the Bank Advice process to identify which BASE24-billpay transactions should be logged to the ITS Transaction Log File (ITLF).
Bank Transaction Summary Table (BKTS)	This table contains daily summary totals required for the BASE24-billpay Billing Application. The BKTS cannot be accessed from a table maintenance screen.
Billing Group Table (BLG)	This table contains general information associated with each billing group, FIID, and statement cycle for the BASE24-billpay Billing Application.
Billing Rate Table (BLRT)	This table contains rate information associated with each billing type, FIID, and effective date range for the BASE24-billpay Billing Application.
Billing Type Table (BLTY)	This table contains general information associated with each billing type and FIID for the BASE24-billpay Billing Application.
Context Table (CNTX)	This table contains information used by the Billpay Server process during inquiry transactions involving multiple requests. The CNTX cannot be accessed from a table maintenance screen.
Cleanup Configuration Table (CLPT)	This table contains configuration information used by the Archive and Cleanup programs.
Customer Billing Summary Table (CBLS)	This table contains the total number of transactions and fee amount for each customer ID, billing group, and statement cycle combination. The CBLS is used for the BASE24-billpay Billing Application and cannot be accessed from a table maintenance screen.

Customer Billpay Table (CBPT)	This table contains information used for online processing of transactions, including the next available reference number and customer vendor number for each customer. Other information in the CBPT is provided for the BASE24-billpay Billing Application.
Customer Vendor Table (CVND)	This table identifies vendors to which a given customer is allowed to make payments.
Customer Vendor Summary Table (CVS)	This table contains the number and total amount of payments made to each customer vendor during the current year and each of as many prior years as the service provider permits. The Remittance Generator process maintains the CVS, which cannot be accessed from a table maintenance screen.
Future Table (FUTR)	This table contains detailed information for each future dated transaction. The Billpay Server process maintains transaction information in the FUTR and the Scheduled Transaction Initiator process generates online transactions from information in the FUTR. The FUTR is maintained using transactions acquired from an RBSI-compliant device or CSR terminal screens. It cannot be accessed from a table maintenance screen.
History Table (HIST)	This table contains one row for each BASE24-billpay financial transaction that has been successfully processed, as well as one row for each failed transaction and log-only nonfinancial transaction.
Loan Application Table (LOAN)	This table contains loan application information entered by a CSR. The LOAN cannot be accessed from a table maintenance screen.
Miscellaneous Request Table (MISC)	This table contains miscellaneous request messages to the financial institution entered by a CSR. The MISC cannot be accessed from a table maintenance screen.
Payment Summary Table (PST)	This table contains the number and total amount of payments made to each master vendor during the current year and each of as many prior years as the service provider permits. The Remittance Generator process maintains the PST, which cannot be accessed from a table maintenance screen.

Rate Table (RATE)	This table identifies the rate group associated with each rate name, transaction source, transaction type, and response code combination for the BASE24-billpay Billing Application.
Rate Group Table (RATG)	This table contains the actual transaction costs associated with a rate group and end range for the BASE24-billpay Billing Application.
Recurring Table (RCUR)	This table contains detailed information for each recurring transaction. The Billpay Server process updates transaction information in the RCUR and the Scheduled Transaction Initiator process generates online transactions from information in the RCUR. The RCUR is maintained using transactions acquired from an RBSI-compliant device or CSR terminal screens. It cannot be accessed from a table maintenance screen.
Remit File	This file contains detailed information for vendor remittance processing, with each record containing information for one remittance. Data in this file is generated by the Remittance Generator process. This file cannot be accessed from a file maintenance screen.
Report Action Table (RPTA)	This table contains information required to generate the Customer Notice Report, New Customer Report, and Customer Vendor List Report. The RPTA cannot be accessed from a table maintenance screen.
Response Text Table (RTXT)	This table contains one row for each action code and reversal reason pair for which either internally displayed or externally displayed text is required.
Retry Table (RTRY)	This table is used by the Scheduled Transaction Initiator process to determine whether transactions in the Future Table that are declined with specific action codes can be retried.
Service Provider Table (SVPT)	This table contains the last customer ID assigned by the service provider. The Billpay Server process uses this table to automatically generate customer IDs for financial institutions that do not use preloaded customer data.

Vendor Table (VNDR)	This table is the master vendor list for the logical network, providing status, control, and customer account verification information for each vendor in the system.
Vendor Branch Table (VNDB)	This table contains information pertinent to a specific vendor branch such as address, phone, contact information, and parameters for verifying the customer account numbers.
Vendor Budget Category Table (VBUD)	This table permits BASE24-billpay system owners to associate vendors with a general category for reporting purposes.
Vendor Mask Table (VNDM)	This table contains edit masks for verifying the account numbers that customers have with vendors.

Base Files Used by BASE24 Remote Banking Products

The BASE24-telebanking product uses the following base files. Other BASE24 products also use these files.

The BASE24-billpay product also uses these base files. However, the majority of these files actually are being used by BASE24-telebanking processes to perform BASE24-billpay transactions. BASE24-billpay processes themselves use only the Logical Network Configuration File (LCONF).

Each file has its own maintenance screens unless noted otherwise. Refer to the ***BASE24 Base Files Maintenance Manual*** for screen illustrations and field descriptions.

External Message File (EMF)	This file specifies which data elements are to be included in the BASE24 external message for incoming and outgoing messages. In the BASE24 Remote Banking products, the BASE24 ISO Host Interface process and the Voice Response Unit (VRU) Interface object use the EMF.
Extract Configuration File (ECF)	This file defines processing parameters for each type of extract an institution might perform in a particular logical network. The ITS Transaction Log File (ITLF) can be extracted.
Host Configuration File (HCF)	This file defines host communications links for transaction message traffic routed to or received from a host authorization system. The BASE24 Remote Banking products share a set of processing parameters and timer limits in the HCF.
Institution Definition File (IDF)	This file contains processing parameters for each institution. BASE24 Remote Banking products have parameters in the IDF for PIN verification, transaction processing, usage accumulation, billing, and reporting.
Key Authorization File (KEYA)	This file contains transaction security information for all BASE24 Authorization processes and objects.

Key File (KEYF)	This file contains transaction security information for the BASE24 Host Interface and Interchange Interface processes. The BASE24 ISO Host Interface process uses this file for BASE24 Remote Banking transactions.
Logical Network Configuration File (LCONF)	This file contains the assigns and params defined for the logical network. The LCONF is accessed using the LCONF Entry program instead of file maintenance screens. Refer to the ***BASE24 Logical Network Configuration File Manual*** for operating instructions.
Positive Balance File (PBF)	This file contains balance information for customer accounts. The PBF contains one record for each account belonging to a BASE24 Remote Banking customer whose account issuer uses the Positive Customer with Balances/History Authorization method (PCBA). BASE24 Remote Banking products also maintain transfer and payment limits and usages in the PBF.
Token File (TKN)	This file specifies which tokens the BASE24 Remote Banking products log to the ITS Transaction Log File (ITLF). The BASE24 Remote Banking products do not generate token information and log only the token information received from the host when processing transactions.

Kernel and Integrated Endpoint Files Used by BASE24 Remote Banking Products

The BASE24-telebanking product uses the following kernel files to configure object-oriented processes. These files are also used by BASE24-telebanking processes to perform BASE24-billpay transactions. BASE24-billpay processes themselves do not use any kernel files. Refer to the ***BASE24 ITS Kernel Files Maintenance Manual*** for screen illustrations, field descriptions, and additional information on the function of each file.

- The Assign File (ASGN) identifies the LCONF assigns that should be listed on the Assign Warmboot screen. For each assign, the ASGN defines the name used on the Assign Warmboot screen, in the Assign/Object Relationship File (AORF), and in the Assign/Process Type Relationship File (APRF).
- The Assign/Object Relationship File (AORF) defines relationships between assign names defined in the ASGN and objects for the Assign Warmboot command.
- The Assign/Process Type Relationship File (APRF) defines relationships between assign names defined in the ASGN and process types for the Assign Warmboot command.
- The EMT/Object Relationship File (EORF) defines relationships between extended memory tables (EMTs) and objects for the EMT Warmboot command.
- The EMT/Process Type Relationship File (EPRF) defines relationships between extended memory tables and process types for the EMT Warmboot command.
- The Event Suppression File (ESF) defines events that are to be suppressed at either the process or network level.
- The Extended Memory Table File (EMT) identifies all extended memory tables used by the BASE24-telebanking product that can be warmbooted.
- The Log Permission Process Partition File (LPF) identifies ranges of subsystem IDs (SSIDs) to be mapped to specific Log Permission processes. The Log Permission processes determine whether an event is to be logged at the network level.
- The Object File (OBJ) defines all objects used in BASE24-telebanking processes.

- The Process File (PRO) defines all BASE24-telebanking processes that contain objects, as well as any procedural processes that can receive process control commands in the ITS environment.
- The Process/Object Relationship File (PORF) defines relationships between processes and objects.
- The Process Type File (PTF) defines all types of BASE24-telebanking processes that use objects, as well as any procedural processes that can receive process control commands in the ITS environment.
- The Server Class Type/Process Relationship File (SPRF) defines relationships between server class types and processes, as well as any procedural processes that can receive process control commands in the ITS environment.
- The Server Class Type File (SCT) defines all server class types. Each server class type uses a different Integrated Server Control File (ISCF) for process control commands.
- The Source Class Map File (SCM) defines relationships between source class values defined for stations and processes in the XPNET system and objects.

The BASE24-telebanking product uses one integrated endpoint file, as well as the kernel files mentioned earlier, to configure object-oriented processes. The Destination File (DEST) defines the destinations for a message based on the source class and message class associated with the message. Refer to the ***BASE24 IEP Configuration Manual*** for the screen illustration, field descriptions, and additional information on the function of the DEST.

Extended Memory Tables

When processing transactions, the BASE24-telebanking product uses tables in extended memory instead of reading many of the disk files. Information is obtained from extended memory more quickly than from disk files. Most of the BASE24-telebanking extended memory tables must be built by the Extended Memory Table Build utility before they can be used by a BASE24-telebanking process. Refer to the ***BASE24 ITS Kernel Configuration Manual*** for information about operating the Extended Memory Table Build utility and making objects start using a revised extended memory table.

The BASE24-telebanking product uses the extended memory tables described below.

Context extended memory table	This table contains context information for current transactions and the objects involved in processing the transactions. The Context Manager object builds this table during initialization and updates it during transaction processing.
Customer extended memory table	This table contains the same information as the Customer Allowed Transaction Table (CATT). The Extended Memory Table Build utility builds this table. The Customer Access object uses this table during transaction processing.
Destination File extended memory table	This table contains the same information as the Destination File (DEST) plus the associated object ID from the Object File (OBJ) for each object name in the DEST. The Extended Memory Table Build utility builds this table. The Destination object uses this table during transaction processing.
Event suppression extended memory table	This table contains the same information as the Event Suppression File (ESF). The Extended Memory Table Build utility builds this table. The Event object uses this table during transaction processing.
Institution Definition File extended memory table	This table contains the same information as the Institution Definition File (IDF). The Extended Memory Table Build utility builds this table. The Financial Institution Access object uses the base, credit line, and BASE24-telebanking segments of the IDF from this table during transaction processing.

Institution routing extended memory table	This table contains the same information as the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF). The Extended Memory Table Build utility builds this table. The Router object uses this table during transaction processing.
Interface extended memory table	This table contains information from the Address Table File (ADRT), External Message File (EMF), Interface Station Table File (IST), and ITS Interface Configuration File (IFCF). The Extended Memory Table Build utility builds this table. The following objects use this table during transaction processing: • Customer Service Interface object • Interface Manager object • Interface Master Services object • Transaction Security object • Voice Response Unit Endpoint Class Handler object • Voice Response Unit Interface object • Voice Response Unit Master object
Interface Security File extended memory table	This table contains the same information as the Interface Security File (ISF). The Transaction Security object builds this table during initialization or warmboot processing. The Transaction Security object uses this table during transaction processing.
Internal Message Translation File extended memory table	This table contains the same information as the Internal Message Translation File (IMTF). The Extended Memory Table Build utility builds this table. The Internal Message Translation Interface object uses this table during transaction processing.
Key Authorization File extended memory table	This table contains the same information as the Key Authorization File (KEYA). The Extended Memory Table Build utility builds this table. The Authorization Security Access object uses this table during transaction processing.

Positive Balance File extended memory table	This table contains the information for one, two, or three Positive Balance File (PBF) records. These are the *from* account, *to* account, and backup account. The Account Access object builds this table at initialization and uses it for each transaction that involves PBF accounts.
Source Class Map File extended memory table	This table contains the same information as the Source Class Map File (SCM). The Extended Memory Table Build utility builds this table. The Class object uses this table during transaction processing.
Source symbolic map extended memory table	This table contains configuration information from the XPNET system. The Extended Memory Table Build utility builds this table. The XPNET Interface object uses this table during transaction processing.
Transaction history file extended memory table	This table contains the information from the Transaction History Configuration File (THCF). The Extended Memory Table Build utility builds this table. The Transaction History object uses this table during transaction processing.
Transaction log file extended memory table	This table contains the information from the Log Configuration File (LCF) that pertains to each acquirer object that is configured in the system. The Extended Memory Table Build utility builds this table. The Transaction object and End-of-Period Process object use this table during transaction processing.

BASE24 Remote Banking Message Types

All transactions are performed using messages exchanged between the originator and authorizer of the transaction. These messages are assigned numbers to identify their message type and function.

Message Functions

Messages can be grouped into the following functional categories.

Request	A request from a transaction originator to a transaction authorizer to authorize a transaction. The transaction may or may not have a financial impact.
Response	A response from a transaction authorizer to a transaction originator approving or declining a transaction request. The transaction request may or may not have a financial impact.
Advice	A message from a transaction originator to a transaction authorizer so the authorizer can update its files. The advice message contains information about a completed transaction of which the authorizer may not be aware. The authorizer must accept the transaction. The BASE24-telebanking product accepts advices from hosts for transactions authorized by the host and sends advices to hosts for transactions authorized by the Integrated Authorization Server process. The Integrated Authorization Server process also accepts advices from the Bank Advice program for logging to the ITLF. The BASE24 Remote Banking products do not accept advices from devices.
Reversal	A message from the transaction originator to the transaction authorizer indicating that any database impacts from an earlier transaction should be reversed because the transaction either could not be completed or was completed incorrectly. The transaction originator can be a remote banking endpoint device or BASE24-billpay. BASE24-billpay is considered the originator for scheduled financial transactions.

Message Types

Message types are used inside and outside of the BASE24 Remote Banking system. The following table identifies the message types used inside the BASE24 Remote Banking system, along with the corresponding message types as they would be sent to, or received from, a host. The internal message types are defined in the table column identified by Int. External message types exchanged between the BASE24 ISO Host Interface process and a host are defined in the table in the column identified by Host.

The external messages exchanged between the Voice Response Unit Interface object and an RBSI-compliant device are ISO messages. However, these message types match the ones shown in the Int column.

Int	Host	Description
1100*	0100	**Authorization Transaction Request —** A request to authorize a nonfinancial transaction.
1101	0100	**Authorization Transaction Request Repeat —** A reissue of the request to authorize a nonfinancial transaction.
1110*	0110	**Authorization Transaction Request Response —** An approval or denial response to a nonfinancial authorization request.
1120†	0120	**Authorization Transaction Advice —** A notification that a nonfinancial transaction has been authorized (approved or declined).
Not used	0121	**Authorization Transaction Advice Repeat —** A reissue of the notification that a nonfinancial transaction has been authorized.
1130	0130	**Authorization Transaction Advice Response —** An acknowledgment of the receipt generated in response to an authorization transaction advice or authorization transaction advice repeat.
1200*	0200	**Financial Transaction Request —** A request to authorize a financial transaction.
1201	0200	**Financial Transaction Request Repeat —** A reissue of the request to authorize a financial transaction.

Int	Host	Description
1210*	0210	**Financial Transaction Request Response —** An approval or denial response to a financial authorization request or financial authorization request repeat.
1220‡	0220	**Financial Transaction Advice —** A notification that a financial transaction has been authorized (approved or declined).
Not used	0221	**Financial Transaction Advice Repeat —** A reissue of the notification that a financial transaction has been authorized.
1230	0230	**Financial Transaction Advice Response —** An acknowledgment of the receipt generated in response to a financial transaction advice or financial transaction advice repeat.
1420*	0420	**Reversal —** A message indicating that a financial or nonfinancial transaction was not completed as authorized.
1421	0421	**Reversal Repeat —** A reissue of the reversal message indicating that a financial transaction was not completed as authorized.
1430	0430	**Reversal Response —** An acknowledgment of the receipt generated in response to a reversal or reversal repeat.

* This message type is used in data communications between the Billpay Server process and Integrated Authorization Server process and between the Scheduled Transaction Initiator process and Integrated Authorization Server process. It is also valid in data communications between a remote banking endpoint device and the Integrated Authorization Server process.

† This message type is used in data communications between the Bank Advice process and the Integrated Authorization Server process; however, it is not valid in data communications between a remote banking endpoint device and the Integrated Authorization Server process.

‡ This message type is not valid in data communications between a remote banking endpoint device and the Integrated Authorization Server process.

Large Message Support

Several BASE24-billpay transactions can use large message support in the Integrated Endpoint process and Integrated Authorization Server process when acquired from an RBSI-compliant device. Some transactions always require large messages, while others can use small or large messages. For the latter group of messages, large messages allow BASE24 Remote Banking to send and receive more data in each message exchanged with an RBSI-compliant device than small messages, thereby improving message efficiency and response time.

Note: Large messages currently are not supported either to or from BASE24 and a host.

The following BASE24-billpay transactions always require large messages:

Transaction Code	Transaction Message Description
3H	Customer vendor list inquiry
3J	Scheduled payments list inquiry
9K	Customer add
9L	Customer inquiry
9M	Customer update
9N	Customer vendor add
9P	Customer vendor deactivate
9Q	Customer vendor update
9T	Customer vendor request mailing
9V	Loan application
9W	Miscellaneous request
B1	Bank list inquiry
B2	Bank branch list inquiry
B4	Miscellaneous requests inquiry
B5	Customer account add

Transaction Code	Transaction Message Description
B6	Customer account activate
B7	Customer account deactivate
B8	Customer account delete
B9	Customer account update
BB	Customer activate
BC	Customer deactivate
BD	Customer vendor activate

The following BASE24-billpay transactions can use small or large messages:

Transaction Code	Transaction Message Description
3K	Scheduled transfers list inquiry response
3L	History inquiry – payments and transfers
3M	Scheduled transactions list inquiry
9R	Master vendor list inquiry
9X	History inquiry – all transactions
B1	Bank list inquiry
B2	Bank branch list inquiry
B4	Miscellaneous requests inquiry
BE	Customer vendor mask verify inquiry

A large-memory model toggle that is set when the Application Network Services object is compiled in integrated server processes, determines whether small messages or large messages are used in your BASE24-telebanking software. If the toggle is set, large messages are used. If the toggle is not set, small messages are used. For more information on this toggle and its impact on message size, refer to the ***BASE24 ITS Kernel Configuration Manual***.

Note: The above option is available for BASE24-telebanking software only. BASE24-billpay uses the same size messages in responses as it receives in requests from BASE24-telebanking.

NET24-XPNET release 2.1 or greater supports the extended features of the Tandem D-series operating system that enable a process to use a large-memory model as well as a small-memory model. Integrated server processes running under control of the XPNET process can use either memory model. Using the small-memory model, objects can allocate space in the upper 64K bytes (32K words) of the data stack and use message buffers of 4096 bytes. Using the large-memory model, objects can allocate space within a flat segment. Flat segments allow memory space to be larger than 65,536 bytes, use message buffers of 32,000 bytes, and require a D30 or later operating system running on a Tandem NonStop Series/RISC machine.

Note: The large-memory model is not available if your ITS environment runs under NET24-XPNET release 2.0 or if your Tandem system is not RISC-based.

Event Messages

BASE24 Remote Banking products generate event messages to assist network operators in performing their daily tasks. Event messages provide information regarding the internal condition of the system. They provide network operators with a primary link to what is actually occurring in the system. Information ranges from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

BASE24 event messages include a product module ID (PMID) and a subsystem ID (SSID) to identify the source of the message, and an event number to identify each event.

Integrated transaction server processes can use SSIDs and event numbers to suppress event messages for the BASE24-telebanking product. Refer to the ***BASE24 ITS Kernel Configuration Manual*** for additional information about event suppression.

Event Management Service (EMS) can use these IDs to filter event messages for both BASE24 Remote Banking products. Refer to the ***NET24-XPNET Event Management Manual*** for information about the XPNET implementation of EMS.

BASE24-telebanking Event Messages

The following table contains the modules and objects that can be used with the BASE24-telebanking product. The PMID and SSID are shown for each module and object.

In addition to identifying the message source, the BASE24-telebanking product also uses the PMID to identify event message documentation files provided on the OK13LOGM subvolume. Each file contains event messages for a specific BASE24 module or object. Refer to the ***BASE24 Log Message Reference Manual*** for a complete explanation of log message documentation.

Product Module or Object Name	PMID	SSID
Account object	ISCOACCT	731
Account Access object	ISCOACAC	774
Application Network Services object	ISKNANS	754

Product Module or Object Name	PMID	SSID
Authorization Combined Positive Customer object	ISCOACPC	743
Authorization Combined Utility object	ISCOACU	734
Authorization Security Access object	ISCOASAC	775
Class object	ISKNCLAS	755
Context Manager object	ISIECTXM	756
Customer object	ISCOCUST	736
Customer Access object	ISCOCSAC	805
Customer Service Interface object	ISCOCSI	735
Destination object	ISIEDEST	757
Endpoint Class Handler Services object	ISIEEPCS	758
End-of-Period Process object	ISCOEOPP	777
Extended Memory Table Build utility	ISKNEMTB	749
Event object	ISKNEVT	759
Financial Institution object	ISCOFI	748
Financial Institution Access object	ISCOFIAC	776
Transaction History object	ISCOHIST	741
Interface Manager object	ISCOIMGR	746
Interface Master Services object	ISCOIMS	750
Internal Message Translation Interface object	ISCOIMTT	751
Internal Message Translation Endpoint Class Handler object	ISIEIMTE	752
Issuer Reporting program	ISTBRPT1	773
Log Permission object	ISKNLOGP	772

Product Module or Object Name	PMID	SSID
Message ISO object	ISIEMISO	760
Multiple Currency object	ISCOMC	745
Process Control object	ISKNPCTL	761
Router object	ISCORTE	744
SQL Refresh	ISCOSQLR	804
Trace object	ISKNTRC	763
Transaction Security object	ISCOTSEC	740
Transaction object	ISCOTXN	742
Voice Response Unit Endpoint Class Handler object	ISCOVRUE	739
Voice Response Unit Master object	ISCOVRUM	737
Voice Response Unit Interface object	ISCOVRUT	738
XPNET Interface object	ISKNXPN	764

BASE24-billpay Event Messages

The following table contains the modules that can be used with the BASE24-billpay product. The PMID and SSID are shown for each module.

In addition to identifying the message source, the BASE24-billpay product also uses the PMID to identify event message documentation files provided on the OK13LOGM subvolume. Each file contains event messages for a specific BASE24-billpay module. Refer to the ***BASE24 Log Message Reference Manual*** for a complete explanation of log message documentation.

Product Module or Object Name	PMID	SSID
Achclear program	TPBPACHC	796
Archive program	TPBPARCH	797
Bank advice process	TPBPBKAV	821
Bank billing program	TPBPBBLG	828
Billing utilities	TPBPBLG	798
Billpay Server process	TPBPBPSV	800
Billpay utilities	TPBPUTL	826
Cleanup program	TPBPCLUP	799
Context utilities	TPBPCTXT	820
Customer billing program	TPBPCBLG	827
Database utilities	TPBPDBUT	822
File and table maintenance	TPBPFM	823
Format utilities	TPBPFX	824
Remittance Generator process and remittance file cleanup routine	TPBPRMTG	801
Report routines	TPBPRPT	802
Scheduled Transaction Initiator process	TPBPSTI	803
Transaction utilities	TPBPTXUT	825

Section 2

BASE24 Remote Banking Transaction Path

A number of processes and processing entities form the BASE24 Remote Banking transaction path—that is, the series of processes and processing entities through which a specific transaction passes in order to be authorized.

The actual path each transaction takes varies based on a number of factors, including the type of transaction being processed, the transaction originator, the transaction authorizer, and the processing set up by the BASE24 processor.

This section discusses the following aspects of the BASE24 Remote Banking transaction path:

- Processes that can be part of the transaction path.
- Linkages between the various processes and processing entities for the purpose of forwarding and receiving transaction messages.
- Service-based routing within the BASE24 Remote Banking products.
- Timers used by BASE24 Remote Banking products to ensure the timely completion of transactions.

Processes in the Transaction Path

Along the transaction path, processing is carried out in stages by several processes, each of which has its own responsibilities. These processes are listed below and described on the following pages. Both BASE24 Remote Banking products use the following processes:

- XPNET process
- Path Server process
- Customer Service Representative Requester process
- Pathway Monitor process
- Integrated Endpoint process
- Integrated Authorization Server process
- Interface Master process
- ISO Host Interface process

The BASE24-billpay product uses the following processes in addition to the processes listed above:

- Pathway Billpay Server process
- XPNET Billpay Server process
- Scheduled Transaction Initiator process
- Remittance Generator process
- Bank Advice process

Refer to section 4 for illustrations of message flows involving BASE24-telebanking transactions and section 5 for illustrations of message flows involving BASE24-billpay transactions.

XPNET Process

The XPNET process provides a common point through which most processes in the BASE24 Remote Banking transaction path communicate.

Path Server Process

The Path Server process provides an interface between the XPNET process and a Pathway Monitor process. The Path Server process passes messages through without performing any translation or evaluation.

Customer Service Representative Requester Processes

The customer service representative (CSR) requester processes are Pathway-based processes that display information on customer service representative (CSR) terminal screens, receive the information entered on terminals, and pass it to the Pathway Monitor process.

There are three CSR requester processes:

- The CSR Control Requester process identifies the customer and selects the BASE24-telebanking or BASE24-billpay process based on the desired transaction. This process is present with BASE24-telebanking configurations that include CSR terminals and with all BASE24-billpay configurations.
- The CSR Telebanking Requester process handles BASE24-telebanking transactions. This process is present with BASE24-telebanking configurations that include CSR terminals and with all BASE24-billpay configurations.
- The CSR Billpay Requester process handles BASE24-billpay transactions, and is present only when the BASE24-billpay product is installed.

Pathway Monitor Process

The Pathway Monitor process is the central controlling process in the Tandem Pathway environment. This process receives messages from the Customer Service Representative Requester processes, then directs BASE24-telebanking transactions to the Path Server process and BASE24-billpay transactions to the Pathway Billpay Server process.

Pathway Billpay Server Process

The Pathway Billpay Server process is one of two Billpay Server processes installed with the BASE24-billpay product. This process authorizes BASE24-billpay transactions other than schedule payment and schedule transfer transactions from CSR terminals without using any XPNET-based processes. It passes schedule payment and schedule transfer transactions to the Path Server process and on to XPNET-based processes for authorization.

Schedule payment and schedule transfer transactions sent from the Pathway Billpay Server process are first routed to the Integrated Authorization Server process for customer and customer account validation. Those transactions that pass validation are then routed to the XPNET Billpay Server process for processing.

Integrated Endpoint Process

The Integrated Endpoint process maintains transaction context and the timer associated with the transaction context for certain transaction phases. It also determines the ultimate destination of the transaction.

When a transaction originates at an RBSI-compliant device, the Integrated Endpoint process receives the transaction first and routes it to the Integrated Authorization Server process.

When a transaction originates at a CSR terminal, the Integrated Endpoint process is not used.

For a detailed description of the Integrated Endpoint process, refer to the ***BASE24 IEP Configuration Manual.***

Integrated Authorization Server Process

The Integrated Authorization Server process authorizes BASE24-telebanking and BASE24-billpay transactions by performing the following tasks:

- Receives transactions from RBSI-compliant devices, the Billpay Server processes, the Scheduled Transaction Initiator process, as well as the following BASE24-billpay transactions from CSR terminals:
    - ❑ Schedule payment – immediate
    - ❑ Schedule payment – future
    - ❑ Schedule payment – recurring
    - ❑ Schedule transfer – immediate

  - ❑ Schedule transfer – future
  - ❑ Schedule transfer – recurring
- Determines the proper authorizer.
- Authorizes the transaction itself or routes the transaction to the XPNET Billpay Server process or a host for authorization.
- Generates advice messages to notify the host of authorized transactions.
- Logs all BASE24-telebanking and some BASE24-billpay transactions to the ITS Transaction Log File (ITLF) or Exception Log File (ELF).

XPNET Billpay Server Process

The XPNET Billpay Server process is one of two Billpay Server processes installed with the BASE24-billpay product. All transactions are received from the Integrated Authorization Server process. The XPNET Billpay Server process authorizes all BASE24-billpay transactions received from RBSI-compliant devices or acquirer hosts, as well as the following BASE24-billpay transactions received from CSR terminals:

- Schedule payment – immediate
- Schedule payment – future
- Schedule payment – recurring
- Schedule transfer – immediate
- Schedule transfer – future
- Schedule transfer – recurring

The BASE24-billpay transactions listed above are nonfinancial transactions. For all of these transactions other than a schedule payment – immediate or a schedule transfer – immediate, the XPNET Billpay Server process authorizes or denies the transaction, inserts a row in the History Table (HIST) for the final disposition of the transaction, inserts a row in the Future Table (FUTR), and inserts a row in the Recurring Table (RCUR), if applicable, for authorized transactions. For authorized schedule payment – immediate and schedule transfer – immediate transactions, the XPNET Billpay Server process generates the corresponding financial transaction requests and sends them to the Integrated Authorization Server process, where they are authorized, logged to the ITLF, and returned to the generating process. For all other authorized schedule payment and transfer transactions, the Scheduled Transaction Initiator process generates the corresponding financial transaction requests and sends them to the Integrated Authorization Server process to be authorized and logged.

For all BASE24-billpay transactions except the schedule payment – immediate and schedule transfer – immediate transactions, the XPNET Billpay Server process also logs the transaction to the History Table (HIST) and forwards a response to the Integrated Authorization Server process for return to the RBSI-compliant device, CSR terminal, or acquirer host. The Integrated Authorization Server process also logs selected transactions to the ITLF. Refer to the "Transaction Logging" discussion in section 3 for a list of these transactions.

Interface Master Process

The Interface Master process manages the stations assigned to remote banking endpoint devices. Tasks include verifying proper station status and performing dynamic key management.

For detailed information on the dynamic key management functions performed by the Interface Master process, refer to the RBSI dynamic key management message flow in section 4 and the ***BASE24 Integrated Server Transaction Security Manual***.

ISO Host Interface Process

The ISO Host Interface process is responsible for communicating with host systems. It becomes involved in processing when the host is the transaction acquirer, the appropriate authorizer, or is the recipient of advice messages for transactions authorized by the Integrated Authorization Server process.

The ISO Host Interface process communicates with hosts using a message format based on the ISO 8583-1987 standard. The ISO Host Interface process does not use the ANSI-based message format used by some BASE24 products. For information on the ISO Host Interface process, refer to the ***BASE24 ISO Host Interface Manual*** and ***BASE24 External Message Manual.***

The ISO Host Interface process performs the following functions:

- Translates messages between the BASE24-telebanking internal message format and the ISO external format recognizable to a host.

- Provides a store-and-forward function, whereby if a host is unavailable, certain types of messages bound for the host can be stored and sent at a later time.
- Sets and stops transaction timers that monitor the time it takes for messages to return to the ISO Host Interface process from another entity.
- Provides for PIN encryption, PIN verification, message authentication, and key management between BASE24 Remote Banking processes and the host. For information on the transaction security functions performed by ISO Host Interface processes, refer to the ***BASE24 Integrated Server Transaction Security Manual***.

Scheduled Transaction Initiator Process

The Scheduled Transaction Initiator process is not part of the initial transaction path; however, this process starts a new transaction path based on information contained in the Future Table (FUTR) for the following BASE24-billpay transactions:

- Schedule payment – future
- Schedule transfer – future
- Schedule payment – recurring
- Schedule transfer – recurring

The Scheduled Transaction Initiator process updates and deletes FUTR entries, adds, updates, and deletes Recurring Table (RCUR) entries, and generates History Table (HIST) entries to reflect transaction results. The Integrated Authorization Server process authorizes transactions initiated by the Scheduled Transaction Initiator process and logs messages to the ITS Transaction Log File (ITLF).

The Scheduled Transaction Initiator process uses the Retry Table (RTRY) to determine whether transactions in the FUTR that are declined with specific action codes can be retried.

For more detailed information on the processing performed by the Scheduled Transaction Initiator process, refer to appendix F.

Remittance Generator Process

The Remittance Generator process, while not actually in the transaction path, plays an important role in transaction processing. The Remittance Generator process identifies successfully completed BASE24-billpay payment transactions in the HIST and writes them to vendor remittance output files. A transaction must reside in the HIST for a specified period of time before it is written to a vendor remittance output file. This allows time for an approved transaction entered at a CSR terminal to be canceled before the vendor remittance is generated. Separate remittance output files are created based on the business date and remittance method.

For more detailed information on the processing performed by the Remittance Generator process, refer to appendix F.

Bank Advice Process

The Bank Advice process is not part of the initial transaction path. This process identifies specific BASE24-billpay transactions in the HIST, generates advice messages, and passes the messages to an Integrated Authorization Server process for logging to the ITLF. The Bank Advice process is used because some BASE24-billpay transactions are logged to the HIST without being logged to the ITLF, and the Super Extract process can extract information from the ITLF but not from the HIST. Refer to the “Transaction Logging” discussion in section 3 for a list of transactions that are logged to the ITLF automatically.

For more detailed information on the processing performed by the Bank Advice process, refer to appendix F.

Process Linkages

Since there are multiple processes involved in transaction processing, transactions must be sent between them for processing. Each link in the transaction path is defined below. Transactions can originate at RBSI-compliant devices, CSR terminals, and hosts. BASE24-billpay processing also includes messages originating at the Scheduled Transaction Initiator process and the Bank Advice process. Links used for transactions depend on the point of origination and whether the transaction involves the BASE24-billpay product. Each group of links is described separately.

RBSI-Compliant Devices

The following links are involved in transactions originating at RBSI-compliant devices. The XPNET process is part of each link except for the one between the Integrated Authorization Server process and the Interface Master process. These two processes communicate directly.

RBSI-Compliant Devices to Integrated Endpoint Process

All RBSI-compliant devices are defined as stations. A station is a physical point in the BASE24 system that is designated to send or receive messages. The XPNET process determines the process that should receive a message using a station table in memory, which it builds at startup from the XPNET node station declarations. Each station declaration contains a DESTINATION attribute and a SOURCECLASS attribute.

The DESTINATION attribute identifies the process to which all messages from the station are to be sent. The DESTINATION attribute for RBSI-compliant device stations should specify an Integrated Endpoint process.

The SOURCECLASS attribute identifies the source class to which the station belongs. The source class is used as a key to the Source Class Map File (SCM). The SCM record identifies which object within the destination process should receive the message first. The SOURCECLASS attribute for RBSI-compliant device stations should contain a value that matches the SCM record for the endpoint-specific endpoint class handler object (e.g., the Voice Response Unit Endpoint Class Handler object).

Integrated Endpoint Process to Integrated Authorization Server Process

The Integrated Endpoint process determines the proper Integrated Authorization Server process to receive a message based on the source class, message class, and destination defined in the Destination File (DEST).

A record must be added to the DEST for each combination of source class and message class that the RBSI-compliant devices can send. The source class defined in the DEST should be the one used for the endpoint station because the RBSI-compliant device, of which an interactive voice response system (IVR) is an example, is the source of the message being routed.

The message class is derived by the Integrated Endpoint process using the product ID, message type, and message variant set in the message from the RBSI-compliant device.

The destination defined in the DEST identifies the process to which all messages with a particular source class and message class are sent. The destination can be the symbolic name of a single Integrated Authorization Server process or the name of a service configured for all Integrated Authorization Server processes. When messages are routed to a service, the XPNET process uses a service-based routing algorithm to distribute messages effectively among the members of the service. To use service-based routing, all Integrated Authorization Server processes must be configured with the SERVICE attribute set to the same value (e.g., TBAUTH). Use of service-based routing is briefly explained later in this section. For detailed information on the service-based routing performed by the XPNET process, refer to the ***NET24-XPNET Network Control Operations Guide***.

Each process declaration in the XPNET system must contain a SOURCECLASS attribute. The source class defined in the SOURCECLASS attribute of the declaration for the Integrated Endpoint process is the value that matches the SCM record for the Integrated Endpoint process, (e.g., IEP).

Integrated Authorization Server Process to Interface Master Process

The Integrated Authorization Server process passes certain data elements from each message it receives to an Interface Master process for station verification. Once the station is verified, the Interface Master process returns the message to the Integrated Authorization Server process. The Integrated Authorization Server process and Interface Master process communicate directly without involving the XPNET process. The Interface Master process used is specified in the IMP SYMBOLIC NAME field on VRU Configuration Data (VCD) screen 1.

The Integrated Authorization Server process also communicates directly with the Interface Master process as needed for dynamic key management.

Integrated Authorization Server Process to XPNET Billpay Server Process

The Integrated Authorization Server process determines whether the message is a BASE24-telebanking or BASE24-billpay message based on information in the Processing Code Definition File (PCDF) and Institution Routing Configuration File (IRCF). BASE24-telebanking transactions are authorized by the Integrated Authorization Server process or sent to a host. The Integrated Authorization Server process passes BASE24-billpay transactions to the XPNET Billpay Server process identified in the Internal Message Translation File (IMTF). The Integrated Authorization Server process converts the ITD into a Telebanking Standard Internal Message (BSTM) and passes the BSTM to the XPNET Billpay Server process. The Integrated Authorization Server process also performs any customer verify, PIN verify, and PIN change transactions required before sending a BASE24-billpay transaction to the XPNET Billpay Server process.

XPNET Billpay Server Process to Integrated Authorization Server Process

The XPNET Billpay Server process authorizes BASE24-billpay transactions and returns the response messages to the Integrated Authorization Server process. In the case of a schedule payment – immediate or schedule transfer – immediate transaction, there is an additional exchange of messages between the XPNET Billpay Server process and the Integrated Authorization Server process so the Integrated Authorization Server process can authorize the withdrawal of funds from the customer’s account for the payment or transfer. The XPNET Billpay Server process obtains the name of the Integrated Authorization Server process from the BP-TELEBANKING param in the Logical Network Configuration File (LCONF).

Integrated Authorization Server Process to ISO Host Interface Process

The Integrated Authorization Server process determines whether a transaction must be sent to a host based on information in the Processing Code Definition File (PCDF) and the Institution Routing Configuration File (IRCF). Based on the authorization level and destination specified in the IRCF, the Integrated Authorization Server process obtains the ISO Host Interface process name and DPC number for the host from the Internal Message Translation File (IMTF), then converts the ITD into a Telebanking Standard Internal Message (BSTM) and passes the BSTM to the ISO Host Interface process. The Integrated Authorization Server process also performs any customer verify or PIN verify transactions if prescreening is required before sending the transaction to a host. The ISO Host Interface process is responsible for converting the BSTM into an ISO external message that is passed to the proper host.

CSR Terminals

The following links are involved in transactions originating at a CSR terminal. The Pathway Monitor process is part of each link between the following processes:

- CSR Control Requester process
- CSR Telebanking Requester process
- CSR Billpay Requester process
- Pathway Billpay Server process
- Path Server process

The XPNET process is part of each link between the following processes:

- Path Server process
- Integrated Authorization Server process
- XPNET Billpay Server process
- ISO Host Interface process

CSR Control Requester Process to Path Server Process

The CSR Control Requester process sends messages for customer verification, PIN verification, and PIN changes to the Path Server process. The Path Server process provides an interface between Pathway-based processes and XPNET-based processes. The CSR Control Requester process uses the Integrated Authorization Server process to perform customer verify, PIN verify, and PIN change transactions. This Path Server process is typically defined in the Pathway Configuration File (PATHCONF) as SERVER-CSI.

CSR Control Requester Process to CSR Telebanking Requester Process or CSR Billpay Requester Process

After any required customer verify, PIN verify, or PIN change transactions are completed, the CSR Control Requester process displays menus containing all available transactions, then sends messages for BASE24-telebanking transactions to the CSR Telebanking Requester process and sends messages for BASE24-billpay transactions to the CSR Billpay Requester process.

CSR Billpay Requester Process to Pathway Billpay Server Process

The CSR Billpay Requester process sends all BASE24-billpay messages to the Pathway Billpay Server process for authorization. This server process is typically defined in the PATHCONF as SV-BP.

Pathway Billpay Server Process to Path Server Process

The Pathway Billpay Server process sends messages for BASE24-billpay schedule payment and schedule transfer transactions to the Path Server process, which provides an interface between Pathway-based processes and XPNET-based processes. XPNET-based processes authorize BASE24-billpay schedule payment and schedule transfer transactions. This Path Server process is defined in the BP-XPNET-CTSE-SERVER-CLASS param in the LCONF.

The Pathway Billpay Server process authorizes all other BASE24-billpay transactions.

Path Server Process to Integrated Authorization Server Process

The Path Server process provides an interface between a Pathway Monitor process and the XPNET process. The Path Server process passes BASE24-telebanking messages through the XPNET process to the Integrated Authorization Server process declared in the station definition, and passes BASE24-billpay schedule payment and schedule transfer transactions to the Integrated Authorization Server process defined in the BP-TELEBANKING param in the LCONF.

Integrated Authorization Server Process to XPNET Billpay Server Process

The Integrated Authorization Server process passes messages for BASE24-billpay transactions to an XPNET Billpay Server process based on information in the Processing Code Definition File (PCDF) and the Institution Routing Configuration File (IRCF). Based on the authorization level and destination specified in the IRCF, the Integrated Authorization Server process obtains the XPNET Billpay Server process name from the Internal Message Translation File (IMTF), then converts the ITD into a Telebanking Standard Internal Message (BSTM) and passes the BSTM to the XPNET Billpay Server process. The specific transaction determines the number of times messages are passed back and forth between the Integrated Authorization Server process and the XPNET Billpay Server process.

Integrated Authorization Server Process to ISO Host Interface Process

The Integrated Authorization Server process determines whether a transaction must be sent to a host based on information in the Processing Code Definition File (PCDF) and the Institution Routing Configuration File (IRCF). Based on the authorization level and destination specified in the IRCF, the Integrated Authorization Server process obtains the ISO Host Interface process name and DPC number for the host from the Internal Message Translation File (IMTF), then converts the ITD into a Telebanking Standard Internal Message (BSTM) and passes the BSTM to the ISO Host Interface process. The ISO Host Interface process is responsible for converting the BSTM into an ISO external message that is passed to the proper host.

Hosts

The following links are involved in transactions received from a host. The XPNET process is part of each link.

ISO Host Interface Process to Integrated Authorization Server Process

The ISO Host Interface process provides an interface between an acquirer host and the Integrated Authorization Server process. The ISO Host Interface process converts the ISO external message into a Telebanking Standard Internal Message (BSTM) and passes it to the Integrated Authorization Server process identified by the value in the AUTH PROCESS field on Host Configuration File (HCF) screen 22.

Integrated Authorization Server Process to XPNET Billpay Server Process

The Integrated Authorization Server process passes messages for BASE24-billpay transactions to an XPNET Billpay Server process based on information in the Processing Code Definition File (PCDF) and the Institution Routing Configuration File (IRCF). Based on the authorization level and destination specified in the IRCF, the Integrated Authorization Server process obtains the XPNET Billpay Server process name from the Internal Message Translation File (IMTF), then converts the ITD into a Telebanking Standard Internal Message (BSTM) and passes the BSTM to the XPNET Billpay Server process. The specific transaction determines the number of times messages are passed back and forth between the Integrated Authorization Server process and the XPNET Billpay Server process.

Scheduled Transaction Initiator Process

The following links are involved in transactions originated by the Scheduled Transaction Initiator process. Each link includes the XPNET process.

Scheduled Transaction Initiator Process to Integrated Authorization Server Process

The Scheduled Transaction Initiator process initiates financial transactions based on information in the Future Table (FUTR). This process passes transactions to the Integrated Authorization Server process defined in the BP-AUTHORIZER param in the LCONF.

Integrated Authorization Server Process to ISO Host Interface Process

The Integrated Authorization Server process determines whether a transaction must be sent to a host based on information in the Processing Code Definition File (PCDF) and the Institution Routing Configuration File (IRCF). Based on the authorization level and destination specified in the IRCF, the Integrated Authorization Server process obtains the ISO Host Interface process name and DPC number for the host from the Internal Message Translation File (IMTF), then converts the ITD into a Telebanking Standard Internal Message (BSTM) and passes the BSTM to the ISO Host Interface process. The ISO Host Interface process is responsible for converting the BSTM into an ISO external message that is passed to the proper host.

Bank Advice Process

The following link is involved in transactions originated by the Bank Advice process. This link includes the XPNET process.

Bank Advice Process to Integrated Authorization Server Process

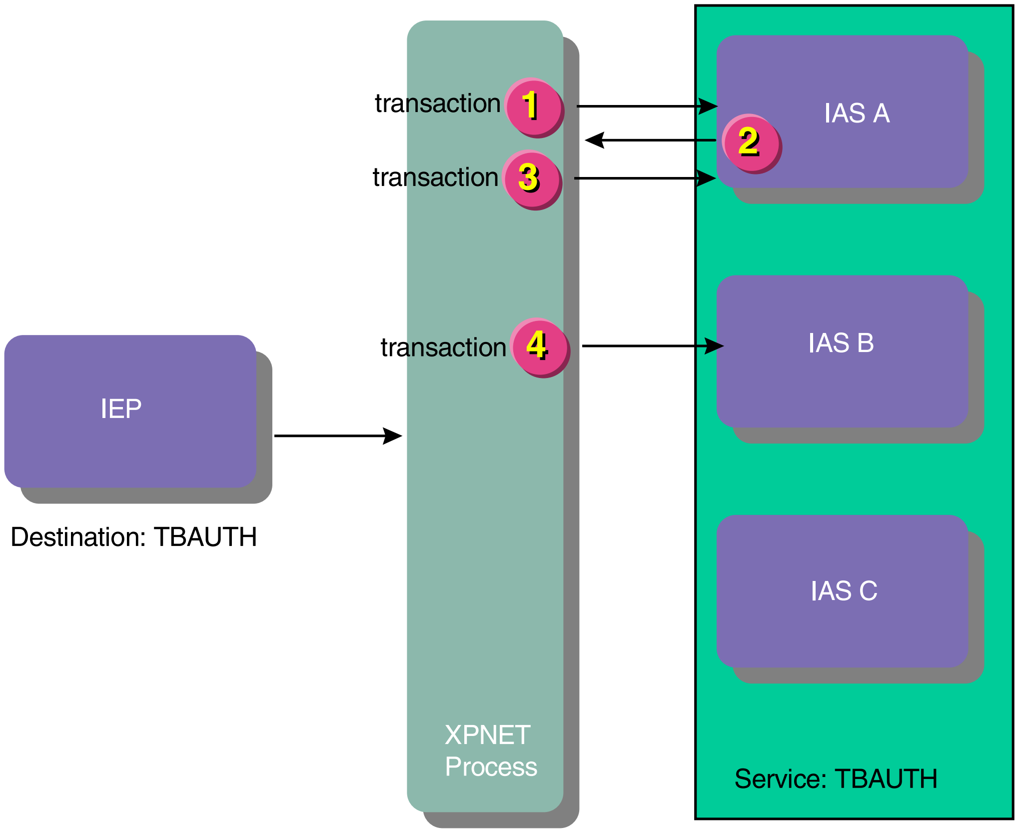
The Bank Advice process initiates advice messages based on information in the History Table (HIST). This process passes advice messages to the Integrated Authorization Server process defined in the BP-TELEBANKING-LOG param in the LCONF.

Service-based Routing

When multiple Integrated Authorization Server processes are in use, transaction requests can be sent to a service instead of directly to a specific Integrated Authorization Server process. When multiple XPNET Billpay Server processes are in use, transaction requests can be sent to a service instead of directly to a specific XPNET Billpay Server process.

This option is designed to avoid the processing inefficiencies that could develop when a specific process is assigned to receive transaction requests. Without service-based routing, the requesting processes assigned to one server process must wait for that server process to complete its transactions even if other server processes are available. With service-based routing, transactions can be distributed efficiently among multiple server processes as they become available.

The following illustration shows how service-based routing functions with the Integrated Endpoint process and Integrated Authorization Server processes. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process
IEP = Integrated Endpoint process

In this example, the XPNET process can send transactions to three Integrated Authorization Server processes that have been associated with the service TBAUTH in the XPNET configuration. The Integrated Endpoint process sends all BASE24-telebanking messages it receives to the TBAUTH destination. The XPNET process sends each transaction it receives to the first available Integrated Authorization Server process for the TBAUTH service, as follows:

1. The XPNET process sends the first transaction to Integrated Authorization Server process A, the first provider available for the TBAUTH service.

2. Integrated Authorization Server process A completes its processing of the first transaction and passes the transaction through the XPNET process to another process for further processing.
3. The XPNET process sends the second transaction to Integrated Authorization Server process A. Because the first transaction processing is complete, process A is still the first available provider for the TBAUTH service.
4. The XPNET process sends the third transaction to Integrated Authorization Server process B because Integrated Authorization Server process A is still processing the second transaction when the XPNET process receives the third transaction for the TBAUTH service.

The XPNET process uses a service-based routing algorithm to determine which provider receives each message sent to a service. Messages sent to a service are distributed on a rotary basis. A provider is considered to be available if it is started and is not currently busy. Providers configured with a startup logic of DEMAND that are stopped, are started only if they are needed to handle the processing load. Once started, they remain in rotation. For a detailed description of the service-based algorithm used by the XPNET process, refer to the ***NET24-XPNET Network Control Operations Guide***.

Processes are configured to a service using the SERVICE attribute. For detailed information on configuring the SERVICE attribute, refer to the ***NET24-XPNET Network Control Command Reference Manual***.

Transaction Timers

Timers enable processes to take action in situations where responses have not been received as expected. The BASE24 Remote Banking products use timers for logical network processing, authorization processing, and host processing. These various timers are set in the Logical Network Configuration File (LCONF), the Host Configuration File (HCF), the ITS Interface Configuration File (IFCF), the Internal Message Translation File (IMTF), and the Key File (KEYF). Fields in the IFCF are set on VRU Configuration Data (VCD) screens.

General Timers

The following general timers are used by multiple BASE24 processes.

Dynamic Key Management Limits — Provide dynamic key management parameters. There are three groups of limits. Each group includes counters as well as timers. Even though this is a discussion of timers, these counters are included because processes use the counters and timers together to determine when a key must be changed.

The Interface Master process uses the following limits contained in the IFCF and set on VCD screen 4 to determine when to change the inbound PIN transport key:

KEY SYNC ERRORS
TIMER VALUE
TRANSACTIONS
ERRORS
CONSECUTIVE ERRORS

The ISO Host Interface process uses the following limits set on KEYF screen 3 to determine when to change the PIN transport key:

PIN KEY TIMER VALUE
PIN KEY TIMER INTERVAL
PIN KEY TRAN
PIN KEY ERROR
CONSECUTIVE PIN KEY ERROR

The ISO Host Interface process uses the following limits set on KEYF screen 3 to determine when to change the message authentication key:

MAC KEY TIMER VALUE
MAC KEY TIMER INTERVAL
MAC KEY TRAN
MAC KEY ERROR
CONSECUTIVE MAC KEY ERROR
KMAC SYNCHRONIZATION ERROR

Security Device Timer Limit — Specifies the maximum amount of time an Integrated Authorization Server process or ISO Host Interface process waits for a response from the security module before the message is considered timed out. This timer limit is set in the SECURE-DEV-TIM-LMT param in the LCONF.

Authorization Timers

The following timers are used by various processes with BASE24-telebanking and BASE24-billpay transactions during authorization processing.

Authorization Request Timer Limit — Specifies the amount of time a remote banking endpoint device waits for a response. This timer limit is contained in the IFCF and is set in the AUTH REQUEST field on VCD screen 18.

Billpay Lag Timer Limit — Specifies the amount of time provided for an authorized BASE24-billpay payment to be reversed before it is processed as a vendor payment. This timer limit is set in the BP-LAG-TIME param in the LCONF and is used by the Bank Advice process and Remittance Generator process.

Billpay STI Lag Timer Limit — Specifies the amount of time a transaction must reside in the Future Table (FUTR) before it can be processed by the Scheduled Transaction Initiator process. This timer limit is set in the BP-STI-LAG-TIME param in the LCONF and is used by the Scheduled Transaction Initiator process.

Billpay STI Final Processing Timer Limit — Specifies the amount of time available to complete processing of FUTR transactions after the Scheduled Transaction Initiator processing window has closed and the STI lag time has expired.

Billpay Transaction Timeout Timer Limit — Specifies the amount of time an XPNET Billpay Server process or Scheduled Transaction Initiator process waits for a response to an authorization request. This timer limit is set in the BP-TXN-TIMEOUT param in the LCONF. The Scheduled Transaction Initiator process uses the limit specified and the XPNET Billpay Server process uses the limit plus five seconds. The value of this limit must be less than the timeout value for the Pathway Billpay Server process set in the Pathway Configuration File (PATHCONF).

Minimum Request Timer Limit — Specifies the minimum amount of time the host needs to authorize a transaction and return a response. This timer limit is set in the MINIMUM REQUEST LIMIT field on IMTF screen 1. This limit must be less than the outbound timer limit for the Integrated Authorization Server process.

If the amount of time remaining to process a transaction (the actual transaction elapsed time subtracted from the authorization time limit established by the message acquirer or the outbound timer limit, whichever is less) is less than the value of this timer, the BASE24 product does not send the message to the host or authorizing process. The BASE24 product either attempts to stand in for the host/external authorizer process or declines the transaction, depending on the value in the AUTH LEVEL field on Institution Routing Configuration File (IRCF) screen 1.

The value in the AUTH REQUEST field on Voice Response Unit Configuration Data (VCD) screen 18 is the authorization time limit for the only message acquirer currently supported by an integrated server application.

Multiple Account Selection Timer Limit — Specifies the amount of time an Integrated Endpoint process retains the message context associated with multiple account selection. The Integrated Endpoint process retains message context while the account information is returned to the RBSI-compliant device and the customer selects an account or requests additional account information. This timer limit is contained in the IFCF and is set in the MULTIPLE ACCOUNT SELECT field on VCD screen 18.

Network Management Timer Limit — Specifies the amount of time an Integrated Endpoint process retains the message context associated with a dynamic key management message. The Integrated Endpoint process retains new key information while the new key information is sent to the RBSI-compliant device. This timer limit is contained in the IFCF and is set in the NETWORK MANAGEMENT field on VCD screen 2.

Outbound Timer Limit — Specifies the maximum amount of time an Integrated Authorization Server process waits for a response from the BASE24 ISO Host Interface process or XPNET Billpay Server process. This defines the maximum time a message can be queued by the XPNET process before it is delivered to the external authorizer or interface process. If the message is queued longer than the time specified in this field, it is not delivered to the external authorizer or interface process. This option is designed to circumvent processing when it becomes apparent that sending the message does not give the external authorizer or interface process enough time to respond before the BASE24 product times out. This timer limit is set in the OUTBOUND LIMIT field on IMTF screen 1.

This field is also used to determine the amount of time remaining to process a transaction. If the actual transaction elapsed time subtracted from the authorization time limit established by the message acquirer is greater than the value of this field, the value of this field is used as the amount of time remaining to process a transaction. This amount is then compared against the value in the MINIMUM REQUEST LIMIT field on IMTF screen 1 to determine whether the message is sent. Refer to the MINIMUM REQUEST LIMIT field description for details.

Queue Timer Limit — Specifies the maximum amount of time that the XPNET process can queue a transaction response before delivering it to another process. The Integrated Authorization Server process and the Integrated Endpoint process use this limit to compute a value that is placed in the message header for use by the XPNET process.

The Integrated Authorization Server process sets the value in the message header for use by the XPNET process directly from this limit without making any adjustments. If the XPNET process cannot deliver the message to the Integrated Endpoint process before the amount of time in the message header passes, the XPNET process returns the transaction to the Integrated Authorization Server process as failed.

The Integrated Endpoint process uses this limit and the remaining authorization request time to set the value in the message header. The Integrated Endpoint process uses the authorization request timer limit and computes the amount of time left for returning the response. The computed time or the queue timer limit, whichever is less, is used to set the value in the message header for use by the XPNET process. If the XPNET process cannot deliver the message to the RBSI-compliant device before the amount of time in the message header passes, the XPNET process returns the transaction to the Integrated Endpoint process as failed.

This timer limit is contained in the IFCF and is set in the QUEUE LIMIT field on VCD screen 18.

ISO Host Interface Process Timers

The following timers are used by ISO Host Interface processes. Not all are directly related to transaction processing; some are related to network management.

Completion Acknowledgment Message Timer Limit — Specifies the amount of time an ISO Host Interface process waits for an acknowledgment message after sending a BASE24-telebanking advice or reversal message to the DPC. This timer limit is set in the COMPLETION ACK field on HCF screen 22.

Extended Network Timer Limit — Specifies the amount of time an ISO Host Interface process waits, once a station has been marked down, before sending an echo-test message. This timer limit is set in the EXTENDED NETWORK field on HCF screen 1.

Inbound Timer Limit — Specifies the amount of time an ISO Host Interface process waits for a response message after sending a BASE24-telebanking transaction request message to the Integrated Authorization Server process. This timer limit is set in the INBOUND LIMIT field on HCF screen 22.

Network Management Timer Limit — Specifies the amount of time a BASE24 ISO Host Interface process waits for a response after sending a network management message (0800 series) to a data processing center (DPC) station (i.e., a host). If a response message (0810) is not received within the time specified, the ISO Host Interface process marks the station down. This timer limit is set in the NETWORK MANAGEMENT field on HCF screen 1.

Outbound Timer Limit — Specifies the amount of time an ISO Host Interface process waits for a response message after sending a BASE24-telebanking transaction request message to the host. This timer limit is set in the OUTBOUND LIMIT field on HCF screen 22. The EXTRN-RQST-TIMEOUT param in the LCONF determines the actions taken when a transaction request times out.

If this param is set to a value of 0 and a request times out, the ISO Host Interface process fails the transaction and returns it to the Integrated Authorization Server process for alternate or stand-in routing. The ISO Host Interface process generates a reversal and sends it to the host if it receives a late response.

If this param is set to a value of 1 and a request times out, the ISO Host Interface process fails the transaction and returns it to the Integrated Authorization Server process for alternate or stand-in routing and also generates a reversal that it sends to the host. If the ISO Host Interface process then receives a late response from the host, it simply drops the message.

Queue Subtract Timer Limit — Specifies the amount of time used to compute the maximum time the XPNET process can wait before sending the BASE24-telebanking message to the host. The ISO Host Interface process subtracts this value from the value in the OUTBOUND TIMER LIMIT field and places the result in the message header.

If the XPNET process cannot send the message to the host within the time passed in the message header, it returns the message to the ISO Host Interface process as failed. This option circumvents processing when it becomes apparent that sending a message will not give a host time to respond before the message times out. This timer limit is set in the QUEUE SUBTRACT field on HCF screen 22.

Store-and-Forward Timer Limit — Specifies the amount of time an ISO Host Interface process waits for an acknowledgment after sending a store-and-forward message to the DPC. This timer limit is set in the STORE AND FORWARD field on HCF screen 1.

Wait-for-Traffic Timer Limit — Specifies the amount of time an ISO Host Interface process waits for traffic on the line to a station. When the timer expires, the ISO Host Interface process determines whether traffic has occurred on the line, then resets the timer if traffic has occurred or sends an echo test message to the station. This timer limit is set in the WAIT FOR TRAFFIC field on HCF screen 1.

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Section 3

BASE24 Remote Banking Transaction Processing Controls

This section describes the basic BASE24 Remote Banking processing controls that are available for the RBSI-compliant device, customer service representative terminal, institution, and host.

Authorization Levels and Methods

The Integrated Authorization Server process performs all customer verification, PIN verification, and can authorize most BASE24-telebanking transactions (some transactions can only be authorized by a host). It works with the XPNET Billpay Server process and Scheduled Transaction Initiator process to authorize a subset of the BASE24-billpay transactions.

This subset includes all BASE24-billpay transactions acquired from a host or an RBSI-compliant device, and BASE24-billpay schedule payment and schedule transfer transactions acquired from a customer service representative (CSR) terminal. These transactions are identified in BASE24 routing configuration files by the special route code value of BP.

The Integrated Authorization Server process also authorizes individual payment and transfer transactions initiated by the Scheduled Transaction Initiator process.

The Integrated Authorization Server process authorizes transactions based on the authorization levels and methods specified by each institution. These levels and methods are described below, along with other authorization functions that the Integrated Authorization Server process performs. Authorization levels and methods do not affect the processing performed by the XPNET Billpay Server process.

Authorization Levels

The authorization level defines the level of participation that an institution has in authorizing each type of transaction. Depending on the type of transaction, a BASE24 Integrated Authorization Server process can perform all authorizations for an institution, or the Integrated Authorization Server process can be set up to forward transactions to a back-end host processor for full or final approval.

The authorization level is specified in the AUTH LEVEL field on Institution Routing Configuration File (IRCF) screen 1. The possible authorization levels are online authorization (level 1), offline authorization (level 2), and online/offline authorization (level 3).

A single institution can process using multiple authorization levels, depending on the number of hosts it uses and the types of transactions it processes. An institution can control processing with multiple authorization levels by defining multiple IRCF records.

Regardless of authorization level, the Customer Table (CSTT) is always required. The Integrated Authorization Server process requires information contained in the CSTT in order to identify the proper transaction set for a customer.

Online Authorization

Online authorization is also referred to as level 1 authorization. At this level, the host performs all transaction authorizations. If the host is offline and, therefore, unable to authorize transactions, the Integrated Authorization Server process declines all transaction requests. Account balance and transaction history files are not maintained in the BASE24 database at this level of authorization.

Although the Integrated Authorization Server process cannot authorize transactions when an institution is using the online authorization level, it can optionally screen, or prescreen, transactions before sending them to the host. If a transaction does not pass the prescreening checks, the Integrated Authorization Server process can decline the transaction. Prescreening is described later in this section.

The online authorization level is required for the following BASE24-telebanking transactions because the data needed to authorize these transactions is only maintained on a host:

- Check stop payment add (A0)
- Statement closing download (3Q)
- Statement download (3R)

The online authorization level is also required for all BASE24-billpay transactions, which must be passed to the XPNET Billpay Server process after being prescreened by the Integrated Authorization Server process. Refer to routing configuration examples later in this section and message flow examples in section 5 for additional information about this requirement.

Offline Authorization

Offline authorization is also referred to as level 2 authorization. At this level, the Integrated Authorization Server process authorizes all BASE24-telebanking transactions and BASE24-billpay transactions without forwarding any authorization requests to the host.

With the offline authorization level, an institution must maintain all needed customer accounts in its BASE24 database. The authorization method determines whether accounts are maintained in the BASE24 database. Refer to the discussion of authorization methods later in this section. Customer accounts are refreshed periodically using information from host files. Transaction log files are periodically extracted to tape or disk and used to update host files.

Online/Offline Authorization

Online/offline authorization is also referred to as level 3 authorization. At this level, the Integrated Authorization Server process optionally screens transactions based on the prescreening settings in the Institution Definition File (IDF) and forwards those transactions that pass the screening checks to the host for final approval. The Integrated Authorization Server process declines transactions that do not pass the screening checks.

The Integrated Authorization Server process also stands in for the host and authorizes transactions when communications to the host are down. The transactions the Integrated Authorization Server process authorizes can be stored and forwarded to the host when communications with the host are restored.

With the online/offline authorization level, an institution must maintain all needed customer account files in its BASE24 database. The authorization method determines whether accounts are maintained in the BASE24 database. Refer to the discussion of authorization methods later in this section. Customer accounts are refreshed periodically using information from host files. Transaction log files are periodically extracted to tape or disk and used to update host files.

Authorization Methods

The authorization method chosen by an institution establishes the file configuration and authorization checks that the Integrated Authorization Server process performs while authorizing a transaction.

Institutions specify the authorization method in the AUTH METHOD field on IRCF screen 1. Possible authorization methods are the Positive Customer Authorization method, the Positive Customer with Accounts Authorization method, and the Positive Customer with Balances/History Authorization method.

A single institution can be set up to process using multiple authorization methods, depending on the BASE24 files and tables it wishes to maintain and the types of BASE24 checks it wishes to perform. An institution can control processing with multiple authorization methods by defining multiple IRCF records.

Positive Customer Authorization Method

With the Positive Customer Authorization method (PCA), the Integrated Authorization Server process verifies the effective date range, status, profile, and identification of the customer without checking any account-level information. The effective date range, which is based on values in the BEGIN DATE and END DATE fields on CSTT screen 1, identifies the period of time a customer ID is available for use in transaction processing. The customer status is based on the value in the VERIFY STATUS field on CSTT screen 1.

The customer profile, which is based on the value in the PROFILE field on CSTT screen 1, identifies the processing codes that are available to the customer. A processing code contains a two-character transaction code and two two-character account type codes. The transaction code identifies the type of transaction (e.g., an inquiry, payment, or transfer). The account type codes identify the type of accounts involved in the transaction (e.g., checking or savings). There are two account type codes because the *from* account and *to* account types must be identified for transfer transactions and two accounts must be identified for dual inquiries.

Each processing code for a particular customer profile is defined in the Customer Allowed Transaction Table (CATT). A financial institution can have its own customer profiles or can share them with other financial institutions. The Integrated Authorization Server process first searches for a customer profile that belongs to the financial institution, then searches for a customer profile that can be used by any financial institution.

Note: For BASE24-billpay nonfinancial transactions (e.g., loan application, customer add, etc.) received from an RBSI-compliant device with a customer ID of all zeros, the CSTT and CATT are not checked.

Customer identification includes a PIN or other information. For most transactions received from a remote banking endpoint device, a PIN is required unless the PIN has already been verified by the device. However, if the customer ID is all zeros, a PIN is not required. The institution can configure whether the PIN is required for transactions originated by customer service representatives. The Integrated Authorization Server process uses the offset value in the PIN VERIFICATION DIGITS field on CSTT screen 1 to perform PIN verification.

Miscellaneous customer identification information
(e.g., mother's maiden name) can be maintained in the CUST INFO fields on CSTT screen 1 or a series of fields on Personal Information Table (PIT) screen 1 and displayed to the customer service representative.

The Positive Customer Authorization method supports the standard BASE24 Remote Banking transaction set, user-defined transactions, and all three authorization levels, with the following restrictions:

1. Balance inquiries and transaction history inquiries must be authorized by the host rather than the BASE24 database because the Positive Customer Authorization method does not access the Positive Balance File (PBF) for balance information or the Transaction History File (THF) for transaction history information. Since this method does not access the PBF, transfer transactions must be authorized by the host or be performed as log-only transactions by the Integrated Authorization Server process.
2. The Integrated Authorization Server process can pass a transaction to the host after checking only the customer status, profile, and identification. One of the other authorization methods must be used if the account status and allowable action need to be checked. For example, check stop payment add transactions and transactions acquired from an OFX client require account status and/or allowable action checks. The Customer/Account Relation Table (CACT) contains this information and the Positive Customer Authorization method does not access the CACT.
3. The Integrated Authorization Server process can pass a BASE24-billpay transaction that does not require account numbers for authorization to the XPNET Billpay Server process. Schedule payment and schedule transfer transactions require account numbers for authorization. An account number is optional for all three scheduled transaction inquiry and both history inquiry transactions. One of the other authorization methods must be used if an account number is required or optionally provided. The Customer/Account Relation Table (CACT) contains this information and the Positive Customer Authorization method does not access the CACT.

Positive Customer with Accounts Authorization Method

With the Positive Customer with Accounts Authorization method (PCAA), the Integrated Authorization Server process verifies the status and allowable actions for the account, as well as the status, profile, and identification of the customer. The account status is based on the value in the STATUS field on CSTT screen 2. The allowable action is based on the value in one of the ACT ALWD fields on CSTT screen 2. The allowable action identifies whether transfer debit (i.e., transfers from an account), bill payment debit, credit (i.e., transfers to an account), or inquiry transactions can be performed on an account. The account status is checked for all transactions that include a customer account, except for the following BASE24-billpay transactions:

- Customer account add
- Customer account deactivate
- Customer account delete
- Customer account update
- Any nonfinancial transactions (e.g., loan application, customer add, etc.) received from an RBSI-compliant device with a customer ID of all zeros

The allowable action indicators are checked for all transactions that include a customer account, except for check stop payment add transactions and the BASE24-billpay transactions listed above.

The customer status, profile, and identification processing for this authorization method is identical to the processing performed for the Positive Customer Authorization method.

The difference between this authorization method and the Positive Customer Authorization method is the CACT. Information from this file is displayed on CSTT screen 2 and contains the information, other than balance and transaction history, for each account associated with a customer.

Positive Customer with Balances/History Authorization Method

With the Positive Customer with Balances/History Authorization method (PCBA), the Integrated Authorization Server process can perform full authorization of all transactions in the standard BASE24 Remote Banking transaction set. With this method, the Integrated Authorization Server process can access the PBF for balance information and access the THF for transaction history information. The Integrated Authorization Server process can update the PBF with transfer and payment transaction results during transaction processing. The Enscribe Refresh process can update the THF and PBF periodically using information from host files.

The customer and account verification processing for this authorization method is identical to the processing performed for the Positive Customer with Accounts Authorization method. This authorization method supports the offline and online/

offline authorization levels. Since this authorization method does not support the online authorization level, it cannot be used with BASE24-billpay transactions that must be passed to the XPNET Billpay Server process or BASE24-telebanking transactions that must be passed to a host.

Transaction Screening

Transaction screening, also referred to as prescreening, is a term used frequently in situations where a BASE24 product shares authorization responsibilities with the host. In such situations, the Integrated Authorization Server process can make preliminary standard checks on the transaction and decline those that do not meet the preliminary requirements. This prevents unnecessary processing by the host.

The Integrated Authorization Server process can perform prescreening only when configured to do so by the institution and when the institution is processing transactions using the online or online/offline authorization level. When the institution is using the offline authorization level, these checks are included automatically as a part of the authorization processing that the Integrated Authorization Server process performs.

Customer Status

The customer status, which is identified by the value in the VERIFY STATUS field on CSTT screen 1, indicates whether the customer initiating a transaction has an active (verified), inactive, or closed customer ID.

The Integrated Authorization Server process can deny a transaction or send it on to the host for authorization, depending on the customer status. A customer cannot perform transactions unless the customer ID is active. Customer service representative (CSRs) can add accounts for the customer as long as the customer ID is active or inactive. CSRs can delete accounts for the customer only when the customer ID is inactive.

The customer status is checked for all BASE24-telebanking transactions and most BASE24-billpay transactions. The customer status is not checked for the following BASE24-billpay transactions:

- Customer account add
- Customer account delete
- Customer activate
- Customer add
- Customer deactivate
- Customer inquiry
- Customer update

- Customer vendor activate
- Customer vendor add
- Customer vendor deactivate
- Customer vendor list inquiry
- Customer vendor mask verify inquiry
- Customer vendor update
- Any nonfinancial transactions (e.g., loan application, customer add, etc.) received from an RBSI-compliant device with a customer ID of all zeros

For BASE24-telebanking transactions, institutions use the CARD STATUS field on IDF screen 2 to indicate whether they want the Integrated Authorization Server process to prescreen the customer status when sending a transaction to a host for authorization. For BASE24-billpay transactions with a nonzero customer ID, the Integrated Authorization Server process always includes a customer status check when processing any transaction it receives, regardless of the value in the CARD STATUS field on IDF screen 2.

PIN Check

Institutions can configure the Integrated Authorization Server process to determine whether a customer entered the correct PIN when initiating a transaction. This is configured in the PIN field on IDF screen 2.

When PINs are included in transaction screening, the checks performed by the Integrated Authorization Server process depend on the transaction type and originator, as follows:

- The PIN verify transaction always requires a PIN.
- The PIN change transaction does not require an existing PIN when the PIN CHANGE REQUIRED field on CSTT screen 1 contains a value of S, indicating the customer has not yet selected their initial PIN.
- The PIN change transaction requires an existing PIN when the PIN CHANGE REQUIRED field on CSTT screen 1 contains a value other than S, indicating the customer has a PIN.

- All other transactions require a PIN when originating from a remote banking endpoint device or a host, unless the transaction (e.g., loan application, customer add, etc.) contains a customer ID of all zeros or indicates that PIN verification was already performed by an RBSI-compliant device.
- The value in the PIN REQUIRED field on IDF screen 40 identifies whether transactions other than PIN change and PIN verify require a PIN when originating from a customer service representative using either a CSR terminal or an RBSI-compliant device. This field is only used for transactions acquired from an RBSI-compliant device if the source code in the transaction is set to a value of IB.

Transaction Security

BASE24 Remote Banking products offer several forms of transaction security, including PIN verification, PIN encryption, PIN changes, message authentication, and dynamic key management. The information presented here summarizes these transaction security features. Refer to the ***BASE24 Integrated Server Transaction Security Manual*** for detailed information about the operation and configuration of each feature.

Two important distinctions exist between the transaction security features of BASE24 Remote Banking products and transaction security features of other BASE24 products. First, the BASE24 Remote Banking products do not use the plastic cards used by other products. As a result, the information typically contained on the magnetic stripe of the plastic card is unavailable for transaction security purposes. This information includes a card prefix, card expiration date, and card verification value.

Second, the PIN offset used by the BASE24 Remote Banking products is not the one used by other BASE24 products. The BASE24 Remote Banking PIN offset appears in the PIN VERIFICATION DIGITS field on CSTT screen 1. The PIN offset used by other BASE24 products appears in the POFST/PVV field on Cardholder Authorization File (CAF) screen 1. Thus, PIN offsets are stored in two BASE24 database locations. These locations can contain the same value or different values.

PIN Verification

PIN verification can be performed for transactions authorized by the Integrated Authorization Server process, authorized by Billpay Server process, or as a prescreening step for transactions authorized by a host. PIN verification can be performed locally at the RBSI-compliant device, by the Integrated Authorization Server process, or by an authorizing host.

If a customer ID of all zeros is provided in a transaction, PIN verification is not performed. Some BASE24-billpay nonfinancial transactions (i.e., loan application, bank list inquiry, and bank branch list inquiry) do not require a customer ID when acquired from an RBSI-compliant device.

The value in the LOCAL PIN VERIFICATION field on VRU Configuration Data (VCD) screen 18 identifies whether an RBSI-compliant device can perform local PIN verification. The RBSI-compliant device sets the PIN capture capability flag in the point of service codes of the request message to a value of S and does not place the PIN in the request message when the PIN is verified locally.

PIN verification by the Integrated Authorization Server process requires a record in the Key Authorization File (KEYA) for each institution performing PIN verification. PIN verification information is maintained on screens 2 through 5 of the KEYA. The value in the PIN CHECK TYPE field on IDF screen 2 and the value in the PIN VERIFICATION GROUP field on IDF screen 40 identify which KEYA record is used. Remaining PIN verification parameters are on IDF screens 2 and 4 and CSTT screen 1.

The Integrated Authorization Server process can use software or security devices to perform PIN verification. Hardware security options include interfaces to various models of Atalla and Racal-Guardata security hardware. Software security options include several methods of PIN encryption and PIN verification.

PIN Encryption

PINs can be sent between the Integrated Authorization Server process and an RBSI-compliant device or a host in encrypted form or in the clear. PINs are encrypted and decrypted within the Integrated Authorization Server process and the Host Interface process.

When a BASE24 system receives a transaction, the PIN may need to be encrypted, decrypted, or translated from one encrypted form to another, depending on how the BASE24 system and the external processor are set up to handle their PINs. The same is true for transactions that the BASE24 system is sending to another entity.

PIN encryption data for messages sent between the RBSI-compliant device and an Integrated Authorization Server process is configured on Interface Security File (ISF) screens 1 and 2. PIN encryption data for messages sent between the Host Interface process and the host is configured on Key File (KEYF) screens 1 and 2.

PIN Changes

The BASE24 Remote Banking transaction set includes PIN change and PIN verify transactions for use by the customer and the customer service representative.

When an institution includes the PIN change transaction in its transaction set, the value in the PIN CHANGE ALLOWED field on IDF screen 40 identifies whether a PIN change can be performed by a customer service representative using either the CSR terminal or an RBSI-compliant device. This option permits an institution to restrict its customer service representatives from changing a PIN when the customer is not present. This field is only used for transactions acquired from an RBSI-compliant device when the source code in the transaction is set to a value of IB.

The flag controlled by the PIN CHANGE REQUIRED field on CSTT screen 1 is associated with PIN changes. This flag, which is set using file maintenance, indicates that the customer must perform a PIN change transaction before he or she can perform any other BASE24 Remote Banking transaction. This flag also controls whether a customer must select an initial PIN or change an existing PIN.

Message Authentication

Message authentication is a method used to monitor the integrity of data flowing between electronic funds transfer (EFT) devices and the transaction authorizer. A message can change as the result of tampering or because of communications difficulties. Message authentication determines whether or not the message has changed.

The ISO Host Interface process supports the generation and verification of message authentication code (MAC) values between it and a host in Racal-Guardata or Atalla hardware and in software.

The Integrated Authorization Server process does not support message authentication between it and an RBSI-compliant device or customer service representative terminal.

Dynamic Key Management

Integrated Authorization Server processes and ISO Host Interface processes use working keys to protect the PINs and message authentication codes (MACs) moving through the BASE24 network. Working keys should be changed regularly to maintain proper security. Monitoring and changing these keys is known as key management, and can be performed manually or automatically. Automated key management is known as dynamic key management.

Dynamic key management requires the use of security modules to perform PIN encryption, PIN verification, and message authentication. It is not supported when these functions are performed in software.

Dynamic key management support is provided between RBSI-compliant devices and the Remote Banking Standard Interface processes (Integrated Endpoint process, Integrated Authorization Server process, and Interface Master process), and between hosts and the ISO Host Interface process. Dynamic key management information for the Remote Banking Standard Interface processes is maintained on VCD screen 4. Dynamic key management information for the Host Interface process is maintained on KEYF screen 3.

Customer Account Selection

A BASE24 Remote Banking customer can have an unlimited number of accounts. Customer information is contained in the CSTT and displayed on CSTT screen 1. Account information other than balances and transaction histories is contained in the CACT and displayed on CSTT screen 2. Account balances are in the PBF, BASE24-telebanking transaction history information is in the THF, and BASE24-billpay transaction history information is in the History Table (HIST).

The configuration of the CACT controls the account selection options available to a customer. This subsection presents the contents of a CACT row, the indexes available for accessing the table, and how the Integrated Authorization Server process uses information in the table during online authorization.

The following combinations can be used to select an account. Two indexes can be used to search for rows in the CACT. The use of these indexes also affects the combinations that are available for selecting an account.

- The customer ID, account number, and account type fully qualify the desired account.
- The customer ID and account number identify the desired account if a customer does not have duplicate account numbers.
- The customer ID and account type identify a group of accounts for multiple account selection when the multiple account index is defined.
- The account number and account type identify the customer ID and desired account when the account number/account type index is defined.
- The account number identifies the customer ID and desired account if the CACT does not have duplicate account numbers and the account number/account type index is defined.
- The customer ID and account qualifier identify the desired account.
- The customer ID is sufficient for PIN verify and PIN change transactions, or user-defined transactions that do not have an account because no account is involved.

Customer/Account Relation Table

The Customer/Account Relation Table (CACT) contains the information described below for each account.

Customer ID	The unique identification number assigned to the owner of this account. This number appears in the CUSTOMER ID fields on CSTT screens 1 and 2.
Account Type	The type of account (e.g., savings, checking, credit). A code that identifies the account type appears in the TYPE field on CSTT screen 2. The codes used for BASE24 Remote Banking account types are based on ISO standards and are configurable. Refer to the topic “Account Type Codes” in section 1 for a list of the default account type codes provided with the BASE24 Remote Banking products. The account number/account type index permits a customer to initiate a transaction by entering the number and type of the desired account without entering a customer ID number. The Integrated Authorization Server process uses the account number and account type in the CACT to locate the customer ID number. When this index is used, each account number and account type combination in the CACT must be unique and be associated with only one customer ID. The multiple account select index permits a customer to include an account type without an account number in the original transaction request, then select the account for a transaction from a list of accounts of the type specified in the original request.
Account Number	The number assigned to the account. This number appears in the ACCOUNT NUMBER field on CSTT screen 2. When the Positive Customer with Balances/History Authorization method is being used, this account number must match the account number in the ACCOUNT NUMBER field on the PBF screens.

Continued on the following page

Account Number *continued*	The account number/account type index permits a customer to initiate a transaction by entering the number and type of the desired account without entering a customer ID number. The Integrated Authorization Server process uses the account number and account type in the CACT to locate the customer ID number. When this index is used, each account number and account type combination in the CACT must be unique and be associated with only one customer ID. The multiple account select index permits a customer to include an account type without an account number in the original transaction request, then select the account for a transaction from a list of accounts of the type specified in the original request.
Account Qualifier	The code that, when used with the customer ID, can identify a specific account without the customer having to enter an account type or account number. This code appears in the QUALIFIER field on CSTT screen 2. Each account qualifier used with one customer ID must be unique. For example, a customer with five accounts must have five different account qualifiers. However, multiple customers can use the same account qualifier because each customer has a unique customer ID. The QUALIFIER field on CSTT screen 2 must contain a unique account qualifier or an account qualifier of blanks for each account associated with a customer ID. The data processing center (DPC) or the file maintenance operator must ensure that an account qualifier is assigned to each account that should have one. The BASE24 Remote Banking table maintenance system cannot enforce this requirement because there is only one CACT. When multiple financial institutions use the CACT, some financial institutions can use account qualifiers even though other financial institutions do not. Using the BASE24 Remote Banking table maintenance system to enforce the account qualifier requirement would require financial institutions that do not use account qualifiers to enter valid qualifiers for each account anyway.

Account Status	The status assigned to the account. A code that identifies the account status appears in the STATUS field on CSTT screen 2. The Integrated Authorization Server process uses this code to determine whether an account is valid. When multiple accounts exist and multiple account selection is supported, the Integrated Authorization Server process excludes invalid accounts from the list of accounts presented to the customer for selection. When multiple accounts exist and multiple account selection is not supported, the Integrated Authorization Server process uses this code to select the appropriate account. Refer to the “Account Selection Hierarchy” topic later in this section for additional information on multiple account selection.
Account Actions Allowed	The type of transactions that can be performed involving the account. The ACT ALWD field on CSTT screen 2 has three code fields—DB (debit), CR (credit), and INQ (inquiry). The code in the DB field identifies whether the Integrated Authorization Server process can perform transactions that debit the account (i.e., transfers or payments from the account). The code in the CR field identifies whether the Integrated Authorization Server process can perform credit transactions (i.e., transfers to the account). The code in the INQ field identifies whether the Integrated Authorization Server process can perform inquiry transactions (e.g., balance inquiries or transaction history inquiries).
Account Description	A text description for the account. This description appears in the DESCRIPTION field on CSTT screen 2.
Account Beginning Date	The date on which the account becomes available for transaction processing. The Integrated Authorization Server process excludes accounts that are not yet available for processing from the list of accounts presented to the customer for selection.

Account Ending Date	The date on which the account is no longer available for transaction processing. The Integrated Authorization Server process excludes accounts that are no longer available for processing from the list of accounts presented to the customer for selection.

Account Selection Hierarchy

The Integrated Authorization Server process uses two procedures to ensure the correct CACT row is used when the request from an RBSI-compliant device or customer service representative terminal is incomplete or inconclusive.

When a processing code does not contain account type information, the Integrated Authorization Server process can modify the account types. The processing code must include account type information so the Integrated Authorization Server process can search the PCDF correctly, as well as use the correct CACT row.

When a customer has multiple accounts with the same account type range and account number, the Integrated Authorization Server process can select the savings, checking, or credit account used for authorization.

Both of these procedures involve multiple account type codes with savings, checking, and credit accounts to identify the same account type. An institution can use the additional account type codes to identify different pricing or other distinguishing factors. However, the Integrated Authorization Server process considers all codes identifying the same account type to be equal when determining how to process a transaction. Account type code ranges are as follows:

Account Type	Code	Additional Codes
Savings	10	1A through 1F
Checking	20	2A through 2H
Credit	30	3A through 3G

Both of these procedures also require that no more than one account of each type be given primary account status (a value of 3, 4, or equivalent alphabetic character in the STATUS field on Customer Table (CSTT) screen 2).

Missing Account Type Information

When the processing code does not include the account type information that typically accompanies a particular transaction code, the Integrated Authorization Server process attempts to update the account type information. For example, a check clearance inquiry, transaction code 3A, should include an account 1 type of DDA or NOW. This procedure is not performed for PIN change and PIN verify transactions because they do not involve accounts.

If the request contains an account qualifier, the Integrated Authorization Server process uses the customer ID and account qualifier to identify the correct account and update the account type information.

When the account type for the *from* account in a transfer or payment transaction is missing, the Integrated Authorization Server process searches the Customer/Account Relation Table for a row that meets one of the following requirements:

1. Matches the customer ID and account number from the transaction, has a valid date range, and has an account status of open primary account.
2. Matches the customer ID and account number from the transaction, has a valid date range, and has an account status of open account.

As soon as a requirement is met, the search ends and the Integrated Authorization Server process uses the account type from the CACT row to update the *from* account portion of the processing code. If the account type in the CACT row is one of the additional codes for savings (1A through 1F), checking (2A through 2H), or credit (3A through 3G), the Integrated Authorization Server process changes the second character of the account type to zero.

If neither requirement is met, the Integrated Authorization Server process leaves the *from* account portion of the processing code set to 00.

When the *to* account type in a transfer transaction or the single account type in any other transaction is missing, the Integrated Authorization Server process searches the CACT for a row that meets one of the following requirements:

1. Matches the customer ID and account number from the transaction, has a valid date range, and has an account status of open primary account.
2. Matches the customer ID and account number from the transaction, has a valid date range, and has an account status of restricted primary account.

3. Matches the customer ID and account number from the transaction, has a valid date range, and has an account status of open account.
4. Matches the customer ID and account number from the transaction, has a valid date range, and has an account status of restricted account.

As soon as a requirement is met, the search ends and the Integrated Authorization Server process uses the account type from the CACT row to update the appropriate portion of the processing code. If the account type in the CACT row is one of the additional codes for savings (1A through 1F), checking (2A through 2H), or credit (3A through 3G), the Integrated Authorization Server process changes the second character of the account type to zero.

If none of these requirements are met, the Integrated Authorization Server process leaves this portion of the processing code set to 00.

Account Numbers with Duplicate Account Type Range

When multiple accounts have the same account type range and account number and the request does not contain an account qualifier, the Integrated Authorization Server process attempts to select the correct CACT row. For example, an account with type 1A and number 1234 and an account with type 1D and number 1234 are handled as accounts with type 10 and number 1234. This procedure is not performed when the request contains an account qualifier because the CACT row can be located using the customer ID and the account qualifier.

The Integrated Authorization Server process can identify the correct CACT row for the following accounts:

- The *from* account for a transfer transaction or payment transaction.
- The *to* account for a transfer transaction.
- The single account for a standard inquiry or history transaction.
- Each account for a user-defined transaction.

Each account has its own set of requirements, which are listed on the following page. As soon as a requirement is met, the search ends and the Integrated Authorization Server process uses the selected account for the transaction. If an account does not meet any of its requirements, the transaction is declined.

When multiple *from* accounts for a transfer or payment transaction have the same account type range and account number, the Integrated Authorization Server process searches the CACT for a row that meets one of the following requirements:

1. Open primary account that has a valid date range and allows transactions that debit the account.
2. Open account that has a valid date range and allows transactions that debit the account.

When multiple *to* accounts for a transfer transaction have the same account type range and account number, the Integrated Authorization Server process searches the CACT for a row that meets one of the following requirements:

1. Open primary account that has a valid date range and allows credit transactions.
2. Restricted primary account that has a valid date range and allows credit transactions.
3. Open account that has a valid date range and allows credit transactions.
4. Restricted account that has a valid date range and allows credit transactions.

When multiple accounts for a standard inquiry or history transaction have the same account type range and account number, the Integrated Authorization Server process searches the CACT for a row that meets one of the following requirements:

1. Open primary account that has a valid date range and allows inquiry transactions.
2. Restricted primary account that has a valid date range and allows inquiry transactions.
3. Open account that has a valid date range and allows inquiry transactions.
4. Restricted account that has a valid date range and allows inquiry transactions.

When multiple accounts for a user-defined transaction have the same account type range and account number, the Integrated Authorization Server process searches the CACT for a row that meets one of the following requirements:

1. Open primary account that has a valid date range.
2. Restricted primary account that has a valid date range.
3. Open account that has a valid date range.
4. Restricted account that has a valid date range.

Transaction Routing

Routing a transaction involves the following:

- Determining whether an institution supports a particular processing code.
- Identifying the authorization level and method, including the host destinations, if applicable.
- Determining whether an advice must be sent to the host when the Integrated Authorization Server process authorizes the transaction.

The information used to perform transaction routing is obtained from the following files:

- Institution Definition File (IDF)
- Processing Code Definition File (PCDF)
- Institution Routing Configuration File (IRCF)
- Internal Message Translation File (IMTF)

The function of each file is described below and on the following pages.

Institution Definition File

The IDF record for each institution using a BASE24 product can include the name of one route profile. The configuration examples that follow in this section contain the following IDF fields:

FIID	The FIID of the financial institution.
ROUTE PROFILE	Part of the key used by the Integrated Authorization Server process to select the IRCF and PCDF records for customers of this financial institution. The default value for this field is the FIID of the financial institution.

Processing Code Definition File

The PCDF contains a record for each processing code in each route profile. For example, if there are five route profiles and each route profile has 250 processing codes, the file contains 1250 records plus the default records provided with each of the BASE24 Remote Banking products.

The default PCDF records are used primarily to load a new route profile. However, one or more of the default values in these records must be changed before the records can be used in actual route profiles. The default PCDF records also are displayed on the ITS Transaction Log File (ITLF) perusal screens, certain customer service representative screens, and are used to generate customer activity reports.

Each PCDF record permits institutions using the route profile to do the following:

- Specify whether the transaction is allowed. This is a yes or no option. The default value is no.
- Assign a route code. The route code is used as part of the key to the IRCF.

  The route code in a PCDF record can be a specific value or a wildcard value. The following table shows the selection hierarchy used by the Integrated Authorization Server process to search for a PCDF record. The default value for the route code in the PCDF is the wildcard value of asterisks.

Selection Hierarchy	PRODUCT ID	ROUTE PROFILE	TRANSACTION CODE	ACCOUNT 1 TYPE	ACCOUNT 2 TYPE
1	Match	Match	Match	Match	Match
2	Match	********	Match	Match	Match

  In the case of BASE24-billpay transactions acquired from hosts, RBSI-compliant devices, and customer service representative terminals, the route code must be a value of BPƀƀ, where ƀ indicates a blank character. This includes transaction codes 3H, 3J–3M, 9A–9N, 9P–9T, 9V–9X, B1–B2, B4–B9, and BB–BE. Transaction codes B0 and B3 are internal transactions and do not need to be configured in the PCDF. Transaction codes BF and BG are never routed, and therefore, also do not need to be configured in the PCDF. Transaction codes 4A, 4B, 50, 5A, and 5B also are BASE24-billpay transactions; however, these transactions are generated by the XPNET Billpay Server process or the Scheduled Transaction Initiator process and do not use the BPƀƀ route code.

- Specify when an advice is required. When the Integrated Authorization Server process authorizes a transaction, it can send an advice to the host for approved transactions, declined transactions, all transactions, or no transactions. The default value is not to send advices for any transactions. In the case of transfer transactions that involve one host for the *from* account and another host for the *to* account, an advice can be sent to each host.
- Assign the transaction description that appears on ITS Transaction Log File (ITLF) perusal screens, certain customer service representative screens, and customer activity reports. Refer to the PCDF section of the ***BASE24 Core Files and Tables Maintenance Manual*** for the transaction descriptions associated with the default records.

The configuration examples that follow in this section contain the following PCDF fields:

PROD ID	A code identifying the BASE24 product for which the processing code is defined. The valid value is 14 (BASE24-telebanking) for both BASE24 Remote Banking products.
ROUTE PROFILE	A code identifying the institution or institutions that use the processing code.
PROCESSING CODE	A three-part code identifying the transaction code and account types involved. In the configuration examples that follow, each part of the code is two characters. The transaction codes used in the configuration examples are used on the PCDF screens. However, the two account types used in the configuration examples are replaced on PCDF screens by 1–6 character codes that are defined in the Account Type Table File (ATT). The “Account Type Codes” discussion in section 1 provides a cross-reference between the two character codes and the 1–6 character codes.
TRANSACTION ALLOWED	A code indicating whether the transaction being defined is allowed.

ROUTE CODE	A value used to group transactions for routing and authorization purposes when multiple destinations must be defined in the IRCF for a processing code or group of processing codes. The reserved value of BPb̸b̸ is used for BASE24-billpay transactions acquired from hosts, RBSI-compliant devices, and customer service representative terminals.
ADVICE REQUIRED	A code identifying the type of transactions for which the Integrated Authorization Server process sends an advice to the host after authorizing a transaction. Valid values are as follows: A = Send advices for approved transactions only. B = Send advices for approved and denied transactions. D = Send advices for denied transactions only. N = Do not send advices.

Institution Routing Configuration File

The IRCF can contain a record for each combination of route profile, route code, and account type used. The route profile is the same one defined in the IDF and used in the PCDF. The route code ties all processing codes with the same route profile and account type to a common set of routing and authorization parameters. The account type identifies the type of account involved.

There can be one or two IRCF records involved in a transaction, depending on the number of accounts involved in the transaction. Transfer transactions use one record for the *from* account and one record for the *to* account. Other transactions use one IRCF record.

The route profile, route code, and account type in IRCF records can be specific values or wildcard values (i.e., asterisks). The following table illustrates the hierarchy used by the Integrated Authorization Server process to search for an IRCF record. The wildcard values in the ACCOUNT TYPE column of this table match the 1–6 character values that must be entered on IRCF screen 1.

Selection Hierarchy	ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE
1	Match	Match	Match
2	Match	Match	******

Selection Hierarchy	ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE
3	Match	****	Match
4	********	Match	Match
5	Match	****	******
6	********	Match	******
7	********	****	Match
8	********	****	******

The number of IRCF records necessary depends on the number of specific route profiles, route codes, and account types used, as well as whether the BASE24-billpay product is installed. For example, if all transactions are authorized the same way, one BASE24-telebanking IRCF record is needed because the route profile, route code, and account type can be given wildcard values. In another example, if all savings accounts go to one host and all other accounts are authorized by the Integrated Authorization Server process, one BASE24-telebanking IRCF record is needed for the savings accounts and one BASE24-telebanking IRCF record is needed for all other accounts. In each of these examples, one more record must be added to the IRCF for routing BASE24-billpay transactions to the XPNET Billpay Server process.

The more detailed the routing requirements become, the more IRCF records are required.

Each IRCF record permits institutions to do the following:

- Specify the authorization level.
- Specify the authorization method.
- Specify the Internal Message Translation File (IMTF) record that identifies the ISO Host Interface process or XPNET Billpay Server process to receive the message from the Integrated Authorization Server process or to send the message to the Integrated Authorization Server process.
- Specify the object used by the Integrated Authorization Server process to communicate with the DPC or XPNET Billpay Server process.

The configuration examples that follow in this section contain the following IRCF fields:

ROUTE PROFILE	A code identifying the institution or institutions that use this configuration.
ROUTE CODE	A value used to group transactions for routing and authorization purposes when multiple destinations must be defined for an institution in the IRCF.
ACCOUNT TYPE	A code identifying the type of account involved in the transaction. The account types used in the configuration examples are replaced on IRCF screens by 1–6 character codes that are defined in the Account Type Table File (ATT). The "Account Type Codes" discussion in section 1 provides a cross-reference between the two character codes and the 1–6 character codes.
AUTH LEVEL	A code identifying the authorization level for transactions that use this configuration. Valid values are as follows: 1 = Online 2 = Offline 3 = Online/offline A value of 1 (online) must be used for transactions authorized by BASE24-billpay or for BASE24-telebanking transactions that can only be authorized on a host.
AUTH METHOD	A code identifying the authorization method for transactions that use this configuration. Valid values are as follows: PCA = Positive Customer Authorization PCAA = Positive Customer with Accounts Authorization PCBA = Positive Customer with Balances/History Authorization

AUTH DESTINATION NAME	The name of the IMTF record that identifies the Host Interface process or XPNET BASE24-billpay Server process to exchange messages with the Integrated Authorization Server process. This value is required for each of the following conditions: • The ROUTE CODE field contains a value of BPb̸b̸. • The AUTH LEVEL field contains a value of 1 or 3. • Advices are to be sent to the DPC for transactions authorized by the Integrated Authorization Server process. This value is optional if the AUTH LEVEL field contains a value of 2.

Internal Message Translation File

The IMTF is used any time the Integrated Authorization Server process needs to communicate with an external interface process or server process that is not object-oriented. Each IMTF record identifies the name of the process that exchanges messages with the Integrated Authorization Server process.

Note: The value in one field on IMTF screen 1—INTERFACE DESTINATION—can be the origination point if the Integrated Authorization Server process is receiving the message or the destination point if the Integrated Authorization Server process is sending the message.

BASE24-telebanking

In a BASE24-telebanking configuration, each ISO Host Interface process is given a unique value, such as HOST1, in the INTERFACE DESTINATION field on IMTF screen 1. The same value is used in the AUTH DESTINATION NAME field on Institution Routing Configuration File (IRCF) screen 1 for each routing configuration that uses the ISO Host Interface process. The PROCESS NAME field on IMTF screen 1 contains the symbolic name, such as P1A^HISF01, assigned to the process in the XPNET system. No other processes or stations are identified in the IMTF.

BASE24-billpay

In a BASE24-billpay configuration, the following processes are identified in the IRCF and IMTF:

- Each ISO Host Interface process is given a unique value, such as HOST1, in the INTERFACE DESTINATION field on IMTF screen 1. The same value is used in the AUTH DESTINATION NAME field on IRCF screen 1 for each routing configuration that uses the ISO Host Interface process.
- Each XPNET Billpay Server process is given a unique value, such as BPSRV, in the INTERFACE DESTINATION field on IMTF screen 1. The same value is used in the AUTH DESTINATION NAME field on IRCF screen 1 for each routing configuration that uses the XPNET Billpay Server process.

The PROCESS NAME field on IMTF screen 1 contains the symbolic name, such as P1A^HISF01 or P1A^BPSV01, assigned to each process in the XPNET system.

Each of the following processes and stations are also identified in the IMTF without being identified in the IRCF:

- XPNET Billpay Server process
- Scheduled Transaction Initiator process
- BASE24-billpay customer service representative terminal station

These processes and stations originate messages bound for the Integrated Authorization Server process. The Integrated Authorization Server process uses these records to validate a station or process when an inbound message is received from a source that is not object-oriented. In the IMTF records for these processes and stations, the INTERFACE DESTINATION field on IMTF screen 1 and the PROCESS NAME field on IMTF screen 1 contain the same value—the symbolic name, such as P1A^BPSV01, P1A^STI01, or S1A^BP01, assigned to the process or station in the XPNET system.

Note: Each XPNET Billpay Server process appears in two IMTF records—one that is identified in the IRCF and one that is not.

Transaction Routing Configuration Examples

The following examples illustrate how the IDF, PCDF, and IRCF can be configured to achieve various transaction routing objectives for three financial institutions. Tables are used to describe the file content and only pertinent fields from each file are included. These fields are described in the "Transaction Routing" discussion presented earlier in this section. The PCDF tables contain only a subset of the BASE24 Remote Banking transaction set. The IMTF is not included in these examples.

Example 1 illustrates a BASE24-telebanking configuration with all transactions handled the same. Example 2 illustrates the changes necessary to add BASE24-billpay to the BASE24-telebanking configuration in example 1. The remaining examples illustrate BASE24-telebanking configurations because the differences from one example to another deal with the authorization performed by the Integrated Authorization Server process without affecting the XPNET Billpay Server process.

Example 1: All Transactions Handled the Same

In this example, all institutions support the same set of processing codes, and advices are not required. All transactions are authorized offline using the Positive Customer with Balances/History Authorization (PCBA) method. BASE24-billpay processing codes cannot be configured in this example because they have to be handled in a different manner than BASE24-telebanking processing codes.

Institution Definition File

The route profile is defined for each institution in the IDF, as shown in the following table. The route profile for each financial institution matches the FIID because the ROUTE PROFILE field on IDF screen 40 uses the FIID as a default value. All three financial institutions could use the same route profile in this configuration.

FIID	ROUTE PROFILE
BNK1	BNK1
BNK2	BNK2
BNK3	BNK3

Processing Code Definition File

One set of PCDF records can be used to route all transactions, as shown in the following table. The wildcard value (asterisks) in the ROUTE PROFILE field matches all route profiles defined in the IDF.

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	********	301000	Y	0000	N
14	********	302000	Y	0000	N
14	********	303000	Y	0000	N
14	********	401020	Y	0000	N
14	********	401030	Y	0000	N
14	********	402010	Y	0000	N
14	********	402030	Y	0000	N
14	********	403010	Y	0000	N
14	********	403020	Y	0000	N

Institution Routing Configuration File

One IRCF record is defined, as shown in the following table. Only one IRCF record is needed because all transactions are handled the same. The authorization destination is not specified since communication with the host is not required.

ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE	AUTH LEVEL	AUTH METHOD	AUTH DESTINATION NAME
********	****	**	2	PCBA	

Example 2: BASE24-billpay Transactions

In this example, all institutions support the same set of processing codes, and advices are not required. All BASE24-telebanking transactions are authorized offline using the Positive Customer with Balances/History Authorization (PCBA) method.

This example is the same as example 1 with the addition of BASE24-billpay transactions. BASE24-billpay processing codes must be configured separately because they are authorized offline by the XPNET Billpay Server process, regardless of how BASE24-telebanking transactions are authorized.

BASE24-billpay transactions can be added to the BASE24-telebanking configurations illustrated in examples 3 through 11 as long as all BASE24-billpay transactions except transaction codes 4A, 4B, 50, 5A, and 5B are configured with the following:

- A route code of BPb̸b̸ in the PCDF
- An entry for route code BPb̸b̸ in the IRCF with authorization level 1 and a destination name that matches the IMTF record for the XPNET Billpay Server process

Refer to the “Internal Message Translation File (IMTF)” topic earlier in this section for BASE24-billpay IMTF configuration requirements.

Institution Definition File

The route profile is defined for each institution in the IDF, as shown in the following table. The route profile for each financial institution matches the FIID because the ROUTE PROFILE field on IDF screen 40 uses the FIID as a default value. All three financial institutions could use the same route profile in this configuration.

FIID	ROUTE PROFILE
BNK1	BNK1
BNK2	BNK2
BNK3	BNK3

Processing Code Definition File

One set of PCDF records can be used to route all transactions, as shown in the following table. The wildcard value (asterisks) in the ROUTE PROFILE field matches all route profiles defined in the IDF. In this example, processing codes with the 9A transaction code are BASE24-billpay nonfinancial transactions that must be routed to the XPNET Billpay Server process. Processing codes with the 50 transaction code are BASE24-billpay financial transactions that are generated by the XPNET Billpay Server process and routed to the Integrated Authorization Server process.

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	********	301000	Y	0000	N
14	********	302000	Y	0000	N
14	********	303000	Y	0000	N
14	********	401020	Y	0000	N

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	********	401030	Y	0000	N
14	********	402010	Y	0000	N
14	********	402030	Y	0000	N
14	********	403010	Y	0000	N
14	********	403020	Y	0000	N
14	********	501000	Y	0000	N
14	********	502000	Y	0000	N
14	********	503000	Y	0000	N
14	********	9A1000	Y	BPb̸b̸	N
14	********	9A2000	Y	BPb̸b̸	N
14	********	9A3000	Y	BPb̸b̸	N

Institution Routing Configuration File

Two IRCF records are defined, as shown in the following table. The second IRCF record is needed because BASE24-billpay transactions require a separate IRCF record with a route code of BPb̸b̸ and an authorization level of 1, regardless of how BASE24-telebanking transactions are handled. The authorization destination for BASE24-telebanking transactions is not specified since communication with the host is not required. The authorization destination for the XPNET Billpay Server process is required for BASE24-billpay transactions.

ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE	AUTH LEVEL	AUTH METHOD	AUTH DESTINATION NAME
********	****	**	2	PCBA	
********	BPb̸b̸	**	1	PCAA	BPSRV

Example 3: Transactions Routed by FIID

In this example, all institutions support the same set of processing codes, advices are not required, and all transactions are authorized online/offline using the Positive Customer with Balances/History Authorization (PCBA) method. Transactions are routed and authorized differently depending on the issuer FIID.

Institution Definition File

The route profile is defined for each institution in the IDF, as shown in the following table. Each institution must have a unique route profile because each institution has a unique authorization destination.

FIID	ROUTE PROFILE
BNK1	BNK1
BNK2	BNK2
BNK3	BNK3

Processing Code Definition File

One set of PCDF records can be used to route all transactions, as shown in the following table. The wildcard value (asterisks) in the ROUTE PROFILE field matches all route profiles defined in the IDF.

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	********	301000	Y	0000	N
14	********	302000	Y	0000	N
14	********	303000	Y	0000	N
14	********	401020	Y	0000	N
14	********	401030	Y	0000	N
14	********	402010	Y	0000	N
14	********	402030	Y	0000	N
14	********	403010	Y	0000	N
14	********	403020	Y	0000	N

Institution Routing Configuration File

One IRCF record must be defined for each route profile because each route profile has a unique authorization destination, as shown in the following table. The authorization name identifies the IMTF record that contains specific host destination information.

If BASE24-billpay transactions are added to the PCDF, only one additional IRCF record—for a route code of BPƀƀ and an authorization level of 1—is needed even though there are three separate hosts. Transactions such as schedule payment and schedule transfer can be routed to one XPNET Billpay Server process because they are forwarded to the Integrated Authorization Server process before being routed to a host. For example, all three financial institutions can use the same PCDF record for transaction code 9A, then use individual PCDF records for transaction code 50. Transaction code 9A goes to the XPNET Billpay Server process and transaction code 50 goes to the specific hosts.

ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE	AUTH LEVEL	AUTH METHOD	AUTH DESTINATION NAME
BNK1	****	**	3	PCBA	HOST1
BNK2	****	**	3	PCBA	HOST2
BNK3	****	**	3	PCBA	HOST3

Example 4: Transactions Routed by Transaction Type

In this example, all institutions support the same set of processing codes and associated options. However, the transactions are routed and authorized differently depending on the type of transaction. For example, the institutions have elected to maintain the THF on the host. As a result, all transactions that access the THF are authorized online using the Positive Customer Authorization (PCA) method and all other transactions are authorized offline using the Positive Customer with Balances/History Authorization (PCBA) method with advices for approved transactions sent to the host. Each institution uses a different host.

Institution Definition File

The route profile is defined for each institution in the IDF, as shown in the following table. Like example 3, each institution must have a unique route profile because each institution has a unique authorization destination.

FIID	ROUTE PROFILE
BNK1	BNK1
BNK2	BNK2
BNK3	BNK3

Processing Code Definition File

One set of PCDF records can be used to route all transactions, as shown in the following table. The route code is used to identify a unique routing configuration for transactions that require access to the THF. The advice required flag is set appropriately depending on the route code.

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	********	301000	Y	0000	A
14	********	302000	Y	0000	A
14	********	303000	Y	0000	A
14	********	3B1000	Y	0001	N
14	********	3B2000	Y	0001	N
14	********	401020	Y	0000	A
14	********	401030	Y	0000	A

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	********	402010	Y	0000	A
14	********	402030	Y	0000	A
14	********	403010	Y	0000	A

Institution Routing Configuration File

Two IRCF records are defined for each institution using their unique route profile, as shown in the following table. The first IRCF record contains a specific route code that is used to route all transactions that require access to the THF. The second IRCF record contains a route code with a wildcard value and is used to route all transactions that do not require access to the THF.

ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE	AUTH LEVEL	AUTH METHOD	AUTH DESTINATION NAME
BNK1	0001	**	1	PCA	HOST1
BNK1	****	**	2	PCBA	HOST1
BNK2	0001	**	1	PCA	HOST2
BNK2	****	**	2	PCBA	HOST2
BNK3	0001	**	1	PCA	HOST3
BNK3	****	**	2	PCBA	HOST3

Example 5: Transactions Routed by Account Type

In this example, all institutions support the same set of processing codes and associated options. However, the transactions are routed and authorized differently depending on the account type involved in the transaction. For example, these institutions have elected to maintain their credit account information on the host. Each institution uses a different host.

As a result, all single-account transactions that involve a credit account are authorized online using the Positive Customer Authorization (PCA) method. All other single-account transactions are authorized offline using the Positive Customer with Balances/History Authorization (PCBA) method with advices for approved transactions sent to the host. For transfer transactions, the account type of each account is used to determine the authorization method used for each portion of the transaction.

Institution Definition File

The route profile is defined for each institution in the IDF, as shown in the following table.

FIID	ROUTE PROFILE
BNK1	BNK1
BNK2	BNK2
BNK3	BNK3

Processing Code Definition File

One set of PCDF records can be used to route all transactions, as shown in the following table. Because the account type is used to route transactions, the route code is set to the default value. The advice required flag is set appropriately depending on the account type. For single-account transactions, the advice required flag must be set for noncredit accounts, but not for credit accounts. For transfer transactions, the advice required flag must be set if either account involved in the transaction is a noncredit account.

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	********	301000	Y	0000	A
14	********	302000	Y	0000	A
14	********	303000	Y	0000	N
14	********	3B1000	Y	0000	A
14	********	3B2000	Y	0000	A
14	********	3B3000	Y	0000	N
14	********	401020	Y	0000	A
14	********	401030	Y	0000	A
14	********	402010	Y	0000	A
14	********	402030	Y	0000	A
14	********	403010	Y	0000	A
14	********	403020	Y	0000	A

Institution Routing Configuration File

Two IRCF records are defined for each institution using their unique route profile, as shown in the following table. The first IRCF record contains a specific credit account type. The second IRCF record contains an account type with a wildcard value and is used to route all transactions that do not involve a credit account type.

ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE	AUTH LEVEL	AUTH METHOD	AUTH DESTINATION NAME
BNK1	****	30	1	PCA	HOST1
BNK1	****	**	2	PCBA	HOST1
BNK2	****	30	1	PCA	HOST2
BNK2	****	**	2	PCBA	HOST2
BNK3	****	30	1	PCA	HOST3
BNK3	****	**	2	PCBA	HOST3

Example 6: Different Processing Codes for Each FIID

In this example, each institution supports a different set of processing codes, but all transactions are authorized offline using the Positive Customer with Balances/History Authorization (PCBA) method. Advices to the host are not required.

Institution Definition File

The route profile is defined for each institution in the IDF, as shown in the following table.

FIID	ROUTE PROFILE
BNK1	BNK1
BNK2	BNK2
BNK3	BNK3

Processing Code Definition File

Three sets of PCDF records are used to route the transactions, as shown in the following table. A set of PCDF records is defined for each institution using their unique route profile.

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	BNK1	301000	Y	0000	N
14	BNK2	302000	Y	0000	N
14	BNK3	303000	Y	0000	N
14	BNK1	3B1000	Y	0000	N
14	BNK2	3B2000	Y	0000	N
14	BNK3	3B3000	Y	0000	N
14	BNK1	401020	Y	0000	N
14	BNK2	401030	Y	0000	N

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	BNK3	402010	Y	0000	N
14	BNK1	402030	Y	0000	N
14	BNK2	403010	Y	0000	N
14	BNK3	403020	Y	0000	N

Institution Routing Configuration File

One IRCF record is defined, as shown in the following table. Only one IRCF record is needed because all transactions are handled the same.

ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE	AUTH LEVEL	AUTH METHOD	AUTH DESTINATION NAME
********	****	**	2	PCBA	

Example 7: Two Sets of Processing Codes Used by Three FIIDs

In this example, institutions 1 and 2 support the same set of processing codes and institution 3 supports a different set of processing codes. All transactions are authorized offline using the Positive Customer with Balances/History Authorization (PCBA) method. Advices to the host are not required.

Institution Definition File

The route profile is defined for each institution in the IDF, as shown in the following table. Institutions 1 and 2 share a common route profile because they share a common transaction set.

FIID	ROUTE PROFILE
BNK1	BNK12
BNK2	BNK12
BNK3	BNK3

Processing Code Definition File

Two sets of PCDF records are used to route the transactions. One set of PCDF records is shared by institutions 1 and 2 using a common route profile. A second set is defined for institution 3 using a unique route profile.

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	BNK12	301000	Y	0000	N
14	BNK3	301000	Y	0000	N
14	BNK12	302000	Y	0000	N
14	BNK3	302000	Y	0000	N
14	BNK12	303000	Y	0000	N
14	BNK3	303000	Y	0000	N
14	BNK12	3B1000	Y	0000	N
14	BNK12	3B2000	Y	0000	N

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	BNK12	3B3000	Y	0000	N
14	BNK12	401020	Y	0000	N
14	BNK12	401030	Y	0000	N
14	BNK12	402010	Y	0000	N
14	BNK12	402030	Y	0000	N
14	BNK12	403010	Y	0000	N
14	BNK12	403020	Y	0000	N

Institution Routing Configuration File

One IRCF record is defined, as shown in the following table. Only one IRCF record is needed because all transactions are handled the same.

ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE	AUTH LEVEL	AUTH METHOD	AUTH DESTINATION NAME
********	****	**	2	PCBA	

Example 8: Additional Processing Codes Used by One FIID

In this example, all institutions support the same basic set of processing codes, but institution 3 has elected to support three additional processing codes. All transactions are authorized offline using the Positive Customer with Balances/ History Authorization (PCBA) method. Advices to the host are not required.

Institution Definition File

The route profile is defined for each institution in the IDF, as shown in the following table. The route profile for each financial institution matches the FIID because the ROUTE PROFILE field on IDF screen 40 uses the FIID as a default value. Institutions 1 and 2 could use the same route profile in this configuration.

FIID	ROUTE PROFILE
BNK1	BNK1
BNK2	BNK2
BNK3	BNK3

Processing Code Definition File

One set of PCDF records can be used to route most transactions, as shown in the following table. The first three records listed are the additional records defined with specific route profile entries used by institution 3.

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	BNK3	3C1000	Y	0000	N
14	BNK3	3C2000	Y	0000	N
14	BNK3	3C3000	Y	0000	N
14	********	301000	Y	0000	N
14	********	302000	Y	0000	N
14	********	303000	Y	0000	N
14	********	3B1000	Y	0000	N
14	********	3B2000	Y	0000	N
14	********	3B3000	Y	0000	N
14	********	401020	Y	0000	N
14	********	401030	Y	0000	N
14	********	402010	Y	0000	N
14	********	402030	Y	0000	N
14	********	403010	Y	0000	N
14	********	403020	Y	0000	N

Institution Routing Configuration File

One IRCF record is defined, as shown in the following table. Only one IRCF record is needed because all transactions are handled the same.

ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE	AUTH LEVEL	AUTH METHOD	AUTH DESTINATION NAME
********	****	**	2	PCBA	

Example 9: Fewer Processing Codes Used by One FIID

In this example, the institutions support the same basic set of processing codes and all transactions are authorized offline using the Positive Customer with Balances/ History Authorization (PCBA) method. However, institutions 1 and 2 support three processing codes not supported by institution 3.

There are several ways this could be configured. Institution 3 could be configured with its own set of processing codes using its own route profile. Institutions 1 and 2 could then share a set of processing codes. Another alternative would be to allow all three institutions to share the common set using the default route profile of asterisks and define the three additional processing codes supported by institutions 1 and 2 in a shared route profile. The second alternative is the one described in the following configuration.

Institution Definition File

The route profile is defined for each institution in the IDF, as shown in the following table. Institutions 1 and 2 share a common route profile because they share a common transaction set.

FIID	ROUTE PROFILE
BNK1	BNK12
BNK2	BNK12
BNK3	BNK3

Processing Code Definition File

One set of PCDF records can be used to route the transactions supported by all three institutions. The first three records listed are the additional records defined with the specific route profile shared by institutions 1 and 2 only.

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	BNK12	3C1000	Y	0000	N
14	BNK12	3C2000	Y	0000	N
14	BNK12	3C3000	Y	0000	N
14	********	301000	Y	0000	N
14	********	302000	Y	0000	N
14	********	303000	Y	0000	N
14	********	3B1000	Y	0000	N
14	********	3B2000	Y	0000	N

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	********	3B3000	Y	0000	N
14	********	401020	Y	0000	N
14	********	401030	Y	0000	N
14	********	402010	Y	0000	N
14	********	402030	Y	0000	N
14	********	403010	Y	0000	N
14	********	403020	Y	0000	N

Institution Routing Configuration File

One IRCF record is defined, as shown in the following table. Only one IRCF record is needed because all transactions are handled the same.

ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE	AUTH LEVEL	AUTH METHOD	AUTH DESTINATION NAME
********	****	**	2	PCBA	

Example 10: Some Processing Codes Authorized Differently by One FIID

In this example, all institutions support the same basic set of processing codes and authorize these transactions offline using the Positive Customer with Balances/History Authorization (PCBA) method. However, institution 3 wants to support a subset of the transfer transactions using the Positive Customer Authorization (PCA) method. Advices to the host are not required.

Institution Definition File

The route profile is defined for each institution in the IDF, as shown in the following table. The route profile for each financial institution matches the FIID because the ROUTE PROFILE field on IDF screen 40 uses the FIID as a default value. Institutions 1 and 2 could use the same route profile in this configuration.

FIID	ROUTE PROFILE
BNK1	BNK1
BNK2	BNK2
BNK3	BNK3

Processing Code Definition File

One set of PCDF records can be used to route most transactions, as shown in the following table. Additional records are defined for the transfer transactions that institution 3 is supporting using the Positive Customer Authorization (PCA) method. These records contain the route profile for institution 3 and a unique route code.

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	BNK3	401030	Y	0003	N
14	BNK3	402030	Y	0003	N
14	BNK3	403010	Y	0003	N
14	BNK3	403020	Y	0003	N
14	********	301000	Y	0000	N
14	********	302000	Y	0000	N
14	********	303000	Y	0000	N
14	********	3B1000	Y	0000	N
14	********	3B2000	Y	0000	N
14	********	3B3000	Y	0000	N
14	********	3C1000	Y	0000	N
14	********	3C2000	Y	0000	N
14	********	3C3000	Y	0000	N
14	********	401020	Y	0000	N
14	********	401030	Y	0000	N
14	********	402010	Y	0000	N
14	********	402030	Y	0000	N
14	********	403010	Y	0000	N
14	********	403020	Y	0000	N

Institution Routing Configuration File

Two IRCF records must be defined, as shown in the following table. The first IRCF record is used to route the special transfer transactions for institution 3. The second IRCF record is used to route all transactions except the subset defined by institution 3.

ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE	AUTH LEVEL	AUTH METHOD	AUTH DESTINATION NAME
BNK3	0003	**	2	PCA	
********	****	**	2	PCBA	

Example 11: Different Processing Codes and Authorization Configuration Used by Each FIID

In this example, each institution supports a unique set of processing codes and a unique routing and authorization configuration, as follows:

- Institution 1 authorizes all transactions offline using the Positive Customer with Balances/History Authorization (PCBA) method. No advices are sent to the host.
- Institution 2 authorizes all transactions involving checking and savings accounts offline using the Positive Customer with Accounts Authorization (PCAA) method and sends advices for both approved and declined transactions to the host. Institution 2 has three different host destinations depending on the account type. Other transactions are authorized online using the Positive Customer Authorization (PCA) method.
- Institution 3 authorizes all transactions involving the THF online using the Positive Customer Authorization (PCA) method. Institution 3 authorizes all other transactions online/offline using the Positive Customer with Balances/History Authorization (PCBA) method and sends advices for approved transactions to the host.

Institution Definition File

The route profile is defined for each institution in the IDF, as shown in the following table. Each institution must have its own route profile because of differences in processing codes and authorization configuration.

FIID	ROUTE PROFILE
BNK1	BNK1
BNK2	BNK2
BNK3	BNK3

Processing Code Definition File

The PCDF must contain a set of records with a unique route profile for each institution, as shown in the following table.

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	BNK1	303000	Y	0000	N
14	BNK1	3B1000	Y	0000	N
14	BNK1	401020	Y	0000	N
14	BNK1	402030	Y	0000	N
14	BNK2	303000	Y	0000	N
14	BNK2	301000	Y	0000	B
14	BNK2	3B2000	Y	0000	B
14	BNK2	401030	Y	0000	B

PROD ID	ROUTE PROFILE	PROCESSING CODE	TRANSACTION ALLOWED	ROUTE CODE	ADVICE REQUIRED
14	BNK2	403010	Y	0000	B
14	BNK3	302000	Y	0000	A
14	BNK3	3B3000	Y	0003	N
14	BNK3	402010	Y	0000	A
14	BNK3	403020	Y	0000	A

Institution Routing Configuration File

IRCF records are defined for each institution using their unique route profile, as shown in the table on the following page. One IRCF record is defined for institution 1 because all transactions are handled the same.

Three IRCF records are defined for institution 2. The first record is used to route transactions involving a savings account (account type 10). The second record is used to route transactions involving a checking account (account type 20). The third record is used to route all transactions not involving a checking or savings account.

Two IRCF records are defined for institution 3. The first record contains a specific route code that is used to route all transactions that require access to the THF. The second record is used to route all transactions that do not require access to the THF.

ROUTE PROFILE	ROUTE CODE	ACCOUNT TYPE	AUTH LEVEL	AUTH METHOD	AUTH DESTINATION NAME
BNK1	****	**	2	PCBA	
BNK2	****	10	2	PCAA	HSAV
BNK2	****	20	2	PCAA	HDDA
BNK2	****	**	1	PCA	HOST2
BNK3	0003	**	1	PCA	HOST3
BNK3	****	**	3	PCBA	HOST3

User-Defined Transactions

The BASE24-telebanking product supports a mechanism that allows customers to define their own transactions. These transactions, called user-defined transactions, are outside the set of standard BASE24-telebanking transactions. User-defined transactions are identified with a transaction code of U*x*, where *x* represents a value from 0 through 9 or A through Z, and cannot be added to the standard BASE24-billpay transaction set.

User-defined transactions can include new types of transactions or they can include variations of standard BASE24-telebanking transactions such as a balance inquiry for a mutual fund account to be authorized on the host. User-defined transactions can be acquired from an RBSI-compliant device, host, or customer service representative terminal and then routed and authorized by the Integrated Authorization Server process without any required BASE24 software modifications. User-defined transactions can also be routed to and authorized on the host. BASE24 authorization of user-defined transactions includes validation of the customer and customer PIN and the optional validation of the account and account status. The PBF and THF are not accessed when authorizing user-defined transactions.

File Maintenance Requirements

The BASE24-telebanking product uses a number of files and tables to configure user-defined transactions. These files and tables and their configuration requirements are described below. The files and tables are listed in alphabetical order. Refer to the ***BASE24 Core Files and Tables Maintenance Manual*** for screen layouts and field descriptions for these files and tables.

Account Type Table File

All account types specified in BASE24-telebanking processing codes must be defined in the Account Type Table File (ATT). The ATT allows definition of an alphanumeric account type with a corresponding account description. The account types defined in the ATT are ISO-based account types. The BASE24-telebanking product includes a default set of ATT records, so only those account types outside of the standard account types must be added to the ATT. The account type description is displayed on the CATT, CSTT, IRCF, PCDF, and ITS Transaction Log File (ITLF) perusal screens. Refer to the ***BASE24 Core Files and Tables Maintenance Manual*** for a complete list of the standard account types.

Customer Table

All BASE24-telebanking transactions, including user-defined transactions, require a row in the Customer Table (CSTT).

Customer/Account Relation Table

The Customer/Account Relation Table (CACT) is accessed for user-defined transactions when all of the following conditions are met:

- The CACT is configured in the system.
- Account information such as the account type, account number, or account qualifier is provided.
- The authorization method is Positive Customer with Accounts Authorization (PCAA) or Positive Customer with Balances/History Authorization (PCBA).

The account status and effective date range are verified when the CACT is accessed for a user-defined transaction. User-defined transactions can be performed against any open, open primary, restricted, or restricted primary account when the transaction date falls within the effective date range defined in the CACT.

The action allowed indicators are not used for user-defined transactions.

Customer Allowed Transaction Table

The Customer Allowed Transaction Table (CATT) is used to define the transaction set supported for a given customer profile. The presence of a row in the CATT indicates the transaction is allowed. One row must be added to the CATT for each FIID and customer profile for which the user-defined transaction is allowed. A row can be added with the default FIID value of asterisks if all institutions support the user-defined transaction.

External Message File

The External Message File (EMF) is used to define the data elements for each message type between the RBSI-compliant device and the Integrated Authorization Server process and also between the Host Interface process and the host. EMF records may require modifications in order to support additional data for user-defined transactions.

The EMF can also be used to map BASE24-telebanking processing codes or transaction codes to a corresponding IMS or CICS transaction code for transactions destined for the host. EMF records require modifications in order to support an IMS or CICS transaction code for user-defined transactions.

Institution Definition File

A route profile is defined for each institution in the Institution Definition File (IDF). The route profile is a mechanism used to group institutions that support the same transaction set and have the same transaction routing requirements. When configuring the route profiles defined in a given logical network, it is important to consider the user-defined transaction set used by each institution.

Institution Routing Configuration File

Institution Routing Configuration File (IRCF) records associated with user-defined transactions contain the authorization level, authorization method, and authorization destination.

The authorization level indicates whether user-defined transactions are authorized online, offline or online/offline. User-defined transactions can be used to access or update information located on the host. The Integrated Authorization Server process can log user-defined transactions to the ITS Transaction Log File (ITLF), but cannot retrieve or update information in the THF or PBF.

The following authorization methods are available for user-defined transactions:

- The Positive Customer Authorization (PCA) method is the same for standard transactions and user-defined transactions.
- The Positive Customer with Accounts Authorization (PCAA) method for user-defined transactions includes all of the processing performed for standard transactions except checking the action allowed flags in the CACT (displayed on CSTT screen 2).
- The Positive Customer with Balances/History Authorization (PCBA) method for user-defined transactions includes the same processing as the Positive Customer with Accounts Authorization method for user-defined transactions.

Processing Code Definition File

All processing codes supported by an institution must be defined in the Processing Code Definition File (PCDF). The transaction code, which represents the first two positions of the processing code, should be defined as U*x*, where *x* represents a value from 0 through 9 or A through Z. User-defined transactions may also contain one or more account types. Positions 3 and 4 of the processing code represent the account 1 type. Positions 5 and 6 of the processing code represent the account 2 type. If account types are specified in a user-defined transaction, they can be used for further processing of the transaction. BASE24 Remote Banking customers must configure a PCDF record for each possible combination of valid transaction code and account type.

One record must be added to the PCDF for each route profile for which the user-defined processing code is allowed. A PCDF record with a route profile value of asterisks should also be added for each user-defined processing code since the default records are used by the Customer Service Representative Requester process and displayed on the ITS Transaction Log File (ITLF) perusal screens. In most cases, the transaction allowed flag in the default record should be set to a value of N. It should be set to a value of Y only if all institutions support the user-defined transaction.

Refer to the PCDF section of the ***BASE24 Core Files and Tables Maintenance Manual*** for descriptions and valid values for the remaining fields to be configured in each record.

Access Methods

User-defined transactions support the same access methods supported for standard BASE24 Remote Banking transactions (e.g., customer ID, customer ID and account type, or customer ID and account qualifier). The following paragraphs describe account type determination and account selection for user-defined transactions.

Account Type Determination

If an account type is not specified in the processing code for a user-defined transaction but an account number or account qualifier is provided as part of the transaction, the Integrated Authorization Server process attempts to determine the account type and update the processing code using the account information provided and the CACT.

If the CACT is not configured in the system or a matching CACT record is not found, the processing code is not updated and the generic form of the processing code (e.g., UA0000) is used for further processing, including searches for PCDF records. If a PCDF record exists for the generic form of the processing code and the authorization method is specified as Positive Customer Authorization (PCA), processing continues because account information is not required to authorize the transaction. However, if a PCDF record does not exist or if a record exists for the generic form of the processing code and the authorization method is specified as Positive Customer with Accounts Authorization (PCAA) or Positive Customer with Balances/History Authorization (PCBA), the transaction is denied.

If a matching CACT row is found, the processing code is updated with the account type from the CACT row (e.g., UA9Z00) and the updated form of the processing code is used for further processing. If the account type in the CACT row is one of the additional codes for savings (1A through 1F), checking (2A through 2H), or credit (3A through 3G), the Integrated Authorization Server process changes the second character of the account type to zero. A PCDF record must exist for the updated form of the processing code in order for processing to continue.

Multiple Account Selection

Multiple account selection is supported for user-defined transactions when all of the following conditions exist:

- The authorization method is Positive Customer with Accounts (PCAA) or Positive Customer with Balances/History (PCBA).
- One or more account types are specified in the processing code.
- The corresponding account numbers or account qualifiers are not specified.
- Multiple accounts of the specified type exist.

Customer Service Representative

User-defined transactions can be performed by a customer service representative using the CSR terminal or using an RBSI-compliant device. From a CSR terminal, the customer service representative selects transactions based on a transaction description displayed on the Transaction Selection screen. Transaction descriptions are configured in the PCDF. From an RBSI-compliant device, the customer service representative selects transactions using the same means provided for standard BASE24 Remote Banking transactions.

User-defined transactions allow entry of the customer ID or account number, PIN, one or two account numbers, one or two account qualifiers, and a transaction amount. If a transaction amount is entered, the user-defined transaction is processed as a financial transaction. If no amount is entered, the user-defined transaction is processed as a nonfinancial transaction. The BASE24-telebanking software does not mandate the presence of any specific data fields (e.g., second account number or transaction amount) for user-defined transactions since the specific data requirements are unknown.

Host Impacts

User-defined transactions can be sent online to the host using the ISO Host Interface process and be included in the ITLF extract that is used by the host. The host must be able to recognize and process these transactions.

Online Transactions

The specific data elements provided in the messages between the ISO Host Interface process and the host for user-defined transactions should be the same as those specified for standard BASE24 Remote Banking transactions. The EMF identifies the mandatory and conditional ISO bits to be sent and received in each ISO message.

The following data elements may be of particular importance for user-defined transactions:

- The Message Type Identifier contains an authorization message type (01xx) for most user-defined requests, although it can contain a financial message type (02xx) if the user-defined transaction has a financial impact on a host file.
- The Processing Code (field 3) contains the user-defined processing code received from the remote banking endpoint device (account types may have been modified by a BASE24-telebanking process).
- The Additional Data, Private Field (field 48) can be used to supply additional data for transaction requests and responses. This field is available to transport user-defined data between BASE24 and the host and is optionally logged to the ITLF. Data in this field is not returned to the customer service representative terminal.

- The Additional Amounts Field (field 54) carries balance information for the accounts involved in the transaction and the remaining amount. This field can be used to return host balances for user-defined account types. These amounts are returned to the RBSI-compliant device and customer service representative terminal.
- The Reserved, Private Field (field 124) can be used to return user-defined token data from the host. The tokens are optionally logged to the ITLF. However, tokens are not returned to the RBSI-compliant device or customer service representative terminal.

ITS Transaction Log File Extract

User-defined transactions are logged to the ITLF and are present on the ITLF extract tape. The host must be able to recognize and process user-defined transactions present on the extract tape.

Reports

User-defined transactions are logged to the ITLF and can be reported on the BASE24-telebanking reports. The BASE24-telebanking reports include all transactions defined in the PCDF. The transaction descriptions displayed on each of the detail and summary reports are configured in the PCDF. All user-defined transactions logged to the ITLF are included in the Daily Customer Activity Detail Report, the Daily Customer Activity Summary Report, and the Monthly Customer Activity Summary Report. Only user-defined transactions processed as financial transactions are included in the Daily Financial Activity Detail Report, the Daily Financial Activity Summary Report, and the Monthly Financial Activity Summary Report.

Reversals

User-defined transactions can be processed as financial or nonfinancial transactions. Reversals are always generated for approved financial transactions that do not successfully complete. Reversals are optionally generated for approved nonfinancial transactions as determined by a configuration parameter at the institution and issuer interface level. Reversals for user-defined transactions are logged to the ITLF and sent to the host if necessary.

RBSI-Compliant Device

User-defined transactions can be initiated at an RBSI-compliant device. BASE24 Remote Banking customers must coordinate support for user-defined transactions with their RBSI-compliant device and software vendor. The specific data elements provided in messages from the RBSI-compliant device for user-defined transactions should be the same as those specified for standard BASE24 Remote Banking transactions.

The following data elements are of particular importance for user-defined transactions:

- The Message Type Identifier contains an authorization message type (01xx) for most user-defined requests, although it can contain a financial message type (02xx) if the user-defined transaction has a financial impact on a host file.
- The Processing Code (field 3) contains the user-defined processing code received from the RBSI-compliant device (account types may have been modified by a BASE24 Remote Banking process).
- The Additional Data, Private Field (field 48) can be used to supply additional data for transaction requests and responses. This field is available to transport user-defined data between the RBSI-compliant device and BASE24 and is optionally logged to the ITLF. Data in this field is not returned to the RBSI-compliant device.
- The Additional Amounts Field (field 54) carries balance information for the accounts involved in the transaction and the remaining amount. This field can be used to return host balances for user-defined account types. These amounts are returned to the RBSI-compliant device.

History Transactions

The BASE24-telebanking standard transaction set includes the history transactions shown in the following table. This table contains a description of each history transaction, the THF key needed to obtain the information, and any additional information that can be used to select the desired transaction.

The transaction code is one part of the processing code used to identify each BASE24-telebanking transaction. The remaining parts of the processing code identify the type of account or accounts involved in the transaction. Refer to appendix C for a complete list of the processing codes included in the standard transaction set.

The THF can have up to four alternate keys in addition to the primary key. Transaction codes 3A and 3C through 3G use one of the alternate keys. These keys are discussed in detail on the following pages. The alternate key specified for a desired transaction code should be configured when the THF is created. If the alternate key is not configured, the THF is searched using the primary key. This results in more THF records being read for a history transaction.

The Selection column indicates whether additional information can be used to define the search of the THF. When the transaction amount is used, only transactions for the specified amount are returned to the RBSI-compliant device or customer service representative terminal. When the start date and end date are used, only transactions that occurred in the specified date range are returned. When the start date is entered without an end date, transactions that occurred from the start date through the current date are returned. The end date cannot be used without a start date.

Code	Description	Key	Selection
3A	Check clearance inquiry.	CQ	Amount
3B	Last count of debit/credit transactions inquiry.	Primary	Date
3C	Last count of account by source transactions inquiry. Transaction sources include ATM, POS device, remote banking endpoint device, and teller terminal.	SC	None
3D	Last count of check transactions inquiry.	DC	Amount
3E	Last count of debit transactions inquiry.	DC	Amount

Code	Description	Key	Selection
3F	Last count of credit transactions inquiry.	DC	Amount
3G	Last count of account transfer transactions inquiry.	TC	None

Transaction History File

The Transaction History File (THF) contains all of the information that the Integrated Authorization Server process can return in response to a history transaction. This file is updated by the BASE24 Enscribe Refresh process only. The Integrated Authorization Server process does not update this file during financial transaction processing.

Record Contents

The THF can contain the following information for a transaction. Each field description identifies information that is returned in response to a history transaction.

Account Number — The number that identifies the account.

Account Type — A code that identifies the type of account. Refer to the “Account Type Codes” discussion in section 1 for the values used in the standard transaction set.

Check Number — The serial number of the check written for the transaction. The presence of a serial number identifies a check transaction. This information is returned in response to a history transaction.

Debit/Credit Indicator — A code that identifies whether a transaction is a debit or credit. This information is returned in response to a history transaction. Valid values are as follows:

C = Credit transaction
D = Debit transaction

FIID — A code that identifies the financial institution responsible for the account.

Host Transaction Code — The code assigned to this transaction for processing on the host. This information is returned in response to a history transaction.

Purge Date — The last date that a record is to be kept in the THF. BASE24 processes do not use this field.

Reference Number — A value that can be assigned by the host to distinguish between two THF records that have the same FIID, account number, account type, and transaction date and time.

Reversal Indicator — A code that identifies whether the transaction had to be reversed. This information is returned in response to a history transaction. Valid values are as follows:

0 = The transaction was not reversed.
1 = The transaction was reversed.

Sufficient Funds Indicator — A code that identifies whether the account had sufficient funds to approve a debit transaction. This information is returned in response to a history transaction. Valid values are as follows:

0 = The account had sufficient funds to approve a debit transaction. This value is also returned for transactions other than debits.
1 = The account did not have sufficient funds to approve a debit transaction.

Transaction Amount — The amount of the transaction. This information is returned in response to a history transaction.

Transaction Code — A code that identifies the type of transaction that has been performed. Refer to the “BASE24 Remote Banking Transactions” discussion in section 1 for the values used in the standard transaction set.

Transaction Date and Time — The date and time that the transaction occurred. This information is returned in response to a history transaction.

Transaction Source — A code that identifies the source of a transaction. This information is returned in response to a history transaction. Any value can be used. However, values reserved for use by BASE24 products are as follows:

A = ATM device
B = Remote banking endpoint device
P = POS device
T = Teller device

Keys

The THF has one primary key and four alternate keys that are used to select the records returned in response to a history transaction. The table of available history transactions earlier in this subsection identifies the key used for each transaction. Regardless of the key used, the same THF fields are returned in response to a history transaction.

The keys and the fields used for each are described in the following table.

Primary key	This key controls the sequence of records in the file. It includes the FIID, account number, account type, transaction date and time, and reference number. Two records cannot have the same primary key. Transaction code 3B uses this key.
Check clearance key (CQ)	This alternate key permits records for an account to be searched for a specific check serial number. Transaction code 3A uses this key. If a transaction amount is entered with this transaction code, only records for the specified amount are returned.
Debit/credit key (DC)	This alternate key permits records for an account to be searched for debit transactions only (transaction code 3E), check transactions only (transaction code 3D), and credit transactions only (transaction code 3F). If a transaction amount is entered with any of these transaction codes, only records for the specified amount are returned.
Source key (SC)	This alternate key permits records for an account to be searched for transactions that originate at a specific source. Transaction code 3C uses this key.

Transaction code key (TC)	This alternate key permits records for an account to be searched for transfer transactions that involve the account. Transaction code 3G uses this key. The value in the Debit/Credit Indicator field identifies whether the record pertains to the *from* or *to* side of the transaction.

Transaction History Configuration File

The number of THFs maintained and the accounts that each file contains are configurable. The Integrated Authorization Server process uses the Transaction History Configuration File (THCF) to determine which THF contains the information needed to authorize a history transaction. The BASE24 Enscribe Refresh process uses the THCF to obtain the names of all THFs. However, each THF must be refreshed individually.

The THCF contains one record for each THF used. A THF-*group* assign must be added to the Logical Network Configuration File (LCONF) for each record in the THCF. The value in the ASSIGN field on THCF screen 1 and the THF-*group* assign must match.

There are several options available for arranging THF records.

- All FIIDs in a refresh group can be in the same THF.
- Each FIID in a refresh group can have its own THF.
- Each FIID in a refresh group can divide their records among multiple THFs, with each THF containing a specific range of account numbers.

Customer Historical Inquiry Transaction Limit

The MAX HISTORY RECORDS field on Customer Table (CSTT) screen 1 controls the maximum number of Transaction History File (THF) records that can be returned in historical inquiry transactions to an RBSI-compliant device for this customer. This field specifies the total number of items that can be returned for the transaction, not just for each response message included in the transaction. This maximum does not apply to historical inquiries requested by a customer service representative (CSR) using the BASE24 CSR terminal screens.

If this field is set to a nonzero value, items in history inquiries – payments and transfers and history inquiries – all transactions acquired from an RBSI-compliant device can only be searched for in descending key order.

Customer Limits and Usage

Institutions can establish amount and usage limits for transfer and payment transactions. These limits are established using the IDF, Bank Table (BANK), Customer Table (CSTT), and PBF. Transactions that exceed the limits as set by the institution are declined.

In addition, accumulator fields in the PBF are used to track transaction activity performed using the BASE24 Remote Banking products during a single usage accumulation period. A usage accumulation period is a set amount of time during which transaction amounts and counts are accumulated. At the end of the usage accumulation period, the amounts and counts are cleared. The transfer and payment transaction usage totals are maintained in the PBF. These accumulator fields are compared to corresponding limits in the IDF and PBF to determine if a transaction should be approved or declined.

BASE24-billpay transfer and payment transactions can use the individual transaction limit established in the BANK and CSTT regardless of authorization method. BASE24 Remote Banking transactions can use the transfer and payment transaction limits and usage fields maintained in the PBF only with the Positive Customer with Balances/History Authorization method.

Institution Definition File Limits

IDF screen 41 contains the BASE24 Remote Banking usage accumulation parameters that are defined at the institution level. There are two sets of usage accumulators so that an institution can track activity over two independent periods of time (e.g., daily and monthly or weekly and monthly). These accumulators, known as periodic and cyclic, operate totally independent of each other. Fields on this screen define the length and dates for these time periods.

Bank Table Limits

BANK screen 1, which is accessed from IDF screens 40, 41, and 42, contains a BASE24-billpay parameter that is defined at the institution level to limit the amount of individual scheduled payment transactions. No accumulator is associated with this parameter. This parameter is also defined at the customer level in the CSTT. If the institution-level and customer-level parameters contain nonzero values, the customer-level parameter overrides the institution-level parameter.

Customer Table Limits

CSTT screen 1 contains a BASE24-billpay parameter that is defined at the customer level to limit the amount of individual scheduled payment transactions. No accumulator is associated with this parameter. This parameter is also defined at the institution level in the Bank Table (BANK). If the institution-level and customer-level parameters contain nonzero values, the customer-level parameter overrides the institution-level parameter.

Positive Balance File Limits

PBF screen 11 contains the BASE24 Remote Banking usage accumulation parameters that are defined at the account level. Periodic and cyclic limits include the number and total amount of transfers and payments that can be made from credit and noncredit accounts. Individual transaction limits include a minimum amount and increment amount so the Integrated Authorization Server process can check the amount of a transfer or payment transaction that withdraws funds from a credit account.

Positive Balance File Totals

PBF screen 11 contains the BASE24 Remote Banking usage totals that are maintained at the account level. Periodic and cyclic totals include the number and amount of transfers and payments that have been made from the account during the current usage accumulation period.

Overdraft Control and Protection

The BASE24 Remote Banking products offer overdraft control and protection. Overdraft control limits the amount an account can be overdrawn while overdraft protection keeps an account from becoming overdrawn by transferring funds to it from another account.

Overdraft Control

The OVERDRAFT LIMIT field on PBF screen 1 contains the amount by which the debit portion of a transfer or payment transaction can exceed the available balance and still be approved. An overdraft limit can be specified for any account. The Integrated Authorization Server process ignores the overdraft limit when an overdraft limit and an overdraft protection account are specified for the same account.

Overdraft Protection

Automatic overdraft protection goes by two names—credit line account or backup account—depending on the type of account used to provide the protection. Credit line indicates that a credit account (i.e., credit card or line of credit) provides the protection while backup account indicates that a noncredit account (i.e., checking, savings, or interest-bearing checking) provides the protection. Overdraft protection can be set up for any type of account, except that a credit line cannot be set up for a credit account. BASE24 Remote Banking transfer and payment transactions can check for automatic overdraft protection.

Each institution can determine whether its PBF records include a segment containing information to identify the account a customer uses for overdraft protection. The CREDIT LINE field on IDF screen 5 controls the presence of overdraft protection account information in the PBF record.

Account Identification

The ACCOUNT TYPE and ACCOUNT NUMBER fields on PBF screen 6 identify the account from which funds are transferred when the customer’s first account does not contain sufficient funds.

An account can have one credit line or backup account. No transfer is made if the credit line or backup account does not have sufficient available funds to make the necessary transfer. Even if the account used as a credit line or backup account has its own credit line or backup account, the second credit line or backup account is not considered when determining whether the transfer can be made.

Transfer Amount

The amount transferred from a credit line or backup account is controlled by values in fields on IDF screen 6. This screen has two TRANSFER METHOD and INCREMENT AMOUNT fields, depending on the type of account used as the credit line or backup account. Credit account types include credit and line of credit. Noncredit account types include checking, savings, and interest-bearing checking. Transfers can be made in an amount that leaves a zero balance in the primary account (i.e., the account that has the overdraft protection) or in the increment amount specified in the IDF.

Using an overdraft of $75.50 as an example, an exact amount transfer is $75.50. With an overdraft of $75.50 and an increment amount of $50, the transfer is $100 because 100 is the smallest multiple of 50 (e.g., 50, 100, 150) that is equal to or greater than the overdraft amount. In this example, a $50 transfer creates a negative balance of $25.50 and a $100 transfer creates a positive balance of $24.50.

Transaction Logging

BASE24 Remote Banking transaction messages are logged to the following files and tables:

- ITS Transaction Log File (ITLF)
- Exception Log File (ELF)
- History Table (HIST)

The Integrated Authorization Server process logs messages for the following transactions to the ITLF or the ELF:

- All BASE24-telebanking transactions.
- Transactions for the following BASE24-billpay transaction codes:
  - ❑ All BASE24-billpay transactions acquired from an RBSI-compliant device.
  - ❑ Transaction codes 3K–3L and 9A–9J for transactions acquired by a host or remote banking device.
  - ❑ Transaction codes 4A, 4B, 50, 5A, and 5B for transactions generated by the XPNET Billpay Server process or Scheduled Transaction Initiator process.
  - ❑ Transaction codes for any other BASE24-billpay transactions acquired by a customer service representative (CSR) terminal, host, or RBSI-compliant device that are identified in the Bank Transaction Advice Table (BKAV) for logging to the ITLF. The Bank Advice process reads these transactions from the HIST and passes them to the Integrated Authorization Server process as advices.

The Pathway Billpay Server, XPNET Billpay Server, and Scheduled Transaction Initiator processes log messages for BASE24-billpay transactions to the History Table (HIST).

ITS Transaction Log File

Messages that contain enough information to be processed properly are logged to the ITLF. The Integrated Authorization Server process is responsible for logging transactions to the ITLF. The ITLF can be perused with the ITLF perusal subsystem and extracted using the BASE24 Super Extract process. However, it cannot be perused from an RBSI-compliant device or customer service representative (CSR) terminal.

The End-of-Period process creates a separate ITLF for each business day and purges outdated ITLFs according to the value of the TLF-FILE-RETENTION param in the Logical Network Configuration File (LCONF). The values of the LGNT-CUTOVER and LGNT-CUTOVER-LEAD-TIME params in the LCONF control the time that the End-of-Period process creates new ITLFs. Refer to the ***BASE24 Remote Banking End-of-Period Processing and Reporting Manual*** for more information on the End-of-Period process.

Message Types Logged

The Integrated Authorization Server process logs the following messages to the ITLF:

Type	Description
1110	Authorization transaction request response
1120	Authorization transaction advice
1210	Financial transaction request response
1420	Reversal
1421	Reversal repeat

Record Format

The Integrated Authorization Server process logs messages to the ITLF from fields in the Internal Transaction Data (ITD). The ITD contains the information the BASE24-telebanking product needs to process a transaction. The Transaction object is responsible for updating ITD fields, providing ITD information to other objects for their use, and logging ITD information to the ITLF.

There are three types of ITD fields, depending on whether the field is logged to the ITLF. These types are mandatory, optional, and not logged. Mandatory fields are always logged to the ITLF, optional fields are logged to the ITLF only when they contain data, and the remaining fields are never logged to the ITLF.

The ITD fields currently used by BASE24 Remote Banking products are presented in appendix A. Each field description includes the following information:

- The tag number that is used to uniquely identify the field in an ITLF record.
- The type of ITD field for ITLF logging purposes.
- The data name for the ITD field when it appears in an ITLF record.
- A description of the ITD field and its contents.

The Super Extract process adds a binary 16 (two-character) field containing the tag number to the beginning of each optional ITD field in an ITLF record. Host programs must use these tag numbers to determine which optional ITD fields an ITLF record contains. There are no eye-catcher characters or other special characters separating ITD fields.

Variable Length Fields

Most ITD fields are fixed in length regardless of the information they contain. However, the length of the ITD fields in the following table depends on the information they contain.

Field Name	Tag Number
Acquirer Dependent	7
Additional Data	120
Advice 1 Dependent	13
Advice 2 Dependent	15
Authorizer Dependent	31
External Acquirer Dependent Data	138

Field Name	Tag Number
External Authorizer Dependent Data	139
Multiple Account Information	141
Reason Code—Exception	85
Refresh Impact	125
Token	118

The length of each of the fields listed above depends on the length of the data in its FDATA field. Each of the fields listed above also contains FLD-LGTH fields for the physical length and logical length of the FDATA field. The Transaction object adjusts the length of the FDATA field and the values in the FLD-LGTH fields to match the data in the FDATA field.

Default Log File

The Transaction object can log transactions to a default ITLF if the ITLF for the business day is unavailable for any reason. However, perusal screens are not available for the default ITLF and the Super Extract process does not extract the default ITLF. Without the default ITLF, the Integrated Authorization Server process declines any messages received when the ITLF for the business day is unavailable.

The TLF-DEFAULT assign in the LCONF identifies the location of the default ITLF. The TLF-*group* assign in the LCONF identifies the location of the ITLF for the business day. Both assigns are required. The file names used in these assigns control whether a default ITLF is used. If the default ITLF and the ITLF for the business day have different file names, the default ITLF can be used if an error occurs when the Transaction object attempts to log a transaction to the ITLF for the business day. If the default ITLF and the ITLF for the business day have the same file name, the default ITLF cannot be used. When the default ITLF and the ITLF for the business day are the same file, any error that occurs when the Transaction object attempts to log a transaction to the ITLF for the business day also occurs when the Transaction object attempts to log the transaction to the default ITLF.

Exception Log File

A record is logged to the ELF only when a message does not contain enough information to be processed properly and be logged to the ITLF. The Interface Manager object and Interface Master Services object are responsible for logging messages to the ELF. File maintenance screens are not available for perusing the ELF and the Super Extract process does not extract the ELF. The End-of-Period process creates a separate ELF for each business day and purges outdated ELFs according to the value of the ELF-FILE-RETENTION param in the LCONF. The values of the LGNT-CUTOVER and LGNT-CUTOVER-LEAD-TIME params in the LCONF control the time that the End-of-Period process creates new ELFs. Refer to the ***BASE24 Remote Banking End-of-Period Processing and Reporting Manual*** for more information on the End-of-Period process.

History Table

The History Table (HIST) is an SQL table used when the BASE24-billpay product is installed. All BASE24-billpay transactions except transaction code 50 are logged to the HIST. The HIST is available for inquiry from customer service representative (CSR) terminals, tables maintenance terminals, and RBSI-compliant devices.

The HIST is not currently extracted using the Super Extract process. The Integrated Authorization Server process always logs a subset of the BASE24-billpay transactions to the ITLF, which can be extracted using the BASE24 Super Extract process. Refer to the “BASE24-billpay Transaction Type Codes” discussion later in this section for a list of these transactions.

The Bank Advice process can select additional BASE24-billpay transactions to be logged to the ITLF, based on parameters defined in the Bank Transaction Advice Table (BKAV). Transactions can be selected for logging to the ITLF based on the following information in the HIST:

- Transaction source (e.g., CSR terminals)
- Transaction code
- Advice type (i.e., approved transactions only, denied transactions only, or both approved and denied transactions)

The Archive program copies and removes old rows from the HIST. This program runs after cutover on a nightly basis, preferably as one of the last steps in the daily cycle. It can operate at a system-wide level or an FIID-specific level, depending on the setting of the BP-ARCHIVE-BY-FIID param in the LCONF. Refer to the ***BASE24 Remote Banking End-of-Period Processing and Reporting Manual*** for additional information about the use of the Archive program.

BASE24-billpay Transaction Type Codes

The following table lists all BASE24-billpay transactions that can be logged to the HIST and ITLF during authorization. Entries in the Transaction Source column identify which transactions are always logged from a particular source. The following codes identify the transaction source:

C = CSR terminal
H = Acquirer host
R = RBSI-compliant device
S = Scheduled Transaction Initiator process
X = XPNET Billpay Server process

A transaction with a check (✓) in the Logged HIST column is always logged to the HIST. A transaction with a source code in the Logged ITLF column is logged to the ITLF only if it originates from the source indicated. A transaction with a check in the Logged HIST column must be configured in the Bank Transaction Advice Table (BKAV) in order to be logged to the ITLF if it originates from a source other than those indicated in the Logged ITLF column. A separate record must be added to the BKAV for each combination of transaction code and transaction source to be logged to the ITLF. Transactions already indicated in the Logged ITLF column should not be configured in the BKAV, since this would cause them to be logged to the ITLF twice.

Transaction			**Logged**	
Description	**Code**	**Source**	**HIST**	**ITLF**
Customer vendor list inquiry	3H	C, R	✓	R
Scheduled payments list inquiry	3J	C, R	✓	R
Scheduled transfers list inquiry	3K	C, H, R	✓	H, R
History inquiry – payments and transfers	3L	C, H, R	✓	H, R

Transaction			Logged	
Description	**Code**	**Source**	**HIST**	**ITLF**
Scheduled transactions list inquiry	3M	C, R	✓	R
Transfer – future	4A	S	✓	S
Transfer – recurring	4B	S	✓	S
Payment – immediate	50	X		X
Payment – future	5A	S	✓	S
Payment – recurring	5B	S	✓	S
Schedule payment – immediate	9A	C, H, R	✓	H, R
Schedule payment – future	9B	C, H, R	✓	H, R
Schedule payment – recurring	9C	C, H, R	✓	H, R
Schedule transfer – future	9D	C, H, R	✓	H, R
Schedule transfer – recurring	9E	C, H, R	✓	H, R
Scheduled payment delete	9F	C, H, R	✓	H, R
Scheduled transfer delete	9G	C, H, R	✓	H, R
Scheduled payment update	9H	C, H, R	✓	H, R
Scheduled transfer update	9J	C, H, R	✓	H, R
Customer add	9K	C, R	✓	R
Customer inquiry	9L	C, R	✓	R
Customer update	9M	C, R	✓	R
Customer vendor add	9N	C, R	✓	R
Customer vendor deactivate	9P	C, R	✓	R
Customer vendor update	9Q	C, R	✓	R
Master vendor list inquiry	9R	C, R	✓	R

Transaction			Logged	
Description	**Code**	**Source**	**HIST**	**ITLF**
Customer vendor request mailing	9T	C, R	✓	R
Loan application	9V	C, R	✓	R
Miscellaneous request	9W	C, R	✓	R
History inquiry – all transactions	9X	C, R	✓	R
Bank list inquiry	B1	C, R	✓	R
Bank branch list inquiry	B2	C, R	✓	R
Miscellaneous request inquiry	B4	C, R	✓	R
Customer account add	B5	C, R	✓	R
Customer account activate	B6	C, R	✓	R
Customer account deactivate	B7	C, R	✓	R
Customer account delete	B8	C, R	✓	R
Customer account update	B9	C, R	✓	R
Customer activate	BB	C, R	✓	R
Customer deactivate	BC	C, R	✓	R
Customer vendor activate	BD	C, R	✓	R
Customer vendor mask verify inquiry	BE	C, R	✓	R

The transaction code and the transaction source determine which BASE24-billpay transactions are logged to the History Table (HIST) and ITS Transaction Log File (ITLF). The Bank Advice process reviews the HIST for additional nonfinancial BASE24-billpay transactions that can be logged to the ITLF following authorization, based on configuration of the BKAV.

All BASE24-billpay transactions except for the internal transfer and payment transactions (4A, 4B, 50, 5A, and 5B) include customer verification. These transactions do not require customer verification because customer verification was performed during the corresponding schedule transfer or schedule payment transactions (9A–9E).

Generating Advices to a Host

Advices notify the host of authorizations made by the Integrated Authorization Server process. The institution controls advice generation for each processing code in the PCDF. The value in the ADVICE REQUIRED field on PCDF screen 1 indicates when the Integrated Authorization Server process should generate an advice message and pass it to the host. There are four options for sending advice messages:

- Do not send advice messages.
- Send advice messages for approved transactions only.
- Send advice messages for declined transactions only.
- Send advice messages for approved and declined transactions.

In the case of transfer transactions that involve one host for the *from* account and another host for the *to* account, an advice can be sent to each host.

Note: Advices are also used by the BASE24-billpay Bank Advice process to notify the Integrated Authorization Server process so the message can be logged to the ITLF. Information configured on Bank Transaction Advice Table (BKAV) screen 1 controls these advices.

Store-and-Forward Processing

When data communications to a host system are down for maintenance or other processing needs, the BASE24-telebanking product can continue to process transactions, storing them in a data file to be transmitted when data communications are reestablished. This function is known as store-and-forward processing. The data file used in this processing is the Store-and-Forward File (SAF).

When data communications are reestablished with a host, the Host Interface process can forward the stored transactions to the host by interspersing the stored transactions with live transactions or by sending the stored transactions consecutively. The method used to send stored transactions to the host is specified in the STORE AND FORWARD PROCESSING FLAGS field on screen 1 of the Host Configuration File (HCF).

HCF screen 1 also contains other fields used for store-and-forward processing with the host. The MAXIMUM OUTSTANDING SAFS field specifies the maximum number of store-and-forward messages that can be outstanding to a multithreaded host at one time. The STORE AND FORWARD field specifies the number of seconds the Host Interface process waits for an acknowledgment after transmitting a stored message to the DPC. The MAX SAF RETRY field specifies the number of times the Host Interface process should attempt to send the same store-and-forward message before logging it to the event log and deleting it from the SAF.

For detailed information on store-and-forward processing to the host system, refer to the ***BASE24 ISO Host Interface Manual.***

Discarding of Nonfinancial Reversals

The values in the DISCARD NON-FINANCIAL REVERSALS field on HCF screen 22 and IDF screen 40 indicate whether or not reversals of nonfinancial transactions can be discarded without being sent to a host or RBSI-compliant device, and without being logged. This field on HCF screen 22 controls whether the ISO Host Interface process discards reversals for nonfinancial transactions without sending them to a host. This field on IDF screen 40 controls whether the Integrated Authorization Server process discards reversals for nonfinancial transactions without logging them to the ITS Log File (ITLF) or sending them to an RBSI-compliant device.

Reversals for some nonfinancial transactions are never discarded, regardless of the value in this field.

For the ISO Host Interface process, reversals for the following nonfinancial transactions to be sent to a host are never discarded. A check (✓) in columns two through four indicates which transaction codes are available for each of the three BASE24 Remote Banking product offerings.

Transaction Code	TB	Enhanced TB	BP	Transaction Description
90	✓			PIN change
9A			✓	Schedule payment – immediate
9B			✓	Schedule payment – future
9C			✓	Schedule payment – recurring
9D		✓	✓	Schedule transfer – future
9E		✓	✓	Schedule transfer – recurring
9F			✓	Scheduled payment delete
9G		✓	✓	Scheduled transfer delete
9H			✓	Scheduled payment update
9J		✓	✓	Scheduled transfer update
A0	✓			Check stop payment add

For the Integrated Authorization Server process, reversals for the following nonfinancial transactions acquired from an RBSI-compliant device or a CSR terminal are never discarded. In this list, the only transaction codes that can be acquired from a CSR terminal that are sent through the Integrated Authorization Server process are 90, 9A–9E, and 9Z.

Transaction Code	TB	Enhanced TB	BP	Transaction Description
90	✓			PIN change
9A			✓	Schedule payment – immediate
9B			✓	Schedule payment – future
9C			✓	Schedule payment – recurring
9D		✓	✓	Schedule transfer – future
9E		✓	✓	Schedule transfer – recurring
9F			✓	Scheduled payment delete
9G		✓	✓	Scheduled transfer delete
9H			✓	Scheduled payment update
9J		✓	✓	Scheduled transfer update
9K		✓	✓	Customer add
9M		✓	✓	Customer update
9N			✓	Customer vendor add
9P			✓	Customer vendor deactivate
9Q			✓	Customer vendor update
9V		✓	✓	Loan application
9W		✓	✓	Miscellaneous request
A0	✓			Check stop payment add
B5		✓	✓	Customer account add
B6		✓	✓	Customer account activate

Transaction Code	TB	Enhanced TB	BP	Transaction Description
B7		✓	✓	Customer account deactivate
B8		✓	✓	Customer account delete
B9		✓	✓	Customer account update
BB		✓	✓	Customer activate
BC		✓	✓	Customer deactivate
BD			✓	Customer vendor activate

Token Handling

The BASE24-atm, BASE24-pos, and BASE24-teller products use tokens to construct internal messages, write messages to log files, and exchange messages with hosts and interchanges. The BASE24-telebanking product does not create tokens during transaction processing or use the information contained in any tokens received from a host. However, it can accept tokens in a message from the host, map the tokens to a field in the Internal Transaction Data (ITD), log the ITD field to the ITLF, and include the ITD field in the ITLF extract.

If the ITD field that contains token information is to be logged to the ITLF, a record must be added to the Token File (TKN) with the information shown in the following table. These fields appear on TKN screen 2.

Field Name	Value
TOKEN GROUP	****
PRODUCT ID	14
TYPE	00
SUBTYPE	**
TRAN LOG	Set to Y (yes) for each token to be logged. Tokens to be logged are placed in the Token field of the ITD. Tokens not to be logged are placed in the Token—Not Logged field of the ITD.

If the ITD field that contains token information is to be extracted from the ITLF, a record must be added to the TKN with the information shown in the following table. These fields appear on TKN screen 3.

Field Name	Value
TOKEN GROUP	**** or the value from the GROUP NAME field on Extract Configuration File (ECF) screen 23.
PRODUCT ID	14
TYPE	00

Field Name	Value
SUBTYPE	**
ORDER FLAG	Y (yes) or N (no), to specify whether the values in the EXTR ORDER fields on this screen are used.
EXTR ORDER	A number identifying the position in which each token should appear in the extract record when the ORDER FLAG field contains a value of Y.

Refer to the ***BASE24 Tokens Manual*** for the procedures necessary for configuring the TKN.

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Section 4

BASE24-telebanking Message Flows

The diagrams in this section illustrate the paths taken by BASE24-telebanking transactions in various types of processing situations. These diagrams are intended to illustrate the transaction path from RBSI-compliant devices and customer service representative (CSR) terminals to the transaction authorizer. This section does not include all possible transaction message flows, only a representative sampling. All flows assume successful authorization of the requested transaction.

This section includes flows for transactions initiated at an interactive voice response system (IVR) and a CSR terminal. In this section, the IVR also represents any other RBSI-compliant device that communicates with BASE24 using the ISO 8583-1993 message format. The flows in this section do not depict processing performed for transactions initiated by a third-party processor.

This section contains illustrations of the following transactions:

- IVR transaction history inquiry authorized on the BASE24 system
- IVR available funds inquiry with multiple account selection authorized on the BASE24 system
- IVR transfer authorized on the BASE24 system with an advice
- IVR transaction history inquiry preauthorized on the BASE24 system and authorized on the host
- IVR transfer preauthorized on the BASE24 system and authorized on the host
- IVR customer and PIN verification authorized on the BASE24 system
- IVR dynamic key management
- CSR customer and PIN verification authorized on the BASE24 system
- CSR available funds inquiry authorized on the BASE24 system
- CSR transfer authorized on the BASE24 system with an advice
- CSR transfer preauthorized on the BASE24 system and authorized on the host

The Integrated Authorization Server process, Integrated Endpoint process, and Interface Master process are object-oriented processes. This section presents descriptions of processing performed by each object in the Integrated Authorization Server process. Refer to the ***BASE24 IEP Objects Reference Manual*** for a description of processing performed by each object in the Integrated Endpoint process. Refer to appendix D for additional information about object-oriented design.

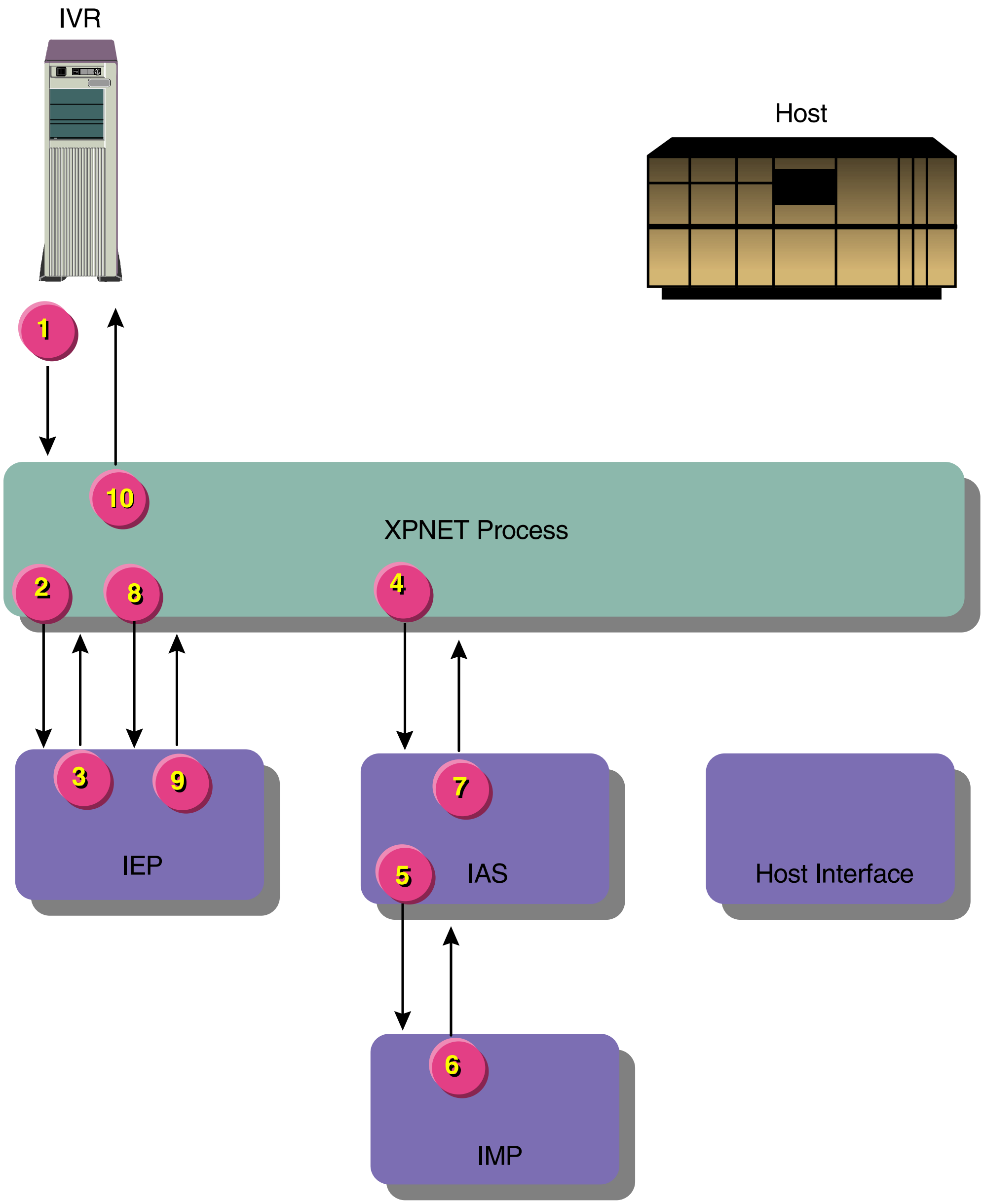
IVR Transaction History Inquiry Authorized on the BASE24 System

The BASE24-telebanking system uses information in the Transaction History File (THF) to process transaction history inquiries. The Positive Customer with Balances/History Authorization method is required to obtain history information from the BASE24 database.

The THF can have four alternate keys. The type of transaction history inquiry requested determines which alternate key is used to select THF records. Some transaction history inquiries can include the transaction amount. If a transaction history request includes an amount, only THF records that have the requested transaction amount are returned. Refer to the "History Transactions" discussion in section 3 for additional information regarding the alternate keys and available inquiry types.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process
IEP = Integrated Endpoint process
IMP = Interface Master process
IVR = Interactive voice response system

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling an IVR. The IVR creates an ISO request message with the transaction request and sends the message to the XPNET process.
2. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the IVR. The XPNET process passes the ISO message to the process named in the station declaration, in this case the Integrated Endpoint process.
3. Using the source class and information from the message, the Integrated Endpoint process identifies the Integrated Authorization Server process as the destination and passes the ISO message to the XPNET process for delivery.
4. The XPNET process passes the ISO message to the Integrated Authorization Server process.
5. The Integrated Authorization Server process obtains station information and requests the Interface Master process to validate the station. See the topic “Step 5 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
6. The Interface Master process validates the station and returns a message to the Interface Manager object in the Integrated Authorization Server process.
7. The Integrated Authorization Server process authorizes the transaction and returns an ISO message to the XPNET process. See the topic “Step 7 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
8. The XPNET process passes the ISO message to the Integrated Endpoint process.
9. The Integrated Endpoint process passes the ISO message with the requested transaction history to the XPNET process for return to the IVR. The Integrated Endpoint process uses the original station name carried in the message to identify the message destination.
10. The XPNET process passes the ISO message to the IVR. The IVR returns the response information to the customer.

Step 5 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the XPNET Interface object and Voice Response Unit Interface object in the order shown.

XPNET Interface Object Processing

The XPNET Interface object performs as follows:

1. Obtains the object ID of the destination object within the Integrated Authorization Server process (i.e., the Voice Response Unit Interface object). The Voice Response Unit Endpoint Class Handler object in the Integrated Endpoint process placed this object ID in the message.
2. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Obtains station information from the interface extended memory table. The interface extended memory table contains information from the ITS Interface Configuration File (IFCF), Interface Station Table File (IST), and Address Table File (ADRT).
2. Calls the Message ISO object to obtain information from the external message fields and updates the Internal Transaction Data (ITD) fields.
3. Calls the Customer object to obtain the FIID. The Customer object calls the Customer Access object to obtain the FIID from the Customer Table (CSTT). The Customer Access object also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contains the account number and not the customer ID. The Customer object calls the Transaction object to update the ITD.
4. Calls the Interface Manager object to validate the station. The Interface Manager object in the Integrated Authorization Server process passes a message to the Interface Master process.

Processing continues with step 6 of the processing summary.

Step 7 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the following objects in the order shown:

- Voice Response Unit Interface object
- Router object
- Authorization Combined Positive Customer object
- Voice Response Unit Interface object
- XPNET Interface object

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Financial Institution object to retrieve and check customer service information from the IDF extended memory table if the source code in the ITD is set to a value of IB.
2. Calls the Transaction Security object to translate the PIN, if necessary. The Transaction Security object obtains security information from the Interface Security File (ISF) extended memory table and calls the Transaction object to update the ITD if PIN translation occurs.
3. Calls the Transaction object to update the ITD with additional information.
4. Passes control of the transaction to the Router object.

Router Object Processing

The Router object performs as follows:

1. Calls the Transaction object to obtain information from the ITD.
2. Calls the Financial Institution object to obtain the route profile for the FIID. The Financial Institution object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution Access object to obtain processing information from the Institution Definition File (IDF) extended memory table.

   c. Calls the Transaction object to update the ITD.

3. Obtains routing configuration information from the institution routing extended memory table. The institution routing extended memory table contains information from the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF).
4. Calls the Transaction object to update configuration information in the ITD.
5. Passes control of the transaction to the Authorization Combined Positive Customer object.

Authorization Combined Positive Customer Object Processing

The Authorization Combined Positive Customer object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Customer object to determine whether the customer is permitted to perform the transaction. The Customer object performs as follows:
   a. Calls the Transaction object to obtain information from the ITD.
   b. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain information from the IDF extended memory table.
   c. Calls the Customer Access object to obtain information from the customer extended memory table, CSTT, and CACT. The customer extended memory table contains the same information as the Customer Allowed Transaction Table (CATT).
   d. Calls the Transaction object to update the ITD.
3. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain authorization and customer service information from the IDF extended memory table.
4. Determines whether a PIN change is required. If a PIN change is required, the transaction is declined.
5. Calls the Transaction Security object to verify the PIN. The Transaction Security object performs as follows:
   a. Calls the Transaction object to obtain PIN information from the ITD.

   b. Checks the point of service information in the ITD to determine whether the PIN needs to be verified. If the PIN capture flag is set to S, processing skips to step 6, otherwise processing continues as follows.
   c. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain PIN verification information from the IDF extended memory table.
   d. Calls the Authorization Security Access object to obtain PIN verification information from the Key Authorization File (KEYA) extended memory table.
   e. Verifies the PIN. The Transaction Security object can verify the PIN in software or call a security device to verify the PIN.
   f. Calls the Transaction object to update the PIN verification results in the ITD and delete the PIN from the ITD.

6. Checks the authorization method. A transaction history inquiry requires the Positive Customer with Balances/History Authorization method.
7. Calls the Transaction History object to obtain history information. The Transaction History object performs as follows:
   a. Calls the Transaction object to obtain the key information needed for the type of history inquiry requested.
   b. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain the refresh group.
   c. Selects the proper THF based on the account number, FIID, and refresh group.
   d. Uses existing ITD information to determine the maximum number of THF records that can be returned per message and per call.
   e. Searches for THF records based on the transaction code and the transaction amount, source code, or check number.
   f. Calls the Transaction object to update last transaction information in the ITD.
   g. Returns information from the THF records to the Authorization Combined Positive Customer object.
8. Calls the Customer object to update PIN tries in the CSTT. The Customer object performs as follows:
   a. Calls the Transaction object to read the ITD.
   b. Calls the Customer Access object to obtain information from the CSTT.

   c. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to access the IDF extended memory table.
   d. Calls the Customer Access object to update the PIN tries in the CSTT, if necessary.
   e. Calls the Transaction object to update the ITD.
9. Sets the action code and calls the Transaction object to update it.
10. Adds the transaction history information to the message that will be returned to the IVR. This information is not placed in the ITD.
11. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Transaction object to obtain information from the ITD, format a log record, and write the log record to the ITS Transaction Log File (ITLF).
2. Calls the Message ISO object to create an ISO message that is returned to the IVR.
3. Passes control of the transaction to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the ISO message to the XPNET process.

Processing continues with step 8 of the processing summary.

IVR Available Funds Inquiry with Multiple Account Selection Authorized on the BASE24 System

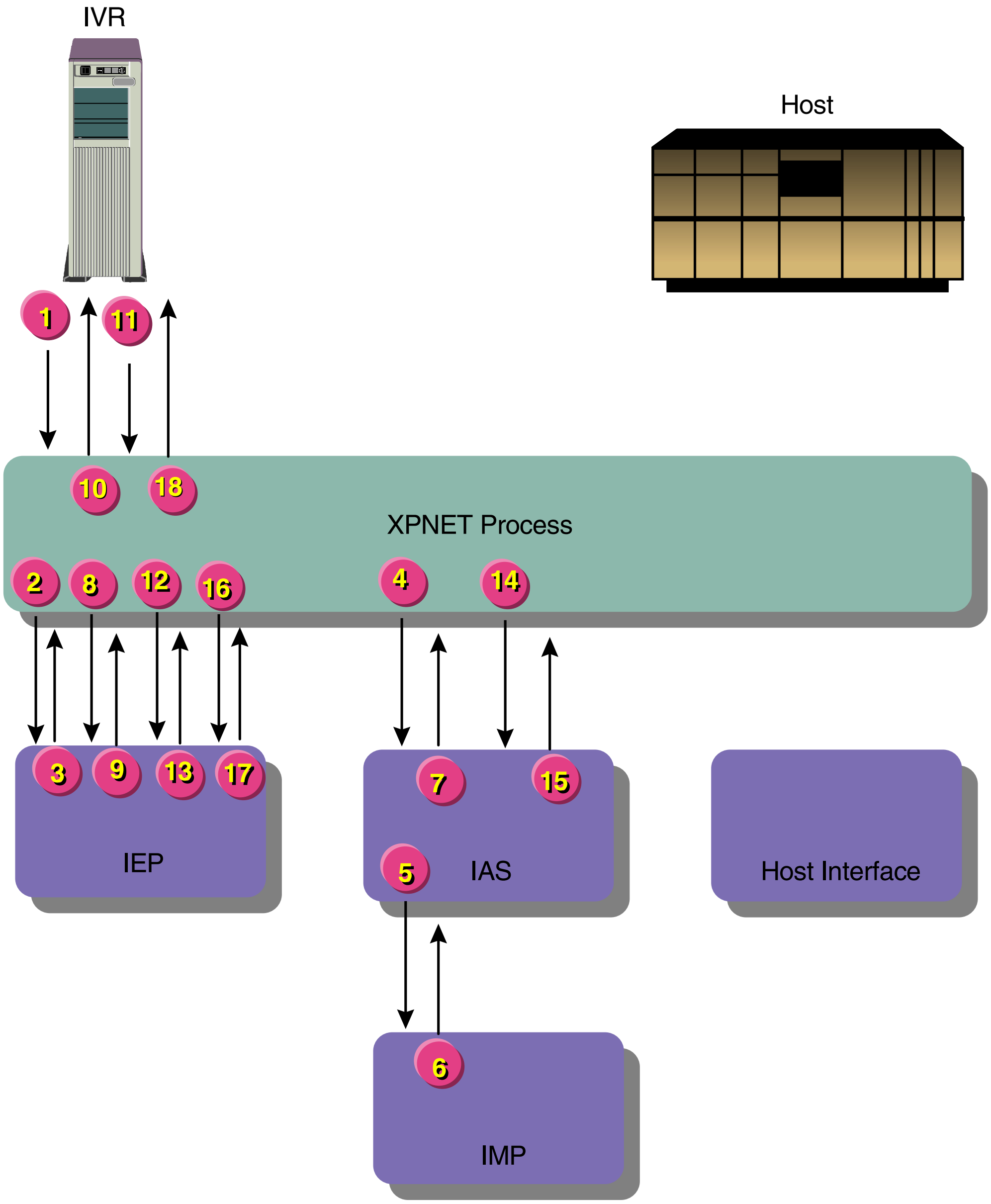
The BASE24-telebanking system obtains information from the Positive Balance File (PBF) to process available funds inquiries. The Positive Customer with Balances/History Authorization method is required to obtain balance information from the BASE24 database.

This message flow also illustrates multiple account selection, which can occur when the following conditions are met:

- The transaction is something other than a PIN change or PIN verify transaction.
- The authorization method is Positive Customer with Accounts or Positive Customer with Balances/History.
- Two or more accounts exist on the BASE24 database with the account number or account type specified by the customer.
- For transactions authorized on the host, multiple account selection must be performed using account information maintained on the BASE24 database before the transaction is sent to the host.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process
IEP = Integrated Endpoint process
IMP = Interface Master process
IVR = Interactive voice response system

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling an IVR. The IVR creates an ISO request message with the transaction request and sends the message to the XPNET process.
2. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the IVR. The XPNET process passes the ISO message to the process named in the station declaration, in this case the Integrated Endpoint process.
3. Using the source class and information from the message, the Integrated Endpoint process identifies the Integrated Authorization Server process as the destination and passes the ISO message to the XPNET process for delivery.
4. The XPNET process passes the ISO message to the Integrated Authorization Server process.
5. The Integrated Authorization Server process obtains station information and requests the Interface Master process to validate the station. See the topic “Step 5 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
6. The Interface Master process validates the station and returns a message to the Interface Manager object in the Integrated Authorization Server process.
7. The Integrated Authorization Server process obtains account selection information and returns it to the XPNET process in an ISO message. See the topic “Step 7 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
8. The XPNET process passes the ISO message with the account selection information to the Integrated Endpoint process.
9. The Integrated Endpoint process stores the message context and passes the ISO message to the XPNET process for return to the IVR. The Integrated Endpoint process uses the original station name carried in the message to identify the message destination.
10. The XPNET process passes the ISO message to the IVR. The IVR returns the response information to the customer.
11. The customer selects the desired account from the account selection information. The IVR passes the account number to the XPNET process in an ISO message.

12. The XPNET process passes the ISO message to the Integrated Endpoint process.
13. The Integrated Endpoint process retrieves the message context and passes it and the ISO message to the XPNET process.
14. The XPNET process passes the message context and ISO message to an Integrated Authorization Server process.
15. The Integrated Authorization Server process authorizes the transaction and returns an ISO message to the XPNET process. See the topic “Step 15 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
16. The XPNET process passes the ISO message to the Integrated Endpoint process.
17. The Integrated Endpoint process passes the ISO message to the XPNET process for return to the IVR. The Integrated Endpoint process uses the original station name carried in the message to identify the message destination.
18. The XPNET process passes the ISO message to the IVR. The IVR returns the response information to the customer.

Step 5 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the XPNET Interface object and Voice Response Unit Interface object in the order shown.

XPNET Interface Object Processing

The XPNET Interface object performs as follows:

1. Obtains the object ID of the destination object within the Integrated Authorization Server process (i.e., the Voice Response Unit Interface object). The Voice Response Unit Endpoint Class Handler object in the Integrated Endpoint process placed this object ID in the message.
2. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Obtains station information from the interface extended memory table. The interface extended memory table contains information from the ITS Interface Configuration File (IFCF), Interface Station Table File (IST), and Address Table File (ADRT).
2. Calls the Message ISO object to obtain information from the external message fields and updates the Internal Transaction Data (ITD).
3. Calls the Customer object to obtain the FIID. The Customer object calls the Customer Access object to obtain the FIID from the Customer Table (CSTT). The Customer Access object also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contains the account number and not the customer ID. The Customer object calls the Transaction object to update the ITD.
4. Calls the Interface Manager object to validate the station. The Interface Manager object in the Integrated Authorization Server process passes a message to the Interface Master process.

Processing continues with step 6 of the processing summary.

Step 7 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the following objects in the order shown:

- Voice Response Unit Interface object
- Router object
- Authorization Combined Positive Customer object
- Voice Response Unit Interface object
- XPNET Interface object

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Financial Institution object to retrieve and check customer service information from the IDF extended memory table if the source code in the ITD is set to a value of IB.
2. Calls the Transaction Security object to translate the PIN, if necessary. The Transaction Security object obtains security information from the Interface Security File (ISF) extended memory table and calls the Transaction object to update the ITD if PIN translation occurs.
3. Calls the Transaction object to update the ITD with additional information.
4. Passes control of the transaction to the Router object.

Router Object Processing

The Router object performs as follows:

1. Calls the Transaction object to obtain information from the ITD.
2. Calls the Financial Institution object to obtain the route profile for the FIID. The Financial Institution object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution Access object to obtain processing information from the Institution Definition File (IDF) extended memory table.
    c. Calls the Transaction object to update the ITD.
3. Obtains routing configuration information from the institution routing extended memory table. The institution routing extended memory table contains information from the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF).
4. Calls the Transaction object to update configuration information in the ITD.
5. Passes control of the transaction to the Authorization Combined Positive Customer object.

Authorization Combined Positive Customer Object Processing

The Authorization Combined Positive Customer object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Customer object to determine whether the customer is permitted to perform the transaction. The Customer object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain information from the IDF extended memory table.
    c. Calls the Customer Account object to access the customer extended memory table, CSTT, and CACT to obtain the accounts to present to the customer for multiple account selection. The customer extended memory table contains the same information as the Customer Allowed Transaction Table (CATT).
    d. Calls the Transaction object to update the ITD.
3. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain authorization and customer service information from the IDF extended memory table.
4. Determines whether a PIN change is required. If a PIN change is required, the transaction is declined.
5. Calls the Transaction Security object to verify the PIN. The Transaction Security object performs as follows:
    a. Calls the Transaction object to obtain PIN information from the ITD.
    b. Checks the point of service information in the ITD to determine whether the PIN needs to be verified. If the PIN capture flag is set to S, processing skips to step 6, otherwise processing continues as follows.
    c. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain PIN verification information from the IDF extended memory table.
    d. Calls the Authorization Security Access object to obtain PIN verification information from the Key Authorization File (KEYA) extended memory table.

   e. Verifies the PIN. The Transaction Security object can verify the PIN in software or call a security device to verify the PIN.
   f. Calls the Transaction object to update the PIN verification results in the ITD and delete the PIN from the ITD.
6. Checks the authorization method. An available funds inquiry requires the Positive Customer with Balances/History Authorization method.
7. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Transaction object to obtain account selection information from the ITD.
2. Calls the Message ISO object to create an ISO message that is returned to the IVR.
3. Passes control of the transaction to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the account selection information in an ISO message to the XPNET process.

Processing continues with step 8 of the processing summary.

Step 15 Processing Detail

The Integrated Authorization Server process controls transaction processing with the following objects in the order shown:

- XPNET Interface object
- Voice Response Unit Interface object
- Authorization Combined Positive Customer object
- Voice Response Unit Interface object
- XPNET Interface object

XPNET Interface Object Processing

The XPNET Interface object performs as follows:

1. Obtains the object ID of the destination object within the Integrated Authorization Server process (i.e., the Voice Response Unit Interface object). The Voice Response Unit Endpoint Class Handler object in the Integrated Endpoint process placed this object ID in the message.
2. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Transaction object to place the context information in the proper ITD fields.
2. Calls the Message ISO object to obtain account information from the external message fields and update the ITD fields.
3. Passes control of the transaction to the Authorization Combined Positive Customer object.

Authorization Combined Positive Customer Object Processing

The Authorization Combined Positive Customer object performs as follows:

1. Calls the Transaction object to read the ITD.
2. Calls the Customer object to determine which account has been selected. The Customer object performs as follows:
    a. Calls the Transaction object to read the ITD.
    b. Calls the Customer Access object to read the CSTT row.
    c. Determines which account has been selected.
3. Checks the authorization method. An available funds inquiry requires the Positive Customer with Balances/History Authorization method.

4. Calls the Account object to obtain the account balance. The Account object performs as follows:
    a. Calls the Transaction object to read the ITD.
    b. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain usage dates and backup account parameters.
    c. Calls the Account Access object to obtain the account balance from the PBF and place the information in the PBF extended memory table.
    d. Calls the Transaction object to update balance information in the ITD.
5. Calls the Customer object to update PIN tries in the CSTT. The Customer object performs as follows:
    a. Calls the Transaction object to read the ITD.
    b. Calls the Customer Access object to read the CSTT.
    c. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to access the IDF extended memory table.
    d. Calls the Customer Access object to update the PIN tries in the CSTT, if necessary.
    e. Calls the Transaction object to update the ITD.
6. Sets the action code.
7. Calls the Transaction object to update the ITD for the completion of authorization processing.
8. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Transaction object to obtain information from the ITD, format a log record, and write the log record to the ITS Transaction Log File (ITLF).
2. Calls the Message ISO object to create an ISO message that is returned to the IVR.
3. Passes control of the transaction to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the ISO message to the XPNET process.

Processing continues with step 16 of the processing summary.

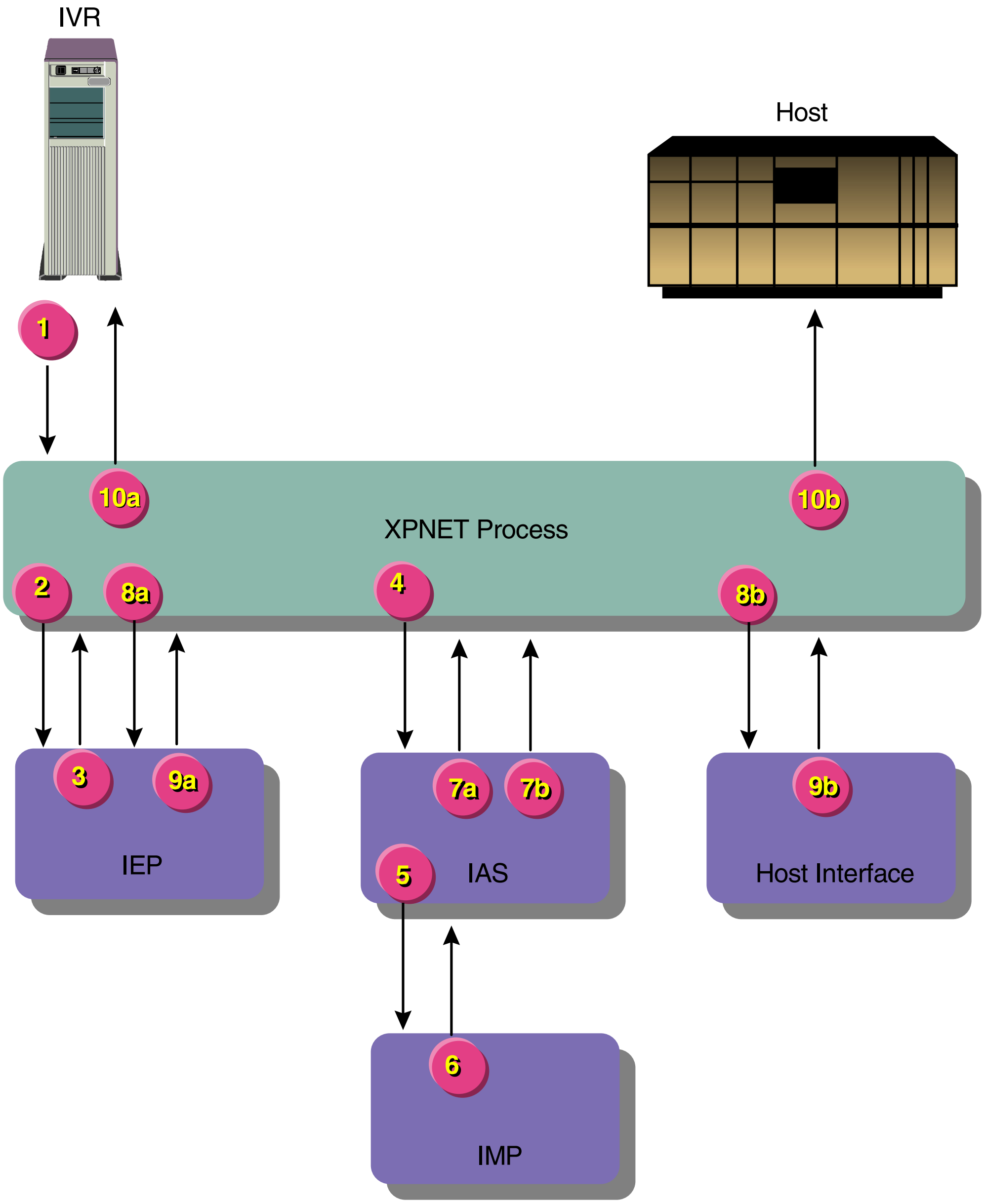
IVR Transfer Authorized on the BASE24 System with an Advice

BASE24-telebanking transfer transactions authorized on the BASE24 system obtain and update information contained in the Positive Balance File (PBF). This message flow assumes the Positive Customer with Balances/History Authorization method for the *from* and *to* accounts. The BASE24-telebanking system also supports one authorization method for the *from* account and another authorization method for the *to* account. However, the Positive Customer with Balances/History Authorization method is required to maintain account balances on the BASE24 system.

This message flow includes an advice to one host for the *from* and *to* accounts. The BASE24-telebanking system also supports sending an advice to one host for the *from* account and sending an advice to a different host for the *to* account.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process
IEP = Integrated Endpoint process
IMP = Interface Master process
IVR = Interactive voice response system

Note: At step 7, the message processing splits into concurrent flows. The message to the IVR is identified by the letter *a* in each step number. The message to the host is identified by the letter *b* in each step number.

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling an IVR. The IVR creates an ISO request message with the transaction request and sends the message to the XPNET process.
2. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the IVR. The XPNET process passes the ISO message to the process named in the station declaration, in this case the Integrated Endpoint process.
3. Using the source class and information from the message, the Integrated Endpoint process identifies the Integrated Authorization Server process as the destination and passes the ISO message to the XPNET process for delivery.
4. The XPNET process passes the ISO message to the Integrated Authorization Server process.
5. The Integrated Authorization Server process obtains station information and requests the Interface Master process to validate the station. See the topic “Step 5 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
6. The Interface Master process validates the station and returns a message to the Interface Manager object in the Integrated Authorization Server process.
7. The Integrated Authorization Server process authorizes the transaction and passes an ISO message and a Telebanking Standard Internal Message (BSTM) to the XPNET process. Refer to the topic “Step 7 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.

   Processing of the ISO message (i.e., the transaction response) continues with step 8a. Processing of the BSTM (i.e., the advice to the host) continues with step 8b.

8a. The XPNET process passes the ISO message with the original station name to the Integrated Endpoint process.

8b. The XPNET process passes the BSTM to the ISO Host Interface process.

9a. The Integrated Endpoint process passes the ISO message to the XPNET process for return to the IVR. The Integrated Endpoint process uses the original station name carried in the message to identify the message destination.

9b. The ISO Host Interface process reformats the BSTM into an ISO external message and passes it to the XPNET process.

10a. The XPNET process passes the ISO message to the IVR. The IVR returns the response information to the customer.

10b. The XPNET process passes the ISO external message to the host.

Step 5 Processing Detail

The Integrated Authorization Server process controls transaction processing with the XPNET Interface object and Voice Response Unit Interface object in the order shown.

XPNET Interface Object Processing

The XPNET Interface object performs as follows:

1. Obtains the object ID of the destination object within the Integrated Authorization Server process (i.e., the Voice Response Unit Interface object). The Voice Response Unit Endpoint Class Handler object in the Integrated Endpoint process placed this object ID in the message.
2. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Obtains station information from the interface extended memory table. The interface extended memory table contains information from the ITS Interface Configuration File (IFCF), Interface Station Table File (IST), and Address Table File (ADRT).
2. Calls the Message ISO object to obtain information from the external message fields and updates the Internal Transaction Data (ITD).

3. Calls the Customer object to obtain the FIID. The Customer object calls the Customer Access object to obtain the FIID from the Customer Table (CSTT). The Customer Access object also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contains the account number and not the customer ID. The Customer object calls the Transaction object to update the ITD.
4. Calls the Interface Manager object to validate the station. The Interface Manager object in the Integrated Authorization Server process passes a message to the Interface Master process.

Processing continues with step 6 of the processing summary.

Step 7 Processing Detail

The Integrated Authorization Server process controls transaction processing with the following objects in the order shown:

- Voice Response Unit Interface object
- Router object
- Authorization Combined Positive Customer object
- Voice Response Unit Interface object
- Internal Message Translation Interface object
- Voice Response Unit Interface object
- XPNET Interface object

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Financial Institution object to retrieve and check customer service information from the IDF extended memory table if the source code in the ITD is set to a value of IB.
2. Calls the Transaction Security object to translate the PIN, if necessary. The Transaction Security object obtains security information from the Interface Security File (ISF) extended memory table and calls the Transaction object to update the ITD if PIN translation occurs.

3. Calls the Transaction object to update the ITD with additional information.
4. Passes control of the transaction to the Router object.

Router Object Processing

The Router object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Financial Institution object to obtain the route profile for the FIID. The Financial Institution object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution Access object to obtain processing information from the Institution Definition File (IDF) extended memory table.
    c. Calls the Transaction object to update the ITD.
3. Obtains routing configuration information for the *from* and *to* accounts from the institution routing extended memory table. The institution routing extended memory table contains information from the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF).
4. Calls the Transaction object to update configuration information in the ITD.
5. Passes control of the transaction to the Authorization Combined Positive Customer object.

Authorization Combined Positive Customer Object Processing

The Authorization Combined Positive Customer object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Customer object to determine whether the customer is permitted to perform the transaction. The Customer object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.

   b. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain information from the IDF extended memory table.
   c. Calls the Customer Access object to access the customer extended memory table, CSTT, and CACT. The customer extended memory table contains the same information as the Customer Allowed Transaction Table (CATT).
   d. Calls the Transaction object to update the ITD.
3. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain authorization and customer service information from the IDF extended memory table.
4. Determines whether a PIN change is required. If a PIN change is required, the transaction is declined.
5. Calls the Transaction Security object to verify the PIN. The Transaction Security object performs as follows:
   a. Calls the Transaction object to obtain PIN information from the ITD.
   b. Checks the point of service information in the ITD to determine whether the PIN needs to be verified. If the PIN capture flag is set to S, processing skips to step 6, otherwise processing continues as follows.
   c. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain PIN verification information from the IDF extended memory table.
   d. Calls the Authorization Security Access object to obtain PIN verification information from the Key Authorization File (KEYA) extended memory table.
   e. Verifies the PIN. The Transaction Security object can verify the PIN in software or call a security device to verify the PIN.
   f. Calls the Transaction object to update the PIN verification results in the ITD and delete the PIN from the ITD.
6. Calls the Account object to obtain account information and perform the transfer. The Account object performs as follows:
   a. Calls the Transaction object to read the ITD.
   b. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain usage dates from the IDF extended memory table.

   c. Calls the Account Access object to obtain balances, holds, limits, and usages for the *from* and *to* accounts from the PBF and place the information in the PBF extended memory table.
   d. Performs the transfer, which includes the following:
      1) Checking for sufficient available funds in the *from* account.
      2) Checking the usage and individual transaction limits for the *from* account.
      3) Calculating the new account balances and usage amounts for the *from* and *to* accounts.
   e. Calls the Account Access object to update the PBF records with the new account balance and usage information.
   f. Calls the Transaction object to update balance and usage information in the ITD.

7. Calls the Customer object to update PIN tries in the CSTT. The Customer object performs as follows:
   a. Calls the Transaction object to read the ITD.
   b. Calls the Customer Access object to read the CSTT.
   c. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to access the IDF extended memory table.
   d. Updates the PIN tries in the CSTT, if necessary.
   e. Calls the Transaction object to update the ITD.
8. Sets the action code.
9. Calls the Transaction object to update the ITD for the completion of authorization processing.
10. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Transaction object to obtain information from the ITD.
2. Determines that an advice is required.

3. Passes control of the transaction to the Internal Message Translation Interface object.

Internal Message Translation Interface Object Processing

The Internal Message Translation Interface object performs as follows:

1. Calls the Transaction object to obtain information for the advice from the ITD.
2. Obtains host routing information from the Internal Message Translation File (IMTF) extended memory table.
3. Creates a BSTM for the ISO Host Interface process.

   Note: The Internal Message Translation Interface object passes the BSTM to the XPNET Interface object immediately after the ISO message for the IVR is sent to the XPNET process.
4. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Transaction object to obtain information from the ITD, format a log record, and write the log record to the ITS Transaction Log File (ITLF).
2. Calls the Message ISO object to create an ISO message that is returned to the IVR.
3. Passes control of the transaction to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object performs as follows:

1. Passes the ISO message containing the transaction response to the XPNET process. This is illustrated on the processing diagram as step 7a.
2. Passes the BSTM (i.e., the advice) to the XPNET process. This is illustrated on the processing diagram as step 7b.

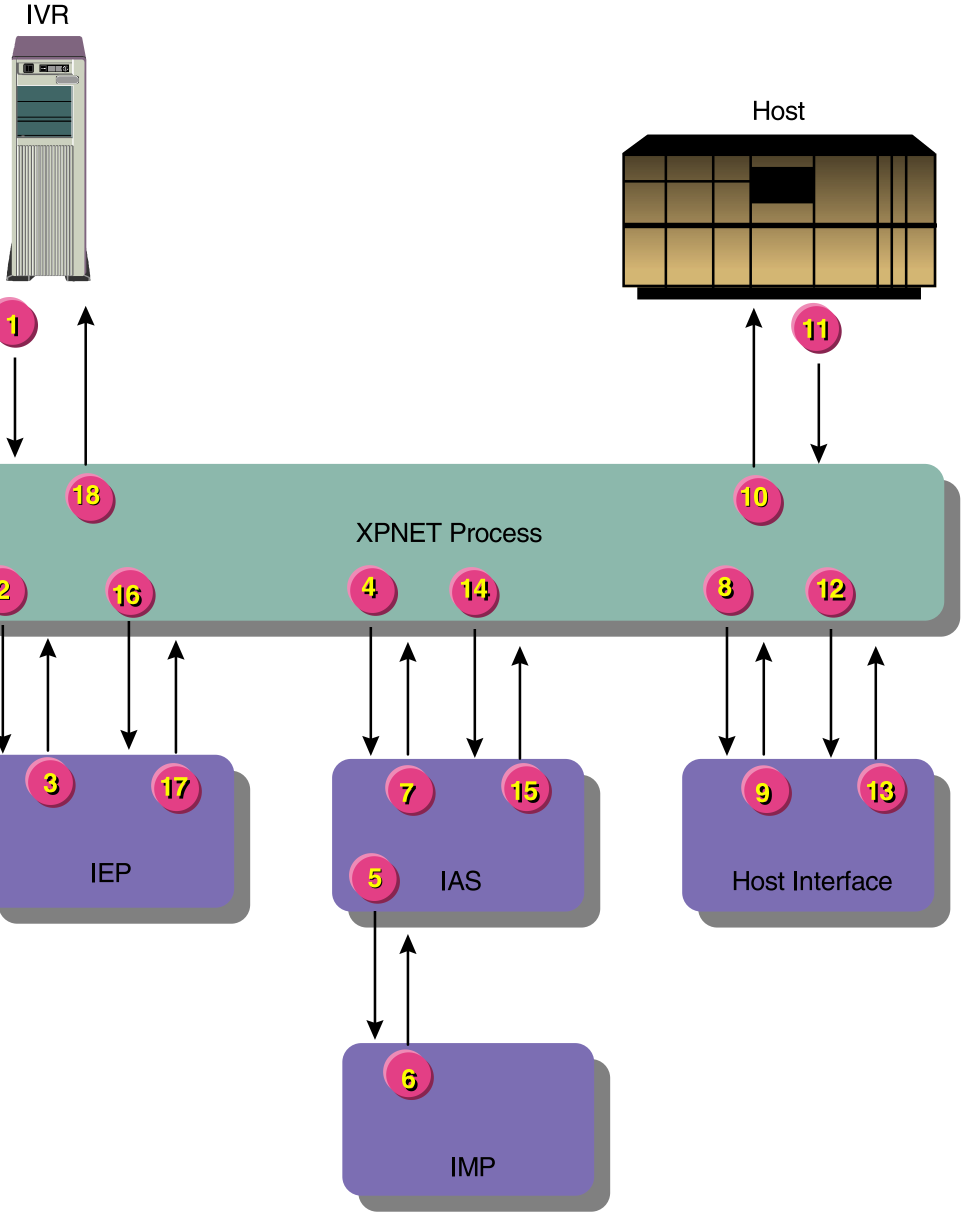
Processing continues with steps 8a and 8b of the processing summary.

IVR Available Funds Inquiry Preauthorized on the BASE24 System and Authorized on the Host

BASE24-telebanking available funds inquiry transactions can be preauthorized on the BASE24 system using the online/offline authorization level and any authorization method. These transactions also can be preauthorized on the BASE24 system using the online authorization level and the Positive Customer and Positive Customer with Account Authorization methods. With the Positive Customer Authorization method, the BASE24-telebanking system can check the customer status and customer PIN. With the Positive Customer with Account Authorization method and the Positive Customer with Balances/History Authorization method, the BASE24-telebanking system can check the customer information and check the account status, including whether the requested transaction can be performed on the account.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process
IEP = Integrated Endpoint process
IMP = Interface Master process
IVR = Interactive voice response system

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling an IVR. The IVR creates an ISO request message with the transaction request and sends the message to the XPNET process.
2. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the IVR. The XPNET process passes the ISO message to the process named in the station declaration, in this case the Integrated Endpoint process.
3. Using the source class and information from the message, the Integrated Endpoint process identifies the Integrated Authorization Server process as the destination and passes the ISO message to the XPNET process for delivery.
4. The XPNET process passes the ISO message to the Integrated Authorization Server process.
5. The Integrated Authorization Server process obtains station information and requests the Interface Master process to validate the station. See the topic “Step 5 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
6. The Interface Master process validates the station and returns a message to the Interface Manager object in the Integrated Authorization Server process.
7. The Integrated Authorization Server process preauthorizes the transaction and passes a Telebanking Standard Internal Message (BSTM) to the XPNET process. Refer to the topic “Step 7 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
8. The XPNET process passes the BSTM to the ISO Host Interface process for the host that is to authorize the request.
9. The ISO Host Interface process reformats the BSTM into an ISO external message and passes it to the XPNET process.
10. The XPNET process passes the ISO external message to the host for authorization.
11. The host authorizes the transaction and returns the ISO external message to the XPNET process.
12. The XPNET process passes the ISO external message to the ISO Host Interface process.

13. The ISO Host Interface process reformats the ISO external message into a BSTM and passes it to the XPNET process for return to the Integrated Authorization Server process.
14. The XPNET process passes the BSTM to the Integrated Authorization Server process.
15. The Integrated Authorization Server process updates the BASE24 database and passes an ISO message to the XPNET process. Refer to the topic “Step 15 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
16. The XPNET process passes the ISO message to the Integrated Endpoint process.
17. The Integrated Endpoint process passes the ISO message with the requested balance information to the XPNET process for return to the IVR. The Integrated Endpoint process uses the original station name carried in the message to identify the message destination.
18. The XPNET process passes the ISO message to the IVR. The IVR returns the response information to the customer.

Step 5 Processing Detail

The Integrated Authorization Server process controls transaction processing with the XPNET Interface object and Voice Response Unit Interface object in the order shown.

XPNET Interface Object Processing

The XPNET Interface object performs as follows:

1. Obtains the object ID of the destination object within the Integrated Authorization Server process (i.e., the Voice Response Unit Interface object). The Voice Response Unit Endpoint Class Handler object in the Integrated Endpoint process placed this object ID in the message.
2. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Obtains station information from the interface extended memory table. The interface extended memory table contains information from the ITS Interface Configuration File (IFCF), Interface Station Table File (IST), and Address Table File (ADRT).
2. Calls the Message ISO object to obtain information from the external message fields and updates the Internal Transaction Data (ITD).
3. Calls the Customer object to obtain the FIID. The Customer object calls the Customer Access object to obtain the FIID from the Customer Table (CSTT). The Customer Access object also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contains the account number and not the customer ID. The Customer object calls the Transaction object to update the ITD.
4. Calls the Interface Manager object to validate the station. The Interface Manager object in the Integrated Authorization Server process passes a message to the Interface Master process.

Processing continues with step 6 of the processing summary.

Step 7 Processing Detail

The Integrated Authorization Server process controls transaction processing with the following objects in the order shown:

- Voice Response Unit Interface object
- Router object
- Internal Message Translation Interface object
- XPNET Interface object

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Financial Institution object to retrieve and check customer service information from the IDF extended memory table if the source code in the ITD is set to a value of IB.

2. Calls the Transaction Security object to translate the PIN, if necessary. The Transaction Security object obtains security information from the Interface Security File (ISF) extended memory table and calls the Transaction object to update the ITD if PIN translation occurs.
3. Calls the Transaction object to update the ITD with additional information.
4. Passes control of the transaction to the Router object.

Router Object Processing

The Router object performs as follows:

1. Calls the Transaction object to obtain information from the ITD.
2. Calls the Financial Institution object to obtain the route profile for the FIID. The Financial Institution object performs as follows:
   a. Calls the Transaction object to obtain information from the ITD.
   b. Calls the Financial Institution Access object to obtain processing information from the Institution Definition File (IDF) extended memory table.
   c. Calls the Transaction object to update the ITD.
3. Obtains routing configuration information from the institution routing extended memory table. The institution routing extended memory table contains information from the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF).
4. Calls the Transaction object to update configuration information in the ITD.
5. Passes control of the transaction to the Internal Message Translation Interface object.

Internal Message Translation Interface Object Processing

The Internal Message Translation Interface object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.

2. Calls the Authorization Combined Positive Customer object to preauthorize the transaction. The Authorization Combined Positive Customer object performs as follows:
    a. Calls the Transaction object to obtain the necessary information from the ITD.
    b. Calls the Customer object to determine whether the customer is permitted to perform the transaction. The Customer object performs as follows:
        1) Calls the Transaction object to read the ITD.
        2) Calls the Financial Institution object. The Financial Institution object calls the Financial Institution Access object to obtain authorization and customer service information from the IDF extended memory table.
        3) Calls the Customer Access object to access the customer extended memory table, CSTT, and CACT when the Positive Customer with Accounts Authorization method or the Positive Customer with Balances/History Authorization method is used. The Customer Access object accesses the customer extended memory table and CSTT without the CACT when the Positive Customer Authorization method is used. The customer extended memory table contains the same information as the Customer Allowed Transaction Table (CATT).
    c. If the CARD STATUS field on IDF screen 2 contains a value of Y, the Authorization Combined Positive Customer object checks the VERIFY STATUS field on CSTT screen 1 for a value of V (verified).
    d. Determines whether a PIN change is required. If a PIN change is required, the transaction is declined.
    e. If the PIN field on IDF screen 2 contains a value of Y, the Authorization Combined Positive Customer object calls the Transaction Security object to verify the PIN. The Transaction Security object performs as follows:
        1) Calls the Transaction object to obtain PIN information from the ITD.
        2) Checks the point of service information in the ITD to determine whether the PIN needs to be verified. If the PIN capture flag is set to S, processing skips to step f, otherwise processing continues as follows.

        3) Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain PIN verification information from the IDF extended memory table.
        4) Calls the Authorization Security Access object to obtain PIN verification information from the Key Authorization File (KEYA) extended memory table.
        5) Verifies the PIN. The Transaction Security object can verify the PIN in software or call a security device to verify the PIN.
        6) Calls the Transaction object to update the PIN verification results in the ITD and delete the PIN from the ITD.
    f. Calls the Transaction object to update the ITD.

3. Obtains the ISO Host Interface process name and DPC number from the Internal Message Translation File (IMTF) extended memory table.
4. Calls the Transaction object to compress the ITD.
5. Creates a BSTM for the ISO Host Interface process.
6. Passes control of the transaction to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the BSTM to the XPNET process.

Processing continues with step 8 of the processing summary.

Step 15 Processing Detail

The Integrated Authorization Server process controls transaction processing with the following objects in the order shown:

- XPNET Interface object
- Internal Message Translation Interface object
- Voice Response Unit Interface object
- XPNET Interface object

XPNET Interface Object Processing

The XPNET Interface object performs as follows:

1. Obtains the object ID of the destination object within the Integrated Authorization Server process (i.e., the Internal Message Translation Interface object).
2. Passes control of the transaction to the Internal Message Translation Interface object.

Internal Message Translation Interface Object Processing

The Internal Message Translation Interface object performs as follows:

1. Calls the Transaction object to expand the compressed ITD contained in the BSTM received from the ISO Host Interface process and place it in the ITD.
2. Calls the Authorization Combined Positive Customer object to update PIN tries and transaction dates. The Authorization Combined Positive Customer object performs as follows:
    a. Calls the Transaction object to obtain the action code, bad PIN count, and processing code from the ITD.
    b. Determines whether the bad PIN count must be changed, based on the action code. If the transaction is approved or the transaction is declined with a bad PIN action code, the Authorization Combined Positive Customer object calls the Customer object. If the bad PIN count does not need to change, processing continues with part c of this step. If the bad PIN count must be changed, the Authorization Combined Positive Customer object calls the Customer object to update PIN processing information in the CSTT. The Customer object performs as follows:
        1) Calls the Transaction object to obtain customer identification information from the ITD.
        2) Calls the Customer Access object to read the CSTT row for the customer.
        3) Calls the Financial Institution object. The Financial Institution object calls the Transaction object to read the ITD, then calls the Financial Institution Access object to obtain PIN processing parameters from the IDF extended memory table.
        4) Calls the Customer Access object to update the BAD PIN COUNT field on CSTT screen 1.

   c. Calls the Transaction object to update the ITD to indicate that the host returned an authorization response.

3. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Transaction object to obtain information from the ITD.
2. Calls the Transaction object to format a log record and write the log record to the ITS Transaction Log File (ITLF).
3. Calls the Message ISO object to create an ISO message that is returned to the IVR.
4. Passes control of the transaction to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the ISO message to the XPNET process.

Processing continues with step 16 of the processing summary.

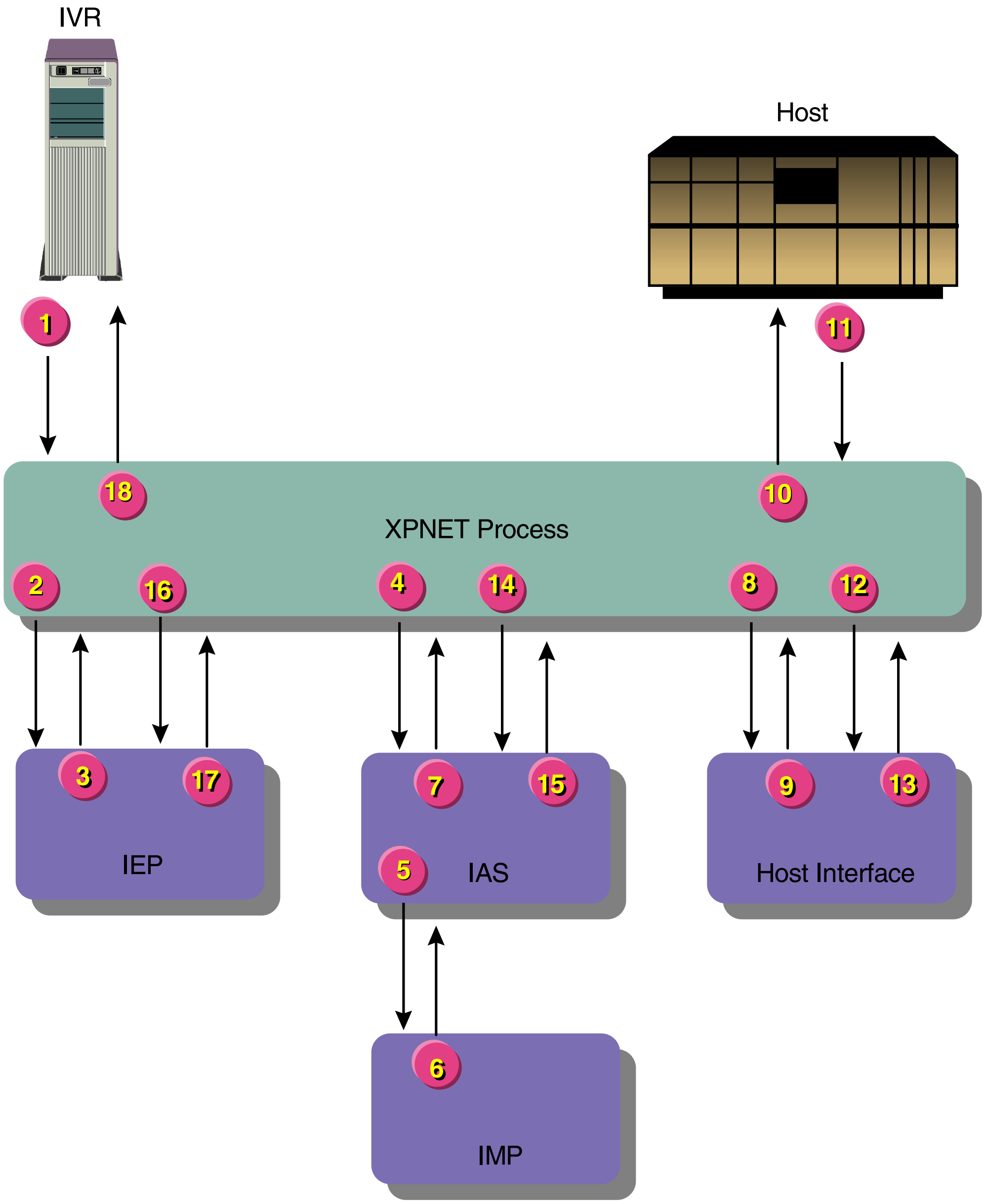
IVR Transfer Preauthorized on the BASE24 System and Authorized on the Host

BASE24-telebanking transfer transactions can be preauthorized on the BASE24 system using the online/offline authorization level and any authorization method. These transactions also can be preauthorized on the BASE24 system using the online authorization level and the Positive Customer and Positive Customer with Account Authorization methods. With the Positive Customer Authorization method, the BASE24-telebanking system can check the customer status, customer profile, and customer PIN. With the Positive Customer with Account Authorization method and the Positive Customer with Balances/History Authorization method, the BASE24-telebanking system can check the customer information and check the account status, including whether the requested transaction can be performed on the account.

Transfer transactions that are authorized on the host can have both accounts on the same host or have each account on a separate host. This message flow illustrates a transfer transaction with both accounts on the same host.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process
IEP = Integrated Endpoint process
IMP = Interface Master process
IVR = Interactive voice response system

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling an IVR. The IVR creates an ISO request message with the transaction request and sends the message to the XPNET process.
2. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the IVR. The XPNET process passes the ISO message to the process named in the station declaration, in this case the Integrated Endpoint process.
3. Using the source class and information from the message, the Integrated Endpoint process identifies the Integrated Authorization Server process as the destination and passes the ISO message to the XPNET process for delivery.
4. The XPNET process passes the ISO message to the Integrated Authorization Server process.
5. The Integrated Authorization Server process obtains station information and requests the Interface Master process to validate the station. See the topic "Step 5 Processing Detail" that follows this summary for a detailed description of the processing performed in this step.
6. The Interface Master process validates the station and returns a message to the Interface Manager object in the Integrated Authorization Server process.
7. The Integrated Authorization Server process preauthorizes the transaction and passes a Telebanking Standard Internal Message (BSTM) to the XPNET process. Refer to the topic "Step 7 Processing Detail" that follows this summary for a detailed description of the processing performed in this step.
8. The XPNET process passes the BSTM to the ISO Host Interface process for the host that is to authorize the request.
9. The ISO Host Interface process reformats the BSTM into an ISO external message and passes it to the XPNET process.
10. The XPNET process passes the ISO external message to the host for authorization.
11. The host authorizes the transaction and returns the ISO external message to the XPNET process.
12. The XPNET process passes the ISO external message to the ISO Host Interface process.
13. The ISO Host Interface process reformats the ISO external message into a BSTM and passes it to the XPNET process.

14. The XPNET process passes the BSTM to the Integrated Authorization Server process.
15. The Integrated Authorization Server process updates the BASE24 database and passes an ISO message to the XPNET process. Refer to the topic “Step 15 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
16. The XPNET process passes the ISO message to the Integrated Endpoint process.
17. The Integrated Endpoint process passes the ISO message to the XPNET process for return to the IVR. The Integrated Endpoint process uses the original station name carried in the message to identify the message destination.
18. The XPNET process passes the ISO message to the IVR.

Step 5 Processing Detail

The Integrated Authorization Server process controls transaction processing with the XPNET Interface object and Voice Response Unit Interface object in the order shown.

XPNET Interface Object Processing

The XPNET Interface object performs as follows:

1. Obtains the object ID of the destination object within the Integrated Authorization Server process (i.e., the Voice Response Unit Interface object). The Voice Response Unit Endpoint Class Handler object in the Integrated Endpoint process placed this object ID in the message.
2. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Obtains station information from the interface extended memory table. The interface extended memory table contains information from the ITS Interface Configuration File (IFCF), Interface Station Table File (IST), and Address Table File (ADRT).

2. Calls the Message ISO object to obtain information from the external message fields and updates the Internal Transaction Data (ITD).
3. Calls the Customer object to obtain the FIID. The Customer object calls the Customer Access object to obtain the FIID from the Customer Table (CSTT). The Customer Access object also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contains the account number and not the customer ID. The Customer object calls the Transaction object to update the ITD.
4. Calls the Interface Manager object to validate the station. The Interface Manager object in the Integrated Authorization Server process passes a message to the Interface Master process.

Processing continues with step 6 of the processing summary.

Step 7 Processing Detail

The Integrated Authorization Server process controls transaction processing with the following objects in the order shown:

- Voice Response Unit Interface object
- Router object
- Internal Message Translation Interface object
- XPNET Interface object

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Financial Institution object to retrieve and check customer service information from the IDF extended memory table if the source code in the ITD is set to a value of IB.
2. Calls the Transaction Security object to translate the PIN, if necessary. The Transaction Security object obtains security information from the Interface Security File (ISF) extended memory table and calls the Transaction object to update the ITD if PIN translation occurs.
3. Calls the Transaction object to update the ITD with additional information.
4. Passes control of the transaction to the Router object.

Router Object Processing

The Router object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Financial Institution object to obtain the route profile for the FIID. The Financial Institution object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution Access object to obtain processing information from the Institution Definition File (IDF) extended memory table.
    c. Calls the Transaction object to update the ITD.
3. Obtains routing configuration information from the institution routing extended memory table. The institution routing extended memory table contains information from the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF).
4. Calls the Transaction object to update configuration information in the ITD.
5. Passes control of the transaction to the Internal Message Translation Interface object.

Internal Message Translation Interface Object Processing

The Internal Message Translation Interface object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Authorization Combined Positive Customer object to preauthorize the transaction. The Authorization Combined Positive Customer object performs as follows:
    a. Calls the Transaction object to obtain the necessary information from the ITD.
    b. Calls the Customer object to determine whether the customer is permitted to perform the transaction. The Customer object performs as follows:
        1) Calls the Transaction object to read the ITD.

    2) Calls the Financial Institution object. The Financial Institution object calls the Financial Institution Access object to obtain authorization and customer service information from the IDF extended memory table.

    3) Calls the Customer Access object to access the customer extended memory table, CSTT, and CACT when the Positive Customer with Accounts Authorization method or the Positive Customer with Balances/History Authorization method is used. The Customer Access object accesses the customer extended memory table and CSTT without the CACT when the Positive Customer Authorization method is used. The customer extended memory table contains the same information as the Customer Allowed Transaction Table (CATT).

c. If the CARD STATUS field on IDF screen 2 contains a value of Y, the Authorization Combined Positive Customer object checks the VERIFY STATUS field on CSTT screen 1 for a value of V (verified).

d. Determines whether a PIN change is required. If a PIN change is required, the transaction is declined.

e. If the PIN field on IDF screen 2 contains a value of Y, the Authorization Combined Positive Customer object calls the Transaction Security object to verify the PIN. The Transaction Security object performs as follows:

    1) Calls the Transaction object to obtain PIN information from the ITD.

    2) Checks the point of service information in the ITD to determine whether the PIN needs to be verified. If the PIN capture flag is set to S, processing skips to step f, otherwise processing continues as follows.

    3) Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain PIN verification information from the IDF extended memory table.

    4) Calls the Authorization Security Access object to obtain PIN verification information from the Key Authorization File (KEYA) extended memory table.

    5) Verifies the PIN. The Transaction Security object can verify the PIN in software or call a security device to verify the PIN.

    6) Calls the Transaction object to update the PIN verification results in the ITD and delete the PIN from the ITD.

f. Calls the Transaction object to update the ITD.

3. Obtains the ISO Host Interface process name and DPC number from the Internal Message Translation File (IMTF) extended memory table.
4. Calls the Transaction object to compress the ITD.
5. Creates a BSTM for the ISO Host Interface process.
6. Passes control of the transaction to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the BSTM to the XPNET process.

Processing continues with step 8 of the processing summary.

Step 15 Processing Detail

The Integrated Authorization Server process controls transaction processing with the following objects in the order shown:

- XPNET Interface object
- Internal Message Translation Interface object
- Voice Response Unit Interface object
- XPNET Interface object

XPNET Interface Object Processing

The XPNET Interface object performs as follows:

1. Obtains the object ID of the destination object within the Integrated Authorization Server process (i.e., the Internal Message Translation Interface object).
2. Passes control of the transaction to the Internal Message Translation Interface object.

Internal Message Translation Interface Object Processing

The Internal Message Translation Interface object performs as follows:

1. Calls the Transaction object to expand the compressed ITD contained in the BSTM received from the ISO Host Interface process and place it in the ITD.
2. Calls the Authorization Combined Positive Customer object to update PIN tries, transaction dates, and account balances. The Authorization Combined Positive Customer object performs as follows:
    a. Calls the Transaction object to obtain the action code, bad PIN count, and processing code from the ITD.

    b. Determines whether the bad PIN count must be changed, based on the action code. If the transaction is approved or the transaction is declined with a bad PIN action code, the Authorization Combined Positive Customer object calls the Customer object. If the bad PIN count does not need to change, processing continues with part c of this step. If the bad PIN count must be changed, the Authorization Combined Positive Customer object calls the Customer object to update PIN processing information in the CSTT. The Customer object performs as follows:
        1) Calls the Transaction object to obtain customer identification information from the ITD.
        2) Calls the Customer Access object to read the CSTT row for the customer.
        3) Calls the Financial Institution object. The Financial Institution object calls the Transaction object to read the ITD, then calls the Financial Institution Access object to obtain PIN processing parameters from the IDF extended memory table.
        4) Updates the BAD PIN COUNT field on CSTT screen 1.

    c. Calls the Account object to update BASE24 account information as a result of the transfer if the Positive Customer with Balances/History Authorization method is being used for the *from* account. Otherwise, processing continues with part d of this step. To update account information, the Account object performs as follows:
        1) Calls the Transaction object to read the ITD.
        2) Calls the Financial Institution object. The Financial Institution object calls the Financial Institution Access object to obtain backup account parameters and the dates used to calculate usage amounts.

        3) Calls the Account Access object to obtain balances, holds, limits, and usages for the account from the Positive Balance File (PBF) and place the information in the PBF extended memory table.

        4) Calculates the new account balance information.

        5) Calls the Account Access object to update the PBF records with the new account balance and usage information.

        6) Calls the Transaction object to update balance and usage information in the ITD.

        Note: The *to* account is included in part c of this step if the Positive Customer with Balances/History Authorization method is used for both the *from* and *to* accounts. The *to* account is not updated at this time if the Positive Customer with Balances/History Authorization method is not used for the *from* account.

    d. Calls the Transaction object to update the ITD to indicate that the host returned an authorization response.

3. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Transaction object to obtain information from the ITD.

2. Obtains advice and impacting information from the ITD. For this transaction, no advices need to be sent and additional impacting is unnecessary.

    No advices need to be sent because the transaction was approved by the host and only one host was involved. If two hosts are involved, the Voice Response Unit Interface object calls the Internal Message Translation Interface object at this time to send an advice to the second host.

    The need for impacting at this time depends on the authorization method used for the *from* and *to* accounts. If the *from* and *to* accounts use an authorization method other than Positive Customer with Balances/History Authorization, impacting is not necessary. If the *from* and *to* accounts use the Positive Customer with Balances/History Authorization method, no additional impacting is necessary at this time because both accounts are updated during the processing performed by the Internal Message Translation Interface object.

If only the *to* account uses the Positive Customer with Balances/History Authorization method, the Voice Response Unit Interface object calls the Authorization Combined Positive Customer object at this time. The Authorization Combined Positive Customer object calls the Account object, which calls the Account Access object to update the PBF for the *to* account.

3. Calls the Transaction object to format a log record and write the log record to the ITS Transaction Log File (ITLF).
4. Calls the Message ISO object to create an ISO message that is returned to the IVR.
5. Passes control of the transaction to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the ISO message to the XPNET process.

Processing continues with step 16 of the processing summary.

IVR Customer and PIN Verification Authorized on the BASE24 System

BASE24-telebanking and BASE24-billpay transactions originating at an IVR may involve customer verification before transactions can be performed by a CSR or branch employee on behalf of a customer. Some transactions may also require PIN verification. Whether a customer or PIN needs to be verified before processing a transaction is at the discretion of the financial institution or service provider.

Customer and PIN verification is not performed for the following BASE24-billpay transactions originating at an IVR when the customer ID is set to a value of all zeros:

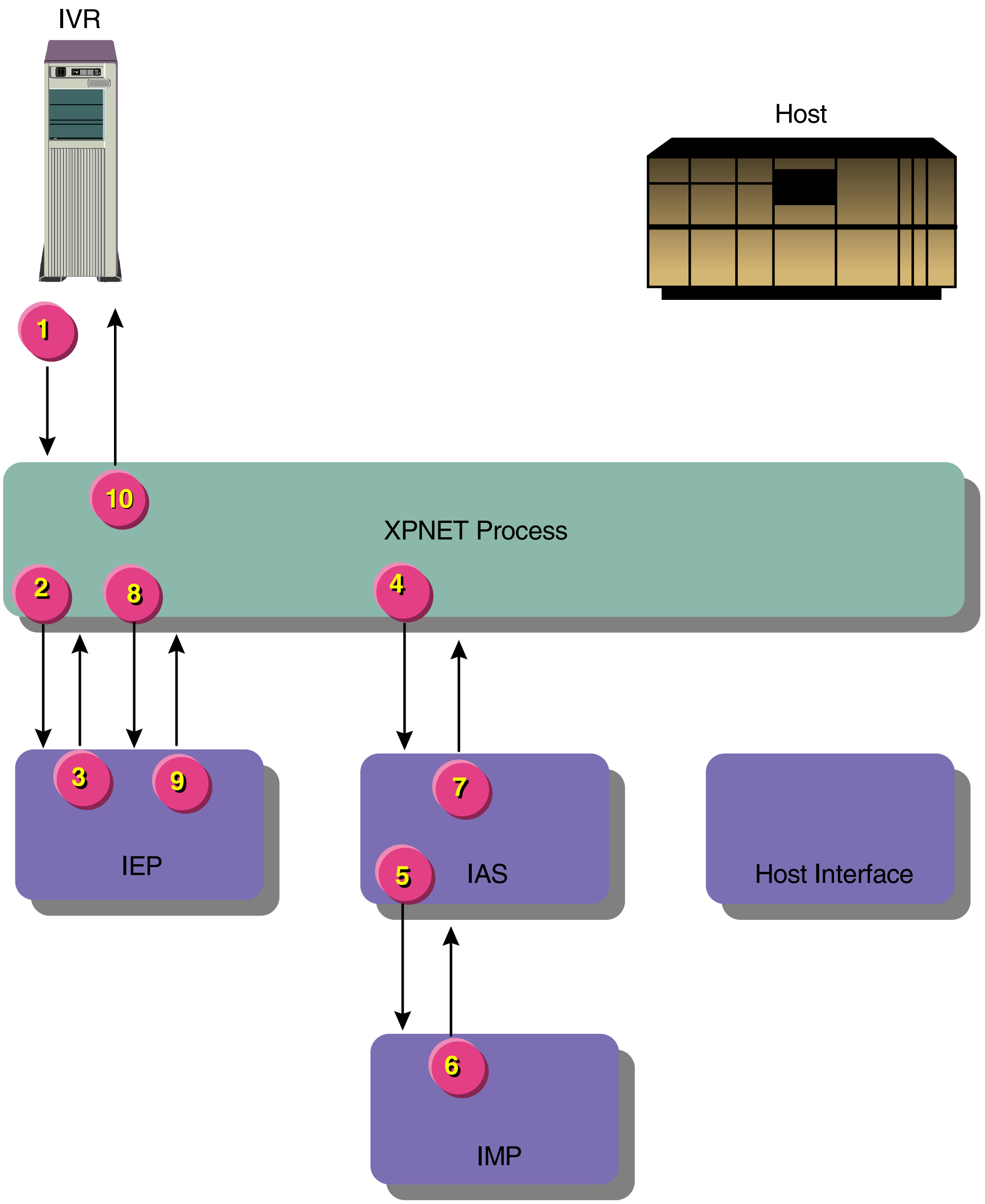
- Bank branch list inquiry
- Bank list inquiry
- Customer add
- Customer inquiry
- Loan application

The value in the PIN REQUIRED field on Institution Definition File (IDF) screen 40 controls whether a PIN is required for CSR transactions. The value in the PIN CHANGE ALLOWED field on IDF screen 40 indicates whether a CSR is allowed to change a PIN on behalf of a customer. These fields are used only if the source code in the incoming request is set to a value of IB (inbound customer service representative).

In this flow, IVR represents an RBSI-compliant device at an inbound CSR call center.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process
IEP = Integrated Endpoint process
IMP = Interface Master process
IVR = Interactive voice response system

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling a CSR. The CSR enters the customer identification information on a PC client screen at an inbound call center. The IVR creates an ISO request message with the transaction request and sends the message to the XPNET process.
2. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the IVR. The XPNET process passes the ISO message to the process named in the station declaration, in this case the Integrated Endpoint process.
3. Using the source class and information from the message, the Integrated Endpoint process identifies the Integrated Authorization Server process as the destination and passes the ISO message to the XPNET process for delivery.
4. The XPNET process passes the ISO message to the Integrated Authorization Server process.
5. The Integrated Authorization Server process obtains station information and requests the Interface Master process to validate the station. See the topic "Step 5 Processing Detail" that follows this summary for a detailed description of the processing performed in this step.
6. The Interface Master process validates the station and returns a message to the Interface Manager object in the Integrated Authorization Server process.
7. The Integrated Authorization Server process formats an ISO response and passes it to the XPNET process. The response indicates whether the customer is verified or contains a customer select request if the customer could not be determined from the information provided.
8. The XPNET process passes the ISO message to the Integrated Endpoint process.
9. The Integrated Endpoint process stores the message context and passes the ISO message to the XPNET process for return to the IVR. The Integrated Endpoint process uses the original station name carried in the message to identify the message destination.
10. The XPNET process passes the ISO message to the IVR.
11. The IVR displays the response information to the CSR. The CSR uses the customer verification response information to select a customer ID or to enter the customer's PIN, if a PIN is required. The customer gives his or her PIN to the CSR. The CSR enters the PIN on a PC client screen. The IVR creates an

ISO request message with the transaction request and sends the message to the XPNET process. For the sake of this message flow, assume that customer selection is not required.

12. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the IVR. The XPNET process passes the ISO message to the process named in the station declaration, in this case the Integrated Endpoint process.
13. The Integrated Endpoint process retrieves the message context and passes it and the ISO message to the XPNET process.
14. The XPNET process passes the message context and ISO message to an Integrated Authorization Server process.
15. The Integrated Authorization Server process authorizes the transaction and returns a response to the XPNET process. See the topic "Step 15 Processing Detail" that follows this summary for a detailed description of the processing performed in this step.
16. The XPNET process passes the ISO message to the Integrated Endpoint process.
17. The Integrated Endpoint process passes the ISO message with the PIN verification information to the XPNET process for return to the IVR. The Integrated Endpoint process uses the original station name carried in the message to identify the message destination.
18. The XPNET process passes the ISO message to the IVR.

Step 5 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the following objects in the order shown:

- XPNET Interface object
- Voice Response Unit Interface object
- XPNET Interface object

XPNET Interface Object Processing

The XPNET Interface object performs as follows:

1. Obtains the object ID of the destination object within the Integrated Authorization Server process (i.e., the Voice Response Unit Interface object). The Voice Response Unit Endpoint Class Handler object in the Integrated Endpoint process placed this object ID in the message.
2. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Obtains station information from the interface extended memory table. The interface extended memory table contains information from the ITS Interface Configuration File (IFCF), Interface Station Table File (IST), and Address Table File (ADRT).
2. Calls the Message ISO object to obtain information from the external message fields and updates the Internal Transaction Data (ITD).
3. Calls the Customer object to obtain the FIID. The Customer object calls the Customer Access object to obtain the FIID from the Customer Table (CSTT). The Customer Access object also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contains the account number and not the customer ID. The Customer object calls the Transaction object to update the ITD.
4. Calls the Interface Manager object to validate the station. The Interface Manager object in the Integrated Authorization Server process passes a message to the Interface Master process.

5. Passes a response message containing verification information to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the message to the XPNET process.

Processing continues with step 6 of the processing summary.

Step 15 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the following objects in the order shown:

- XPNET Interface object
- Voice Response Unit Interface object
- Router object
- Authorization Combined Positive Customer object
- Voice Response Unit Interface object
- XPNET Interface object

XPNET Interface Object Processing

The XPNET Interface object passes the message to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Financial Institution object to retrieve and check customer service information from the IDF extended memory table if the source code in the ITD is set to a value of IB.
2. Calls the Transaction Security object to translate the PIN, if necessary. The Transaction Security object obtains security information from the Interface Security File (ISF) extended memory table and calls the Transaction object to update the ITD if PIN translation occurs.

3. Calls the Transaction object to update the ITD with additional information.
4. Passes control of the transaction to the Router object.

Router Object Processing

The Router object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Financial Institution object to obtain the route profile for the FIID. The Financial Institution object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution Access object to obtain processing information from the IDF extended memory table.
    c. Calls the Transaction object to update the ITD.
3. Obtains routing configuration information from the institution routing extended memory table. The institution routing extended memory table contains information from the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF).
4. Calls the Transaction object to update configuration information in the ITD.
5. Passes control of the transaction to the issuer object, which is the Authorization Combined Positive Customer object.

Authorization Combined Positive Customer Object Processing

The Authorization Combined Positive Customer object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Customer object to determine whether the customer is permitted to perform the transaction. The Customer object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.

   b. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain information from the IDF extended memory table.
   c. Calls the Customer Access object to access the customer extended memory table, CSTT, and CACT. The customer extended memory table contains the same information as the Customer Allowed Transaction Table (CATT).
   d. Calls the Transaction object to update the ITD.
3. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain authorization and customer service information from the IDF extended memory table.
4. Calls the Transaction Security object to verify the PIN. The Transaction Security object performs as follows:
   a. Calls the Transaction object to obtain PIN information from the ITD.
   b. Checks the point of service information in the ITD to determine whether the PIN needs to be verified. If the PIN capture flag is set to S, processing skips to step 5, otherwise processing continues as follows.
   c. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain PIN verification information from the IDF extended memory table.
   d. Calls the Authorization Security Access object to obtain PIN verification information from the Key Authorization File (KEYA) extended memory table.
   e. Verifies the PIN. The Transaction Security object can verify the PIN in software or call a security device to verify the PIN.
   f. Calls the Transaction object to update the PIN verification results in the ITD and delete the PIN from the ITD.
5. Calls the Customer object to update PIN tries in the CSTT. The Customer object performs as follows:
   a. Calls the Transaction object to read the ITD.
   b. Calls the Customer Access object to read the CSTT.
   c. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to access the IDF extended memory table.

   d. Calls the Customer Access object to update the PIN tries in the CSTT, if necessary because of a bad PIN try.
   e. Calls the Transaction object to update the ITD.
6. Sets the action code.
7. Calls the Transaction object to update the ITD for the completion of authorization processing.
8. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Transaction object to obtain information from the ITD, format a log record, and write the log record to the ITS Transaction Log File (ITLF).
2. Passes a message containing ITD information to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the message to the XPNET process.

Processing continues with step 16 of the processing summary.

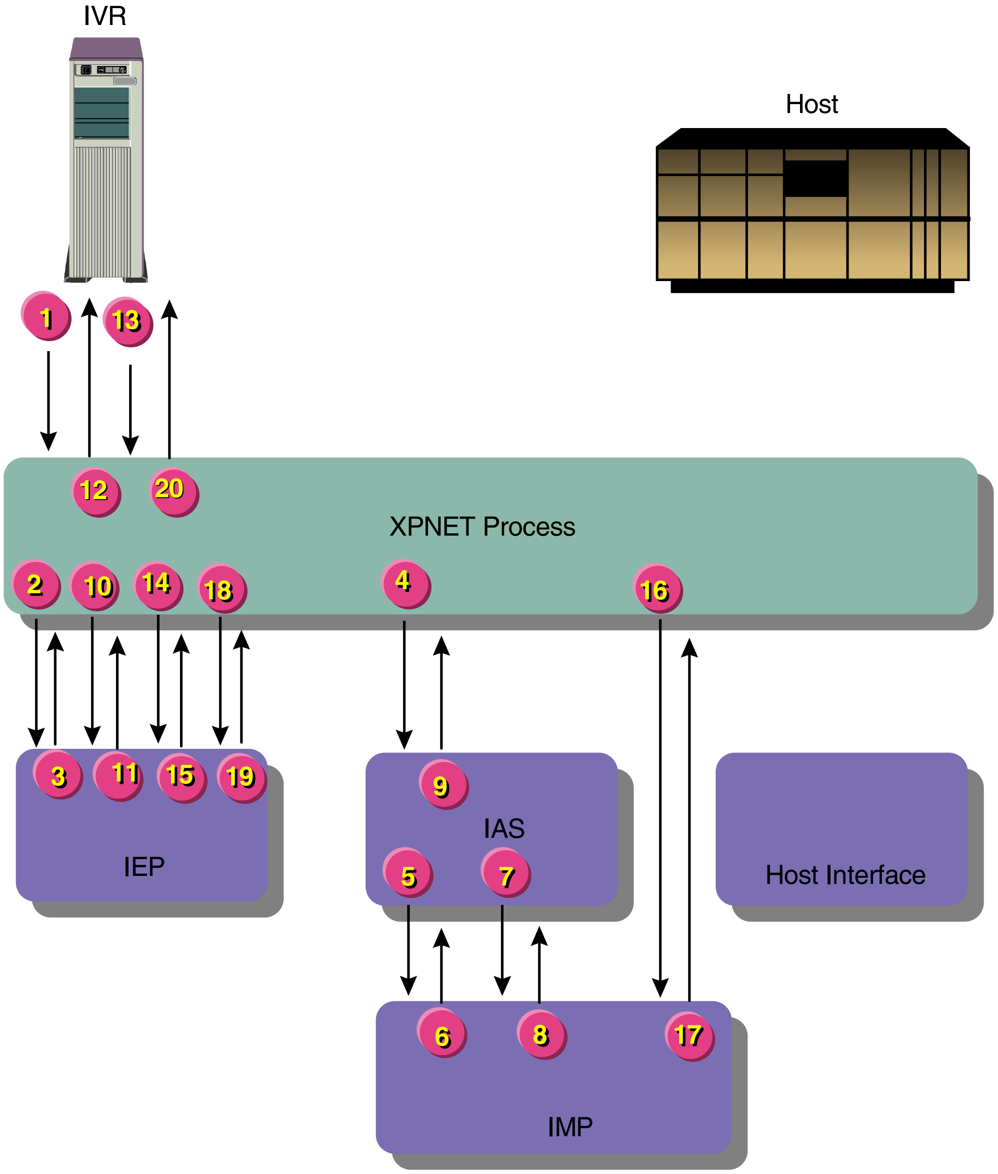
IVR Dynamic Key Management Message Flow

BASE24-telebanking can perform dynamic key management of inbound PIN encryption keys for RBSI-compliant devices. The following message flow depicts the object-level processing involved when a dynamic key management threshold is reached and the inbound PIN encryption key is changed.

This message flow assumes that the interface instance for the device is properly configured to support dynamic key management. For detailed information on configuring dynamic key management, refer to the ***BASE24 Integrated Server Transaction Security Manual***.

Processing Diagram

The following diagram illustrates how messages are used when a new key is passed from BASE24 Remote Banking to an RBSI-compliant device when a dynamic key management threshold is reached. A key to the abbreviations follows the illustration.



Key

IAS	=	Integrated Authorization Server process
IEP	=	Integrated Endpoint process
IMP	=	Interface Master process
IVR	=	Interactive voice response system

Processing Summary

Dynamic key management is performed as follows:

1. A customer initiates a transaction by calling an IVR. The IVR creates an ISO request message with the transaction request and sends the message to the XPNET process. The transaction includes the customer's PIN.
2. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the IVR. The XPNET process passes the ISO message to the process named in the station declaration, in this case the Integrated Endpoint process.
3. Using the source class and information from the message, the Integrated Endpoint process identifies the Integrated Authorization Server process as the destination and passes the ISO message to the XPNET process for delivery.
4. The XPNET process passes the ISO message to the Integrated Authorization Server process.
5. The Integrated Authorization Server process obtains station information and requests the Interface Master process to validate the station. See the topic "Step 5 Processing Detail" that follows this summary for a detailed description of the processing performed in this step.
6. The Voice Response Unit Master object in the Interface Master process validates the station on which the request was received. No dynamic key management counters are increased at this time. The Voice Response Unit Master object returns a message to the Integrated Authorization Server process to indicate that the station is valid.
7. The Integrated Authorization Server process receives a message from the Interface Master process indicating that the station is valid and proceeds with authorization of the transaction, including processing of the PIN. See the topic "Step 7 Processing Detail" that follows this summary for a detailed description of the processing performed in this step.
8. The Interface Master process receives a message from the Integrated Authorization Server process to update dynamic key management counters. See the topic "Step 8 Processing Detail" that follows this summary for a detailed description of the processing performed in this step.
9. The Voice Response Unit Interface object performs the following:
    a. Calls the Transaction object to log the denied transaction to the ITS Log File (ITLF).
    b. Moves the new key data into the external message response.

   c. Calls the Message ISO object to create an ISO message that is returned to the IVR.

   d. Calls the XPNET Interface object to send the response to the Integrated Endpoint process with a Message Context Store request. In this request, the symbolic name of the Interface Master process is specified as the destination for a subsequent network management message from the IVR or for the context if it times out.

10. The XPNET process passes the ISO message response to the Integrated Endpoint process.

11. The Endpoint Class Handler Services object calls the Context Manager object to store the message context and set a timer (based on the value of the NETWORK MANAGEMENT field on VCD screen 2). The Endpoint Class Handler Services object then sends the response to the station on which the request was initially received.

12. The XPNET process passes the ISO message response to the IVR.

13. The IVR processes the key change and sends a network management message (message type 1804) back to BASE24 Remote Banking. For the sake of this message flow, assume that the new key was validated and changed successfully.

14. The XPNET process passes the 1804 network management message to the Integrated Endpoint process.

15. The Voice Response Unit Endpoint Class Handler object in the Integrated Endpoint process receives the 1804 message within the network management message time limit and calls the Endpoint Class Handler Services object, which performs the following:

   a. Calls the Context Manager object to retrieve the stored context. The context key used to retrieve the context includes the name of the interface instance and the transaction sequence number.

   b. Sends the 1804 message and the retrieved context to the Voice Response Unit Master object in the Interface Master process.

   c. Calls the Destination object to perform message-based routing. Ordinarily, routing is configured such that message classes with a message type of 1804 and a message variant of *normal* are routed to the Interface Master process. However, since the Endpoint Class Handler Services object already sent the 1804 message due to context routing in step b, it does not send the message a second time.

16. The XPNET process passes the 1804 network management message and retrieved context to the Interface Master process.

17. The Interface Master process processes the 1804 network management message. See the topic “Step 17 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
18. The XPNET process passes the 1814 network management message to the Integrated Endpoint process.
19. The Integrated Endpoint process routes the 1814 network management response to the IVR.
20. The XPNET process delivers the 1814 network management response to the IVR.

Step 5 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the XPNET Interface object and Voice Response Unit Interface object in the order shown.

XPNET Interface Object Processing

The XPNET Interface object performs as follows:

1. Obtains the object ID of the destination object within the Integrated Authorization Server process (i.e., the Voice Response Unit Interface object). The Voice Response Unit Endpoint Class Handler object in the Integrated Endpoint process placed this object ID in the message.
2. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Obtains station information from the interface extended memory table. The interface extended memory table contains information from the ITS Interface Configuration File (IFCF), Interface Station Table File (IST), and Address Table File (ADRT).
2. Calls the Message ISO object to obtain information from the external message fields and updates the Internal Transaction Data (ITD).

3. Calls the Customer object to obtain the FIID. The Customer object calls the Customer Access object to obtain the FIID from the Customer Table (CSTT). The Customer Access object also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contains the account number and not the customer ID. The Customer object calls the Transaction object to update the ITD.
4. Calls the Interface Manager object to validate the station. The Interface Manager object in the Integrated Authorization Server process passes a message to the Interface Master process.

Processing continues with step 6 of the processing summary.

Step 7 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the following objects in the order shown:

- Voice Response Unit Interface object
- Transaction Security object
- Router object
- Authorization Combined Positive Customer object
- Transaction Security object
- Authorization Combined Positive Customer object
- Voice Response Unit Interface object
- Interface Manager object

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Financial Institution object to retrieve and check customer service information from the IDF extended memory table if the source code in the ITD is set to a value of IB.
2. Calls the Transaction Security object to translate the PIN, if necessary. The Transaction Security object obtains security information from the Interface Security File (ISF) extended memory table and calls the Transaction object to update the ITD if PIN translation occurs.

3. Calls the Transaction Security object to process the PIN contained in the authorization request. The Transaction Security object moves the external PIN block and encryption key into the ITD.
4. Passes control of the transaction to the Router object.

Router Object Processing

The Router object performs as follows:

1. Calls the Transaction object to obtain information from the ITD.
2. Calls the Financial Institution object to obtain the route profile for the FIID. The Financial Institution object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution Access object to obtain processing information from the Institution Definition File (IDF) extended memory table.
    c. Calls the Transaction object to update the ITD.
3. Obtains routing configuration information from the institution routing extended memory table. The institution routing extended memory table contains information from the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF).
4. Calls the Transaction object to update configuration information in the ITD.
5. Passes control of the transaction to the Authorization Combined Positive Customer object.

Authorization Combined Positive Customer Object Processing

The Authorization Combined Positive Customer object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Customer object to determine whether the customer is permitted to perform the transaction. The Customer object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.

   b. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain information from the IDF extended memory table.
   c. Calls the Customer Access object to obtain information from the customer extended memory table, CSTT, and CACT. The customer extended memory table contains the same information as the Customer Allowed Transaction Table (CATT).
   d. Calls the Transaction object to update the ITD.
3. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain authorization and customer service information from the IDF extended memory table.
4. Determines whether a PIN change is required. If a PIN change is required, the transaction is declined.
5. Calls the Transaction Security object to verify the PIN. The Transaction Security object performs as follows:
   a. Calls the Transaction object to obtain PIN information from the ITD.
   b. Checks the point of service information in the ITD to determine whether the PIN needs to be verified. For the purpose of this flow, assume that the PIN has not yet been verified.
   c. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain PIN verification information from the IDF extended memory table.
   d. Calls the Authorization Security Access object to obtain PIN verification information from the Key Authorization File (KEYA) extended memory table.
   e. Attempts to verify the PIN. The Transaction Security object can verify the PIN in software or call a security device to verify the PIN. For the purpose of this message flow, assume that a PIN length error occurred.
6. Calls the Transaction object to update the ITD with action code 127 to indicate that the transaction was denied because the PIN could not be verified because of its length.
7. Passes control of the transaction to the Voice Response Unit Interface object.

Voice Response Unit Interface Object Processing

The Voice Response Unit Interface object performs as follows:

1. Calls the Interface Manager object to validate the station. This station validation request is made to update the dynamic key management counters in the interface extended memory table.
2. The Interface Manager object in the Integrated Authorization Server process passes a message to the Interface Master process.

Processing continues with step 8 of the processing summary.

Step 8 Processing Detail

The Interface Master process controls transaction processing in this step with the following objects in the order shown:

- Voice Response Unit Master object
- Interface Master Services object
- Transaction Security object
- Voice Response Unit Master object

Voice Response Unit Master Object Processing

The Voice Response Unit Master object receives the station validation request and performs as follows:

1. Calls the Interface Master Services object with a Key Counter Manage request.
2. The Interface Master Services object increases the error and consecutive error counters in the Interface extended memory table by one for the PIN length error encountered. The object then compares these counts to the corresponding limits in the table. For the sake of this message flow, assume that the error counter has reached its limit. The object sets a “key change required” flag in its response and returns control to the Voice Response Unit Master object.
3. Calls the Transaction Security object to generate new key data.

Transaction Security Object Processing

The Transaction Security object performs as follows:

1. Calls the security module to generate a random PIN encryption key. The security module generates the key and returns it to the Transaction Security object.
2. Writes the new key to the Interface Security File (ISF) in place of the old key. It does not toggle the current key index in the ISF at this time. Thus, the previous “new” key is still the current key.
3. Returns the new key data, including a hexadecimal counter for synchronization between the RBSI-compliant device and BASE24 Remote Banking, to the Voice Response Unit Master object.

Voice Response Unit Master Object Processing

The Voice Response Unit Master object performs as follows:

1. Moves the new key data into the station validation response, sets the “key change in progress” flag in the Interface extended memory table to true, and calls the XPNET Interface object.
2. The XPNET Interface object delivers the station validation response to the Integrated Authorization Server process.

Step 17 Processing Detail

The Interface Master process controls transaction processing in this step with the following objects in the order shown:

- Voice Response Unit Master object
- Transaction Security object
- Voice Response Unit Master object

Voice Response Unit Master Object Processing

The Voice Response Unit Master object performs as follows:

1. Locates the associated Station table entry in the Interface extended memory table.
2. Determines whether the IVR accepted the new inbound PIN encryption key. For the sake of this message flow, assume that the new key was accepted.
3. Calls the Transaction Security to put the new key into use.

Transaction Security Object Processing

The Transaction Security object performs as follows:

1. Toggles the current key index in the ISF to place the new key into service and writes the ISF record to disk. The new key was already placed in the ISF earlier.
2. Formats a Key Update request and sends it to the Transaction Security objects in all the Integrated Authorization Server processes that communicate with this Interface Master process. This request includes the new index, the latest pair of keys, and the name of the interface instance.

Voice Response Unit Master Object Processing

The Voice Response Unit Master object performs as follows:

1. Formats an 1814 network management response message and sends the message to the Endpoint Class Handler Services object in the Integrated Endpoint process.
2. Sets the “key change required” and “key change in progress” flags in the Interface extended memory table to false for the interface instance.

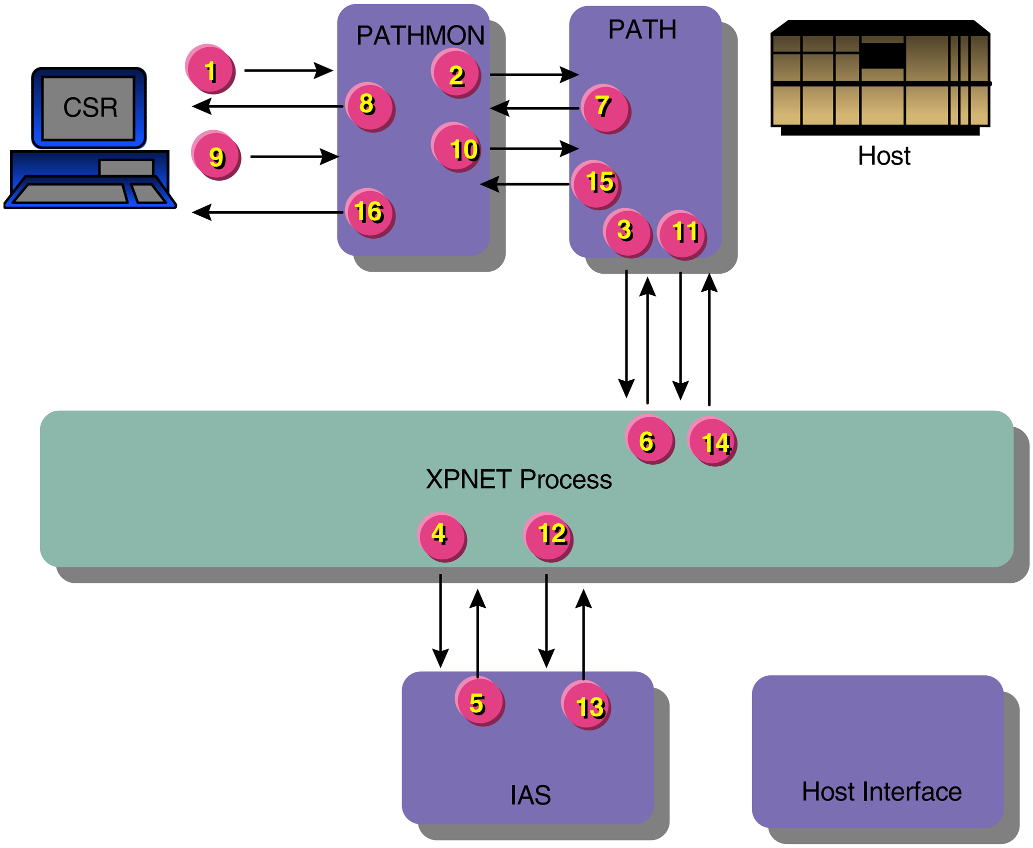
CSR Customer and PIN Verification Authorized on the BASE24 System

All BASE24-telebanking transactions originating at a CSR terminal involve customer and optional PIN verification before a transaction menu is displayed. All but two BASE24-billpay transactions originating at a CSR terminal also involve customer and PIN verification before a transaction menu is displayed. These two exceptions are customer add and loan application transactions.

The value in the PIN REQUIRED field on Institution Definition File (IDF) screen 40 controls whether a PIN is required for CSR transactions. The value in the PIN CHANGE REQUIRED field on Customer Table (CSTT) screen 1 controls whether a PIN change is required before a transaction menu is displayed.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

CSR = Customer Service Representative Requester process
IAS = Integrated Authorization Server process
PATH = Path Server process
PATHMON = Pathway Monitor process

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling a CSR. The CSR enters the customer verification information at the CSR terminal. The Customer Service Representative (CSR) Control Requester process validates the data entered and passes it to the Pathway Monitor process.
2. The Pathway Monitor process passes the message to a Path Server process.
3. The Path Server process passes the message to the XPNET process.
4. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the Path Server process station. The XPNET process passes the message to the process named in the station declaration, in this case the Integrated Authorization Server process.
5. The Integrated Authorization Server process verifies the customer and returns a response to the XPNET process. See the topic “Step 5 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
6. The XPNET process passes the response message to the Path Server process.
7. The Path Server process passes the response message to the Pathway Monitor process.
8. The Pathway Monitor process passes the response message with customer verification information to the CSR Control Requester process, which displays the information on the CSR terminal. The CSR uses the customer verification response screen to enter the customer’s PIN, if a PIN is required.
9. If a PIN is required, the customer gives his or her PIN to the CSR. The CSR enters the PIN at the CSR terminal. The CSR Control Requester process validates the data entered, places it in the Internal Transaction Data (ITD), and passes it to the Pathway Monitor process.

10. The Pathway Monitor process passes the message with the ITD to a Path Server process.
11. The Path Server process passes the message to the XPNET process.
12. The XPNET process passes the message to the Integrated Authorization Server process. The XPNET process uses the original station name carried in the message to identify the message destination.
13. The Integrated Authorization Server process authorizes the transaction and returns a response to the XPNET process. See the topic “Step 13 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
14. The XPNET process passes the response message to the Path Server process.
15. The Path Server process passes the response message to the Pathway Monitor process.
16. The Pathway Monitor process passes the response message with PIN verification information to the CSR Control Requester process, which displays the information on the CSR terminal. The CSR uses the PIN verification response screen to select the transaction menu.

Step 5 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the following objects in the order shown:

- XPNET Interface object
- Customer Service Interface object
- XPNET Interface object

XPNET Interface Object Processing

The XPNET Interface object passes the message to the Customer Service Interface object.

Customer Service Interface Object Processing

The Customer Service Interface object performs as follows:

1. Calls the Customer object to obtain the FIID. The Customer object calls the Customer Access object to obtain the FIID from the CSTT. The Customer Access object also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contains the account number and not the customer ID.
2. Passes a response message containing verification information to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the message to the XPNET process.

Processing continues with step 6 of the processing summary.

Step 13 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the following objects in the order shown:

- XPNET Interface object
- Customer Service Interface object
- Router object
- Authorization Combined Positive Customer object
- Customer Service Interface object
- XPNET Interface object

XPNET Interface Object Processing

The XPNET Interface object passes the message to the Customer Service Interface object.

Customer Service Interface Object Processing

The Customer Service Interface object performs as follows:

1. Calls the Transaction Security object to translate the PIN, if necessary. The Transaction Security object obtains security information from the Interface Security File (ISF) extended memory table and calls the Transaction object to update the ITD if PIN translation occurs.
2. Calls the Transaction object to update the ITD with additional information.
3. Passes control of the transaction to the Router object.

Router Object Processing

The Router object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Financial Institution object to obtain the route profile for the FIID. The Financial Institution object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution Access object to obtain processing information from the IDF extended memory table.
    c. Calls the Transaction object to update the ITD.
3. Obtains routing configuration information from the institution routing extended memory table. The institution routing extended memory table contains information from the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF).
4. Calls the Transaction object to update configuration information in the ITD.
5. Passes control of the transaction to the issuer object, which is the Authorization Combined Positive Customer object.

Authorization Combined Positive Customer Object Processing

The Authorization Combined Positive Customer object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Customer object to determine whether the customer is permitted to perform the transaction. The Customer object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain information from the IDF extended memory table.
    c. Calls the Customer Access object to access the customer extended memory table, CSTT, and CACT. The customer extended memory table contains the same information as the Customer Allowed Transaction Table (CATT).
    d. Calls the Transaction object to update the ITD.
3. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain authorization and customer service information from the IDF extended memory table.
4. Calls the Transaction Security object to verify the PIN. The Transaction Security object performs as follows:
    a. Calls the Transaction object to obtain PIN information from the ITD.
    b. Checks the point of service information in the ITD to determine whether the PIN needs to be verified. If the PIN capture flag is set to S, processing skips to step 5, otherwise processing continues as follows.
    c. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain PIN verification information from the IDF extended memory table.
    d. Calls the Authorization Security Access object to obtain PIN verification information from the Key Authorization File (KEYA) extended memory table.
    e. Verifies the PIN. The Transaction Security object can verify the PIN in software or call a security device to verify the PIN.

    f. Calls the Transaction object to update the PIN verification results in the ITD and delete the PIN from the ITD.

5. Calls the Customer object to update PIN tries in the CSTT. The Customer object performs as follows:
    a. Calls the Transaction object to read the ITD.
    b. Calls the Customer Access object to read the CSTT.
    c. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to access the IDF extended memory table.
    d. Calls the Customer Access object to update the PIN tries in the CSTT, if necessary because of a bad PIN try.
    e. Calls the Transaction object to update the ITD.
6. Sets the action code.
7. Calls the Transaction object to update the ITD for the completion of authorization processing.
8. Passes control of the transaction to the Customer Service Interface object.

Customer Service Interface Object Processing

The Customer Service Interface object performs as follows:

1. Calls the Transaction object to obtain information from the ITD, format a log record, and write the log record to the ITS Transaction Log File (ITLF).
2. Passes a message containing ITD information to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the message to the XPNET process.

Processing continues with step 14 of the processing summary.

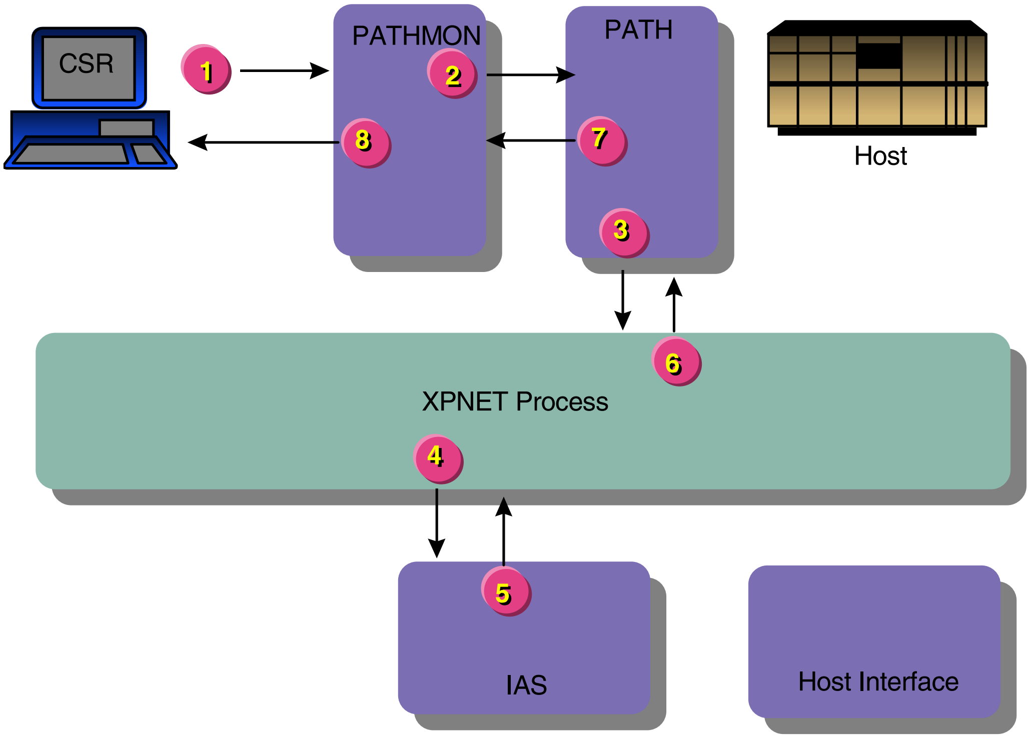
CSR Available Funds Inquiry Authorized on the BASE24 System

The BASE24-telebanking system obtains information from the Positive Balance File (PBF) to process available funds inquiries. The Positive Customer with Balances/History Authorization method is required to obtain balance information from the BASE24 database.

This transaction assumes that the customer and the customer’s PIN have been successfully verified.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

CSR = Customer Service Representative Requester process
IAS = Integrated Authorization Server process
PATH = Path Server process
PATHMON = Pathway Monitor process

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling a CSR. The CSR enters the transaction information at the CSR terminal after verifying the customer and PIN. The Customer Service Representative (CSR) Telebanking Requester process validates the data entered and places it in fields of the Internal Transaction Data (ITD). The ITD is part of the message that is passed to the Pathway Monitor process.
2. The Pathway Monitor process passes the message with the ITD to a Path Server process.
3. The Path Server process passes the message to the XPNET process.
4. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the Path Server process station. The XPNET process passes the message to the process named in the station declaration, in this case the Integrated Authorization Server process.
5. The Integrated Authorization Server process authorizes the transaction and returns a response to the XPNET process. See the topic “Step 5 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
6. The XPNET process passes the response message to the Path Server process.
7. The Path Server process passes the response message to the Pathway Monitor process.
8. The Pathway Monitor process passes the response message to the CSR Telebanking Requester process, which displays the information on the CSR terminal. The CSR uses the information to advise the customer.

Step 5 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the following objects in the order shown:

- XPNET Interface object
- Customer Service Interface object
- Router object
- Authorization Combined Positive Customer object

- Customer Service Interface object
- XPNET Interface object

XPNET Interface Object Processing

The XPNET Interface object passes the message to the Customer Service Interface object.

Customer Service Interface Object Processing

The Customer Service Interface object performs as follows:

1. Calls the Customer object to obtain the FIID. The Customer object calls the Customer Access object to obtain the FIID from the Customer Table (CSTT). The Customer Access object also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contains the account number and not the customer ID. The Customer object calls the Transaction object to update the ITD.
2. Passes control of the transaction to the Router object.

Router Object Processing

The Router object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Financial Institution object to obtain the route profile for the FIID. The Financial Institution object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution Access object to obtain processing information from the Institution Definition File (IDF) extended memory table.
    c. Calls the Transaction object to update the ITD.
3. Obtains routing configuration information from the institution routing extended memory table. The institution routing extended memory table contains information from the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF).

4. Calls the Transaction object to update configuration information in the ITD.
5. Passes control of the transaction to the issuer object, which is the Authorization Combined Positive Customer object.

Authorization Combined Positive Customer Object Processing

The Authorization Combined Positive Customer object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Customer object to determine whether the customer is permitted to perform the transaction. The Customer object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain information from the IDF extended memory table.
    c. Calls the Customer Access object to access the customer extended memory table, CSTT, and CACT. The customer extended memory table contains the same information as the Customer Allowed Transaction Table (CATT).
    d. Calls the Transaction object to update the ITD.
3. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain authorization and customer service information from the IDF extended memory table.
4. Checks the authorization method. An available funds inquiry requires the Positive Customer with Balances/History Authorization method.
5. Calls the Account object to obtain the account balance. The Account object performs as follows:
    a. Calls the Transaction object to read the ITD.
    b. Calls the Financial Institution object. The Financial Institution object calls the Financial Institution Access object to obtain usage dates and backup account parameters.
    c. Calls the Account Access object to obtain the account balance from the PBF and place the information in the PBF extended memory table.

   d. Calls the Transaction object to update balance and usage information in the ITD.

6. Sets the action code.
7. Calls the Transaction object to update the ITD for the completion of authorization processing.
8. Passes control of the transaction to the Customer Service Interface object.

Customer Service Interface Object Processing

The Customer Service Interface object performs as follows:

1. Calls the Transaction object to obtain information from the ITD, format a log record, and write the log record to the ITS Transaction Log File (ITLF).
2. Passes a message containing ITD information to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the message to the XPNET process.

Processing continues with step 6 of the processing summary.

CSR Transfer Authorized on the BASE24 System with an Advice

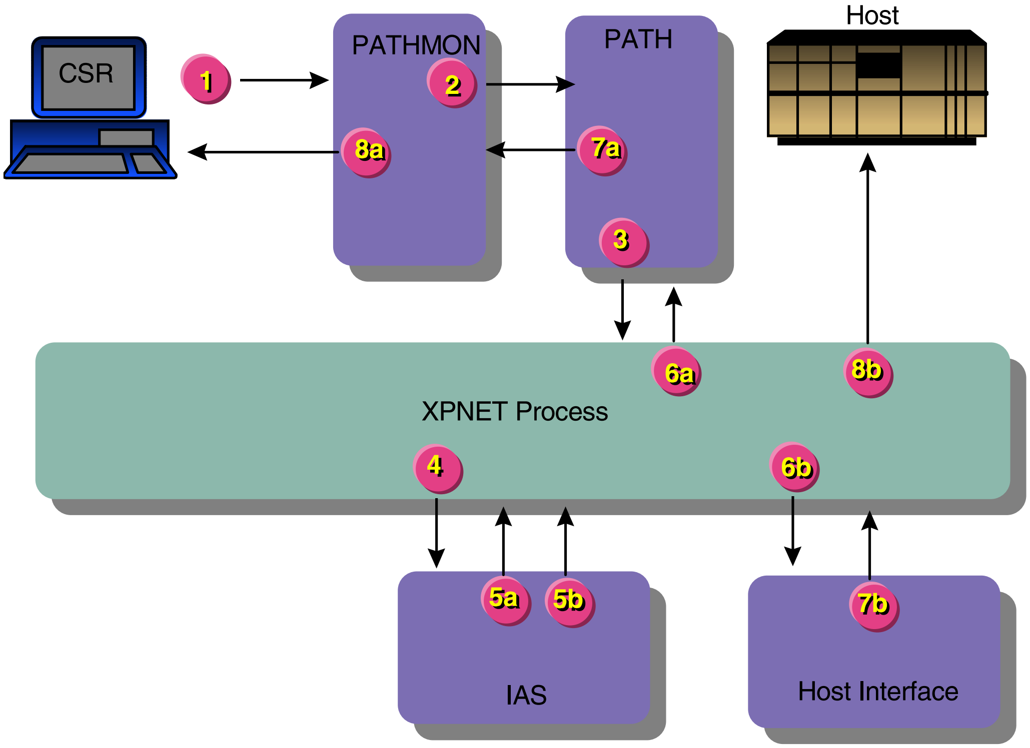
BASE24-telebanking transfer transactions authorized on the BASE24 system obtain and update information contained in the Positive Balance File (PBF). The Positive Customer with Balances/History Authorization method is required to maintain account balances on the BASE24 system.

This message flow includes an advice to one host for the *from* and *to* accounts. The BASE24-telebanking system also supports sending an advice to one host for the *from* account and sending an advice to a different host for the *to* account.

This transaction assumes that the customer and the customer's PIN have been successfully verified.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

CSR = Customer Service Representative Requester process
IAS = Integrated Authorization Server process
PATH = Path Server process
PATHMON = Pathway Monitor process

Note: At step 5, the message processing splits into concurrent flows. The response to the CSR terminal is identified by steps 5a through 8a. The advice to the host is identified by steps 5b through 8b.

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling a CSR. The CSR enters the transaction information at the CSR terminal after verifying the customer and PIN. The Customer Service Representative (CSR) Telebanking Requester process validates the data entered and places it in fields of the Internal Transaction Data (ITD). The ITD is part of the message that is passed to the Pathway Monitor process.
2. The Pathway Monitor process passes the message with the ITD to a Path Server process.
3. The Path Server process passes the message to the XPNET process.
4. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the Path Server process station. The XPNET process passes the message to the process named in the station declaration, in this case the Integrated Authorization Server process.
5. The Integrated Authorization Server process authorizes the transaction and returns two messages to the XPNET process—a response for the CSR terminal and an advice for the host. The advice message is formatted as a Telebanking Standard Internal Message (BSTM). See the topic “Step 5 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.

   Processing of the response continues with step 6a. Processing of the advice continues with step 6b.

6a. The XPNET process passes the message to the Path Server process.

6b. The XPNET process passes the BSTM to the ISO Host Interface process.

7a. The Path Server process passes the message to the Pathway Monitor process.

7b. The ISO Host Interface process reformats the BSTM into an ISO external message and passes it to the XPNET process.

8a. The Pathway Monitor process passes the response message to the CSR Telebanking Requester process, which displays the information on the CSR terminal. The CSR uses the information to advise the customer.

8b. The XPNET process passes the ISO external message to the host.

Step 5 Processing Detail

The Integrated Authorization Server process controls transaction processing in this step with the following objects in the order shown:

- XPNET Interface object
- Customer Service Interface object
- Router object
- Authorization Combined Positive Customer object
- Customer Service Interface object
- Internal Message Translation Interface object
- XPNET Interface object

XPNET Interface Object Processing

The XPNET Interface object passes the message to the Customer Service Interface object.

Customer Service Interface Object Processing

The Customer Service Interface object performs as follows:

1. Calls the Customer object to obtain the FIID. The Customer object calls the Customer Access object to obtain the FIID from the Customer Table (CSTT). The Customer Access object also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contains the account number and not the customer ID. The Customer object calls the Transaction object to update the ITD.
2. Passes control of the transaction to the Router object.

Router Object Processing

The Router object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.

2. Calls the Financial Institution object to obtain the route profile for the FIID. The Financial Institution object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution Access object to obtain processing information from the Institution Definition File (IDF) extended memory table.
    c. Calls the Transaction object to update the ITD.
3. Obtains routing configuration information from the institution routing extended memory table. The institution routing extended memory table contains information from the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF).
4. Calls the Transaction object to update configuration information in the ITD.
5. Passes control of the transaction to the issuer object, which is the Authorization Combined Positive Customer object.

Authorization Combined Positive Customer Object Processing

The Authorization Combined Positive Customer object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Customer object to determine whether the customer is permitted to perform the transaction. The Customer object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.
    b. Calls the Financial Institution object. The Financial Institution object calls the Transaction object to obtain information from the ITD, then calls the Financial Institution Access object to obtain information from the IDF extended memory table.
    c. Calls the Customer Access object to access the customer extended memory table, CSTT, and CACT. The customer extended memory table contains the same information as the Customer Allowed Transaction Table (CATT).
    d. Calls the Transaction object to update the ITD.
3. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain authorization and customer service information from the IDF extended memory table.

4. If the CARD STATUS field on IDF screen 2 contains a value of Y, the Authorization Combined Positive Customer object checks the VERIFY STATUS field on CSTT screen 1 for a value of V (verified).
5. Calls the Account object to obtain account information and perform the transfer. The Account object performs as follows:
    a. Calls the Transaction object to read the ITD.
    b. Calls the Financial Institution object, which calls the Financial Institution Access object to obtain usage dates from the IDF extended memory table.
    c. Calls the Account Access object to obtain balances, holds, limits, and usages for the *from* and *to* accounts from the PBF and place the information in the PBF extended memory table.
    d. Performs the transfer, which includes the following:
        1) Checking for sufficient available funds in the *from* account.
        2) Checking the usage and individual transaction limits for the *from* account.
        3) Calculating the new account balances and usage amounts for the *from* and *to* accounts.
    e. Calls the Account Access object to update the PBF records with the new account balance and usage information.
    f. Calls the Transaction object to update balance and usage information in the ITD.
6. Sets the action code.
7. Calls the Transaction object to update the ITD for the completion of authorization processing.
8. Passes control of the transaction to the Customer Service Interface object.

Customer Service Interface Object Processing

The Customer Service Interface object performs as follows:

1. Calls the Transaction object to read the ITD.
2. Determines that an advice is required.

3. Calls the Internal Message Translation Interface object. The Internal Message Translation Interface object performs as follows:
    a. Calls the Transaction object to obtain information for the advice from the ITD.
    b. Obtains host routing information from the Internal Message Translation File (IMTF) extended memory table.
    c. Creates a BSTM for the ISO Host Interface process.

       Note: The Internal Message Translation Interface object passes the BSTM to the XPNET Interface object immediately after the message for the CSR terminal is sent to the Pathway Monitor process. This is illustrated on the processing diagram as step 5b. Processing of this message continues with step 6b of the processing summary.
4. Calls the Transaction object to obtain information from the ITD, format a log record, and write the log record to the ITS Transaction Log File (ITLF).
5. Passes a message containing ITD information to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the message to the XPNET process. This is illustrated on the processing diagram as step 5a. Processing of this message continues with step 6a of the processing summary.

CSR Transfer Preauthorized on the BASE24 System and Authorized on the Host

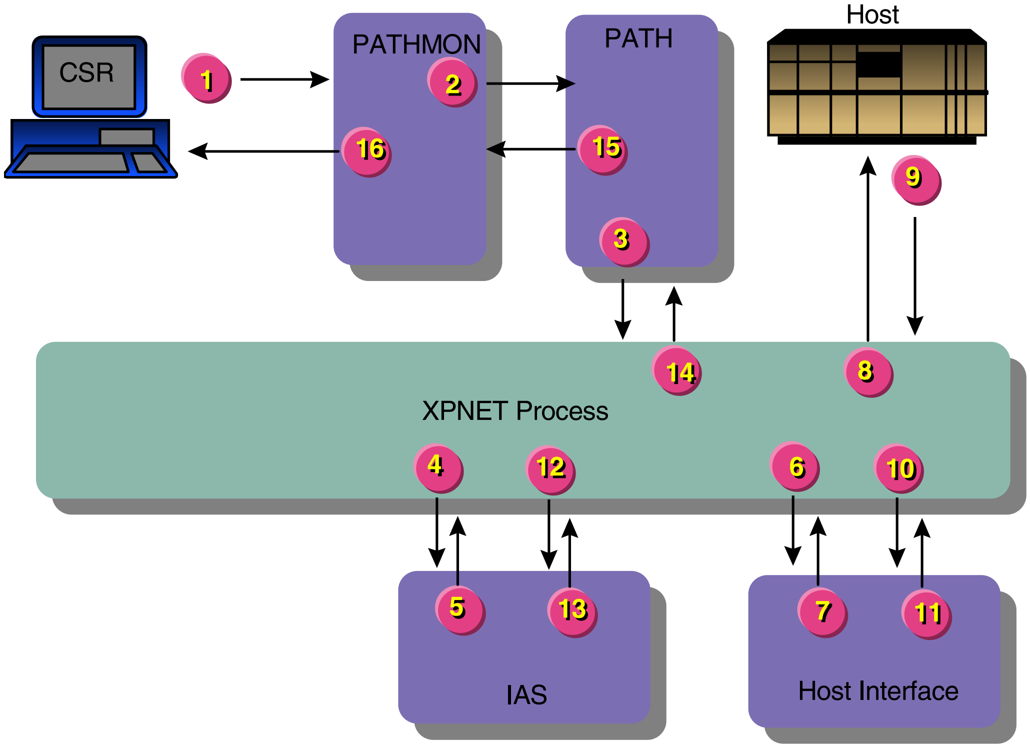
BASE24-telebanking transfer transactions can be preauthorized on the BASE24 system using the online/offline authorization level and any authorization method. These transactions also can be preauthorized on the BASE24 system using the online authorization level and the Positive Customer and Positive Customer with Account Authorization methods. With the Positive Customer Authorization method, the BASE24-telebanking system can check the customer status and customer PIN. With the Positive Customer with Account Authorization method and the Positive Customer with Balances/History Authorization method, the BASE24-telebanking system can check the customer information and check the account status, including whether the requested transaction can be performed on the account.

Transfer transactions that are authorized on the host can have both accounts on the same host or have each account on a separate host. This message flow illustrates a transfer transaction with both accounts on the same host.

This transaction assumes that the customer and the customer’s PIN have been successfully verified.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

CSR = Customer Service Representative Requester process
IAS = Integrated Authorization Server process
PATH = Path Server process
PATHMON = Pathway Monitor process

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling a CSR. The CSR enters the transaction information at the CSR terminal after verifying the customer and PIN. The Customer Service Representative (CSR) Telebanking Requester process validates the data entered and places it in fields of the Internal Transaction Data (ITD). The ITD is part of the message that is passed to the Pathway Monitor process.

2. The Pathway Monitor process passes the message with the ITD to a Path Server process.
3. The Path Server process passes the message to the XPNET process.
4. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the Path Server process station. The XPNET process passes the message to the process named in the station declaration, in this case the Integrated Authorization Server process.
5. The Integrated Authorization Server process preauthorizes the transaction and passes a Telebanking Standard Internal Message (BSTM) to the XPNET process. See the topic “Step 5 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
6. The XPNET process passes the BSTM to the ISO Host Interface process for the host that is to authorize the request.
7. The ISO Host Interface process reformats the BSTM into an ISO external message and passes it to the XPNET process.
8. The XPNET process passes the ISO external message to the host for authorization.
9. The host authorizes the transaction and returns the ISO external message to the XPNET process.
10. The XPNET process passes the ISO external message to the ISO Host Interface process.
11. The ISO Host Interface process reformats the ISO external message into a BSTM and passes it to the XPNET process.
12. The XPNET process passes the BSTM to the Integrated Authorization Server process.
13. The Integrated Authorization Server process updates the BASE24 database and returns a response to the XPNET process. Refer to the topic “Step 13 Processing Detail” that follows this summary for a detailed description of the processing performed in this step.
14. The XPNET process passes the message to the Path Server process.
15. The Path Server process passes the response to the Pathway Monitor process.
16. The Pathway Monitor process creates a response message from the ITD and passes the response message to the CSR Telebanking Requester process, which displays the information on the CSR terminal. The CSR uses the information to advise the customer.

Step 5 Processing Detail

The Integrated Authorization Server process controls transaction processing with the following objects in the order shown:

- XPNET Interface object
- Customer Service Interface object
- Router object
- Internal Message Translation Interface object

XPNET Interface Object Processing

The XPNET Interface object passes the message to the Customer Service Interface object.

Customer Service Interface Object Processing

The Customer Service Interface object performs as follows:

1. Calls the Customer object to obtain the FIID. The Customer object calls the Customer Access object to obtain the FIID from the Customer Table (CSTT). The Customer Access object also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contains the account number and not the customer ID. The Customer object calls the Transaction object to update the ITD.
2. Passes control of the transaction to the Router object.

Router Object Processing

The Router object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Financial Institution object to obtain the route profile for the FIID. The Financial Institution object performs as follows:
    a. Calls the Transaction object to obtain information from the ITD.

   b. Calls the Financial Institution Access object to obtain processing information from the Institution Definition File (IDF) extended memory table.
   c. Calls the Transaction object to update the ITD.

3. Obtains routing configuration information from the institution routing extended memory table. The institution routing extended memory table contains information from the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF).
4. Calls the Transaction object to update configuration information in the ITD.
5. Passes control of the transaction to the issuer object, which is the Internal Message Translation Interface object.

Internal Message Translation Interface Object Processing

The Internal Message Translation Interface object performs as follows:

1. Calls the Transaction object to obtain the necessary information from the ITD.
2. Calls the Authorization Combined Positive Customer object to preauthorize the transaction. The Authorization Combined Positive Customer object performs as follows:
   a. Calls the Transaction object to obtain the necessary information from the ITD.
   b. Calls the Customer object to determine whether the customer is permitted to perform the transaction. The Customer object performs as follows:
      1) Calls the Transaction object to read the ITD.
      2) Calls the Financial Institution object. The Financial Institution object calls the Financial Institution Access object to obtain authorization and customer service information from the IDF extended memory table.
      3) Calls the Customer Access object to access the customer extended memory table, CSTT, and CACT when the Positive Customer with Accounts Authorization method or the Positive Customer with Balances/History Authorization method is used. The Customer Access object accesses the customer extended memory table and CSTT without the CACT when the Positive Customer

Authorization method is used. The customer extended memory table contains the same information as the Customer Allowed Transaction Table (CATT).

   c. If the CARD STATUS field on IDF screen 2 contains a value of Y, the Authorization Combined Positive Customer object checks the VERIFY STATUS field on CSTT screen 1 for a value of V (verified).

   d. Calls the Transaction object to update the ITD.

3. Obtains the ISO Host Interface process name and DPC number from the Internal Message Translation File (IMTF) extended memory table.
4. Calls the Transaction object to compress the ITD.
5. Creates a BSTM for the ISO Host Interface process.
6. Calls the XPNET Interface object to pass the BSTM to the XPNET process.

Processing continues with step 6 of the processing summary.

Step 13 Processing Detail

The Integrated Authorization Server process controls transaction processing with the following objects in the order shown:

- XPNET Interface object
- Internal Message Translation Interface object
- Customer Service Interface object
- XPNET Interface object

XPNET Interface Object Processing

The XPNET Interface object passes the external message to the Internal Message Translation Interface object.

Internal Message Translation Interface Object Processing

The Internal Message Translation Interface object performs as follows:

1. Calls the Transaction object to expand the compressed ITD contained in the BSTM received from the ISO Host Interface process and place it in the ITD.
2. Calls the Authorization Combined Positive Customer object to update transaction dates and account balances. The Authorization Combined Positive Customer object performs as follows:
    a. Calls the Transaction object to obtain the action code and processing code from the ITD.

    b. If the Positive Customer with Balances/History Authorization method is used for the *from* account, calls the Account object to update BASE24 account balances and usages as a result of the transfer. Otherwise, processing continues with part c of this step. To update account balances and usages, the Account object performs as follows:
        1) Calls the Transaction object to read the ITD.
        2) Calls the Financial Institution object. The Financial Institution object calls the Financial Institution Access object to obtain backup account parameters and the dates used to calculate usage amounts.
        3) Calls the Account Access object to obtain balances for the account from the Positive Balance File (PBF) and place the information in the PBF extended memory table.
        4) Calculates the new account balance and usage amounts.
        5) Calls the Account Access object to update the PBF records with the new account balance and usage information.
        6) Calls the Transaction object to update balance and usage information in the ITD.

    Note: The *to* account is included in part b of this step if the Positive Customer with Balances/History Authorization method is used for both the *from* and *to* accounts. The *to* account is not updated at this time if the Positive Customer with Balances/History Authorization method is not used for the *from* account.

    c. Calls the Transaction object to update the ITD to indicate that the host returned an authorization response.
3. Passes control of the transaction to the Customer Service Interface object.

Customer Service Interface Object Processing

The Customer Service Interface object performs as follows:

1. Calls the Transaction object to obtain information from the ITD.
2. Obtains advice and impacting information from the ITD. For this transaction, no advices need to be sent and additional impacting is unnecessary.

   No advices need to be sent because the transaction was approved by the host and only one host was involved. If two hosts are involved, the Customer Service Interface object calls the Internal Message Translation Interface object at this time to send an advice to the second host.

   The need for impacting at this time depends on the authorization method used for the *from* and *to* accounts. If the *from* and *to* accounts use an authorization method other than Positive Customer with Balances/History Authorization, impacting is not necessary. If the *from* and *to* accounts use the Positive Customer with Balances/History Authorization method, no additional impacting is necessary at this time because both accounts are updated during the processing performed by the Internal Message Translation Interface object.

   If only the *to* account uses the Positive Customer with Balances/History Authorization method, the Customer Service Interface object calls the Authorization Combined Positive Customer object at this time. The Authorization Combined Positive Customer object calls the Account object, which calls the Account Access object to update the PBF for the *to* account.
3. Calls the Transaction object to format a log record for the transfer transaction and write the log record to the ITLF.
4. Passes a message containing ITD information to the XPNET Interface object.

XPNET Interface Object Processing

The XPNET Interface object passes the message to the XPNET process. Processing of this message continues with step 14 of the processing summary.

Section 5

BASE24-billpay Message Flows

The diagrams in this section illustrate the paths taken by BASE24-billpay transactions in various types of processing situations. These diagrams are intended to illustrate the transaction path from RBSI-compliant devices and customer service representative (CSR) terminals to the transaction authorizer. This section does not include all possible transaction message flows, only a representative sampling. All transactions assume successful authorization of the requested transaction.

This section includes flows for transactions initiated at an interactive voice response system (IVR) and a customer service representative (CSR) terminal. In this section, the IVR also represents any other RBSI-compliant device that communicates with BASE24 using the ISO 8583-1993 message format. The flows in this section do not depict processing performed for transactions initiated by a third-party processor.

This section contains illustrations of the following transactions:

- IVR transfer – immediate authorized on the BASE24 system
- IVR payment – immediate authorized on the BASE24 system
- IVR payment – immediate authorized on the host
- IVR payment – recurring
- CSR transaction authorized by the Pathway Billpay Server process
- CSR payment – immediate authorized on the BASE24 system
- CSR payment – future
- STI payment – recurring authorized on the BASE24 system
- STI payment – future authorized on the host
- Bank Advice log message

BASE24-billpay transactions originating at an IVR may involve customer verification before transactions can be performed by a CSR on behalf of a customer. Some transactions may also require PIN verification. Whether a customer or PIN needs to be verified before processing a transaction is at the discretion of the financial institution or service provider. Refer to the “IVR Customer and PIN Verification Authorization on the BASE24 System” message flow in section 4 for an illustration of this processing.

Customer and PIN verification is not performed for the following BASE24-billpay transactions originating at an IVR when the customer ID is set to a value of all zeros:

- Bank branch list inquiry
- Bank list inquiry
- Customer add
- Customer inquiry
- Loan application

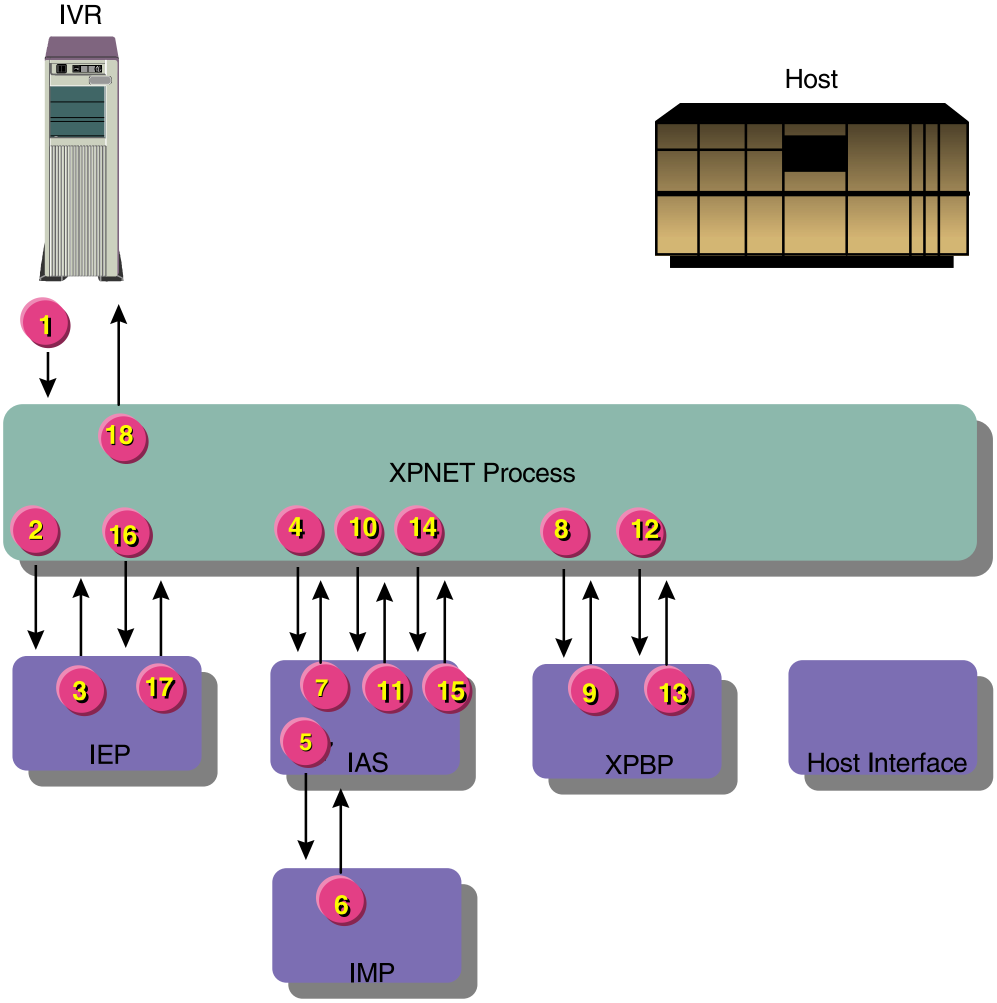
All BASE24-billpay transactions originating from a CSR terminal, except for loan application and customer add transactions, involve customer and PIN verification before a transaction menu is displayed. Refer to the “CSR Customer and PIN Verification Authorization on the BASE24 System” message flow in section 4 for an illustration of this processing.

IVR Payment – Immediate Authorized on the BASE24 System

Immediate payments involve a schedule payment – immediate transaction that automatically generates a payment – immediate transaction. The schedule payment – immediate transaction is a nonfinancial transaction and the payment – immediate transaction is a financial transaction. This message flow assumes the payment is made from an account maintained in the Positive Balance File (PBF). The Positive Customer with Balances/History Authorization (PCBA) method is required when account balances are maintained on the BASE24 system.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process
IEP = Integrated Endpoint process
IMP = Interface Master process
IVR = Interactive voice response system
XPBP = XPNET Billpay Server process

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling an IVR. The IVR creates an ISO request message with the transaction request and sends the message to the XPNET process. The transaction code is 9A (schedule payment – immediate).
2. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the IVR. The XPNET process passes the ISO message to the process named in the station declaration, in this case the Integrated Endpoint process.
3. Using the source class and information from the message, the Integrated Endpoint process identifies the Integrated Authorization Server process as the destination and passes the ISO message to the XPNET process for delivery.
4. The XPNET process passes the ISO message to the Integrated Authorization Server process.
5. The Integrated Authorization Server process obtains station information from the interface extended memory table, obtains the FIID from the Customer Table (CSTT), and requests the Interface Master process to validate the station. The Integrated Authorization Server process also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contained the account number and not the customer ID.
6. The Interface Master process validates the station and returns a message to the Integrated Authorization Server process.
7. The Integrated Authorization Server process determines whether the customer can perform the requested transaction, verifies the PIN, converts the message to a Telebanking Standard Internal Message (BSTM), and passes the BSTM request to the XPNET process.
8. The XPNET process passes the BSTM request to the XPNET Billpay Server process.

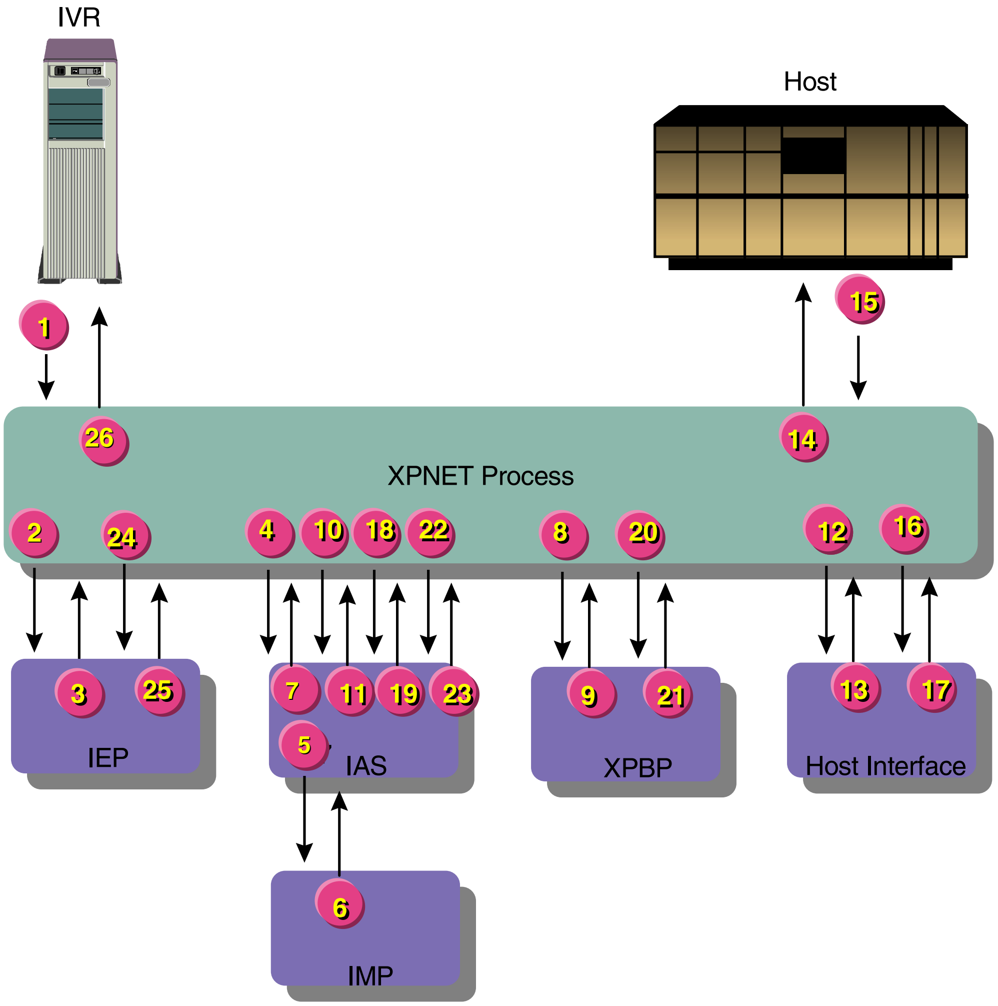
9. The XPNET Billpay Server process inserts a row for transaction code 9A in the History Table (HIST), verifies the vendor and scheduled payment information, and passes a BSTM request with transaction code 50 (payment – immediate) to the XPNET process.
10. The XPNET process passes the BSTM request with transaction code 50 to the Integrated Authorization Server process.
11. The Integrated Authorization Server process authorizes the transaction code 50 amount, updates the customer's PBF account, logs a record for transaction code 50 to the ITS Transaction Log File (ITLF), and passes the BSTM response to the XPNET process.
12. The XPNET process passes the BSTM response for transaction code 50 to the XPNET Billpay Server process.
13. The XPNET Billpay Server process updates the BSTM for transaction code 9A with the results of transaction code 50, then updates the row for transaction code 9A in the HIST and returns the BSTM response for transaction code 9A to the XPNET process. The BSTM for transaction code 50 is not inserted in a HIST row.
14. The XPNET process passes the BSTM response for transaction code 9A to the Integrated Authorization Server process.
15. The Integrated Authorization Server process converts the BSTM response to an ISO response message, logs a record for transaction code 9A to the ITLF, and passes the ISO message to the XPNET process.
16. The XPNET process passes the ISO message to the Integrated Endpoint process.
17. The Integrated Endpoint process passes the ISO message to the XPNET process for return to the IVR. The Integrated Endpoint process uses the original station name carried in the message to identify the message destination.
18. The XPNET process passes the ISO message to the IVR. The IVR returns the response information to the customer.

IVR Payment – Immediate Authorized on the Host

Immediate payments involve a schedule payment – immediate transaction that automatically generates a payment – immediate transaction. The schedule payment – immediate transaction is a nonfinancial transaction and the payment – immediate transaction is a financial transaction. This message flow assumes the payment is made from an account maintained on the host. A minimum of the Positive Customer with Accounts Authorization (PCAA) method is required because account numbers must be available to perform BASE24-billpay payment transactions.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process
IEP = Integrated Endpoint process
IMP = Interface Master process
IVR = Interactive voice response system
XPBP = XPNET Billpay Server process

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling an IVR. The IVR creates an ISO request message with the transaction request and sends the message to the XPNET process. The transaction code is 9A (schedule payment – immediate).
2. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the IVR. The XPNET process passes the ISO message to the process named in the station declaration, in this case the Integrated Endpoint process.
3. Using the source class and information from the message, the Integrated Endpoint process identifies the Integrated Authorization Server process as the destination and passes the ISO message to the XPNET process for delivery.
4. The XPNET process passes the ISO message to the Integrated Authorization Server process.
5. The Integrated Authorization Server process obtains station information from the interface extended memory table, obtains the FIID from the Customer Table (CSTT), and requests the Interface Master process to validate the station. The Integrated Authorization Server process also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contained the account number and not the customer ID.
6. The Interface Master process validates the station and returns a message to the Integrated Authorization Server process.
7. The Integrated Authorization Server process determines whether the customer can perform the requested transaction, verifies the PIN, converts the message to a Telebanking Standard Internal Message (BSTM), and passes the BSTM request to the XPNET process.
8. The XPNET process passes the BSTM request to the XPNET Billpay Server process.

9. The XPNET Billpay Server process inserts a row for transaction code 9A in the History Table (HIST), verifies the vendor and scheduled payment information, and passes a BSTM request with transaction code 50 (payment – immediate) to the XPNET process.
10. The XPNET process passes the BSTM request with transaction code 50 to the Integrated Authorization Server process.
11. The Integrated Authorization Server process determines the transaction is to be authorized by the host and passes the BSTM request to the XPNET process.
12. The XPNET process passes the BSTM request to the ISO Host Interface process for the host that is to authorize the request.
13. The ISO Host Interface process reformats the BSTM request into an ISO external request message and passes it to the XPNET process.
14. The XPNET process passes the ISO external request message to the host for authorization.
15. The host authorizes the transaction and returns the ISO external response message to the XPNET process.
16. The XPNET process passes the ISO external response message to the ISO Host Interface process.
17. The ISO Host Interface process reformats the ISO external response message into a BSTM response and passes it to the XPNET process.
18. The XPNET process passes the BSTM response to the Integrated Authorization Server process.
19. The Integrated Authorization Server process logs a record for transaction code 50 to the ITS Transaction Log File (ITLF) and passes the BSTM response to the XPNET process.
20. The XPNET process passes the BSTM response for transaction code 50 to the XPNET Billpay Server process.
21. The XPNET Billpay Server process updates the BSTM for transaction code 9A with the results of transaction code 50, then updates the row for transaction code 9A in the HIST and returns the BSTM response for transaction code 9A to the XPNET process. The BSTM for transaction code 50 is not inserted in a HIST row.
22. The XPNET process passes the BSTM response for transaction code 9A to the Integrated Authorization Server process.

23. The Integrated Authorization Server process converts the BSTM response to an ISO response message, logs a record for transaction code 9A to the ITLF, and passes the ISO message to the XPNET process.
24. The XPNET process passes the ISO message to the Integrated Endpoint process.
25. The Integrated Endpoint process passes the ISO message to the XPNET process for return to the IVR. The Integrated Endpoint process uses the original station name carried in the message to identify the message destination.
26. The XPNET process passes the ISO message to the IVR. The IVR returns the response information to the customer.

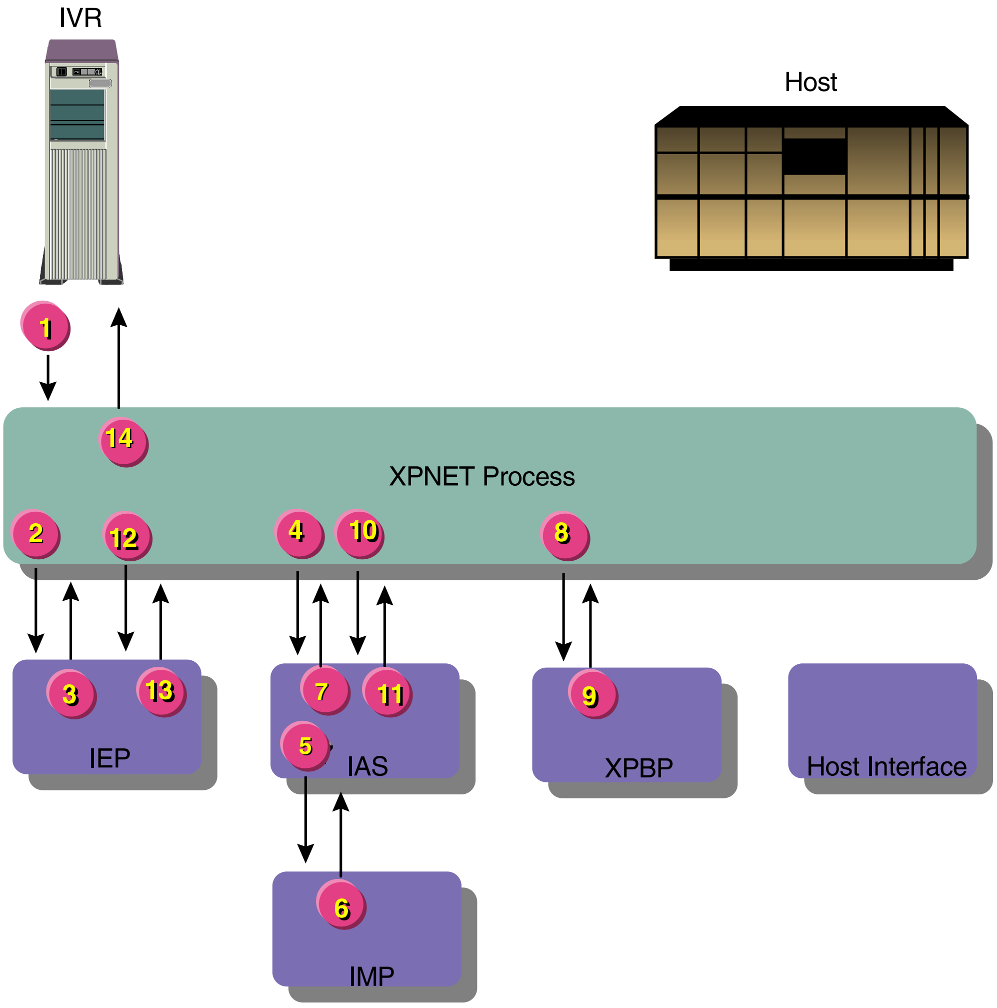
IVR Payment – Recurring

BASE24-billpay payment – recurring transactions involve scheduling a series of payments to be performed in the future. A minimum of the Positive Customer with Accounts Authorization (PCAA) method must be used for this transaction because account numbers must be available to perform BASE24-billpay payment transactions.

There is no difference in processing between the Positive Customer with Accounts Authorization (PCAA) method and the Positive Customer with Balances/History Authorization (PCBA) method during this transaction flow because the account balance is not checked until the payment is initiated by the Scheduled Transaction Initiator process on the scheduled date. The account balances can be maintained on the BASE24 system or on the host.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS	=	Integrated Authorization Server process
IEP	=	Integrated Endpoint process
IMP	=	Interface Master process
IVR	=	Interactive voice response system
XPBP	=	XPNET Billpay Server process

Processing Summary

The transaction is processed as follows:

1. A customer initiates a transaction by calling an IVR. The IVR creates an ISO request message with the transaction request and sends the message to the XPNET process. The transaction code is 9C (schedule payment – recurring).
2. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the IVR. The XPNET process passes the ISO message to the process named in the station declaration, in this case the Integrated Endpoint process.
3. Using the source class and information from the message, the Integrated Endpoint process identifies the Integrated Authorization Server process as the destination and passes the ISO message to the XPNET process for delivery.
4. The XPNET process passes the ISO message to the Integrated Authorization Server process.
5. The Integrated Authorization Server process obtains station information from the interface extended memory table, obtains the FIID from the Customer Table (CSTT), and requests the Interface Master process to validate the station. The Integrated Authorization Server process also determines the customer ID from the Customer/Account Relation Table (CACT), if the request contained the account number and not the customer ID.
6. The Interface Master process validates the station and returns a message to the Integrated Authorization Server process.
7. The Integrated Authorization Server process determines whether the customer can perform the requested transaction, verifies the PIN, converts the message to a Telebanking Standard Internal Message (BSTM), and passes the BSTM request to the XPNET process.
8. The XPNET process passes the BSTM request to the XPNET Billpay Server process.
9. The XPNET Billpay Server process verifies the vendor and scheduled payment information, adds the payment to the Future Table (FUTR) and Recurring Table (RCUR) for use by the Scheduled Transaction Initiator process, inserts a row for the transaction in the History Table (HIST), and returns the BSTM response to the XPNET process.
10. The XPNET process passes the BSTM response to the Integrated Authorization Server process.

11. The Integrated Authorization Server process converts the BSTM response to an ISO response message, logs a record for the transaction to the ITLF, and passes the ISO message to the XPNET process.
12. The XPNET process passes the ISO message to the Integrated Endpoint process.
13. The Integrated Endpoint process passes the ISO message to the XPNET process for return to the IVR. The Integrated Endpoint process uses the original station name carried in the message to identify the message destination.
14. The XPNET process passes the ISO message to the IVR. The IVR returns the response information to the customer.

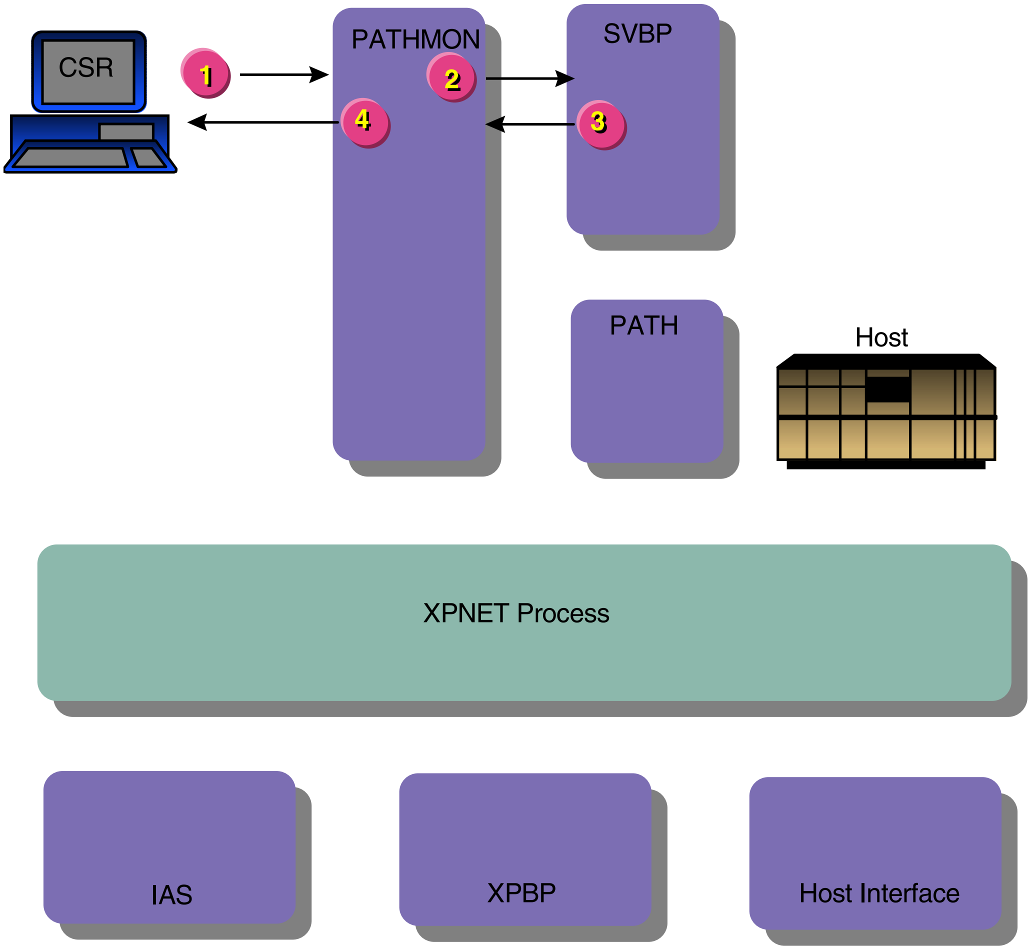
CSR Transaction Authorized by the Pathway Billpay Server Process

This message flow illustrates all BASE24-billpay CSR transactions that can be authorized by the Pathway Billpay Server process without using the XPNET Billpay Server process. This includes all BASE24-billpay transactions initiated at a CSR terminal except transaction codes 9A through 9E. Transaction codes 9A through 9E schedule payments and transfers, and must involve the XPNET Billpay Server process.

This message flow assumes the customer and customer’s PIN have already been verified. Refer to the “CSR Customer and PIN Verification Authorization on the BASE24 System” message flow in section 4 for an illustration of customer and PIN verification processing. All BASE24-billpay CSR transactions except loan application and customer signup involve customer verification and PIN verification before a transaction menu is displayed.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

CSR = Customer Service Representative Requester process
IAS = Integrated Authorization Server process
PATH = Path Server process
PATHMON = Pathway Monitor process
SVBP = Pathway Billpay Server process
XPBP = XPNET Billpay Server process

Processing Summary

The transaction is processed as follows:

1. The CSR enters the transaction information at the CSR terminal after verifying the customer and PIN. The Customer Service Representative (CSR) Billpay Requester process validates the data entered and passes the message to the Pathway Monitor process.
2. The Pathway Monitor process passes the message to the Pathway Billpay Server process.
3. The Pathway Billpay Server process validates the message format and content, performs the requested transaction, inserts a row in the History Table (HIST), and returns the response message to the Pathway Monitor process.
4. The Pathway Monitor process passes the response message to the CSR Billpay Requester process, which displays the information on the CSR terminal. The CSR uses the information to advise the customer.

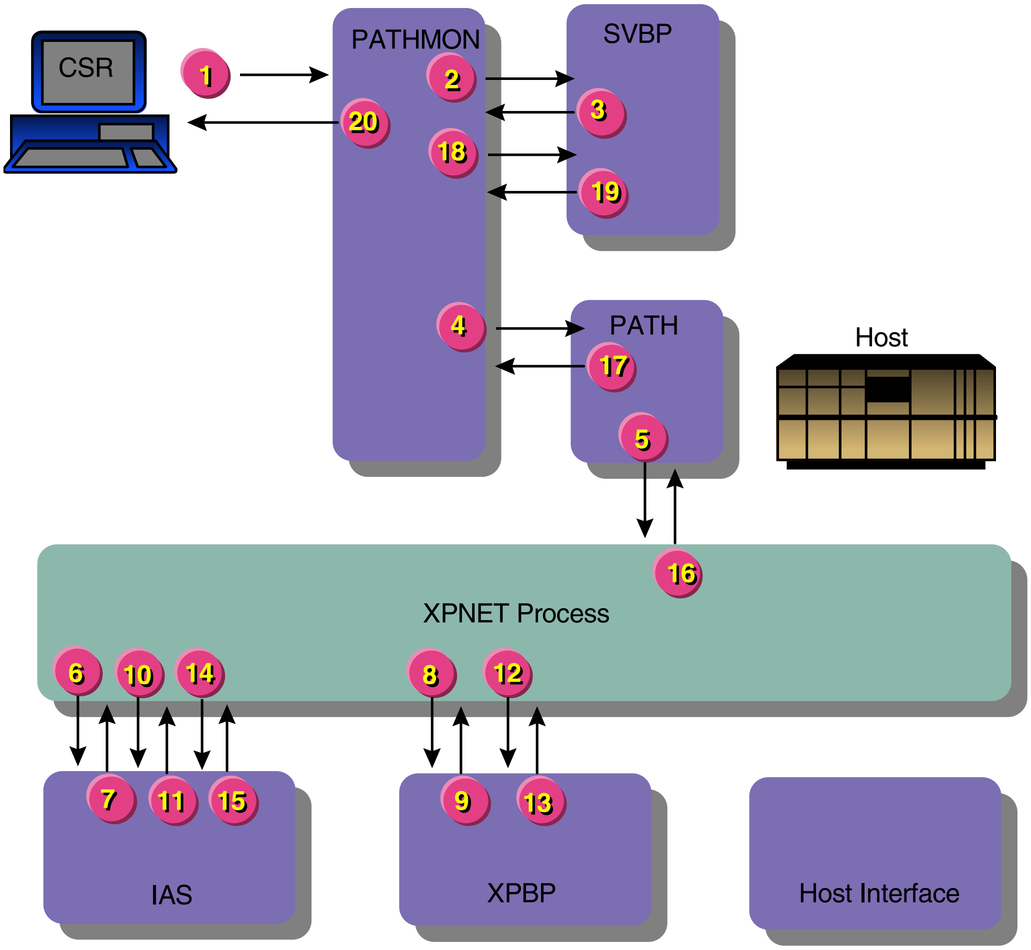
CSR Payment – Immediate Authorized on the BASE24 System

Immediate payments involve a schedule payment – immediate transaction that automatically generates a payment – immediate transaction. The schedule payment – immediate transaction is a nonfinancial transaction and the payment – immediate transaction is a financial transaction. This message flow assumes the payment is made from an account maintained in the Positive Balance File (PBF). The Positive Customer with Balances/History Authorization method is required when account balances are maintained on the BASE24 system.

This message flow assumes the customer and customer’s PIN have already been verified. Refer to the “CSR Customer and PIN Verification Authorization on the BASE24 System” message flow in section 4 for an illustration of customer and PIN verification processing. All BASE24-billpay CSR transactions except loan application and customer signup involve customer verification and PIN verification before a transaction menu is displayed.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

CSR = Customer Service Representative Requester process
IAS = Integrated Authorization Server process
PATH = Path Server process
PATHMON = Pathway Monitor process
SVBP = Pathway Billpay Server process
XPBP = XPNET Billpay Server process

Processing Summary

The transaction is processed as follows:

1. The CSR enters the transaction information at the CSR terminal after verifying the customer and PIN. The Customer Service Representative (CSR) Billpay Requester process validates the data entered and passes the message to the Pathway Monitor process.
2. The Pathway Monitor process passes the message to the Pathway Billpay Server process.
3. The Pathway Billpay Server process validates the message format and content, determines the transaction code is 9A (schedule payment – immediate), creates a BSTM request, and passes it to the Pathway Monitor process.
4. The Pathway Monitor process passes the BSTM request to a Path Server process.
5. The Path Server process passes the message to the XPNET process.
6. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the Path Server process. The XPNET process passes the message to the process named in the station declaration, in this case the Integrated Authorization Server process.
7. The Integrated Authorization Server process identifies the FIID, determines whether the customer can perform the requested transaction, and passes the BSTM request to the XPNET process.
8. The XPNET process passes the BSTM request to the XPNET Billpay Server process.
9. The XPNET Billpay Server process inserts a row for transaction code 9A in the History Table (HIST), verifies the vendor and scheduled payment information, and passes a BSTM request with transaction code 50 (payment – immediate) to the XPNET process.
10. The XPNET process passes the BSTM request with transaction code 50 to the Integrated Authorization Server process.
11. The Integrated Authorization Server process authorizes the transaction code 50 amount, updates the customer's PBF account, logs a record for transaction code 50 to the ITS Transaction Log File (ITLF), and passes the BSTM response to the XPNET process.

12. The XPNET process passes the BSTM response for transaction code 50 to the XPNET Billpay Server process.
13. The XPNET Billpay Server process updates the BSTM for transaction code 9A with the results of transaction code 50, then updates the row for transaction code 9A in the HIST and returns the BSTM response for transaction code 9A to the XPNET process. The BSTM for transaction code 50 is not inserted in a HIST row.
14. The XPNET process passes the BSTM response for transaction code 9A to the Integrated Authorization Server process.
15. The Integrated Authorization Server process logs a record for transaction code 9A to the ITLF and passes the message to the XPNET process.
16. The XPNET process passes the BSTM response to the Path Server process.
17. The Path Server process passes the BSTM response to the Pathway Monitor process.
18. The Pathway Monitor process passes the BSTM response to the Pathway Billpay Server process.
19. The Pathway Billpay Server process passes the response message to the Pathway Monitor process.
20. The Pathway Monitor process passes the response message to the CSR Billpay Requester process, which displays the information on the CSR terminal. The CSR uses the information to advise the customer.

CSR Payment – Future

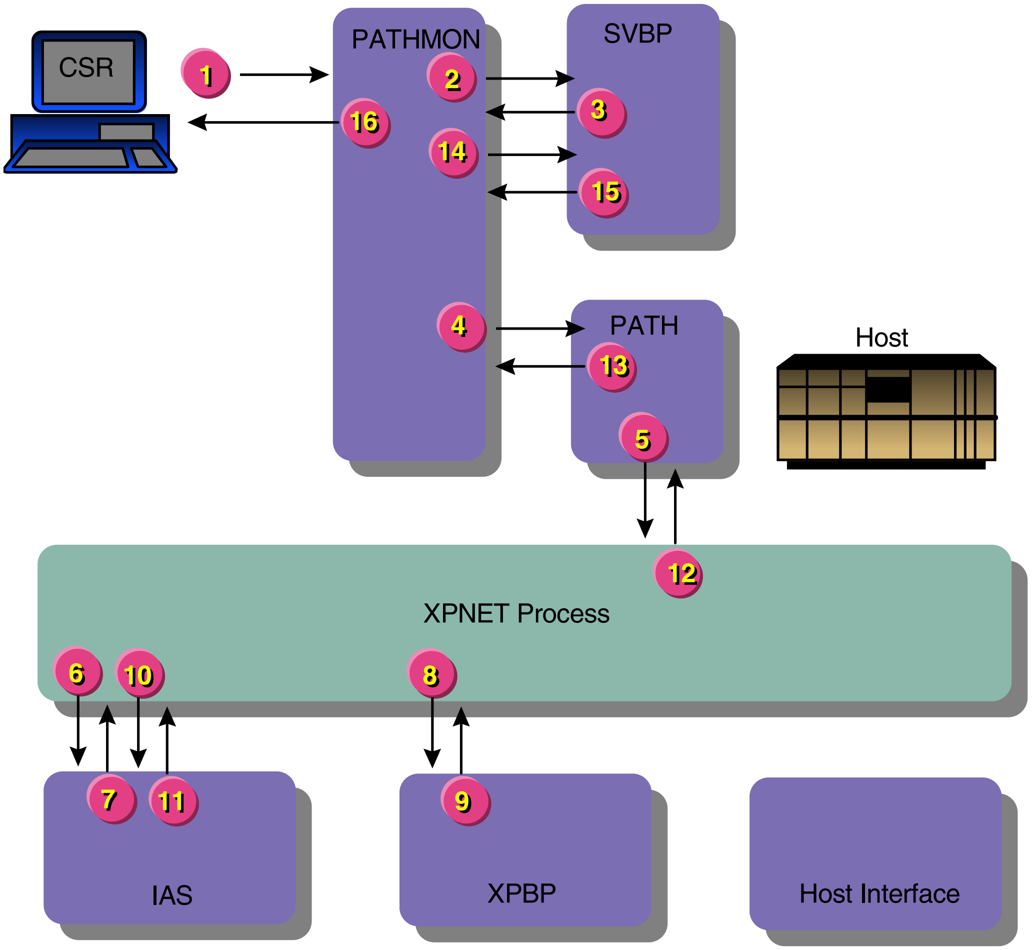
BASE24-billpay payment – future transactions involve scheduling a payment to be performed in the future. Either the Positive Customer with Accounts Authorization (PCAA) method or the Positive Customer with Balances/History Authorization (PCBA) method must be used for this transaction. The Positive Customer Authorization (PCA) method cannot be used for this transaction because account numbers must be available to schedule BASE24-billpay payment transactions.

There is no difference in processing between the Positive Customer with Accounts Authorization (PCAA) method and the Positive Customer with Balances/History Authorization (PCBA) method during this transaction flow because the account balance is not checked until the payment is initiated by the Scheduled Transaction Initiator process on the scheduled date. The account balances can be maintained on the BASE24 system or on the host.

This message flow assumes the customer and customer's PIN have already been verified. Refer to the "CSR Customer and PIN Verification Authorization on the BASE24 System" message flow in section 4 for an illustration of customer and PIN verification processing. All BASE24-billpay CSR transactions except loan application and customer signup involve customer verification and PIN verification before a transaction menu is displayed.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

CSR = Customer Service Representative Requester process
IAS = Integrated Authorization Server process
PATH = Path Server process
PATHMON = Pathway Monitor process
SVBP = Pathway Billpay Server process
XPBP = XPNET Billpay Server process

Processing Summary

The transaction is processed as follows:

1. The CSR enters the transaction information at the CSR terminal after verifying the customer and PIN. The Customer Service Representative (CSR) Billpay Requester process validates the data entered and passes the message to the Pathway Monitor process.
2. The Pathway Monitor process passes the message to the Pathway Billpay Server process.
3. The Pathway Billpay Server process validates the message format and content, determines the transaction code is 9B (schedule payment – future), creates a BSTM request, and passes it to the Pathway Monitor process.
4. The Pathway Monitor process passes the BSTM request to a Path Server process.
5. The Path Server process passes the BSTM request to the XPNET process.
6. The XPNET process retrieves the destination for the message from the DESTINATION attribute in the XPNET station declaration for the Path Server process. The XPNET process passes the message to the process named in the station declaration, in this case the Integrated Authorization Server process.
7. The Integrated Authorization Server process identifies the FIID, determines whether the customer can perform the requested transaction, and passes the BSTM request to the XPNET process.
8. The XPNET process passes the BSTM request to the XPNET Billpay Server process.
9. The XPNET Billpay Server process verifies the vendor and scheduled payment information, adds the payment to the Future Table (FUTR) for use by the Scheduled Transaction Initiator process, inserts a row for the transaction in the History Table (HIST), and returns the BSTM response to the XPNET process.
10. The XPNET process passes the BSTM response to the Integrated Authorization Server process.
11. The Integrated Authorization Server process logs a record to the ITLF and passes the message to the XPNET process.
12. The XPNET process passes the BSTM response to the Path Server process.
13. The Path Server process passes the BSTM response to the Pathway Monitor process.

14. The Pathway Monitor process passes the BSTM response to the Pathway Billpay Server process.
15. The Pathway Billpay Server process passes the response message to the Pathway Monitor process.
16. The Pathway Monitor process passes the response message to the CSR Billpay Requester process, which displays the information on the CSR terminal. The CSR uses the information to advise the customer.

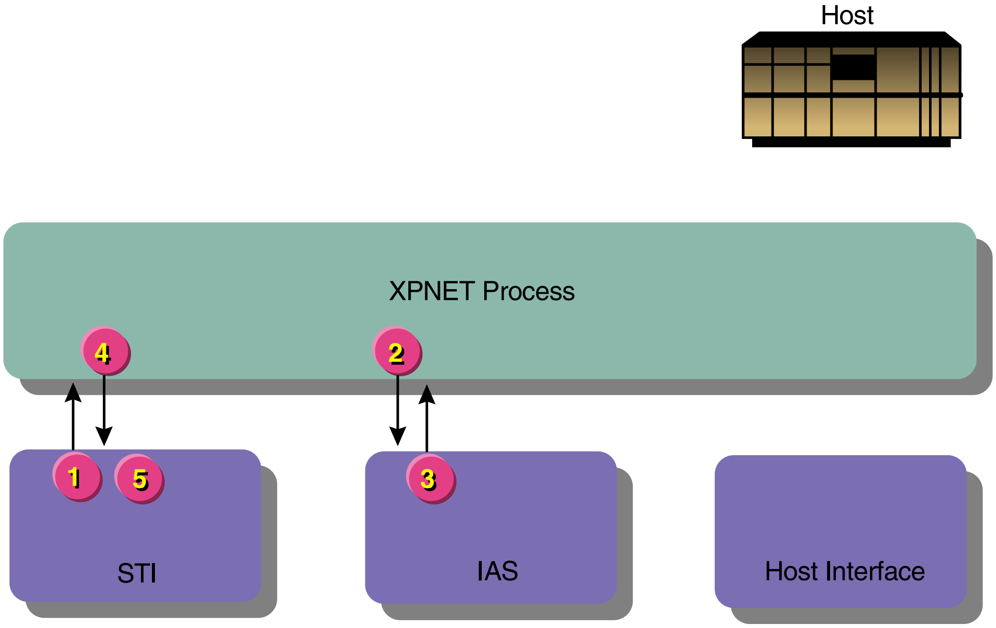
STI Payment – Recurring Authorized on the BASE24 System

The Scheduled Transaction Initiator process starts a new transaction path based on information contained in the Future Table (FUTR) for each transfer and payment transaction to be processed on the current date. This example illustrates a payment – recurring transaction (transaction code 9C).

While not shown in this example, the Scheduled Transaction Initiator process uses information in the Retry Table (RTRY) to determine whether transactions in the FUTR that are declined with specific action codes can be retried.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process
STI = Scheduled Transaction Initiator process

Processing Summary

The transaction is processed as follows:

1. The Scheduled Transaction Initiator process locates a transaction with transaction code 9C (schedule payment – recurring) and the current date in the FUTR. It then updates the Recurring Table (RCUR) row to reflect the payment is being processed, creates a BSTM request with transaction code 5B (payment – recurring) from information in the FUTR row, and passes the BSTM request to the XPNET process.
2. The XPNET process passes the BSTM to the Integrated Authorization Server process.
3. The Integrated Authorization Server process authorizes the payment amount, updates the customer's PBF account, logs a record for transaction code 5B to the ITS Transaction Log File (ITLF), and passes the BSTM response to the XPNET process.
4. The XPNET process passes the BSTM response to the Scheduled Transaction Initiator process.
5. The Scheduled Transaction Initiator process performs the following:
   a. Inserts the 5B transaction in a History Table (HIST) row.
   b. Removes the Future Table row for the 9C transaction just completed.
   c. Updates the Recurring Table (RCUR) row to reflect the payment just completed, as follows:
      - If no payments remain in the RCUR row, it removes RCUR row.
      - If payments remain in the RCUR row, it updates the row and inserts a new Future Table (FUTR) row for the next payment.

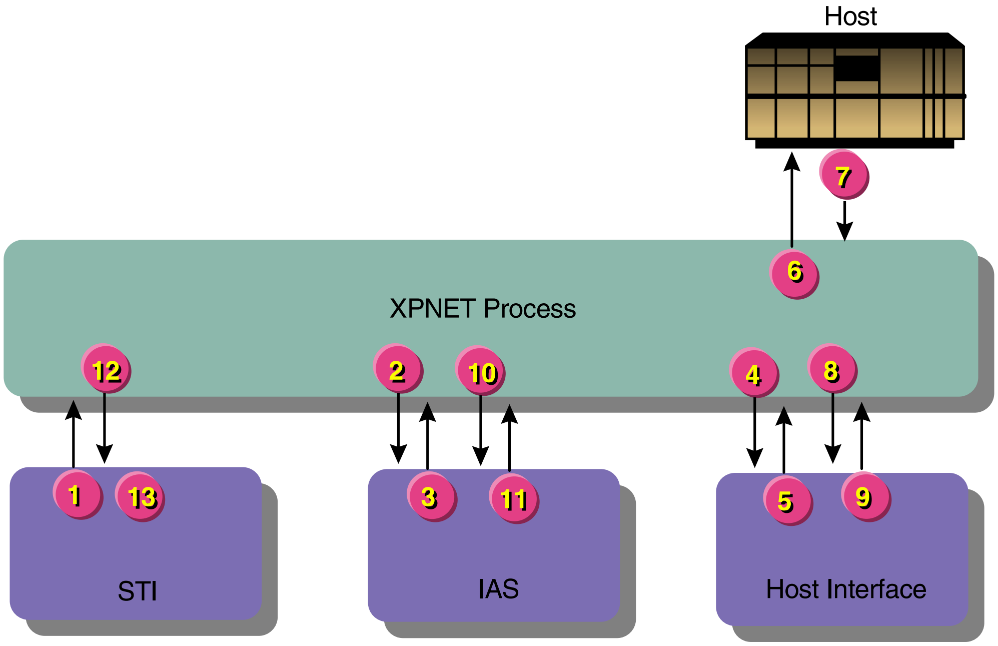
STI Payment – Future Authorized on the Host

The Scheduled Transaction Initiator process starts a new transaction path based on information contained in the Future Table (FUTR) for each transfer and payment transaction to be processed on the current date. This example illustrates a payment – future transaction (transaction code 9B).

While not shown in this example, the Scheduled Transaction Initiator process uses information in the Retry Table (RTRY) to determine whether transactions in the FUTR that are declined with specific action codes can be retried.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process
STI = Scheduled Transaction Initiator process

Processing Summary

The transaction is processed as follows:

1. The Scheduled Transaction Initiator process locates a transaction with transaction code 9B (schedule payment – future) and the current date in the Future Table (FUTR). It then updates the FUTR row to reflect the payment is being processed, creates a BSTM request with transaction code 5A (payment – future), and passes the BSTM to the XPNET process.
2. The XPNET process passes the BSTM to the Integrated Authorization Server process.
3. The Integrated Authorization Server process determines the transaction is to be authorized by the host and passes the BSTM request to the XPNET process.
4. The XPNET process passes the BSTM request to the ISO Host Interface process for the host that is to authorize the request.
5. The ISO Host Interface process reformats the BSTM request into an ISO external request message and passes it to the XPNET process.
6. The XPNET process passes the ISO external request message to the host for authorization.
7. The host authorizes the transaction and returns the ISO external response message to the XPNET process.
8. The XPNET process passes the ISO external response message to the ISO Host Interface process.
9. The ISO Host Interface process reformats the ISO external response message into a BSTM response and passes it to the XPNET process.
10. The XPNET process passes the BSTM response to the Integrated Authorization Server process.
11. The Integrated Authorization Server process logs a record for transaction code 5A to the ITS Transaction Log File (ITLF) and passes the BSTM response to the XPNET process.
12. The XPNET process passes the BSTM response to the Scheduled Transaction Initiator process.
13. The Scheduled Transaction Initiator process removes the FUTR row for the 9B transaction just completed and inserts the 5A transaction in a History Table (HIST) row.

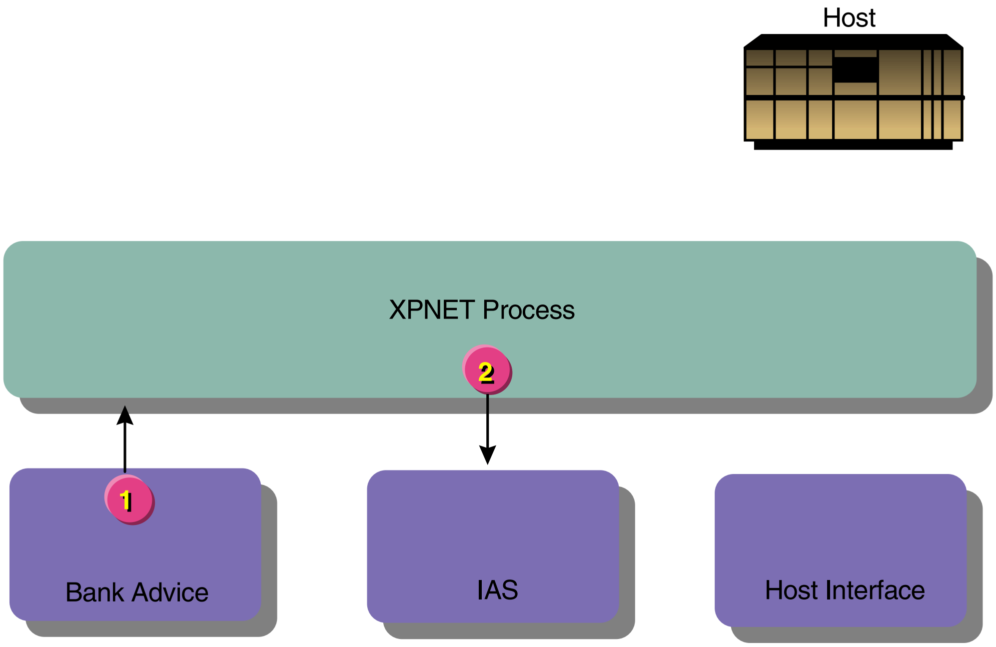
Bank Advice Log Message

The Bank Advice process identifies specific BASE24-billpay CSR transactions in the History Table (HIST), generates advice messages, and passes the messages to an Integrated Authorization Server process for logging to the ITS Transaction Log File (ITLF). The Bank Advice process uses information in the Bank Transaction Advice Table (BKAV) to identify which HIST rows should be logged to the ITLF.

The advice message is generated based on a row added to the HIST during an earlier transaction. It is not part of the original transaction.

Processing Diagram

The following diagram illustrates the flow of the transaction through the BASE24 system. A key to the abbreviations follows the illustration.



Key

IAS = Integrated Authorization Server process

Processing Summary

The transaction is processed as follows:

1. The Bank Advice process creates a BSTM advice message based on information contained in a HIST row and passes it to the XPNET process.
2. The XPNET process passes the message to the Integrated Authorization Server process. The Integrated Authorization Server process logs the advice to the ITLF without returning a response.

Section 6

BASE24 Remote Banking Transaction Impacts on the Database

This section describes the impact of BASE24 Remote Banking transactions on the balance and usage accumulation totals in the Customer Table (CSTT), Interface Security File (ISF), and Positive Balance File (PBF). It also describes how BASE24 Remote Banking transactions impact data other than balances or counts in the BASE24 Remote Banking database.

BASE24 Database Balance and Usage Impacts

This subsection describes the impact of BASE24 Remote Banking transactions on balance and usage accumulation totals in the Customer Table (CSTT), Interface Security File (ISF), and Positive Balance File (PBF).

The BAD PIN COUNT field on Customer Table (CSTT) screen 1 is increased by one for any transaction received that contains an incorrect PIN. The KEY COUNTER field on Interface Security File (ISF) screen 2 for the inbound PIN encryption key is increased by one each time the key is changed due to dynamic key management processing.

Payment and transfer transactions are the only BASE24 Remote Banking transactions that have any impact on balances in the BASE24 database, and the only file that is affected by these transactions is the Positive Balance File (PBF). The table provided later in this subsection, referred to as the impact table, shows the impacts of approved transactions on PBF balances.

The following table contains a list of balance totals that can be impacted by BASE24 Remote Banking transactions. The captions for these totals are included in the impact table beginning on the next page and match the field names of the corresponding totals on PBF screen 1. The updated PBF totals have different meanings depending on whether the PBF record represents a credit or noncredit account.

Positive Balance File Totals	
Noncredit	**Credit**
AVAILABLE BALANCE	AVAILABLE CREDIT
LEDGER BALANCE	CREDIT LIMIT
AMOUNT ON HOLD	CREDIT BALANCE

In the balance columns of the impact table, a plus sign (+) indicates the transaction amount is added to the PBF field and a minus sign (–) indicates the transaction amount is subtracted from the PBF field.

A check (✓) in columns two through four of the impact table identify which transactions are available for each of the three BASE24 Remote Banking product offerings. The BASE24-billpay transactions in the impact table are the transaction codes generated to update PBF accounts. However, different transaction codes are used to initiate the BASE24-billpay transactions, as shown in the following table:

Transaction Codes Used to Update		**Transaction Codes Used to Initiate**	
Code	**Description**	**Code**	**Description**
4A	Transfer – future	9D	Schedule transfer – future
4B	Transfer – recurring	9E	Schedule transfer – recurring
50	Payment – immediate	9A	Schedule payment – immediate
5A	Payment – future	9B	Schedule payment – future
5B	Payment – recurring	9C	Schedule payment – recurring

The following transactions impact the Positive Balance File (PBF):

Tran Code	**TB**	**Enhanced TB**	**BP**	**Transaction Description**		**Amount on Hold / Credit Balance**	**Ledger Balance / Credit Limit**	**Available Balance / Available Credit**
40	✓			Transfer – immediate	from savings account 1		–	–
					to savings account 2		+	+
40	✓			Transfer – immediate	from savings account		–	–
					to checking account		+	+
40	✓			Transfer – immediate	from savings account		–	–
					to credit account	–		+

Tran Code	TB	Enhanced TB	BP	Transaction Description		Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
40	✓			Transfer – immediate	from savings account		–	–
					to line of credit account	–		+
40	✓			Transfer – immediate	from savings account		–	–
					to interest-bearing checking account		+	+
40	✓			Transfer – immediate	from savings account		–	–
					to commercial loan	–		+
40	✓			Transfer – immediate	from savings account		–	–
					to installment loan	–		+
40	✓			Transfer – immediate	from savings account		–	–
					to mortgage loan	–		+
40	✓			Transfer – immediate	from checking account		–	–
					to savings account		+	+
40	✓			Transfer – immediate	from checking account 1		–	–
					to checking account 2		+	+
40	✓			Transfer – immediate	from checking account		–	–
					to credit account	–		+
40	✓			Transfer – immediate	from checking account		–	–
					to line of credit account	–		+

Tran Code	TB	Enhanced TB	BP	Transaction Description		Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
40	✓			Transfer – immediate	from checking account		–	–
					to interest-bearing checking account		+	+
40	✓			Transfer – immediate	from checking account		–	–
					to commercial loan	–		+
40	✓			Transfer – immediate	from checking account		–	–
					to installment loan	–		+
40	✓			Transfer – immediate	from checking account		–	–
					to mortgage loan	–		+
40	✓			Transfer – immediate	from credit account	+		–
					to savings account		+	+
40	✓			Transfer – immediate	from credit account	+		–
					to checking account		+	+
40	✓			Transfer – immediate	from credit account	+		–
					to interest-bearing checking account		+	+
40	✓			Transfer – immediate	from line of credit account	+		–
					to savings account		+	+
40	✓			Transfer – immediate	from line of credit account	+		–
					to checking account		+	+

Tran Code	TB	Enhanced TB	BP	Transaction Description		Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
40	✓			Transfer – immediate	from line of credit account	+		–
					to interest-bearing checking account		+	+
40	✓			Transfer – immediate	from interest-bearing checking account		–	–
					to savings account		+	+
40	✓			Transfer – immediate	from interest-bearing checking account		–	–
					to checking account		+	+
40	✓			Transfer – immediate	from interest-bearing checking account		–	–
					to credit account	–		+
40	✓			Transfer – immediate	from interest-bearing checking account		–	–
					to line of credit account	–		+
40	✓			Transfer – immediate	from interest-bearing checking account 1		–	–
					to interest-bearing checking account 2		+	+
40	✓			Transfer – immediate	from interest-bearing checking account		–	–
					to commercial loan	–		+

Tran Code	TB	Enhanced TB	BP	Transaction Description		Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
40	✓			Transfer – immediate	from interest-bearing checking account		–	–
					to installment loan	–		+
40	✓			Transfer – immediate	from interest-bearing checking account		–	–
					to mortgage loan	–		+
4A		✓	✓	Transfer – future	from savings account 1		–	–
					to savings account 2		+	+
4A		✓	✓	Transfer – future	from savings account		–	–
					to checking account		+	+
4A		✓	✓	Transfer – future	from savings account		–	–
					to credit account	–		+
4A		✓	✓	Transfer – future	from savings account		–	–
					to line of credit account	–		+
4A		✓	✓	Transfer – future	from savings account		–	–
					to interest-bearing checking account		+	+
4A		✓	✓	Transfer – future	from savings account		–	–
					to commercial loan	–		+
4A		✓	✓	Transfer – future	from savings account		–	–
					to installment loan	–		+

Tran Code	TB	Enhanced TB	BP	Transaction Description		Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
4A		✓	✓	Transfer – future	from savings account		–	–
					to mortgage loan	–		+
4A		✓	✓	Transfer – future	from checking account		–	–
					to savings account		+	+
4A		✓	✓	Transfer – future	from checking account 1		–	–
					to checking account 2		+	+
4A		✓	✓	Transfer – future	from checking account		–	–
					to credit account	–		+
4A		✓	✓	Transfer – future	from checking account		–	–
					to line of credit account	–		+
4A		✓	✓	Transfer – future	from checking account		–	–
					to interest-bearing checking account		+	+
4A		✓	✓	Transfer – future	from checking account		–	–
					to commercial loan	–		+
4A		✓	✓	Transfer – future	from checking account		–	–
					to installment loan	–		+
4A		✓	✓	Transfer – future	from checking account		–	–
					to mortgage loan	–		+

Tran Code	TB	Enhanced TB	BP	Transaction Description		Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
4A		✓	✓	Transfer – future	from credit account	+		–
					to savings account		+	+
4A		✓	✓	Transfer – future	from credit account	+		–
					to checking account		+	+
4A		✓	✓	Transfer – future	from credit account	+		–
					to interest-bearing checking account		+	+
4A		✓	✓	Transfer – future	from line of credit account	+		–
					to savings account		+	+
4A		✓	✓	Transfer – future	from line of credit account	+		–
					to checking account		+	+
4A		✓	✓	Transfer – future	from line of credit account	+		–
					to interest-bearing checking account		+	+
4A		✓	✓	Transfer – future	from interest-bearing checking account		–	–
					to savings account		+	+
4A		✓	✓	Transfer – future	from interest-bearing checking account		–	–
					to checking account		+	+

Tran Code	TB	Enhanced TB	BP	Transaction Description		Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
4A		✓	✓	Transfer – future	from interest-bearing checking account		–	–
					to credit account	–		+
4A		✓	✓	Transfer – future	from interest-bearing checking account		–	–
					to line of credit account	–		+
4A		✓	✓	Transfer – future	from interest-bearing checking account 1		–	–
					to interest-bearing checking account 2		+	+
4A		✓	✓	Transfer – future	from interest-bearing checking account		–	–
					to commercial loan	–		+
4A		✓	✓	Transfer – future	from interest-bearing checking account		–	–
					to installment loan	–		+
4A		✓	✓	Transfer – future	from interest-bearing checking account		–	–
					to mortgage loan	–		+
4B		✓	✓	Transfer – recurring	from savings account 1		–	–
					to savings account 2		+	+
4B		✓	✓	Transfer – recurring	from savings account		–	–
					to checking account		+	+

Tran Code	TB	Enhanced TB	BP	Transaction Description		Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
4B		✓	✓	Transfer – recurring	from savings account		–	–
					to credit account	–		+
4B		✓	✓	Transfer – recurring	from savings account		–	–
					to line of credit account	–		+
4B		✓	✓	Transfer – recurring	from savings account		–	–
					to interest-bearing checking account		+	+
4B		✓	✓	Transfer – recurring	from savings account		–	–
					to commercial loan	–		+
4B		✓	✓	Transfer – recurring	from savings account		–	–
					to installment loan	–		+
4B		✓	✓	Transfer – recurring	from savings account		–	–
					to mortgage loan	–		+
4B		✓	✓	Transfer – recurring	from checking account		–	–
					to savings account		+	+
4B		✓	✓	Transfer – recurring	from checking account 1		–	–
					to checking account 2		+	+
4B		✓	✓	Transfer – recurring	from checking account		–	–
					to credit account	–		+

Tran Code	TB	Enhanced TB	BP	Transaction Description		Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
4B		✓	✓	Transfer – recurring	from checking account		–	–
					to line of credit account	–		+
4B		✓	✓	Transfer – recurring	from checking account		–	–
					to interest-bearing checking account		+	+
4B		✓	✓	Transfer – recurring	from checking account		–	–
					to commercial loan	–		+
4B		✓	✓	Transfer – recurring	from checking account		–	–
					to installment loan	–		+
4B		✓	✓	Transfer – recurring	from checking account		–	–
					to mortgage loan	–		+
4B		✓	✓	Transfer – recurring	from credit account	+		–
					to savings account		+	+
4B		✓	✓	Transfer – recurring	from credit account	+		–
					to checking account		+	+
4B		✓	✓	Transfer – recurring	from credit account	+		–
					to interest-bearing checking account		+	+
4B		✓	✓	Transfer – recurring	from line of credit account	+		–
					to savings account		+	+

Tran Code	TB	Enhanced TB	BP	Transaction Description		Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
4B		✓	✓	Transfer – recurring	from line of credit account	+		–
					to checking account		+	+
4B		✓	✓	Transfer – recurring	from line of credit account	+		–
					to interest-bearing checking account		+	+
4B		✓	✓	Transfer – recurring	from interest-bearing checking account		–	–
					to savings account		+	+
4B		✓	✓	Transfer – recurring	from interest-bearing checking account		–	–
					to checking account		+	+
4B		✓	✓	Transfer – recurring	from interest-bearing checking account		–	–
					to credit account	–		+
4B		✓	✓	Transfer – recurring	from interest-bearing checking account		–	–
					to line of credit account	–		+
4B		✓	✓	Transfer – recurring	from interest-bearing checking account 1		–	–
					to interest-bearing checking account 2		+	+

Tran Code	TB	Enhanced TB	BP	Transaction Description		Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
4B		✓	✓	Transfer – recurring	from interest-bearing checking account		–	–
					to commercial loan	–		+
4B		✓	✓	Transfer – recurring	from interest-bearing checking account		–	–
					to installment loan	–		+
4B		✓	✓	Transfer – recurring	from interest-bearing checking account		–	–
					to mortgage loan	–		+
50			✓	Payment – immediate from savings account			–	–
50			✓	Payment – immediate from checking account			–	–
50			✓	Payment – immediate from credit account		+		–
50			✓	Payment – immediate from line of credit account		+		–
50			✓	Payment – immediate from interest-bearing checking account			–	–
5A			✓	Payment – future from savings account			–	–
5A			✓	Payment – future from checking account			–	–
5A			✓	Payment – future from credit account		+		–

Tran Code	TB	Enhanced TB	BP	Transaction Description	Amount on Hold / Credit Balance	Ledger Balance / Credit Limit	Available Balance / Available Credit
5A			✓	Payment – future from line of credit account	+		–
5A			✓	Payment – future from interest-bearing checking account		–	–
5B			✓	Payment – recurring from savings account		–	–
5B			✓	Payment – recurring from checking account		–	–
5B			✓	Payment – recurring from credit account	+		–
5B			✓	Payment – recurring from line of credit account	+		–
5B			✓	Payment – recurring from interest-bearing checking account		–	–

Other Data Impacts

This subsection describes the impact of BASE24 Remote Banking transactions on data other than balances or counts in the Customer Table (CSTT), Customer/Account Relation Table (CACT), Customer Billpay Table (CBPT), Customer Vendor Table (CVND), Loan Application Table (LOAN), Miscellaneous Request Table (MISC), Personal Information Table (PIT), Vendor Branch Table (VNDB), and Vendor Table (VNDR).

Customer Table (CSTT)

The CSTT is impacted by the PIN change, customer add, customer update, customer activate, and customer deactivate transactions as follows:

- The PIN change transaction updates the PIN VERIFICATION DIGITS field on CSTT screen 1.
- The customer add transaction adds a new row to the CSTT.
- The customer update transaction can update the following fields on CSTT screen 1:

  ALTERNATE CONTACT
  BRANCH ID
  PROFILE
  TYPE
  CUST INFO (3 fields)
  DEFAULT ACCT TYPE
  DEFAULT ACCT NUM
  SERVICE FEE ACCT TYPE
  SERVICE FEE ACCT NUM

- The customer activate transaction changes the value of the VERIFY STATUS field on CSTT screen 1 from I (initial) to V (verified), sets the BEGIN DATE field to the current calendar date, sets the END DATE field to the value defined in the default customer data for the customer’s financial institution, and sets the LAST UPDATE field for the customer’s verify status to the current calendar date. If the value of the VERIFY STATUS field is already set to V (verified), then only the BEGIN DATE and END DATE fields are changed.
- The customer deactivate transaction sets the END DATE field on CSTT screen 1 to the current calendar date minus one day.

Customer/Account Relation Table (CACT)

The CACT is impacted by the customer account add, customer account activate, customer account deactivate, customer account delete, and customer account update transactions as follows:

- The customer account add transaction adds a new row to the CACT.
- The customer account activate transaction updates the BEG DATE field on CSTT screen 2 to the current date and sets the END DATE field to the value defined in the default Customer Account Relation Table (CACT) row for the customer's financial institution.
- The customer deactivate transaction updates the END DATE field on CSTT screen 2 to the current calendar date minus one day.
- The customer account delete transaction deletes an existing row from the CACT.
- The customer account update transaction can update the following fields on CSTT screen 2:

  QUALIFIER
  DESCRIPTION
  ACT ALWD: DB
  ACT ALWD: CR
  ACT ALWD: INQ

Customer Billpay Table (CBPT)

The CBPT is impacted by the customer add and customer vendor add transactions as follows:

- The customer add transaction causes a new row to be added to the CBPT the first time the customer ID is used to perform a BASE24-billpay transaction.
- The customer vendor add transaction updates the Last Customer Vendor Number field on CBPT screen 1 by one.

Customer Vendor Table (CVND)

The CVND is impacted by the customer vendor add, customer vendor activate, customer vendor deactivate, and customer vendor update as follows:

- The customer vendor add transaction adds a new row to the CVND.
- The customer vendor activate transaction updates the value of the Account Verify: Status field on CVND screen 1 from I (initial) to V (verified) and sets the Account Verify: Date field to the current date and time. The transaction also sets the Account Begin Date field to the current calendar date and the Account End Date field to the maximum date allowed. If the account verify status is already verified, only the account begin and end dates are changed.
- The customer vendor deactivate transaction updates the Account End Date field on CVND screen 1 to the current calendar date minus one day.
- The customer vendor update transaction can update the following fields on CVND screen 1:

  Account With Vendor
  Budget Category
  Customer Vendor Short Name
  Info 1
  Info 2

Loan Application Table (LOAN)

The loan application transaction adds a new row to the LOAN.

Miscellaneous Request Table (MISC)

The miscellaneous request transaction adds a new row to the MISC.

Personal Information Table (PIT)

The PIT is impacted by the customer add and customer update transactions as follows:

- The customer add transaction adds a new row to the PIT.
- The customer update transaction can update any of the following fields on PIT screen 1, except for the PERSONAL ID field.

Vendor Branch Table (VNDB)

The VNDB can be impacted by the customer vendor add and customer vendor update transactions as follows:

- The customer vendor add transaction can cause a new row to be added to the VNDB if the vendor branch is not already defined and the system allows unregistered vendors to be added to the database.
- For public vendors with a vendor verify status of verified, the customer vendor update transaction can update all of the fields on VNDB screens 1 and 2, except for the key fields (Vendor Number and Branch Number).
- The customer vendor update transaction can cause a new row to be added to the VNDB if the vendor branch number is set to zero in the request and an existing row does not already exist in the VNDB for the vendor branch name, address 1, and address 2 values provided in the request.

Vendor Table (VNDR)

The VNDR can be impacted by the customer vendor add and customer vendor update transactions as follows:

- The customer vendor add transaction can cause a new row to be added to the VNDR if the vendor is not already defined and the system allows unregistered vendors to be added to the database.
- For public vendors with a vendor verify status of verified, the customer vendor update transaction can update all fields on VNDR screens 1, 2 and 3, except for the key field (Vendor Number).

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Appendix A

BASE24 Remote Banking Messages

The Internal Transaction Data (ITD) contains the information the BASE24 Remote Banking products need to process a transaction. The Transaction object is responsible for updating ITD fields, providing ITD information to other objects for their use, and logging ITD information to the ITS Transaction Log File (ITLF).

The ITD is placed in the Telebanking Standard Internal Message (BSTM) for movement between the BASE24-telebanking Integrated Authorization Server process and the following processes:

- Bank Advice process
- ISO Host Interface process
- Pathway Billpay Server process
- Scheduled Transaction Initiator process
- XPNET Billpay Server process

This appendix presents the ITD fields and BSTM format used by the BASE24 Remote Banking products.

Telebanking Standard Internal Message

The Telebanking Standard Internal Message (BSTM) contains the following fields:

Position	Level	Field Name and Description	Data Type
1–4	02	MTI The type of message described by the BSTM. Valid values are as follows: 1100 = Authorization Request 1101 = Authorization Request Repeat 1110 = Authorization Request Response 1120 = Authorization Advice 1200 = Financial Request 1201 = Financial Request Repeat 1210 = Financial Request Response 1220 = Financial Advice 1420 = Reversal 1421 = Reversal Repeat	PIC 9(4)
5–6	02	PROD-ID The BASE24 product originating this transaction. For BASE24-telebanking and BASE24-billpay transactions, this field is set to 14.	PIC X (2)
7–22	02	SRC-SYM-NAM The symbolic name of the process from which the message originated.	PIC X(16)
23–38	02	DEST-SYM-NAM The symbolic name of the process to which the message is to be sent.	PIC X(16)
39–42	02	TAG The sequence number associated with the transaction.	TYPE BINARY 32

Position	Level	Field Name and Description	Data Type
43–44	02	FAILURE-FLG A code indicating whether this is a failed message. Valid values are as follows: 0 = No failure 1 = Failure	TYPE BINARY 16
45–46	02	DPC-NUM The data processing center (DPC) number identifying the DPC to which the transaction is to be sent.	TYPE BINARY 16
47–48	02	LAST-X-OFST The offset into the DATA field of the transaction data carried in the BSTM structure that is not carried in ITD fields or tokens.	TYPE BINARY 16
49–50	02	LAST-X-LGTH The length of the transaction data carried in the BSTM structure that is not carried in ITD fields or tokens.	TYPE BINARY 16
51–52	02	TKN-OFST The offset into the DATA field of the token data carried in the BSTM structure.	TYPE BINARY 16
53–54	02	TKN-LGTH The length of the token data carried in the BSTM structure.	TYPE BINARY 16
55–56	02	ITD-OFST The offset into the DATA field of the internal transaction data (ITD) carried in the BSTM structure.	TYPE BINARY 16
57–58	02	ITD-LGTH The length of the ITD carried in the BSTM structure.	TYPE BINARY 16

Position	Level	Field Name and Description	Data Type
59–8064 or 59–3697	02	DATA A buffer that contains transaction data, tokens, and ITD. If your system is compiled to support large messages, the size of this field is the sum of the lengths of the largest possible transaction data (4096 bytes), largest possible token buffer (400 bytes), and all ITD fields that can used by the BASE24-telebanking and BASE24-billpay products (3510 bytes). If your system is not compiled to support large messages, the size of this field is the sum of the lengths of the largest possible transaction data (933 bytes), largest possible token buffer (400 bytes), and all ITD fields that can used by the BASE24-telebanking and BASE24-billpay products (2306 bytes).	PIC X(8006) PIC X(3639)

ITD Field Types

There are three types of ITD fields, depending on whether the field is logged to the ITLF. These types are mandatory, optional, and not logged. Mandatory fields are always logged to the ITLF, optional fields are logged to the ITLF only when they contain data, and the remaining fields are never logged to the ITLF.

An ITLF record is arranged as follows:

- Header fields
- Mandatory ITD fields
- Optional ITD fields that contain data

Header fields and mandatory ITD fields are arranged in a fixed format for all ITLF records. These fixed formats are part of the record layouts in the TLFDS and TLFXDS files on the OC13TXN subvolume.

The ITD fields currently used by the BASE24 Remote Banking products are presented in this appendix in alphabetical order by field name. Each field description includes the following information:

- The tag number that is used to arrange optional fields in an ITLF record.
- The type of ITD field for ITLF logging purposes.
- The data name for the ITD field when it appears in an ITLF record.
- A description of the ITD field and its contents.

Note: The ITD contains some optional fields that are reserved for future BASE24 Remote Banking features or for future use by other BASE24 products. Unused optional fields are not described in this appendix. These fields will be added to this appendix when the BASE24 Remote Banking products start to use them.

ITD Tag Numbers

Each ITD field has a unique identification number called a tag number. These tag numbers are part of the ITD field definitions in the ITDDS file on the OC13DDL subvolume. The standard definitions used in the ITD field definitions are provided in the CDEFSGDS file on the OC13DDL subvolume.

The Super Extract process adds a binary 16 (two-character) field containing the tag number to the beginning of each optional ITD field in an ITLF record. Host programs must use these tag numbers to determine which optional ITD fields an ITLF record contains. Optional ITD fields that contain data are arranged in ascending order by tag number. There are no eye-catcher characters or other special characters separating ITD fields.

The ITD field descriptions presented in this appendix are arranged in tag number order. The following table presents the ITD field names for each tag number and indicates which fields are logged to the ITLF. The type codes used in the second column of this table are as follows:

M = Mandatory. Always logged to the ITLF in a fixed format.
N = Not logged. Never logged to the ITLF.
O = Optional. Logged to the ITLF only when the field contains data.

Tag	Type	Internal Transaction Data (ITD) Field Name
1	M	Account 1
2	N	Account 1 number
3	M	Account 2
4	N	Account 2 number
6	M	Acquirer
7	O	Acquirer dependent
8	N	Acquirer identifier
10	N	Acquirer/issuer relationship
11	M	Acquirer object identifier
13	O	Advice 1 dependent
14	O	Advice 1 required
15	O	Advice 2 dependent
16	O	Advice 2 required
21	N	Amount remaining
22	O	Transaction amount—original
23	O	Transaction amount
24	O	Transaction amount—cardholder billing
25	O	Transaction amount—cardholder billing 2
28	O	Approval code
29	N	Approval code length

Tag	Type	Internal Transaction Data (ITD) Field Name
30	M	Authorizer
31	O	Authorizer dependent
33	N	Authorization hold usage
34	M	Authorizer indicator
35	M	Authorizer object identifier
36	N	Authorization time limit
37	N	Account balance
38	O	Backup account information
39	N	Balance option
46	M	Issuer
51	N	Destination
52	M	Discriminant
54	O	Impact 1 object identifier
55	O	Impact 2 object identifier
56	M	Local transaction
57	N	Multiple account selection count
58	N	Multiple account selection display
63	M	Message substate
64	M	Message type identifier
65	O	Original data elements
66	M	Message type identifier—original
67	N	Primary account number
70	N	PIN data
71	M	Capture date
73	M	Processing code
74	M	Product identifier
75	M	Point of service
76	O	PIN verification digits
82	M	Action
85	O	Reason code—exception
86	O	Reason code—negative
87	O	Reason code—message
88	M	Route code
89	N	Routing data
90	M	Route floor
91	M	Route type
92	M	Sequence number
102	M	Terminal identifier
105	M	Entry timestamp
106	O	External issuer time
107	M	Log timestamp
108	N	TMF transaction identifier
111	M	Transaction originator

Tag	Type	Internal Transaction Data (ITD) Field Name
112	M	Transaction responder
114	N	Account 1 qualifier
115	N	Account 2 qualifier
117	O	Function code
118	O	Token
119	N	Token—not logged
120	O	Additional data
121	O	External acquirer time
122	O	Transaction amount—original transaction
123	N	Last transaction allowed count
124	M	Refresh indicator
125	O	Refresh impact
126	N	TMF abort 1 notify flag
127	N	TMF abort 2 notify flag
128	O	PIN change required
129	O	Action date
130	O	Payee
131	O	Payment date
132	O	Recurring transaction data
133	O	Payee description
134	O	Customer reference number
135	O	Source code
138	O	External acquirer dependent data
139	O	External authorization dependent data
140	N	Acquirer dependent—not logged
141	O	Multiple account information
142	O	Message format type ID
143	N	Last (x) area

Account 1

Tag Number: 1

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.BCD.ACCT1

The following fields identify account 1, which is the *from* account involved in a transfer transaction, the first account involved in a dual inquiry, or the single account involved in transactions other than a transfer.

The account number and account type can be entered by the customer or determined from an account qualifier entered by the customer.

Position	Level	Field Name and Description	Data Type
1–18		ITD-ACCT1	
1–16	02	NUM-BCD The following fields contain the account number in binary-coded decimal (BCD) format for account 1.	
1–14	03	NUM The account number for account 1.	PIC X(14)
15–16	03	LGTH The length of the account number for account 1.	TYPE BINARY 16
17–18	02	TYP A code identifying the type of account 1. Default values provided by ACI are as follows: 00 = Default 10, 1A–1F = Savings account 20, 2A–2H = Checking account 30, 3A–3G = Credit account	PIC X(2)

Position	Level	Field Name and Description	Data Type
		38 = Line of credit 58 = Certificate 59 = Retirement account 90 = Interest-bearing checking account 9A = Commercial loan 9B = Installment loan 9C = Mortgage loan Other alphanumeric values are user defined.	

Account 1 Number

Tag Number: 2

ITLF Status: Not logged

The following field contains the account number for account 1, which is the *from* account involved in a transfer transaction, the first account involved in a dual inquiry, user-defined transaction, or the single account involved in transactions other than a transfer. This number is stored in ASCII format.

The account number can be entered by the customer or determined from an account qualifier entered by the customer.

Position	Level	Field Name and Description	Data Type
1–28		ITD-ACCT1-NUM	PIC X(28)

Account 2

Tag Number: 3

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.BCD.ACCT2

The following fields identify account 2, which is the *to* account involved in a transfer transaction, the second account involved in a dual available funds inquiry transaction, or the second account involved in a user-defined transaction.

The account number and account type can be entered by the customer or determined from an account qualifier entered by the customer.

Position	Level	Field Name and Description	Data Type
1–18		ITD-ACCT2	
1–16	02	NUM-BCD The following fields contain the account number in binary-coded decimal (BCD) format for account 2.	
1–14	03	NUM The account number for account 2.	PIC X(14)
15–16	03	LGTH The length of the account number for account 2.	TYPE BINARY 16
17–18	02	TYP A code identifying the type of account 2. Default values provided by ACI are as follows: 00 = Default 10, 1A–1F = Savings account 20, 2A–2H = Checking account 30, 3A–3G = Credit account	PIC X(2)

Position Level Field Name and Description Data Type

38 = Line of credit
58 = Certificate
59 = Retirement account
90 = Interest-bearing checking account
9A = Commercial loan
9B = Installment loan
9C = Mortgage loan

Other alphanumeric values are user defined.

Account 2 Number

Tag Number: 4

ITLF Status: Not logged

The following field contains the account number for account 2, which is the *to* account involved in a transfer transaction, the second account involved in a dual available funds inquiry transaction, or the second account involved in a user-defined transaction. This number is stored in ASCII format.

The account number can be entered by the customer or determined from an account qualifier entered by the customer.

Position	Level	Field Name and Description	Data Type
1–28		ITD-ACCT2-NUM	PIC X(28)

Acquirer

Tag Number: 6

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.ACQ

The following fields contain information about the acquiring institution (i.e., the financial institution that controls the RBSI-compliant device or customer service representative terminal).

The information in these fields is obtained from the external message, ITS Interface Configuration File (IFCF), and Address Table File (ADRT).

Position	Level	Field Name and Description	Data Type
1–22		ITD-ACQ	
1–4	02	FIID The financial institution identifier.	PIC X(4)
5–7	02	INST-CNTRY-CDE A code identifying the country where the institution is located. Country codes are based on the ISO 3166-1-1997 standard, ***Codes for the Representation of Names of Countries.***	PIC X(3)
8–18	02	ID-CDE The routing and transit number or other identification number for the institution. In the United States, this field can contain the routing and transit number of nine characters that should be right-justified and zero filled to the left.	PIC X(11)

Position	Level	Field Name and Description	Data Type
19–22	02	LGNT The logical network associated with the terminal where the transaction was performed.	PIC X(4)

Acquirer Dependent

Tag Number: 7

ITLF Status: Optional

Data Name: TLF.BUS-OLF.ACQ-DPNT

The following fields are user defined and specific to the acquirer object of this transaction.

Position	Level	Field Name and Description	Data Type
1–254		ITD-ACQ-DPNT	
1–4	02	FLD-LGTH The following fields define the length of data contained in the FDATA field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16
5–254	02	FDATA Information specific to the acquirer object of this transaction. The structure and content of this field is defined by the acquirer. This is a variable-length field. The actual length of this field is adjusted to match the length of the data it contains.	PIC X(250)

Acquirer Identifier

Tag Number: 8

ITLF Status: Not logged

The following fields contain information that identifies the process or object that acquired the message.

Position	Level	Field Name and Description	Data Type
1–20		ITD-ACQ-ID	
1–19	02	FLD A value identifying the message acquirer, (e.g., merchant ID, FIID, or interface name).	PIC X(19)
20	02	XFILL (NOT USED)	PIC X(1)

Acquirer/Issuer Relationship

Tag Number: 10

ITLF Status: Not logged

The following fields contain information about the relationship between the transaction acquirer and the account issuer.

The Customer object always sets the relationship flag to on-us.

Position	Level	Field Name and Description	Data Type
1–2		ITD-ACQ-ISS-RELN	
1	02	FLG A code identifying the relationship between the transaction acquirer and the account issuer. Valid values are as follows: F = Foreign. Switch or unknown prefix (not used). N = Not-on-us. Known prefix where acquiring and issuing institutions are different, but in the same logical network (not used). O = On-us. Known prefix where acquiring and issuing institutions are the same.	PIC X(1)
2	02	XFILL (NOT USED)	PIC X(1)

Acquirer Object Identifier

Tag Number: 11

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.BCD.ACQ-OBJ-ID

The following field contains a code that identifies the object that acquired the transaction from an outside entity.

Position	Level	Field Name and Description	Data Type
1–2		ITD-ACQ-OBJ-ID Valid values are as follows: 1004 = Customer Service Interface object 1013 = Internal Message Translation Interface object 1023 = Voice Response Unit Interface object	TYPE BINARY 16

Advice 1 Dependent

Tag Number: 13

ITLF Status: Optional

Data Name: TLF.BUS-OLF.ADVC1-DPNT

The following fields contain additional data for the object that receives an advice for account 1. These fields can be used when the BASE24 product authorizes a transaction and the host that maintains account 1 requires an advice. The object receiving the advice defines the structure and content of these fields.

Position	Level	Field Name and Description	Data Type
1–254		ITD-ADVC1-DPNT	
1–4	02	FLD-LGTH The following fields define the length of data contained in the FDATA field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16
5–254	02	FDATA Information specific to the object that receives an advice for this transaction. This is a variable-length field. The actual length of this field is adjusted to match the length of the data it contains.	PIC X(250)

Advice 1 Required

Tag Number: 14

ITLF Status: Optional

Data Name: TLF.BUS-OLF.ADVC1-REQ

The following fields contain information that is required to send an advice for account 1. These fields can be used when the BASE24 product authorizes a transaction and the host that maintains account 1 requires an advice.

Position	Level	Field Name and Description	Data Type
1–20		ITD-ADVC1-REQ	
1–2	02	OBJ-ID A code defining the type of object. The valid value is 1013, which identifies the Internal Message Translation Interface object.	TYPE BINARY 16
3–18	02	NAM The symbolic name of the primary authorizer.	PIC X(16)
19	02	FLG A code identifying when advices are necessary. Valid values are as follows: A = Send advices for approved transactions only. B = Send advices for approved and denied transactions. D = Send advices for denied transactions only. N = Do not send advices.	PIC X(1)
20	02	XFILL (NOT USED)	PIC X(1)

Advice 2 Dependent

Tag Number: 15

ITLF Status: Optional

Data Name: TLF.BUS-OLF.ADVC2-DPNT

The following fields contain additional data for the object that receives an advice for account 2. These fields can be used when the BASE24 product authorizes a transaction and the host that has account 2 requires an advice. The object receiving the advice defines the structure and content of these fields.

Position	Level	Field Name and Description	Data Type
1–254		ITD-ADVC2-DPNT	
1–4	02	FLD-LGTH The following fields define the length of data contained in the FDATA field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16
5–254	02	FDATA Information specific to the object that receives an advice for this transaction. This is a variable-length field. The actual length of this field is adjusted to match the length of the data it contains.	PIC X(250)

Advice 2 Required

Tag Number: 16

ITLF Status: Optional

Data Name: TLF.BUS-OLF.ADVC2-REQ

The following fields contain information that is required to send an advice for account 2. These fields can be used when the BASE24 product authorizes a transaction and the host that has account 2 requires an advice.

Position	Level	Field Name and Description	Data Type
1–20		ITD-ADVC2-REQ	
1–2	02	OBJ-ID A code defining the type of object. The valid value is 1013, which identifies the Internal Message Translation Interface object.	TYPE BINARY 16
3–18	02	NAM The symbolic name for the primary authorizer.	PIC X(16)
19	02	FLG A code identifying when advices are necessary. Valid values are as follows: A = Send advices for approved transactions only. B = Send advices for approved and denied transactions. D = Send advices for denied transactions only. N = Do not send advices.	PIC X(1)
20	02	XFILL (NOT USED)	PIC X(1)

Amount Remaining

Tag Number: 21

ITLF Status: Not logged

The following field contains the maximum amount available for the customer to transfer when insufficient funds exist in the account or when the original amount requested for a transfer would cause the BASE24-telebanking transfer limits in the Positive Balance File (PBF) to be exceeded.

Position	Level	Field Name and Description	Data Type
1–8		ITD-AMT-REMAIN	TYPE BINARY 64

Transaction Amount—Original

Tag Number: 22

ITLF Status: Optional

Data Name: TLF.BUS-OLF.AMT-TXN-ORIG

The following fields contain the original amount of the transaction, expressed in various currencies. This information may be present in authorization, financial, or reversal transactions.

There are six amount fields, one of which is not used. However, the fields that are used contain the same information because the BASE24-telebanking product does not support multiple currencies.

Position	Level	Field Name and Description	Data Type
1–48		ITD-AMT-TXN-ORIG	
1–8	02	ACQ The original amount of the transaction, expressed in the currency of the financial institution that acquired the transaction.	TYPE BINARY 64
9–16	02	CHB The original amount of the transaction, expressed in the currency of the customer’s account.	TYPE BINARY 64
17–24	02	CRD-ACCPT The original amount of the transaction, expressed in the currency of the card acceptor that acquired the transaction. The BASE24-telebanking product does not use this field.	TYPE BINARY 64

Position	**Level**	**Field Name and Description**	**Data Type**
25–32	02	ISS The original amount of the transaction, expressed in the currency of the financial institution that authorized the transaction.	TYPE BINARY 64
33–40	02	LGNT The original amount of the transaction, expressed in the currency of the logical network that acquired the transaction.	TYPE BINARY 64
41–48	02	TERM The original amount of the transaction, expressed in the currency of the remote banking endpoint or customer service representative terminal where the transaction was initiated.	TYPE BINARY 64

Transaction Amount

Tag Number: 23

ITLF Status: Optional

Data Name: TLF.BUS-OLF.AMT-TXN

In a request, the following fields contain the amounts requested by the customer, expressed in various currencies. In a response, these fields contain the amounts approved by the issuer (i.e., unchanged from the request if approved, zero if denied). In a reversal, these fields contain the amount reversed. Currency codes are based on the ISO 4217-1995 standard, ***Codes for the Representation of Currencies and Funds.***

There are four sets of amount and currency code fields. However, all four sets of fields contain the same information because the BASE24-telebanking product does not support multiple currencies.

Position	Level	Field Name and Description	Data Type
1–40		ITD-AMT-TXN	
1–10	02	ACQ The following fields contain the amount of funds requested by the customer, expressed in the currency of the financial institution that acquired the transaction. The acquiring institution owns the remote banking endpoint or customer service representative terminal.	
1–8	03	AMT The amount of funds requested by the customer.	TYPE BINARY 64
9–10	03	CRNCY-CDE A code identifying the currency.	TYPE BINARY 16

Position	**Level**	**Field Name and Description**	**Data Type**
11–20	02	ISS The following fields contain the amount of funds requested by the customer, expressed in the currency of the financial institution that authorized the transaction.	
11–18	03	AMT The amount of funds requested by the customer.	TYPE BINARY 64
19–20	03	CRNCY-CDE A code identifying the currency.	TYPE BINARY 16
21–30	02	LGNT The following fields contain the amount of funds requested by the customer, expressed in the currency of the logical network that acquired the transaction.	
21–28	03	AMT The amount of funds requested by the customer.	TYPE BINARY 64
29–30	03	CRNCY-CDE A code identifying the currency.	TYPE BINARY 16
31–40	02	TXN The following fields contain the amount of funds requested by the customer, expressed in the currency of the remote banking endpoint or customer service representative terminal where the transaction was initiated.	

Position	Level	Field Name and Description	Data Type
31–38	03	AMT The amount of funds requested by the customer.	TYPE BINARY 64
39–40	03	CRNCY-CDE A code identifying the currency.	TYPE BINARY 16

Transaction Amount—Cardholder Billing

Tag Number:	24
ITLF Status:	Optional
Data Name:	TLF.BUS-OLF.AMT-TXN-CHB

In a request, the following fields contain the amounts requested by the customer, expressed in the currency of the customer's account. In a response, these fields contain the amounts approved by the issuer (i.e., unchanged from the request if approved, zero if denied). In a reversal, these fields contain the amount reversed. Currency codes are based on the ISO 4217-1995 standard, ***Codes for the Representation of Currencies and Funds.***

Position	Level	Field Name and Description	Data Type
1–10		ITD-AMT-TXN-CHB	
1–8	02	AMT The amount of funds requested by the customer.	TYPE BINARY 64
9–10	02	CRNCY-CDE A code identifying the currency.	TYPE BINARY 16

Transaction Amount—Cardholder Billing 2

Tag Number: 25

ITLF Status: Optional

Data Name: TLF.BUS-OLF.AMT-TXN-CHB2

These fields contain information for transfer transactions only. In a request, the following fields contain the amounts requested by the customer, expressed in the currency of the customer's *to* account. In a response, these fields contain the amounts approved by the issuer (i.e., unchanged from the request if approved, zero if denied). In a reversal, these fields contain the amount reversed. Currency codes are based on the ISO 4217-1995 standard, ***Codes for the Representation of Currencies and Funds.***

Position	Level	Field Name and Description	Data Type
1–10		ITD-AMT-TXN-CHB2	
1–8	02	AMT The amount of funds requested by the customer.	TYPE BINARY 64
9–10	02	CRNCY-CDE A code identifying the currency.	TYPE BINARY 16

Approval Code

Tag Number: 28

ITLF Status: Optional

Data Name: TLF.BUS-OLF.APPRV

The following fields contain the approval code assigned to the transaction by the transaction authorizer.

Position	Level	Field Name and Description	Data Type
1–10		ITD-APPRV	
1–9	02	CDE The approval code assigned to the transaction by the transaction authorizer.	PIC X(9)
10	02	XFILL (NOT USED)	PIC X(1)

Approval Code Length

Tag Number: 29

ITLF Status: Not logged

The following field contains the maximum length of the approval code supported by the transaction acquirer.

Position	Level	Field Name and Description	Data Type
1–2		ITD-APPRV-CDE-LGTH The maximum length of the approval code supported by the transaction acquirer. Valid values are 0 through 9.	TYPE BINARY 16

Authorizer

Tag Number: 30

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.AUTH

The following fields contain information about the transaction authorizer.

Position	Level	Field Name and Description	Data Type
1–22		ITD-AUTH	
1–16	02	NAM The symbolic name of the interface instance that identifies the Internal Message Translation File (IMTF) record used to route the message.	PIC X(16)
17–22	02	STAN The system trace audit number (STAN) assigned by the issuer interface (e.g., the ISO Host Interface process) to uniquely identify the message.	PIC 9(6)

Authorizer Dependent

Tag Number: 31

ITLF Status: Optional

Data Name: TLF.BUS-OLF.AUTH-DPNT

A variable-length area containing fields specific to the authorizer object of the transaction. The authorizer defines the structure and content of these fields.

Position	Level	Field Name and Description	Data Type
1–254		ITD-AUTH-DPNT	
1–4	02	FLD-LGTH The following fields define the length of data contained in the FDATA field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16
5–254	02	FDATA Information specific to the authorizer object of the transaction. This is a variable-length field. The actual length of this field is adjusted to match the length of the data it contains.	PIC X(250)

Authorization Hold Usage

Tag Number: 33

ITLF Status: Not logged

The following fields contain values that indicate whether preauthorization holds are supported and whether a card sequence number is used to retrieve authorization hold information.

Position	Level	Field Name and Description	Data Type
1–2		ITD-AUTH-HLD-USG	
1	02	ALWD A flag indicating whether preauthorization holds are supported. This field is hard coded with a value of Y, indicating that preauthorization holds are supported. The BASE24-telebanking product uses preauthorization hold information contained in the Positive Balance File (PBF).	PIC X(1)
2	02	CRD-SEQ-NUM-FLG A flag indicating whether card sequence numbers are used to obtain preauthorization hold information. This field is hard coded with a value of 0, indicating that card sequence numbers are not used. The BASE24-telebanking product does not use cards.	PIC X(1)

Authorizer Indicator

Tag Number: 34

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.AUTH-IND

The following fields contain information identifying the authorizer that authorized the transaction.

Position	Level	Field Name and Description	Data Type
1–2		ITD-AUTH-IND	
1	02	FLG A code indicating which authorizer authorized the transaction. Valid values are as follows: P = Primary 1 = Alternate 1 2 = Alternate 2 D = Default or denial authorization	PIC X(1)
2	02	XFILL (NOT USED)	PIC X(1)

Authorizer Object Identifier

Tag Number: 35

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.BCD.AUTH-OBJ-ID

The following field contains a code defining the type of object that authorized the transaction.

Position	Level	Field Name and Description	Data Type
1–2		ITD-AUTH-OBJ-ID	TYPE BINARY 16

A code that defines the type of object that authorized the transaction. Valid values are as follows:

1006 = Denial Authorization object (part of the Authorization Combined Utility object)

1013 = Internal Message Translation Interface object

1017 = Positive Customer Authorization method performed by the Authorization Combined Positive Customer object

1018 = Positive Customer with Accounts Authorization method performed by the Authorization Combined Positive Customer object

1019 = Positive Customer with Balances/History Authorization method performed by the Authorization Combined Positive Customer object

Authorization Time Limit

Tag Number: 36

ITLF Status: Not logged

The following field contains the Julian timestamp of the time by which a response must be returned to the transaction acquirer.

Position	Level	Field Name and Description	Data Type
1–8		ITD-AUTH-TIM-LMT	TYPE BINARY 64

Account Balance

Tag Number: 37

ITLF Status: Not logged

The following fields contain balance information for account 1 and account 2, including the new balances resulting from a transfer transaction and current balances for inquiry transactions.

Position	Level	Field Name and Description	Data Type
1–74		ITD-BAL	
1–2	02	CNT The number of balances contained in the INFO field. Valid values are 1 through 4.	TYPE BINARY 16
3–74	02	INFO	OCCURS 4 TIMES

Each occurrence of the following fields contains one balance for an account involved in the transaction. For example, a transfer of funds between two noncredit accounts has four occurrences. Each account has two balances that must be updated, as shown in the following table.

Occurrence	Account	Amount Type
1	1	Ledger balance
2	1	Available balance
3	2	Ledger balance
4	2	Available balance

Each occurrence is 18 positions in length.

Note: When a transaction involves two accounts, the first two occurrences of this field can contain balance information for account 1 or account 2. If balance information is available for both accounts, the first two occurrences contain information

Position Level Field Name and Description Data Type

for account 1. However, if balance information is available for account 2 only, the first two occurrences contain that information.

03 AMT-TYP PIC X(2)

A code indicating the type of balance information returned. Valid values are as follows:

01 = Ledger balance of account 1 (noncredit account)
02 = Available balance of account 1 (noncredit account)
03 = Credit balance (credit account)
04 = Amount due (not currently supported)
05 = Available credit balance (credit account)
16 = Ledger balance of account 2 (noncredit account)
17 = Available balance of account 2 (noncredit account)

03 AMT TYPE BINARY 64

The balance for the account.

03 CRNCY-CDE TYPE BINARY 16

A code identifying the currency used to express the account balance. Currency codes are based on the ISO 4217-1995 standard, ***Codes for the Representation of Currencies and Funds.***

03 DAT

The date of the account balance.

04 YY PIC X(2)
04 MM PIC X(2)
04 DD PIC X(2)

Backup Account Information

Tag Number: 38

ITLF Status: Optional

Data Name: TLF.BUS-OLF.BACKUP

If a backup account was used because sufficient funds were not available in account 1, the following fields contain information concerning the backup account portion of the transaction.

Position	Level	Field Name and Description	Data Type
1–64		ITD-BACKUP	
1–18	02	ACCT The following fields identify the backup account.	
1–16	03	NUM-BCD The following fields contain the backup account number in binary-coded decimal (BCD) format.	
1–14	04	NUM The account number.	PIC X(14)
15–16	04	LGTH The length of the account number.	TYPE BINARY 16
17–18	03	TYP A code identifying the type of backup account. Valid values provided by ACI are as follows: 00 = Default 10, 1A–1F = Savings account 20, 2A–2H = Checking account	PIC X(2)

Position	Level	Field Name and Description	Data Type
		30, 3A–3G = Credit account 38 = Line of credit 58 = Certificate 59 = Retirement account 90 = Interest-bearing checking account 9A = Commercial loan 9B = Installment loan 9C = Mortgage loan Other alphanumeric values are user defined.	
19–46	02	ACCT-NUM The backup account number in ASCII format.	PIC X(28)
47–54	02	AMT-XFER The amount of funds transferred from the backup account to account 1, expressed in the currency of the backup account.	TYPE BINARY 64
55–62	02	AMT-XFER-CHB The amount of funds transferred from the backup account to account 1, expressed in the currency of account 1.	TYPE BINARY 64
63–64	02	CRNCY-CDE The currency code of the backup account.	TYPE BINARY 16

Balance Option

Tag Number: 39

ITLF Status: Not logged

The following fields contain codes that indicate how balance information is presented to customers.

Position	Level	Field Name and Description	Data Type
1–2		ITD-BAL-OPT	
1	02	INFO A code that identifies which balances, if any, the institution prefers to have reported to the customer upon completion of a transaction. Valid values are as follows: 0 = No balance information. 1 = Ledger balance only. 2 = Available balance only. 3 = Ledger and available balances, ledger balance preferred. 4 = Ledger and available balances, available balance preferred. This value is obtained from the Institution Definition File (IDF).	PIC X(1)
2	02	PRSTN The BASE24-telebanking product does not use this field.	PIC X(1)

Issuer

Tag Number: 46

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.ISS

The following fields contain information pertaining to the account issuer.

Position	Level	Field Name and Description	Data Type
1–20		ITD-ISS	
1–4	02	FIID The FIID of the financial institution that issued the account.	PIC X(4)
5–15	02	ID-CDE A code identifying the financial institution that issued the account. In the United States, this field can contain the routing and transit number.	PIC X(11)
16–19	02	LGNT The logical network associated with the financial institution that issued the account.	PIC X(4)
20	02	DISCRD-NON-FNCL-RVSL A code indicating whether the transaction authorizer is to discard reversals of nonfinancial transactions other than PIN change transactions.	PIC X(1)

Position Level Field Name and Description Data Type

Valid values are as follows:

Y = Yes, discard reversals of nonfinancial transactions other than PIN change transactions. Do not log them or forward them to the issuer.
N = No, do not discard reversals of nonfinancial transactions. Log them and forward them to the issuer.

This value is obtained from the Institution Definition File (IDF) or the Host Configuration File (HCF).

Destination

Tag Number: 51

ITLF Status: Not logged

The following field contains the name or object ID that identifies the interface instance to which a request is routed.

Position	Level	Field Name and Description	Data Type
1–16		ITD-DEST	PIC X(16)

Discriminant

Tag Number: 52

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ALT-KEY1.DISCRIM

The following field contains a value used by interface objects for duplicate transaction checking.

Position	Level	Field Name and Description	Data Type
1–2		ITD-DISCRIM	TYPE BINARY 16

Impact 1 Object Identifier

Tag Number: 54

ITLF Status: Optional

Data Name: TLF.BUS-OLF.IMPACT1-OBJ-ID

The following field contains a code identifying the authorization method used to update the BASE24 database for the first account in a transaction when the BASE24 product is not the authorizer. The BASE24 product could be preauthorizing the transaction, standing in for the host, or receiving a response from an external issuer.

Position	Level	Field Name and Description	Data Type
1–2		ITD-IMPACT1-OBJ-ID A code identifying the authorization method used to update the BASE24 database for the first account in a transaction when the BASE24 product is not the authorizer. Valid values are as follows: 1017 = Positive Customer Authorization method performed by the Authorization Combined Positive Customer object 1018 = Positive Customer with Accounts Authorization method performed by the Authorization Combined Positive Customer object 1019 = Positive Customer with Balances/History Authorization method performed by the Authorization Combined Positive Customer object	TYPE BINARY 16

Impact 2 Object Identifier

Tag Number: 55

ITLF Status: Optional

Data Name: TLF.BUS-OLF.IMPACT2-OBJ-ID

The following field contains a code identifying the authorization method used to update the BASE24 database for the second account in the transaction when the BASE24 product is not the authorizer. The BASE24 product could be preauthorizing the transaction, standing in for the host, or receiving a response from an external issuer.

Position	Level	Field Name and Description	Data Type
1–2		ITD-IMPACT2-OBJ-ID	TYPE BINARY 16

A code identifying the authorization method used to update the BASE24 database for the second account in a transaction when the BASE24 product is not the authorizer. Valid values are as follows:

1017 = Positive Customer Authorization method performed by the Authorization Combined Positive Customer object

1018 = Positive Customer with Accounts Authorization method performed by the Authorization Combined Positive Customer object

1019 = Positive Customer with Balances/History Authorization method performed by the Authorization Combined Positive Customer object

Local Transaction

Tag Number: 56

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.LOCAL-TXN

The following fields contain the date and time that the original transaction was performed.

Position	Level	Field Name and Description	Data Type
1–12		ITD-LOCAL-TXN	
1–6	02	DAT The calendar date on which the transaction was performed.	
1–2	03	YY	PIC X(2)
3–4	03	MM	PIC X(2)
5–6	03	DD	PIC X(2)
7–12	02	TIM The time, in hhmmss format, when the transaction was performed.	PIC X(6)

Multiple Account Selection Count

Tag Number: 57

ITLF Status: Not logged

The following field contains the maximum number of accounts that can be returned to the RBSI-compliant device or customer service representative terminal in a single response.

Position	Level	Field Name and Description	Data Type
1–2		ITD-MAS-ALWD-CNT The maximum number of accounts that can be returned to the RBSI-compliant device or customer service representative terminal in a single response. Valid values are as follows: 0 = Multiple account selection is not supported. 2–15 = The maximum number of accounts that can be presented. This value is obtained from the IFCF.	TYPE BINARY 16

Multiple Account Selection Display

Tag Number: 58

ITLF Status: Not logged

The following field contains a code that identifies which information the institution prefers to have displayed on customer service representative (CSR) terminal screens when a multiple account selection situation is encountered.

Position	Level	Field Name and Description	Data Type
1–2		ITD-MAS-DSPY	
1	02	OPT A code that identifies which information the institution prefers to have displayed when a multiple account selection situation is encountered. Valid values are as follows: 1 = Display account descriptions only 2 = Display account numbers only 3 = Display account descriptions and numbers, account descriptions preferred 4 = Display account descriptions and numbers, account numbers preferred This value is obtained from the Institution Definition File (IDF).	PIC X(1)
2	02	XFILL (NOT USED)	PIC X(1)

Message Substate

Tag Number: 63

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.BCD.MSG-SUBSTATE

The following field contains a code used by interface objects to identify message processing results.

Position	Level	Field Name and Description	Data Type
1–2		ITD-MSG-SUBSTATE	TYPE BINARY 16
		A code used by interface objects to identify message processing results. Valid values are as follows: 0 = Normal 1 = Duplicate transaction 2 = Not sent to authorizer 3 = Timeout 4 = System error 5 = Format error 6 = Failed preauthorization 7 = Reversal not found 8 = Responder down 9 = Advice 1 not sent 10 = Advice 2 not sent	

Message Type Identifier

Tag Number: 64

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.MTI

The following field contains the type of message logged to the ITLF.

Position	Level	Field Name and Description	Data Type
1–4		ITD-MTI	PIC X(4)

The type of message logged to the ITLF. Valid values are as follows:

1100 = Authorization request to an Integrated Authorization Server process
1101 = Authorization request repeat to an Integrated Authorization Server process
1110 = Authorization request response from an Integrated Authorization Server process
1120 = Authorization advice from an Integrated Authorization Server process
1121 = Authorization advice repeat from an Integrated Authorization Server process
1200 = Financial request to an Integrated Authorization Server process
1201 = Financial request repeat to an Integrated Authorization Server process
1210 = Financial request response from an Integrated Authorization Server process
1220 = Financial advice from an Integrated Authorization Server process
1221 = Financial advice repeat from an Integrated Authorization Server process
1420 = Reversal advice to an Integrated Authorization Server process
1421 = Reversal advice repeat to an Integrated Authorization Server process
1430 = Reversal advice response from an Integrated Authorization Server process

Original Data Elements

Tag Number: 65

ITLF Status: Optional

Data Name: TLF.BUS-OLF.ORIG-DATA

The following fields contain the original transaction information for a reversal transaction. Otherwise, they are blank.

Position	**Level**	**Field Name and Description**	**Data Type**
1–34		ITD-ORIG-DATA	
1–4	02	MTI The type of message that was reversed. Valid values are as follows: 1110 = Authorization request response from an Integrated Authorization Server process 1210 = Financial request response from an Integrated Authorization Server process	PIC X(4)
5–10	02	STAN The sequence number originally assigned by the message initiator to uniquely identify the message.	PIC 9(6)
11–22	02	LOCAL-TXN-DAT-TIM The date and time, in YYMMDDhhmmss format, the original transaction was processed by the BASE24 product.	PIC 9(12)
23–33	02	ACQ-INST-ID-CDE A code identifying the acquiring institution. In the United States, this is the routing and transit number.	PIC X(11)
34	02	XFILL (NOT USED)	PIC X(1)

Message Type Identifier—Original

Tag Number: 66

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.MTI-ORIG

The following field contains the type of message originally received by the BASE24 product.

Position	Level	Field Name and Description	Data Type
1–4		ITD-MTI-ORIG The type of message originally received by the BASE24 product. Valid values are as follows: 1100 = Authorization request to an Integrated Authorization Server process 1101 = Authorization request repeat to an Integrated Authorization Server process 1200 = Financial request to an Integrated Authorization Server process 1201 = Financial request repeat to an Integrated Authorization Server process	PIC X(4)

Primary Account Number

Tag Number: 67

ITLF Status: Not logged

The following fields contain customer ID information for the customer who initiated the transaction. This information is expressed in ASCII format.

Position	Level	Field Name and Description	Data Type
1–28		ITD-PAN	
1–28	02	NUM The customer identification number.	PIC X(28)

PIN Data

Tag Number: 70

ITLF Status: Not logged

The following fields contain the current PIN, new PIN, and PIN encryption information for the transaction.

Position	Level	Field Name and Description	Data Type
1–132		ITD-PIN-DATA	
1–16	02	PIN The current PIN received from the RBSI-compliant device or customer service representative terminal.	PIC X(16)
17–32	02	KPT The current PIN transport key.	PIC X(16)
33–48	02	KPT-ALT The previous PIN transport key.	PIC X(16)
49–64	02	KDT The BASE24-telebanking product does not use this field.	PIC X(16)
65–80	02	KDT-ALT The BASE24-telebanking product does not use this field.	PIC X(16)

Position	Level	Field Name and Description	Data Type
81	02	PBT A code identifying the PIN block format of the PIN in external messages. Valid values are as follows: 0 = Clear PINs 1 = ANSI (PIN/PAN) PIN block 3 = PIN/PAD PIN block This value is obtained from the Key File (KEYF).	PIC X(1)
82	02	PAD-CHAR The pad character used in the formation of the PIN/PAD PIN block placed in the external message.	PIC X(1)
83	02	ENCRYPT-TYP A code identifying the type of PIN encryption used by the BASE24 product. Valid values are as follows: H = Hardware PIN encryption N = None (clear PINs) S = Software PIN encryption This value is obtained from the Key File (KEYF).	PIC X(1)
84	02	ENCRYPT-LVL A code identifying the level of encryption used for PINs moving between an external entity and the BASE24 product. The valid value is 1 (single).	PIC X(1)
85–100	02	XOR-VAL The BASE24-telebanking product does not use this field.	PIC X(16)

Position	Level	Field Name and Description	Data Type
101–116	02	PIN-NEW-1 The first entry of the new PIN for a PIN change transaction.	PIC X(16)
117–132	02	PIN-NEW-2 The second entry of the new PIN for a PIN change transaction.	PIC X(16)

Capture Date

Tag Number: 71

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.CAPTR-DAT

The following fields contain the date the transaction was processed by the BASE24 product.

Position	Level	Field Name and Description	Data Type
1–6		ITD-CAPTR-DAT	
1–2	02	YY	PIC X(2)
3–4	02	MM	PIC X(2)
5–6	02	DD	PIC X(2)

Processing Code

Tag Number: 73

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.PROC-CDE

The following fields contain the processing code for the transaction. The processing code identifies the type of transaction performed and the type of accounts involved.

Position	Level	Field Name and Description	Data Type
1–6		ITD-PROC-CDE	
1–2	02	TXN-CDE	PIC X(2)

A code identifying the type of transaction performed. Default values provided by ACI are shown in the following table. In this table, *count* refers to the number of transactions that can be returned in one message. Other alphanumeric values are user defined.

30 = Available funds inquiry
3A = Check clearance inquiry
3B = Last count of debit/credit transactions inquiry
3C = Last count of account by source transactions inquiry
3D = Last count of check transactions inquiry
3E = Last count of debit transactions inquiry
3F = Last count of credit transactions inquiry
3G = Last count of account transfer transactions inquiry
3H = Customer vendor list inquiry
3J = Scheduled payments list inquiry
3K = Scheduled transfers list inquiry
3L = History inquiry – payments and transfers
3M = Scheduled transactions list inquiry
3N = Customer account list inquiry
3P = Multiple account balance inquiry
3Q = Statement closing download
3R = Statement download
40 = Transfer – immediate

Position Level Field Name and Description Data Type

4A = Transfer – future
4B = Transfer – recurring
50 = Payment – immediate
5A = Payment – future
5B = Payment – recurring
90 = PIN change
91 = PIN verify
9A = Schedule payment – immediate
9B = Schedule payment – future
9C = Schedule payment – recurring
9D = Schedule transfer – future
9E = Schedule transfer – recurring
9F = Scheduled payment delete
9G = Scheduled transfer delete
9H = Scheduled payment update
9J = Scheduled transfer update
9K = Customer add
9L = Customer inquiry
9M = Customer update
9N = Customer vendor add
9P = Customer vendor deactivate
9Q = Customer vendor update
9R = Master vendor list inquiry
9S = Master vendor add
9T = Customer vendor request mailing
9V = Loan application
9W = Miscellaneous request
9X = History inquiry – all transactions
A0 = Check stop payment add
B0 = Customer account list inquiry (Internal use only)
B1 = Bank list inquiry
B2 = Bank branch list inquiry
B3 = Cancel immediate payment (Internal use only)
B4 = Miscellaneous requests inquiry
B5 = Customer account add
B6 = Customer account activate
B7 = Customer account deactivate
B8 = Customer account delete
B9 = Customer account update
BB = Customer activate
BC = Customer deactivate
BD = Customer vendor activate

Position	Level	Field Name and Description	Data Type
		BE = Customer vendor mask verify inquiry BF = Customer verify BG = Customer select	
3–4	02	ACCT1-TYP A code identifying the account type for account 1. Account 1 is the *from* account in a transfer transaction, the first account in a dual balance inquiry or user-defined transaction, or the only account involved in other transactions. Default values provided by ACI are as follows: 00 = None 10 = Savings account 20 = Checking account 30 = Credit account 38 = Line of credit 58 = Certificate 59 = Retirement account 90 = Interest-bearing checking account 9A = Commercial loan 9B = Installment loan 9C = Mortgage loan Other alphanumeric values are user defined.	PIC X(2)
5–6	02	ACCT2-TYP A code identifying the account type for account 2. Account 2 is the *to* account in a transfer transaction or the second account in a dual available funds inquiry or user-defined transaction. Refer to the ACCT1-TYP field description for a list of valid values.	PIC X(2)

Product Identifier

Tag Number: 74

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.BCD.PROD-ID

The following field contains a code that identifies the BASE24 product involved in the transaction.

Position	Level	Field Name and Description	Data Type
1–2		ITD-PROD-ID A code that identifies the BASE24 product involved in the transaction. A value of 14 identifies the BASE24-telebanking product.	BINARY 16

Point of Service

Tag Number: 75

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.PT-SVC

The following fields contain a series of codes intended to identify terminal capability, terminal environment, and presentation of security data. These fields are used to indicate specific conditions present at the time a transaction occurs. Valid values are defined in the ISO 8583-1993 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specification.***

Position	Level	Field Name and Description	Data Type
1–14		ITD-PT-SVC	
1	02	CRD-DATA-INPUT-CAP A code indicating the primary means of getting the information on the card into the terminal. The BASE24-telebanking product does not use this field.	PIC X(1)
2	02	CRDHLDR-AUTH-CAP A code indicating the primary means of verifying the cardholder at this terminal. The BASE24-telebanking product does not use this field.	PIC X(1)
3	02	CRD-CAPTR-CAP A code indicating whether the terminal has the ability to capture a card. The BASE24-telebanking product does not use this field.	PIC X(1)
4	02	OPER-ENVIRON A code indicating whether the terminal was attended by the transaction acceptor and where the terminal is located. The BASE24-telebanking product uses this code to indicate	PIC X(1)

Position	Level	Field Name and Description	Data Type
		whether the transaction is initiated through a RBSI-compliant device or a customer service representative terminal. Valid values are as follows: 0 = No terminal was used. 1 = Attended terminal located on the premises of the transaction acceptor. For the BASE24-telebanking product, this code identifies a customer service representative terminal. 2 = Unattended terminal located on the premises of the transaction acceptor. 3 = Attended terminal located off the premises of the transaction acceptor. 4 = Unattended terminal located off the premises of the transaction acceptor. 5 = Unattended terminal located on the premises of the accountholder. For the BASE24-telebanking product, this code identifies an RBSI-compliant device. 6–7 = Reserved for ISO use. 8 = Reserved for national use. 9 = Reserved for private use. A–I = Reserved for ISO use. J–R = Reserved for national use. S–Z = Reserved for private use.	
5	02	CRDHLDR-PRSN	PIC X(1)
		A code indicating whether the cardholder was present at the point of service and, if not, why not. The BASE24-telebanking product does not use this field.	
6	02	CRD-PRSN	PIC X(1)
		A code indicating whether the card was present at the point of service. The BASE24-telebanking product does not use this field.	

Position	Level	Field Name and Description	Data Type
7	02	CRD-DATA-INPUT-MDE A code indicating the method used to input the information from the card to the terminal. The BASE24-telebanking product passes this value to the host.	PIC X(1)
8	02	CRDHLDR-AUTH-METHOD A code indicating the method used for verifying the identity of the cardholder. The BASE24-telebanking product does not use this field.	PIC X(1)
9	02	CRDHLDR-AUTH-ENTITY A code indicating the entity verifying the identity of the cardholder. The BASE24-telebanking product does not use this field.	PIC X(1)
10	02	CRD-DATA-OUTPUT-CAP A code indicating the ability of the terminal to update the card. The BASE24-telebanking product does not use this field.	PIC X(1)
11	02	TERM-OUTPUT-CAP A code indicating the ability of the terminal to print or display messages. The BASE24-telebanking product does not use this field.	PIC X(1)
12	02	PIN-CAPTR-CAP A code indicating the length of PIN that the acquirer endpoint is capable of capturing and whether PIN verification is necessary. The BASE24-telebanking product uses the values C and S in this field during PIN verification. Valid values are as follows: 0 = No PIN capture capability exists. 1 = The PIN capture capability for device is unknown. 2–3 = Reserved for ISO use. 4 = The terminal can capture a four-character PIN.	PIC X(1)

Position	Level	Field Name and Description	Data Type
		5 = The terminal can capture a five-character PIN. 6 = The terminal can capture a six-character PIN. 7 = The terminal can capture a seven-character PIN. 8 = The terminal can capture an eight-character PIN. 9 = The terminal can capture a nine-character PIN. A = The terminal can capture a 10-character PIN. B = The terminal can capture an 11-character PIN. C = The terminal can capture a 12-character PIN. For BASE24-telebanking transactions, this value means PIN verification has not been performed. D–I = Reserved for ISO use. J–R = Reserved for national use. S = Reserved for private use. For BASE24-telebanking transactions, this value means PIN verification has already been performed. T–Z = Reserved for private use.	
13	02	PIN-PRSN-FLG A code indicating whether the customer entered a PIN at the point of service device. The BASE24-telebanking product uses the following values in this field: Y = Yes, the customer entered a PIN. N = No, the customer did not enter a PIN.	PIC X(1)
14	02	XFILL (NOT USED)	PIC X(1)

PIN Verification Digits

Tag Number: 76

ITLF Status: Optional

Data Name: TLF.BUS-OLF.PVD

The following fields contain the old and new PIN verification digits (PVD) for a PIN change transaction.

Position	Level	Field Name and Description	Data Type
1–32		ITD-PVD	
1–16	02	NEW The new PIN verification digits (PVD) for a PIN change transaction.	PIC X(16)
17–32	02	OLD The old PIN verification digits (PVD) for a PIN change transaction.	PIC X(16)

Action

Tag Number: 82

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.ACT

The following fields contain information about the action taken or to be taken and the account to which the information applies.

Position	Level	Field Name and Description	Data Type
1–4		ITD-ACT	
1–3	02	CDE A code identifying the action taken or to be taken and the reason for the action. Action codes are described in appendix B. **Note:** Procedural BASE24 products call this code the response code.	PIC X(3)
4	02	IND A code indicating the account to which an action code applies. This code is used for action code 114 (no account of type requested) to identify the account type. Valid values are as follows: 1 = Account 1 (single account or *from* account) 2 = Account 2 (second account or *to* account) B = Backup account	PIC X(1)

Reason Code—Exception

Tag Number: 85

ITLF Status: Optional

Data Name: TLF.BUS-OLF.RSN-CDE-EXCPT

If an exception condition exists indicating the transaction should not have been approved, or that does not allow the transaction to be fully posted to the database, information in the following fields indicates the problem within the BASE24 product.

The INFO field can occur up to five times. The actual length of the Reason Code—Exception field is based on the number of times the INFO field occurs. If more than five exception conditions exist, these fields contain the first five that occur.

Position	Level	Field Name and Description	Data Type
1–36		ITD-RSN-CDE-EXCPT	
1–4	02	FLD-LGTH Defines the length of data contained in the INFO field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16
5–6	02	CNT The number of INFO fields that contain data.	TYPE BINARY 16

Position	Level	Field Name and Description	Data Type
7–36	02	INFO The action code, posting flag, object identifier, and record type associated with an error. Each occurrence is 8 positions in length.	OCCURS 5 TIMES
	03	CDE The action code that would have been returned as a result of the transaction posting attempt. Refer to appendix B for valid values.	PIC X(3)
	03	POST-FLG A code identifying the results of the transaction posting attempt. Valid values are as follows: N = The transaction was not posted to the BASE24 database. P = The transaction was partially posted to the BASE24 database. Y = The transaction was posted to the BASE24 database or posting was not applicable (i.e., the code is set for a condition that does not involve updating the database).	PIC X(1)
	03	OBJ-ID A code defining the type of object that encountered the error. Valid values are as follows: 1001 = Account object 1005 = Customer object 1006 = Denial Authorization object (part of the Authorization Combined Utility object) 1013 = Internal Message Translation Interface object 1017 = Positive Customer Authorization method performed by the Authorization Combined Positive Customer object	TYPE BINARY 16

Position	Level	Field Name and Description	Data Type

1018 = Positive Customer with Accounts Authorization method performed by the Authorization Combined Positive Customer object

1019 = Positive Customer with Balances/History Authorization method performed by the Authorization Combined Positive Customer object

1020 = Router object

1025 = Customer Access object

	03	REC-TYP	PIC X(1)

A code identifying the account being updated when the posting error occurred. Valid values are as follows:

1 = Account 1 (single account or *from* account)
2 = Account 2 (*to* account)
B = Backup account

	03	XFILL (NOT USED)	PIC X(1)

Reason Code—Negative

Tag Number: 86

ITLF Status: Optional

Data Name: TLF.BUS-OLF.RSN-CDE-NEG

The following field contains a code that identifies the reason this customer could not perform this transaction.

Position	Level	Field Name and Description	Data Type
1–2		ITD-RSN-CDE-NEG A code identifying the reason this customer could not perform this transaction. Valid values are as follows: 00 = Customer ID has a verify status of issued but inactive 02 = Customer ID has a verify status of closed This value is obtained from the Customer Table (CSTT).	PIC X(2)

Reason Code—Message

Tag Number: 87

ITLF Status: Optional

Data Name: TLF.BUS-OLF.RSN-CDE-MSG

The following field contains a code that provides the receiver of a request, advice, or notification message with the reason for that message. For original authorizations and financial transactions, it identifies why the type of message was sent (i.e., why an advice instead of a request). If the transaction is a reversal or adjustment, this indicates the reason the reversal or adjustment was generated. Valid values are listed in the ISO 8583-1993 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specification.***

Position	Level	Field Name and Description	Data Type
1–4		ITD-RSN-CDE-MSG	PIC X(4)

Route Code

Tag Number: 88

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.RTE-CDE

The following field contains the route code used to route the transaction. This value is used to group transactions for routing and authorization purposes when multiple destinations or authorization methods must be defined for an institution in the Institution Routing Configuration File (IRCF).

Position	Level	Field Name and Description	Data Type
1–4		ITD-RTE-CDE	PIC X(4)

Routing Data

Tag Number: 89

ITLF Status: Not logged

The following fields contain information used by the Router object to determine the correct routing destination for a message.

Position	Level	Field Name and Description	Data Type
1–62		ITD-RTE-DATA	
1–2	02	AUTH-IDX A code identifying the current routing destination. Valid values are as follows: 1 = Destination identified in the PRI fields 2 = Destination identified in the ALT1 fields 3 = Destination identified in the ALT2 fields 4 = Default authorizer	TYPE BINARY 16
3–20	02	PRI The following fields identify the primary routing destination.	
3–18	03	NAM The symbolic name of the authorization destination.	PIC X(16)

Position	Level	Field Name and Description	Data Type
19–20	03	OBJ-ID A code defining the type of object. Valid values are as follows: IMTT = Internal Message Translation Interface object PCA = Positive Customer Authorization method performed by the Authorization Combined Positive Customer object PCAA = Positive Customer with Accounts Authorization method performed by the Authorization Combined Positive Customer object PCBA = Positive Customer with Balances/History Authorization method performed by the Authorization Combined Positive Customer object	TYPE BINARY 16
21–38	02	ALT1 The following fields identify the first alternate routing destination.	
21–36	03	NAM The symbolic name of the authorization destination. The BASE24-telebanking product does not use this field.	PIC X(16)

Position	Level	Field Name and Description	Data Type
37–38	03	OBJ-ID A code defining the type of object. The BASE24-telebanking Integrated Authorization Server process uses this field when the authorization level is online/offline. Valid values are as follows: PCA = Positive Customer Authorization method performed by the Authorization Combined Positive Customer object PCAA = Positive Customer with Accounts Authorization method performed by the Authorization Combined Positive Customer object PCBA = Positive Customer with Balances/History Authorization method performed by the Authorization Combined Positive Customer object	TYPE BINARY 16
39–56	02	ALT2 The following fields identify the second alternate routing destination. The BASE24-telebanking product does not use these fields.	
39–54	03	NAM The symbolic name of the authorization destination.	PIC X(16)

Position	Level	Field Name and Description	Data Type
55–56	03	OBJ-ID A code defining the type of object. Valid values are as follows: PCA = Positive Customer Authorization method performed by the Authorization Combined Positive Customer object PCAA = Positive Customer with Accounts Authorization method performed by the Authorization Combined Positive Customer object PCBA = Positive Customer with Balances/History Authorization method performed by the Authorization Combined Positive Customer object	TYPE BINARY 16
57–60	02	DFLT-AUTH-LMT The maximum amount, in whole currency units (e.g., dollars), of a transaction authorized using the action specified in the DFLT-AUTH-ACT field. If the transaction amount exceeds the amount in this field, the transaction will be referred. The BASE24-telebanking product does not use the value in this field.	TYPE BINARY 32
61	02	DFLT-AUTH-ACT A code identifying the default authorization method to take for transactions meeting this configuration if the authorizer specified in the AUTH DESTINATION NAME field is unavailable. The BASE24-telebanking product does not use the value in this field.	PIC X(1)
62	02	XFILL (NOT USED)	PIC X(1)

Route Floor

Tag Number: 90

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.RTE-FLR

The following fields contain a code indicating the set of destinations used to route the transaction. The BASE24-telebanking product uses over-the-floor destinations only.

Position	Level	Field Name and Description	Data Type
1–2		ITD-RTE-FLR	
1	02	FLG A code indicating the set of destinations used to route the transaction. Valid values are as follows: O = Over-the-floor destinations were used U = Under-the-floor destinations were used	PIC X(1)
2	02	XFILL (NOT USED)	PIC X(1)

Route Type

Tag Number: 91

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.RTE-TYP

The following fields contain a code indicating the routing table used to route the transaction.

Position	Level	Field Name and Description	Data Type
1–2		ITD-RTE-TYP	
1	02	FLG A code indicating the routing table used to route the transaction. The valid value is P (Institution Routing Configuration File).	PIC X(1)
2	02	XFILL (NOT USED)	PIC X(1)

Sequence Number

Tag Number: 92

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ALT-KEY1.SEQ-NUM

The following field contains the sequence number assigned to the transaction by a BASE24 product. For the BASE24-telebanking product, the RBSI-compliant device must assign this sequence number and place it in the message.

Position	Level	Field Name and Description	Data Type
1–12		ITD-SEQ-NUM	PIC X(12)

Terminal Identifier

Tag Number: 102

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.TERM-ID

The following field contains the terminal ID of the terminal where the transaction was performed.

Position	Level	Field Name and Description	Data Type
1–16		ITD-TERM-ID	PIC X(16)

Entry Timestamp

Tag Number: 105

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.BCD.TIM-ENTRY-TS

The following field contains the time, expressed as a Julian timestamp, that the transaction entered a BASE24 Integrated Authorization Server process. This timestamp is used to calculate performance statistics.

Position	Level	Field Name and Description	Data Type
1–8		ITD-TIM-ENTRY-TS	TYPE BINARY 64

External Issuer Time

Tag Number: 106

ITLF Status: Optional

Data Name: TLF.BUS-OLF.TIM-EXTRN-ISS

The following field contains the amount of time, in microseconds, taken by an external issuer to authorize the transaction and return it to the BASE24 product.

Position	Level	Field Name and Description	Data Type
1–8		ITD-TIM-EXTRN-ISS	TYPE BINARY 64

Log Timestamp

Tag Number:	107
ITLF Status:	Mandatory
Data Name:	TLF.BUS-MLF.BCD.TIM-LOG-TS

The following field contains the date and time, expressed as a Julian timestamp, that the transaction was logged to the ITLF.

Position	Level	Field Name and Description	Data Type
1–8		ITD-TIM-LOG-TS	TYPE BINARY 64

TMF Transaction Identifier

Tag Number: 108

ITLF Status: Not logged

The following field contains the transaction identifier that was assigned to the transaction by the TMF process when the BEGINTRANSACTION procedure was called.

Position	Level	Field Name and Description	Data Type
1–8		ITD-TMF-TXN-ID	TYPE BINARY 64

Transaction Originator

Tag Number: 111

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.TXN-ORIGINATOR

The following fields contain a code indicating where the transaction originated.

Position	Level	Field Name and Description	Data Type
1–2		ITD-TXN-ORIGINATOR	
1	02	FLG A code indicating where the transaction originated. Valid values are as follows: 0 = Terminal (not used currently) 1 = Terminal object (not used currently) 2 = Remote banking device 3 = Voice Response Unit Interface object 4 = Host 5 = ISO Host Interface process 6 = Authorization object (Authorization Combined Positive Customer object or Authorization Combined Utility object) 7 = XPNET or Pathway Billpay Server process 8 = Scheduled Transaction Initiator process	PIC X(1)
2	02	XFILL (NOT USED)	PIC X(1)

Transaction Responder

Tag Number: 112

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.ASCII.TXN-RESPONDER

The following fields contain a code indicating where the transaction was authorized.

Position	Level	Field Name and Description	Data Type
1–2		ITD-TXN-RESPONDER	
1	02	FLG A code indicating where the transaction was authorized. Valid values are as follows: 0 = Terminal (not used currently) 1 = Terminal object (not used currently) 2 = Remote banking device 3 = Voice Response Unit Interface object 4 = Host 5 = ISO Host Interface process 6 = Authorization object (Authorization Combined Positive Customer object or Authorization Combined Utility object) 7 = XPNET or Pathway Billpay Server process	PIC X(1)
2	02	XFILL (NOT USED)	PIC X(1)

Account 1 Qualifier

Tag Number: 114

ITLF Status: Not logged

The following field contains a code used in place of the account number and account type to identify account 1, which is the *from* account involved in a transfer transaction, the first account involved in a dual inquiry, user-defined transaction, or the single account involved in transactions other than a transfer.

The account qualifier can be entered by the customer.

Position	Level	Field Name and Description	Data Type
1–4		ITD-ACCT1-QUAL	PIC X(4)

Account 2 Qualifier

Tag Number: 115

ITLF Status: Not logged

The following field contains a code used in place of the account number and account type to identify account 2, which is the *to* account involved in a transfer transaction, the second account involved in a dual available funds inquiry transaction, or the second account involved in a user-defined transaction.

The account qualifier can be entered by the customer.

Position	Level	Field Name and Description	Data Type
1–4		ITD-ACCT2-QUAL	PIC X(4)

Function Code

Tag Number: 117

ITLF Status: Optional

Data Name: TLF.BUS-OLF.FNCT

The following fields contain a code indicating the specific purpose of the message within its message class.

Position	Level	Field Name and Description	Data Type
1–4		ITD-FNCT	
1–3	02	CDE A code indicating the specific purpose of the message within its message class. Valid values are listed in the ISO 8583-1993 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specification.*** Private use codes defined for BASE24-telebanking multiple account selection are as follows: 970 = More accounts exist for account 1. 971 = More accounts exist for account 2. 972 = No more accounts exist.	PIC X(3)
4	02	XFILL (NOT USED)	PIC X(1)

Token

Tag Number: 118

ITLF Status: Optional

Data Name: TLF.BUS-OLF.TKN

The following fields contain token data returned from the host that must be logged to the ITLF. The BASE24-telebanking product does not use any token data in its processing.

Position	Level	Field Name and Description	Data Type
1–604		ITD-TKN	
1–4	02	FLD-LGTH Defines the length of data contained in the FDATA field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16
5–604	02	FDATA Token data returned from the host that must be logged to the ITLF. This is a variable-length field. The actual length of this field is adjusted to match the length of the data it contains.	PIC X(400)

Token—Not Logged

Tag Number: 119

ITLF Status: Not logged

The following fields contain token data returned from the host that is not logged to the ITLF. The BASE24-telebanking product does not use any token data in its processing.

Position	Level	Field Name and Description	Data Type
1–604		ITD-TKN-NL	
1–4	02	FLD-LGTH Defines the length of data contained in the FDATA field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16
5–604	02	FDATA Token data returned from the host that is not logged to the ITLF. This is a variable-length field. The actual length of this field is adjusted to match the length of the data it contains.	PIC X(400)

Additional Data

Tag Number: 120

ITLF Status: Optional

Data Name: TLF.BUS-OLF.ADNL-DATA

The following fields are used when the data accompanying a transaction does not have a field dedicated to carry it. This data can include the following:

- The key data required to reposition in the Transaction History File (THF) or History Table (HIST) for an inquiry transaction
- Search and positioning information for statement closing download and statement download transactions
- Check information for inquiry transactions or check stop payment add transactions
- Vendor-specific data for schedule payment transactions

These fields can also be used for user-defined transactions.

Position	Level	Field Name and Description	Data Type
1–204		ITD-ADNL-DATA	
1–4	02	FLD-LGTH The following fields define the length of data contained in the FDATA field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16

Position	Level	Field Name and Description	Data Type
5–204	02	FDATA Additional data that needs to accompany the transaction, but is not defined in a specific field. This is a variable-length field. The actual length of this field is adjusted to match the length of the data it contains.	PIC X(200)
5–204	02	CHK-STOP-PMNT Check information for a check stop payment add transaction. This is a variable-length field. The actual length of this field is adjusted to match the length of the data it contains.	PIC X(200)

External Acquirer Time

Tag Number: 121

ITLF Status: Optional

Data Name: TLF.BUS-OLF.TIM-EXTRN-ACQ

The following field contains the amount of time, in microseconds, taken outside of a BASE24 process to gather information from the customer for an interactive transaction (e.g., multiple account selection).

Position	Level	Field Name and Description	Data Type
1–8		ITD-TIM-EXTRN-ACQ	TYPE BINARY 64

Transaction Amount—Original Transaction

Tag Number: 122

ITLF Status: Optional

Data Name: TLF.BUS-OLF.AMT-TXN-ORIG-TXN

The following field contains the original amount of the transaction, expressed in the currency of the remote banking endpoint or customer service representative terminal where the transaction was initiated. This information may be present in authorization, financial, or reversal transactions.

Position	Level	Field Name and Description	Data Type
1–8		ITD-AMT-TXN-ORIG-TXN	TYPE BINARY 64

Last Transaction Allowed Count

Tag Number: 123

ITLF Status: Not logged

The following fields contain the values necessary to determine the number of items that can be returned in a single inquiry transaction response message.

Position	Level	Field Name and Description	Data Type
1–4		ITD-LAST-TXN-ALWD-CNT	
1–2	02	ACQ-MAX-CNT	TYPE BINARY 16

The maximum number of items that can be returned to the RBSI-compliant device (e.g., interactive voice response system) or customer service representative (CSR) terminal in a single response. Valid values are 0 through 19. A value of 0 indicates that the transaction is not supported. The maximum number that can be returned in each response depends on the type of transaction, and whether small or large messages are being used as follows:

	Maximum Number Returned	
Transaction Type	**Small Message**	**Large Message**
Bank list inquiry	4	19
Bank branch list inquiry	4	16
Customer account list inquiry	15	15
Customer vendor list inquiry	1	4
Customer vendor mask inquiry	2	8
History inquiry – all transactions	1	7

Position Level Field Name and Description Data Type

Transaction Type	Maximum Number Returned	
	Small Message	**Large Message**
History inquiry – payments and transfers	1	7
Last *count* of inquiry transactions	15	15
Master vendor list inquiry	1	5
Miscellaneous requests inquiry	1	5
Multiple account balance inquiry	13	13
Scheduled payments list inquiry	2	9
Scheduled transfers list inquiry	2	10
Scheduled transactions list inquiry	2	8
Statement download	3	3
Statement closing download	16	16

This value is obtained from the ITS Interface Configuration File (IFCF).

3–4 02 CUST-MAX-CNT TYPE BINARY 16

The maximum number of Transaction History File (THF) records that can be returned in a series of sequential responses for this customer. This maximum does not apply to historical inquiries acquired from a BASE24 CSR terminal. Valid values are as follows:

0 = No maximum is enforced.
1–9999 = The maximum number of THF records or HIST rows that can be returned.

This value is obtained from the Customer Table (CSTT) or an external request.

Refresh Indicator

Tag Number: 124

ITLF Status: Mandatory

Data Name: TLF.BUS-MLF.BCD.REFR

The following fields indicate the BASE24-telebanking files that can be refreshed and what the current refresh indicator is for each file.

Position	Level	Field Name and Description	Data Type
1–22		ITD-REFR	
1–2	02	CNT The number of entries contained in the FILE fields.	TYPE BINARY 16
3–22	02	FILE The type and refresh indicator for each file. Each occurrence is two positions in length.	OCCURS 10 TIMES
	03	TYP A code indicating to the Enscribe Refresh process the type of file associated with the value in the FILE.IND field. Valid values are as follows: 1 = Positive Balance File 1 2 = Positive Balance File 2 3 = Positive Balance File 3	PIC X(1)
	03	IND A code indicating to the Enscribe Refresh process the version of the file when this transaction occurred.	PIC X(1)

Refresh Impact

Tag Number: 125

ITLF Status: Optional

Data Name: TLF.BUS-OLF.REFR-IMPACT

The following fields contain information the Enscribe Refresh process uses to impact the database for a financial transaction.

The INFO field can occur up to ten times. The actual length of the Refresh Impact field is based on the number of times the INFO field occurs.

Position	Level	Field Name and Description	Data Type
1–106		ITD-REFR-IMPACT	
1–4	02	FLD-LGTH The following fields define the length of data contained in the INFO field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16
5–6	02	CNT The number of occurrences of the INFO field.	TYPE BINARY 16
7–106	02	INFO Information the Enscribe Refresh process uses to impact the database for a financial transaction. Each occurrence is 10 positions in length.	OCCURS 10 TIMES

Position	Level	Field Name and Description	Data Type
	03	AMT-TYP A code identifying which balance the Enscribe Refresh process is to impact. Valid values are as follows: A = Available balance/available credit H = Amount on hold/credit balance I = Cash in L = Ledger balance O = Cash out	PIC X(1)
	03	ACCT-IND A code identifying which account the Enscribe Refresh process is to impact. Valid values are as follows: 1 = Account 1 2 = Account 2 B = Backup account	PIC X(1)
	03	AMT The amount the Enscribe Refresh process uses to impact the database.	TYPE BINARY 64

TMF Abort 1 Notify Flag

Tag Number: 126

ITLF Status: Not logged

The following field contains a flag indicating whether the Positive Balance File (PBF) that contains the *from* account in a payment or transfer transaction (account 1) has been updated. Core objects must update files that are not audited by the Transaction Monitoring Facility (TMF) process when a transaction cannot be completed. The TMF process automatically updates the files that it audits when a transaction cannot be completed.

Position	Level	Field Name and Description	Data Type
1–2		ITD-TMF-ABORT1-NTFY-FLG A flag indicating whether the PBF that contains the *from* account in a payment or transfer transaction (account 1) has been updated. Valid values are as follows: 0 = The PBF has not been updated. 1 = The PBF has been updated.	TYPE BINARY 16

TMF Abort 2 Notify Flag

Tag Number: 127

ITLF Status: Not logged

The following field contains a flag indicating whether the PBF that contains the *to* account in a transfer transaction (account 2) has been updated. Core objects must update files that are not audited by the TMF process when a transaction cannot be completed. The TMF process automatically updates the files that it audits when a transaction cannot be completed.

Position	Level	Field Name and Description	Data Type
1–2		ITD-TMF-ABORT2-NTFY-FLG A flag indicating whether the PBF that contains the *to* account in a transfer transaction (account 2) has been updated. Valid values are as follows: 0 = The PBF has not been updated. 1 = The PBF has been updated.	TYPE BINARY 16

PIN Change Required

Tag Number: 128

ITLF Status: Optional

Data Name: TLF.BUS-OLF.PIN-CHNG-REQ

The following fields contain a code indicating whether a PIN change transaction must be performed.

Position	Level	Field Name and Description	Data Type
1–2		ITD-PIN-CHNG-REQ	
1	02	FLG A code indicating whether a PIN change transaction must be performed. Valid values are as follows: Y = Yes, a PIN change transaction is required. N = No, a PIN change transaction is not required. S = Yes, a PIN change transaction is required to select an initial PIN. This value is obtained from the Customer Table (CSTT).	PIC X(1)
2	02	XFILL (NOT USED)	PIC X(1)

Action Date

Tag Number: 129

ITLF Status: Optional

Data Name: TLF.BUS-OLF.ACT-DAT

The following field contains a date, in YYMMDD format, for a future or recurring transfer or payment transaction.

Position	Level	Field Name and Description	Data Type
1–6		ITD-ACT-DAT	PIC X(6)

Payee

Tag Number: 130

ITLF Status: Optional

Data Name: TLF.BUS-OLF.PAYEE

The following fields contain the number of a customer vendor to receive a customer payment.

Position	Level	Field Name and Description	Data Type
1–26		ITD-PAYEE	
1–25	02	FLD The number of a customer vendor to receive a customer payment.	PIC X(25)
26	02	XFILL (NOT USED)	PIC X(1)

Payment Date

Tag Number: 131

ITLF Status: Optional

Data Name: TLF.BUS-OLF.PMNT-DAT

The following field contains the date, in YYYYMMDD format, that a payment or transfer transaction will be performed.

Position	Level	Field Name and Description	Data Type
1–8		ITD-PMNT-DAT	PIC X(8)

Recurring Transaction Data

Tag Number: 132

ITLF Status: Optional

Data Name: TLF.BUS-OLF.RECUR-TXN-DATA

The following fields contain information used to schedule recurring payments or transfers.

Position	Level	Field Name and Description	Data Type
1–8		ITD-RECUR-TXN-DATA	
1–2	02	PRD-TYP A code identifying how often a recurring transfer or payment is performed. Valid values are as follows: WK = Weekly 2W = Biweekly (every two weeks) MN = Monthly Q = Quarterly S = Semiannually A = Annually	PIC X(2)
3–6	02	NUM-PRD The remaining number of times a recurring transaction is scheduled to be initiated.	PIC X(4)
7–8	02	SKIP-NXT-PMNT-IND The number of scheduled recurring payment or scheduled recurring transfers that should be skipped. Valid values are as follows: 0 = Do not skip any payments or transfers. 1 = Skip one payment or transfer.	TYPE BINARY 16

Payee Description

Tag Number: 133

ITLF Status: Optional

Data Name: TLF.BUS-OLF.PAYEE-DESCR

The following fields contain information about the vendor to receive a customer payment.

Position	Level	Field Name and Description	Data Type
1–60		ITD-PAYEE-DESCR	
1–28	02	CUST-ACCT-NUM-VNDR The customer account with the vendor to receive the customer payment.	PIC X(28)
29–60	02	VNDR-NAM The name of the vendor to receive the customer payment.	PIC X(32)

Customer Reference Number

Tag Number: 134

ITLF Status: Optional

Data Name: TLF.BUS-OLF.CUST-REF-NUM

The following field contains a value generated by the XPNET or Pathway Billpay Server process for identification purposes when it processes a transaction.

Position	Level	Field Name and Description	Data Type
1–6		ITD-CUST-REF-NUM	PIC X(6)

Source Code

Tag Number: 135

ITLF Status: Optional

Data Name: TLF.BUS-OLF.SRC-CDE

The following field contains a code indicating the type of device or program that originated the transaction. The value in this field is either set in the transaction message by the acquirer or is set from the value of the DEFAULT SOURCE CODE field on VRU Configuration Data (VCD) screen 18 if a source code is not provided in the message. For inbound customer service representative transactions acquired from an RBSI-compliant device, this field must be set to a value of IB for customer service processing information to be retrieved from the Institution Data File (IDF).

Position	Level	Field Name and Description	Data Type
1–2		ITD-SRC-CDE	PIC X(2)

A code indicating the type of device or program that originated the transaction. The code may be a reserved value or may be user defined. The reserved values are as follows:

AD = Audio device (e.g., interactive voice response system)
BL = BASE24-billpay Billing application
IB = Inbound customer service representative (using the BASE24 CSR terminal or an external RBSI-compliant device)
PC = Personal computer
SP = Screen phone

External Acquirer Dependent Data

Tag Number: 138

ITLF Status: Optional

Data Name: TLF.BUS-OLF.ACQ-DPNT-EXTRN

The following fields are user defined and specific to the external acquirer of this transaction.

Position	Level	Field Name and Description	Data Type
1–104		ITD-ACQ-DPNT-EXTRN	
1–4	02	FLD-LGTH The following fields define the length of data contained in the FDATA field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16
5–104	02	FDATA Information specific to the external acquirer of this transaction (e.g., the ISO Host Interface process). The structure and content of this field is defined by the acquirer. This is a variable-length field. The actual length of this field is adjusted to match the length of the data it contains.	PIC X(100)

External Authorization Dependent Data

Tag Number: 139

ITLF Status: Optional

Data Name: TLF.BUS-OLF.AUTH-DPNT-EXTRN

The following fields are user defined and specific to the external authorizer of this transaction.

Position	Level	Field Name and Description	Data Type
1–104		ITD-AUTH-DPNT-EXTRN	
1–4	02	FLD-LGTH The following fields define the length of data contained in the FDATA field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16
5–104	02	FDATA Information specific to the external authorizer of this transaction (e.g., the host). The structure and content of this field is defined by the authorizer. This is a variable-length field. The actual length of this field is adjusted to match the length of the data it contains.	PIC X(100)

Acquirer Dependent—Not Logged

Tag Number: 140

ITLF Status: Not logged

The following fields are user defined and specific to the external process acquiring this transaction.

Position	Level	Field Name and Description	Data Type
1–454		ITD-ACQ-DPNT-NL	
1–4	02	FLD-LGTH The following fields define the length of data contained in the FDATA field.	
1–2	03	MAX The maximum physical length of data the field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the field.	TYPE BINARY 16
5–454	02	FDATA Information specific to the external process acquiring this transaction. The structure and content of this field is defined by the acquirer. Data in this field must be returned in the response to the acquirer but does not need to be logged. This is a variable-length field. The actual length of this field is adjusted to match the length of the data it contains.	PIC X(450)

Multiple Account Information

Tag Number: 141

ITLF Status: Optional

Data Name: TLF.BUS-OLF.MULT-ACCT-INFO

The following fields contain the account number, account type, and account action code for all of the accounts in a multiple accounts balance inquiry transaction.

The INFO field can occur up to 13 times. The actual length of the Multiple Account Information field is based on the number of times the INFO field occurs.

Position	Level	Field Name and Description	Data Type
1–292		ITD-MULT-ACCT-INFO	
1–4	02	FLD-LGTH The following fields define the length of data contained in the INFO field.	
1–2	03	MAX The maximum physical length of data the INFO field can contain.	TYPE BINARY 16
3–4	03	LOGL The logical length of the data contained in the INFO field.	TYPE BINARY 16
5–6	02	CNT The number of accounts contained in the INFO field. Valid values are 1 through 13.	TYPE BINARY 16
7–292	02	INFO Each occurrence of the following fields contains information for one account. Each occurrence is 22 positions in length.	OCCURS 13 TIMES

Position	Level	Field Name and Description	Data Type
	03	NUM-BCD The following fields contain account number information.	
	04	NUM The account number.	PIC X(14)
	04	LGTH The length of the account number.	TYPE BINARY 16
	03	TYP A code identifying the type of account. Default values provided by ACI are as follows: 00 = None 10 = Savings account 20 = Checking account 30 = Credit account 38 = Line of credit 58 = Certificate 59 = Retirement account 90 = Interest-bearing checking account 9A = Commercial loan 9B = Installment loan 9C = Mortgage loan	PIC X(2)
	03	CDE The action code returned as a result of performing the inquiry. Refer to appendix B for valid values.	PIC X(3)
	03	USER-FLD (NOT USED)	PIC X(1)

Message Format Type ID

Tag Number: 142

ITLF Status: Optional

Data Name: TLF.BUS-OLF.FRMT-TYP-ID

This field is reserved for future use. It is intended to be used when different message formats are required.

Position	Level	Field Name and Description	Data Type
1–8		ITD-FRMT-TYP-ID	
1–4	02	PROD-TYP This field is reserved for product use.	PIC X(4)
5–8	02	OTHER-TYP This field can be used as required.	PIC X(4)

Last (x) Area

Tag Number: 143

ITLF Status: Not logged

The following fields contain the address of the external message block for large messages and the length of the external message block.

Position	Level	Field Name and Description	Data Type
1–6		ITD-LAST-X-AREA	
1–2	02	LAST-X-LGTH The length, in bytes, of the external message block for large messages in the allocated flat segment of memory for the Integrated Authorization Server process.	TYPE BINARY 16 SIGNED
3–6	02	LAST-X-ADDR The address of the external message block for large messages in the allocated flat segment of memory for the Integrated Authorization Server process.	TYPE BINARY 32 SIGNED

Appendix B

BASE24 Remote Banking Action Codes

This appendix contains descriptions of the authorization action codes supported by BASE24 Remote Banking products. Transaction authorizers use action codes to indicate how transactions should be handled. BASE24 Remote Banking products carry action codes in the ITD-ACT.CDE and ITD-RSN-CDE-EXCPT.INFO.CDE fields of the Internal Transaction Data (ITD).

The following table provides a description of each action code supported by the BASE24 Remote Banking products. Action codes used by the BASE24-billpay product are identified by a check (✓) in the BP column. Some of these action codes are also used when the Enhanced Telebanking Application is installed. Action codes are generated by XPNET Billpay Server and Pathway Billpay Server processes, generated by core objects in the Integrated Authorization Server process, and translated from external response codes by the ISO Host Interface process. The description of each action code generated by core objects includes the circumstances under which the action code is generated and the core objects that generate it. The description of each action code translated by the ISO Host Interface process includes the external response code that is associated with it.

Refer to the BASE24-telebanking/ISO conversion tables in the ***BASE24 External Message Manual*** for descriptions of the external response codes that are translated into BASE24 Remote Banking action codes.

Code	BP	Description
000	✓	APPROVED The Authorization Combined Positive Customer object generates this action code when the transaction is successful and no backup or overdraft account is involved. The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when the transaction is successful. The ISO Host Interface process translates this action code from external message response code 00 if the message class is 01 or 02.
001		HONOR WITH IDENTIFICATION The ISO Host Interface process translates this action code from external message response code 08.
002		APPROVED FOR PARTIAL AMOUNT The ISO Host Interface process translates this action code from external message response code 10.
003		APPROVED (VIP) The ISO Host Interface process translates this action code from external message response code 11.
004		APPROVED, UPDATE TRACK 3 The ISO Host Interface process translates this action code from external message response code 16.
080		APPROVED WITH BACKUP ACCOUNT The Authorization Combined Positive Customer object generates this action code when the Account object uses a backup account to complete the transaction. The ISO Host Interface process translates this action code from external message response code N0.

Code	BP	Description
081		APPROVED WITH OVERDRAFT The Authorization Combined Positive Customer object generates this action code when the Account object uses an overdraft account to complete the transaction. The ISO Host Interface process translates this action code from external message response code N1.
082	✓	APPROVED, CUSTOMER REACTIVATED The Pathway or XPNET Billpay Server process generates this action code when it processes a customer activate transaction for a customer ID that is already active. When this action code is returned, the customer activate transaction reactivated the customer ID. When action code 000 is returned for a customer activate transaction, the transaction activated a customer ID that was not already active.
100		DENY, DO NOT HONOR The Authorization Combined Positive Customer object generates this action code when the VERIFY STATUS field on Customer Table (CSTT) screen 1 is set to the value C (closed). The ISO Host Interface process translates this action code from external message response codes 05 or 06. The ISO Host Interface process also uses this action code when it does not recognize the response code provided by the host.
101		EXPIRED CARD The ISO Host Interface process translates this action code from external message response code 54.
102		SUSPECTED FRAUD The ISO Host Interface process translates this action code from external message response code 59.
103		CARD ACCEPTOR CONTACT ACQUIRER The ISO Host Interface process translates this action code from external message response code 60.

Code	BP	Description
104		RESTRICTED CARD The ISO Host Interface process translates this action code from external message response code 62.
105		CARD ACCEPTOR CALL ACQUIRER’S SECURITY DEPARTMENT The ISO Host Interface process translates this action code from external message response code 66.
106		DENY, PIN TRIES EXCEEDED The Authorization Combined Positive Customer object generates this action code when the Transaction Security object indicates that the number of PIN tries exceeds the limit set in the MAX PIN TRIES field on Institution Definition File (IDF) screen 2. The ISO Host Interface process translates this action code from external message response code 75.
107		REFER TO CARD ISSUER The ISO Host Interface process translates this action code from external message response code 01.
108		REFER TO CARD ISSUER, SPECIAL CONDITIONS The ISO Host Interface process translates this action code from external message response code 02.
109		INVALID MERCHANT The ISO Host Interface process translates this action code from external message response code 03.

Code	BP	Description
110	✓	DENY, AMOUNT INVALID The Authorization Combined Positive Customer object generates this action code when the Account object determines that the transaction amount is incorrect. The amount of a transfer from a credit account must meet the following requirements: • Be equal to or greater than a nonzero value in the MINIMUM CASH ADVANCE AMOUNT field on Positive Balance File (PBF) screen 11. • Be a multiple of the value in the CASH ADVANCE INCREMENT field on PBF screen 11. The Scheduled Transaction Initiator process or XPNET Billpay Server process generates this action code if the amount of a payment transaction in the Future Table (FUTR) exceeds the payment high limit in the Bank Table (BANK) or Customer Table (CSTT). The ISO Host Interface process translates this action code from external message response codes 13 or 64.
111		DENY, CARD NUMBER/ID INVALID The ISO Host Interface process translates this action code from external message response code 14.
112		DENY, PIN IS REQUIRED The Authorization Combined Positive Customer object generates this action code when the Transaction Security object indicates that the ITD does not contain a PIN when one is required.
113		UNACCEPTABLE FEE The ISO Host Interface process translates this action code from external message response code 23.

Code	BP	Description
114	✓	DENY, NO ACCOUNT OF TYPE The Authorization Combined Positive Customer object generates this action code when the Account object cannot find a PBF record that matches the entered account number or when the Customer Access object cannot find a Customer/Account Relation Table (CACT) row that matches the entered customer account number. The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when the account specified in a transaction cannot be found in the CACT. The ISO Host Interface process translates this action code from external message response codes 39, 42, 44, 52, or 53.
115	✓	REQUESTED FUNCTION NOT SUPPORTED The Pathway Billpay Server process and XPNET Billpay Server process generate this action code for the following reasons: • A customer add transaction is received for a financial institution that uses preloaded customer data. • An attempt is made to delete a customer account when the customer verify status is not I (Initial). The ISO Host Interface process translates this action code from external message response code 40.

Code	BP	Description
116		DENY, NOT SUFFICIENT FUNDS The Authorization Combined Positive Customer object generates this action code when the Account object identifies one of the following conditions: • The transfer amount is greater than the account 1 balance and account 1 (the *from* account) does not have a backup account or overdraft limit. • The transfer amount is greater than the account 1 balance and the backup account is the same as account 2 (the *to* account). • The transfer amount is greater than the sum of the account 1 balance and the backup account balance. • The transfer amount is greater than the sum of the account 1 balance and the unused portion of the overdraft limit, and account 1 does not have a backup account. **Note:** The Account object ignores any account 1 overdraft limit when account 1 has a backup account. The ISO Host Interface process translates this action code from external message response code 51.
117		DENY, INCORRECT PIN The Authorization Combined Positive Customer object generates this action code when the Transaction Security object indicates that the PIN entered by the customer does not match the information maintained in the BASE24 database or an error occurred when encrypting the validation data in software. The ISO Host Interface process translates this action code from external message response code 55.

Code	BP	Description
118	✓	DENY, CARD/ID NOT FOUND The following objects generate this action code when the Customer Access object cannot locate a Customer Table (CSTT) row for a specified customer identification number: • Authorization Combined Positive Customer object • Authorization Combined Utility object • Customer Service Interface object • Voice Response Unit Interface object The Pathway Billpay Server process, XPNET Billpay Server process, and Scheduled Transaction Initiator process generate this action code when they cannot locate a CSTT row to process a transaction. The ISO Host Interface process translates this action code from external message response code 56.
119	✓	TRANSACTION NOT PERMITTED TO CARDHOLDER The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when an attempt is made to reactivate a customer ID and reactivation is not allowed. The ISO Host Interface process translates this action code from external message response code 57.
120		DENY, TXN NOT ALLOWED AT TERM The Authorization Combined Positive Customer object generates this action code when a PIN change transaction is initiated by a customer service representative and the PIN CHANGE ALLOWED field on IDF screen 40 is set to the value N (transaction not allowed). The ISO Host Interface process translates this action code from external message response code 58.

Code	BP	Description
121	✓	DENY, WDL AMT LIMIT EXCEEDED The Authorization Combined Positive Customer object generates this action code when the Account object identifies one or more of the following conditions for account 1 (the *from* account): • The amount in the PERIODIC USAGE AMOUNT field on PBF screen 11 is greater than or equal to the amount in the PERIODIC LIMIT AMOUNT field on PBF screen 11. • The sum of the amount in the PERIODIC USAGE AMOUNT field on PBF screen 11 and the transfer or payment amount is greater than or equal to the amount in the PERIODIC LIMIT AMOUNT field on PBF screen 11. • The amount in the CYCLIC USAGE AMOUNT field on PBF screen 11 is greater than or equal to the amount in the CYCLIC LIMIT AMOUNT field on PBF screen 11. • The sum of the amount in the CYCLIC USAGE AMOUNT field on PBF screen 11 and the transfer or payment amount is greater than or equal to the amount in the CYCLIC LIMIT AMOUNT field on PBF screen 11. The Pathway or XPNET Billpay Server processes generate this action code when the amount of a payment exceeds the payment high limit defined in the Bank Table (BANK) or Customer Table (CSTT). The ISO Host Interface process translates this action code from external message response code 61.
122		DENY, SECURITY VIOLATION The Voice Response Unit Interface object generates this action code when the external message received from the RBSI-compliant device does not contain a PIN and the device cannot perform local PIN verification. The ISO Host Interface process translates this action code from external message response code 63.

Code	BP	Description
123		DENY, WDL FREQ LIMIT EXCEEDED The Authorization Combined Positive Customer object generates this action code when the Account object identifies either of the following conditions for account 1 (the *from* account): • The value in the PERIODIC USAGE COUNT field on PBF screen 11 is greater than or equal to the value in the PERIODIC LIMIT COUNT field on PBF screen 11. • The value in the CYCLIC USAGE COUNT field on PBF screen 11 is greater than or equal to the value in the CYCLIC LIMIT COUNT field on PBF screen 11. The ISO Host Interface process translates this action code from external message response code 65.
124		VIOLATION OF LAW The ISO Host Interface process translates this action code from external message response code 93.
125		DENY, CARD/ID NOT EFFECTIVE The Authorization Combined Positive Customer object generates this action code when the VERIFY STATUS field on Customer Table (CSTT) screen 1 is set to the value I (issued but inactive).
127		DENY, PIN LENGTH ERROR The Authorization Combined Positive Customer object generates this action code when the Transaction Security object indicates that a PIN cannot be verified because of its length.
128		DENY, PIN KEY SYNCH ERROR The Authorization Combined Positive Customer object generates this action code when the Transaction Security object indicates that a PIN cannot be verified because of a key synchronization error.

Code	BP	Description
180		DENY, AMOUNT NOT FOUND The Authorization Combined Positive Customer object generates this action code when the Transaction History object cannot locate a Transaction History File (THF) record that has the requested transaction amount. The ISO Host Interface process translates this action code from external message response code O2.
181		DENY, PIN CHANGE REQUIRED The Authorization Combined Positive Customer object generates this action code for transactions other than PIN Change when the PIN CHANGE REQUIRED field on Customer Table (CSTT) screen 1 is set to the value Y or S. The ISO Host Interface process translates this action code from external message response code O3.
182		DENY, NEW PIN INVALID The following objects generate this action code when the Transaction Security object indicates that the two new PIN entries for a PIN change transaction do not match: • Authorization Combined Positive Customer object • Customer Service Interface object • Voice Response Unit Interface object The ISO Host Interface process translates this action code from external message response code O4.
183	✓	DENY, BANK NOT FOUND The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when the FIID that has been referenced is not configured in the Bank Table. The ISO Host Interface process translates this action code from external message response code O5.

Code	BP	Description
184	✓	DENY, BANK NOT EFFECTIVE The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when an attempt is made to add a vendor to a customer’s vendor list outside the effective date window in the Bank Table. The ISO Host Interface process translates this action code from external message response code O6.
185	✓	DENY, CUSTOMER VENDOR NOT FOUND The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when the vendor requested is not associated with the customer requesting the transaction. The ISO Host Interface process translates this action code from external message response code O7.
186	✓	DENY, CUSTOMER VENDOR NOT EFFECTIVE The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when the vendor associated with the customer requesting a payment is unavailable. A vendor can be unavailable because it has an account verify status of any value other than verified or has an effective date window that does not include the current date. The ISO Host Interface process translates this action code from external message response code O8.
187	✓	DENY, CUSTOMER VENDOR ACCOUNT INVALID The Pathway Billpay Server process and XPNET Billpay Server process generate this action code if the entered customer vendor account number fails the account number mask check when added or updated. The ISO Host Interface process translates this action code from external message response code O9.

Code	BP	Description
188	✓	DENY, VENDOR NOT FOUND The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when the vendor data cannot be found in the Vendor Table (VNDR). The ISO Host Interface process translates this action code from external message response code P0.
189	✓	DENY, VENDOR NOT EFFECTIVE The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when the master vendor is unavailable. A master vendor can be unavailable because it has an account verify status of any value other than verified or has an effective date window that does not include the current date. The ISO Host Interface process translates this action code from external message response code P1.
190	✓	DENY, VENDOR DATA INVALID The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when vendor-specific information required by the vendor has not been entered. The ISO Host Interface process translates this action code from external message response code P2.
191	✓	DENY, PAYMENT DATE INVALID The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when the date entered for the beginning date for a recurring payment or the payment date of a future payment has already occurred. The ISO Host Interface process translates this action code from external message response code P3.

Code	BP	Description
192	✓	DENY, PERSONAL ID NOT FOUND The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when they cannot obtain the proper Personal Information Table (PIT) row when activating, deactivating, or inquiring about a customer. The ISO Host Interface process translates this action code from external message response code P4.
193	✓	DENY, SCHEDULED TRANSACTIONS EXIST The Pathway Billpay Server process and XPNET Billpay Server process generate this action code when an attempt is made to deactivate a customer ID that has scheduled transactions. The ISO Host Interface process translates this action code from external message response code P5.
200		DO NOT HONOR The ISO Host Interface process translates this action code from external message response codes 04 or 67.
201		EXPIRED CARD The ISO Host Interface process translates this action code from external message response code 33.
202		SUSPECTED FRAUD The ISO Host Interface process translates this action code from external message response code 34.
203		CARD ACCEPTOR CONTACT ACQUIRER The ISO Host Interface process translates this action code from external message response code 35.
204		RESTRICTED CARD The ISO Host Interface process translates this action code from external message response code 36.

Code	BP	Description
205		CARD ACCEPTOR CALL ACQUIRER'S SECURITY DEPARTMENT The ISO Host Interface process translates this action code from external message response code 37.
206		ALLOWABLE PIN TRIES EXCEEDED The ISO Host Interface process translates this action code from external message response code 38.
207		SPECIAL CONDITIONS The ISO Host Interface process translates this action code from external message response code 07.
208		LOST CARD The ISO Host Interface process translates this action code from external message response code 41.
209		STOLEN CARD The ISO Host Interface process translates this action code from external message response code 43.
300		SUCCESSFUL The ISO Host Interface process translates this action code from external message response code 00 if the message class is 03.
301		NOT SUPPORTED BY RECEIVER The ISO Host Interface process translates this action code from external message response code 24.
302		UNABLE TO LOCATE RECORD ON FILE The ISO Host Interface process translates this action code from external message response code 25.
303		DUPLICATE RECORD, OLD RECORD REPLACED The ISO Host Interface process translates this action code from external message response code 26.

Code	BP	Description
304		FIELD EDIT ERROR The ISO Host Interface process translates this action code from external message response code 27.
305		FILE LOCKED OUT The ISO Host Interface process translates this action code from external message response code 28.
306		NOT SUCCESSFUL The ISO Host Interface process translates this action code from external message response code 29.
307		FORMAT ERROR The XPNET Billpay Server process generates this action code when it receives a standard internal message (BSTM) that contains an incorrectly formatted field. The ISO Host Interface process translates this action code from external message response code 30.
308		DUPLICATE, NEW RECORD REJECTED The ISO Host Interface process translates this action code from the external message response code Q3.
400		ACCEPTED The Authorization Combined Positive Customer object places this action code in the Exception Reason Code field of the ITD when a transaction is reversed successfully. The ISO Host Interface process translates this action code from external message response code 00 if the message class is 04.
500		ACCEPTED The ISO Host Interface process translates this action code from external message response code 00 if the message class is 05.
600		ACCEPTED The ISO Host Interface process translates this action code from external message response code 00 if the message class is 06.

Code	BP	Description
800		NMM OK The Voice Response Unit Master object generates this action code when it responds to a network management request message from the RBSI-compliant device. The ISO Host Interface process translates this action code from external message response code 00 if the message class is 08.
901		ADVICE ACKNOWLEDGED, NO FINANCIAL LIABILITY ACCEPTED The ISO Host Interface process translates this action code from external message response code 09.

Code	BP	Description
902	✓	DENY, TRANSACTION INVALID The Authorization Combined Positive Customer object generates this action code in the following conditions: • The transaction is a THF inquiry or an available funds inquiry and the authorization method is not Positive Customer with Balances/History Authorization. • The transaction is a check stop payment add transaction, which can only be authorized on a host. For this transaction, the authorization level must be set to 1 and the authorization method must be Positive Customer with Balances/History Authorization. • The Transaction History object receives a transaction other than a THF inquiry. • The Customer object cannot find an entry in the customer extended memory table for the customer profile and processing code received in the message. The customer profile and processing code information in the customer extended memory table is obtained from the Customer Allowed Transaction Table (CATT). The Authorization Combined Utility object generates this action code when the Router object cannot find an entry in the institution routing extended memory table for the requested processing code. Processing code information in the institution routing extended memory table is obtained from the Processing Code Definition File (PCDF). The Pathway Billpay Server process and XPNET Billpay Server process generate this action code for the following reasons: • An attempt is made to cancel a previous transaction and cancellation is not allowed. • The customer accounts specified in a scheduled transfer transaction are the same. The Scheduled Transaction Initiator process generates this action code when a Future Table (FUTR) row contains an invalid transaction type. The ISO Host Interface process translates this action code from external message response code 12.

Code	BP	Description
903		REENTER TRANSACTION The ISO Host Interface process translates this action code from external message response code 19.
904	✓	DENY, DATABASE ERROR The Pathway Billpay Server process and XPNET Billpay Server process generates this action code when an error occurs while accessing a database table or file.
905		ACQUIRER NOT SUPPORTED BY SWITCH The ISO Host Interface process translates this action code from external message response code 31.
906		CUTOVER IN PROCESS The ISO Host Interface process translates this action code from external message response code 90.
907		ISSUER OR SWITCH INOPERATIVE The ISO Host Interface process translates this action code from external message response code 91.
908		DENY, ROUTE DEST NOT FOUND The Authorization Combined Utility object generates this action code in the following situations: • The Financial Institution and Financial Institution Access objects cannot obtain the institution route profile for the Router object. The Router object uses the route profile, which is defined in the ROUTE PROFILE field on IDF screen 40, to locate the proper information in the institution routing extended memory table. • The Router object cannot find an entry in the institution routing extended memory table for the institution routing configuration information. This information in the institution routing extended memory table is obtained from the Institution Routing Configuration File (IRCF). The ISO Host Interface process translates this action code from external message response codes 15 or 92.

Code	BP	Description
909	✓	DENY, SYSTEM MALFUNCTION The Authorization Combined Positive Customer object generates this action code when it cannot communicate properly with the Account object. The Customer Service Interface object generates this action code when it cannot communicate properly with the Customer object, Transaction Security object, or Transaction object. The Internal Message Translation Interface object generates this action code when it cannot communicate properly with the Router object to reroute a transaction. The Voice Response Unit Interface object generates this action code when it receives a late response from the device or cannot communicate properly with one of the following objects: • Authorization Combined Positive Customer object • Customer object • Transaction object • Transaction Security object The Pathway Billpay Server process, XPNET Billpay Server process, and Scheduled Transaction Initiator process generate this action code when an internal system error other than a formatting error occurs. The ISO Host Interface process translates this action code from external message response code 96.
912		DENY, ISSUER UNAVAILABLE The Authorization Combined Positive Customer object generates this action code when BASE24 stands in for the host, the transaction is an available funds inquiry or a Transaction History File (THF) inquiry, and the authorization method is not Positive Customer with Balances/History Authorization. The Authorization Combined Utility object generates this action code when the authorization level is online and the host is unavailable.

Code	BP	Description
913		DENY, DUPLICATE TRANSMISSION The Customer Service Interface object generates this action code when it receives a reversal advice for an ITS Transaction Log File (ITLF) record that has already been reversed. The ISO Host Interface process translates this action code from external message response code 94.
914		NOT ABLE TO TRACE BACK TO ORIGINAL TXN The Voice Response Unit Master object generates this action code when it responds to a late network management request message from the RBSI-compliant device.
915		RECONCILIATION CUTOVER OR CHECKPOINT ERROR The ISO Host Interface process translates this action code from external message response code 95.
919		KEY SYNC ERROR The following objects generate this action code when a key synchronization error occurs during the translation of the PIN from the external PIN block to the internal PIN block: • Customer Service Interface object • Voice Response Unit Interface object
921		DENY, SECURITY ERROR The Authorization Combined Positive Customer object generates this action code when the Transaction Security object returns an error other than database error, key synchronization error, invalid PIN, invalid PIN length, or PIN tries exceeded.
923	✓	DENY, REQUEST IN PROGRESS The XPNET Billpay Server process generates this action code when an attempt is made to cancel a payment or transfer and it is already being processed by the Scheduled Transaction Initiator process.

Code	BP	Description
940	✓	DENY, DATABASE ERROR The Account object generates this action code when the Account Access object cannot access or update the PBF. The Customer Service Interface object generates this action code when the Financial Institution and Financial Institution Access objects cannot locate the customer service interface control parameters. These parameters are maintained in the IDF. The Authorization Combined Positive Customer object generates this action code in the following situations: • The Customer Access object cannot read the Customer Table (CSTT) or Customer/Account Relation Table (CACT). • The Transaction History object cannot obtain needed information from the Transaction History File (THF). • The Transaction Security object cannot obtain needed information from the database. The Pathway Billpay Server and XPNET Billpay Server processes generate this action code when they cannot retrieve, insert, or delete a row from a database table. The ISO Host Interface process translates this action code from external message response code O0.
941		DENY, INVALID CURRENCY CODE The following objects generate this action code when currency codes for the transaction, acquirer, and logical network are not the same: • Customer Service Interface object • Voice Response Unit Interface object The ISO Host Interface process translates this action code from external message response code O1.

Code	BP	Description
942	✓	DENY, AMOUNT FORMAT ERROR The XPNET Billpay Server process generates this action code when it receives a standard internal message (BSTM) that contains an incorrectly formatted amount field. The ISO Host Interface process translates this action code from external message response code P6.
943	✓	DENY, CUSTOMER VENDOR FORMAT ERROR The XPNET Billpay Server process generates this action code when it receives a standard internal message (BSTM) that contains an incorrectly formatted customer vendor number. The ISO Host Interface process translates this action code from external message response code P7.
944	✓	DENY, DATE FORMAT ERROR The XPNET Billpay Server process generates this action code when it receives a standard internal message (BSTM) that contains one or more incorrectly formatted dates. The ISO Host Interface process translates this action code from external message response code P8.
945	✓	DENY, NAME FORMAT ERROR The XPNET Billpay Server process generates this action code when it receives a standard internal message (BSTM) that contains one or more incorrectly formatted names. The ISO Host Interface process translates this action code from external message response code P9.
946	✓	DENY, ACCOUNT FORMAT ERROR The XPNET Billpay Server process generates this action code when it receives a standard internal message (BSTM) that contains invalid account information, which can include default account pair, service fee pair, *to* account data, or *from* account data. The ISO Host Interface process translates this action code from external message response code Q0.

Code	BP	Description
947	✓	DENY, RECURRING DATA ERROR The XPNET Billpay Server process generates this action code when it receives a standard internal message (BSTM) that contains incorrect required data for a recurring transaction. The ISO Host Interface process translates this action code from external message response code Q1.
948	✓	DENY, UPDATE NOT ALLOWED The XPNET Billpay Server process generates this action code when attempts are made to update information on recurring payments and transfers that cannot be updated once the first transaction has taken place. The ISO Host Interface process translates this action code from external message response code Q2.

Appendix C

BASE24 Remote Banking Default Processing Codes

The BASE24 Remote Banking products support both standard and nonstandard transactions. Standard transactions are defined by a default set of processing codes that is provided with the standard BASE24 Remote Banking products. A complete list of these default transaction codes are presented in this appendix.

Nonstandard transactions permit the customer or ACI to define transactions in addition to those defined in the set of default processing codes. Nonstandard transactions are categorized based on who is responsible for them, as follows:

- User-defined transactions can be defined by the customer. These transactions have a transaction code of U*x,* where *x* represents a value of 0 through 9 or A through Z. Refer to the topic "Configuring User-Defined Transactions" in section 3 for examples of user-defined transactions.
- Custom development transactions can be defined as part of customer-specific development work (i.e., RPQs) done by ACI personnel. These transactions can have a transaction code of P*x,* Q*x,* or R*x,* where *x* represents a value of 0 through 9 or A through Z.
- Distributor transactions can be defined by the ACI regional marketing groups. These transactions can have a transaction code of X*x,* Y*x,* or Z*x,* where *x* represents a value of 0 through 9 or A through Z.
- ACI-use-only transactions are reserved for use with products other than BASE24 Remote Banking. These transactions can have a transaction code of W*x,* where *x* represents a value of 0 through 9 or A through Z.

Each standard and nonstandard transaction is identified by a six-character processing code. The processing code contains a two-character transaction code, a two-character account type code for the first account involved in the transaction, and a two-character account type code for the second account involved in the transaction.

The following table lists the default processing codes provided with the BASE24 Remote Banking products. These processing codes appear in the Customer Allowed Transaction Table (CATT) and the Processing Code Definition File (PCDF). Processing codes available for each of the three BASE24 Remote Banking product offerings are identified with a check (✓) in the fourth through sixth columns of the table. Most transaction codes appear multiple times to include various account type combinations.

Values in the Account 1 and Account 2 columns of the table are based on default values defined in the Account Type Table File (ATT) ACI provides with the BASE24 Remote Banking products. There are two values on each line in these columns. The top value is used on file maintenance screens and reports. The bottom value is stored in BASE24 Remote Banking files and tables, is used by BASE24 Remote Banking processes, and appears in internal and external messages. The bottom values are provided for your convenience because the configuration examples in section 3 use them.

Processing Code						
Transaction Code	**Account 1 Type**	**Account 2 Type**	**TB**	**Enhanced TB**	**BP**	**Transaction Description**
30	NONE 00	NONE 00	✓			Available funds inquiry from an unspecified account type.
30	SAV 10	NONE 00	✓			Available funds inquiry from a savings account.
30	SAV 10	DDA 20	✓			Available funds inquiry from a savings account and a checking account.
30	DDA 20	NONE 00	✓			Available funds inquiry from a checking account.
30	DDA 20	SAV 10	✓			Available funds inquiry from a checking account and a savings account.
30	CR 30	NONE 00	✓			Available funds inquiry from a credit account.
30	LOCR 38	NONE 00	✓			Available funds inquiry from a line of credit account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
30	CD 58	NONE 00	✓			Available funds inquiry from a certificate account.
30	IRA 59	NONE 00	✓			Available funds inquiry from a retirement account.
30	NOW 90	NONE 00	✓			Available funds inquiry from an interest-bearing checking account.
30	CMRCL 9A	NONE 00	✓			Available funds inquiry from a commercial loan.
30	INSTL 9B	NONE 00	✓			Available funds inquiry from an installment loan.
30	MRTGL 9C	NONE 00	✓			Available funds inquiry from a mortgage loan.
3A	NONE 00	NONE 00	✓			Check clearance inquiry from an unspecified account type.
3A	DDA 20	NONE 00	✓			Check clearance inquiry from a checking account.
3A	NOW 90	NONE 00	✓			Check clearance inquiry from an interest-bearing checking account.
3B	NONE 00	NONE 00	✓			Last count of debit/credit transactions inquiry from an unspecified account type.
3B	SAV 10	NONE 00	✓			Last count of debit/credit transactions inquiry to a savings account.
3B	DDA 20	NONE 00	✓			Last count of debit/credit transactions inquiry for a checking account.
3B	CR 30	NONE 00	✓			Last count of debit/credit transactions inquiry for a credit account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	**Account 1 Type**	**Account 2 Type**				
3B	LOCR 38	NONE 00	✓			Last count of debit/credit transactions inquiry for a line of credit account.
3B	CD 58	NONE 00	✓			Last count of debit/credit transactions inquiry for a certificate account.
3B	IRA 59	NONE 00	✓			Last count of debit/credit transactions inquiry for a retirement account.
3B	NOW 90	NONE 00	✓			Last count of debit/credit transactions inquiry for an interest-bearing checking account.
3B	CMRCL 9A	NONE 00	✓			Last count of debit/credit transactions inquiry for a commercial loan.
3B	INSTL 9B	NONE 00	✓			Last count of debit/credit transactions inquiry for an installment loan.
3B	MRTGL 9C	NONE 00	✓			Last count of debit/credit transactions inquiry for a mortgage loan.
3C	NONE 00	NONE 00	✓			Last count of account by source transactions inquiry for an unspecified account type.
3C	SAV 10	NONE 00	✓			Last count of account by source transactions inquiry for a savings account.
3C	DDA 20	NONE 00	✓			Last count of account by source transactions inquiry for a checking account.
3C	CR 30	NONE 00	✓			Last count of account by source transactions inquiry for a credit account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
3C	LOCR 38	NONE 00	✓			Last count of account by source transactions inquiry for a line of credit account.
3C	CD 58	NONE 00	✓			Last count of account by source transactions inquiry for a certificate account.
3C	IRA 59	NONE 00	✓			Last count of account by source transactions inquiry for a retirement account.
3C	NOW 90	NONE 00	✓			Last count of account by source transactions inquiry for an interest-bearing checking account.
3C	CMRCL 9A	NONE 00	✓			Last count of account by source transactions inquiry for a commercial loan.
3C	INSTL 9B	NONE 00	✓			Last count of account by source transactions inquiry for an installment loan.
3C	MRTGL 9C	NONE 00	✓			Last count of account by source transactions inquiry for a mortgage loan.
3D	NONE 00	NONE 00	✓			Last count of check transactions inquiry for an unspecified account type.
3D	DDA 20	NONE 00	✓			Last count of check transactions inquiry for a checking account.
3D	NOW 90	NONE 00	✓			Last count of check transactions inquiry for an interest-bearing checking account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
3E	NONE 00	NONE 00	✓			Last count of debit transactions inquiry for an unspecified account type.
3E	SAV 10	NONE 00	✓			Last count of debit transactions inquiry for a savings account.
3E	DDA 20	NONE 00	✓			Last count of debit transactions inquiry for a checking account.
3E	CR 30	NONE 00	✓			Last count of debit transactions inquiry for a credit account.
3E	LOCR 38	NONE 00	✓			Last count of debit transactions inquiry for a line of credit account.
3E	CD 58	NONE 00	✓			Last count of debit transactions inquiry for a certificate account.
3E	IRA 59	NONE 00	✓			Last count of debit transactions inquiry for a retirement account.
3E	NOW 90	NONE 00	✓			Last count of debit transactions inquiry for an interest-bearing checking account.
3F	NONE 00	NONE 00	✓			Last count of credit transactions inquiry for an unspecified account type.
3F	SAV 10	NONE 00	✓			Last count of credit transactions inquiry for a savings account.
3F	DDA 20	NONE 00	✓			Last count of credit transactions inquiry for a checking account.
3F	CR 30	NONE 00	✓			Last count of credit transactions inquiry for a credit account.
3F	LOCR 38	NONE 00	✓			Last count of credit transactions inquiry for a line of credit account.

Processing Code			**TB**	**Enhanced TB**	**BP**	**Transaction Description**
Transaction Code	**Account 1 Type**	**Account 2 Type**				
3F	CD 58	NONE 00	✓			Last count of credit transactions inquiry for a certificate account.
3F	IRA 59	NONE 00	✓			Last count of credit transactions inquiry for a retirement account.
3F	NOW 90	NONE 00	✓			Last count of credit transactions inquiry for an interest-bearing checking account.
3F	CMRCL 9A	NONE 00	✓			Last count of credit transactions inquiry for a commercial loan.
3F	INSTL 9B	NONE 00	✓			Last count of credit transactions inquiry for an installment loan.
3F	MRTGL 9C	NONE 00	✓			Last count of credit transactions inquiry for a mortgage loan.
3G	NONE 00	NONE 00	✓			Last count of account transfer transactions inquiry for an unspecified account type.
3G	SAV 10	NONE 00	✓			Last count of account transfer transactions inquiry for a savings account.
3G	DDA 20	NONE 00	✓			Last count of account transfer transactions inquiry for a checking account.
3G	CR 30	NONE 00	✓			Last count of account transfer transactions inquiry for a credit account.
3G	LOCR 38	NONE 00	✓			Last count of account transfer transactions inquiry for a line of credit account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
3G	NOW 90	NONE 00	✓			Last count of account transfer transactions inquiry for an interest-bearing checking account.
3H	NONE 00	NONE 00			✓	Customer vendor list inquiry.
3J	NONE 00	NONE 00			✓	Scheduled payments list inquiry.
3K	NONE 00	NONE 00		✓	✓	Scheduled transfers list inquiry.
3L	NONE 00	NONE 00		✓	✓	History inquiry – payments and transfers.
3M	NONE 00	NONE 00		✓	✓	Scheduled transactions list inquiry.
3N	NONE 00	NONE 00	✓			Customer account list inquiry.
3P	NONE 00	NONE 00	✓			Multiple account balance inquiry.
3Q	NONE 00	NONE 00		✓	✓	Statement closing download for an unspecified account type.
3Q	SAV 10	NONE 00		✓	✓	Statement closing download for a savings account.
3Q	DDA 20	NONE 00		✓	✓	Statement closing download for a checking account.
3Q	CR 30	NONE 00		✓	✓	Statement closing download for a credit account.
3Q	LOCR 38	NONE 00		✓	✓	Statement closing download for a line of credit account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
3Q	NOW 90	NONE 00		✓	✓	Statement closing download for an interest-bearing checking account.
3Q	CMRCL 9A	NONE 00		✓	✓	Statement closing download for a commercial loan.
3Q	INSTL 9B	NONE 00		✓	✓	Statement closing download for an installment loan.
3Q	MRTGL 9C	NONE 00		✓	✓	Statement closing download for a mortgage loan.
3R	NONE 00	NONE 00		✓	✓	Statement download for an unspecified account type.
3R	SAV 10	NONE 00		✓	✓	Statement download for a savings account.
3R	DDA 20	NONE 00		✓	✓	Statement download for a checking account.
3R	CR 30	NONE 00		✓	✓	Statement download for a credit account.
3R	LOCR 38	NONE 00		✓	✓	Statement download for a line of credit account.
3R	NOW 90	NONE 00		✓	✓	Statement download for an interest-bearing checking account.
3R	CMRCL 9A	NONE 00		✓	✓	Statement download for a commercial loan.
3R	INSTL 9B	NONE 00		✓	✓	Statement download for an installment loan.
3R	MRTGL 9C	NONE 00		✓	✓	Statement download for a mortgage loan.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	**Account 1 Type**	**Account 2 Type**				
40	NONE 00	NONE 00	✓			Transfer – immediate between unspecified account types.
40	SAV 10	SAV 10	✓			Transfer – immediate from a savings account to a savings account.
40	SAV 10	DDA 20	✓			Transfer – immediate from a savings account to a checking account.
40	SAV 10	CR 30	✓			Transfer – immediate from a savings account to a credit account.
40	SAV 10	LOCR 38	✓			Transfer – immediate from a savings account to a line of credit account.
40	SAV 10	NOW 90	✓			Transfer – immediate from a savings account to an interest-bearing checking account.
40	SAV 10	CMRCL 9A	✓			Transfer – immediate from a savings account to a commercial loan.
40	SAV 10	INSTL 9B	✓			Transfer – immediate from a savings account to an installment loan.
40	SAV 10	MRTGL 9C	✓			Transfer – immediate from a savings account to a mortgage loan.
40	DDA 20	SAV 10	✓			Transfer – immediate from a checking account to a savings account.
40	DDA 20	DDA 20	✓			Transfer – immediate from a checking account to a checking account.
40	DDA 20	CR 30	✓			Transfer – immediate from a checking account to a credit account.
40	DDA 20	LOCR 38	✓			Transfer – immediate from a checking account to a line of credit account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
40	DDA 20	NOW 90	✓			Transfer – immediate from a checking account to an interest-bearing checking account.
40	DDA 20	CMRCL 9A	✓			Transfer – immediate from a checking account to a commercial loan.
40	DDA 20	INSTL 9B	✓			Transfer – immediate from a checking account to an installment loan.
40	DDA 20	MRTGL 9C	✓			Transfer – immediate from a checking account to a mortgage loan.
40	CR 30	SAV 10	✓			Transfer – immediate from a credit account to a savings account.
40	CR 30	DDA 20	✓			Transfer – immediate from a credit account to a checking account.
40	CR 30	NOW 90	✓			Transfer – immediate from a credit account to an interest-bearing checking account.
40	LOCR 38	SAV 10	✓			Transfer – immediate from a line of credit account to a savings account.
40	LOCR 38	DDA 20	✓			Transfer – immediate from a line of credit account to a checking account.
40	LOCR 38	NOW 90	✓			Transfer – immediate from a line of credit account to an interest-bearing checking account.
40	NOW 90	SAV 10	✓			Transfer – immediate from an interest-bearing checking account to a savings account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
40	NOW 90	DDA 20	✓			Transfer – immediate from an interest-bearing checking account to a checking account.
40	NOW 90	CR 30	✓			Transfer – immediate from an interest-bearing checking account to a credit account.
40	NOW 90	LOCR 38	✓			Transfer – immediate from an interest-bearing checking account to a line of credit account.
40	NOW 90	NOW 90	✓			Transfer – immediate from an interest-bearing checking account to an interest-bearing checking account.
40	NOW 90	CMRCL 9A	✓			Transfer – immediate from an interest-bearing checking account to a commercial loan.
40	NOW 90	INSTL 9B	✓			Transfer – immediate from an interest-bearing checking account to an installment loan.
40	NOW 90	MRTGL 9C	✓			Transfer – immediate from an interest-bearing checking account to a mortgage loan.
4A	NONE 00	NONE 00		✓	✓	Transfer – future between unspecified account types.
4A	SAV 10	SAV 10		✓	✓	Transfer – future from a savings account to a savings account.
4A	SAV 10	DDA 20		✓	✓	Transfer – future from a savings account to a checking account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
4A	SAV 10	CR 30		✓	✓	Transfer – future from a savings account to a credit account.
4A	SAV 10	LOCR 38		✓	✓	Transfer – future from a savings account to a line of credit account.
4A	SAV 10	NOW 90		✓	✓	Transfer – future from a savings account to an interest-bearing checking account.
4A	SAV 10	CMRCL 9A		✓	✓	Transfer – future from a savings account to a commercial loan.
4A	SAV 10	INSTL 9B		✓	✓	Transfer – future from a savings account to an installment loan.
4A	SAV 10	MRTGL 9C		✓	✓	Transfer – future from a savings account to a mortgage loan.
4A	DDA 20	SAV 10		✓	✓	Transfer – future from a checking account to a savings account.
4A	DDA 20	DDA 20		✓	✓	Transfer – future from a checking account to a checking account.
4A	DDA 20	CR 30		✓	✓	Transfer – future from a checking account to a credit account.
4A	DDA 20	LOCR 38		✓	✓	Transfer – future from a checking account to a line of credit account.
4A	DDA 20	NOW 90		✓	✓	Transfer – future from a checking account to an interest-bearing checking account.
4A	DDA 20	CMRCL 9A		✓	✓	Transfer – future from a checking account to a commercial loan.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	**Account 1 Type**	**Account 2 Type**				
4A	DDA 20	INSTL 9B		✓	✓	Transfer – future from a checking account to an installment loan.
4A	DDA 20	MRTGL 9C		✓	✓	Transfer – future from a checking account to a mortgage loan.
4A	CR 30	SAV 10		✓	✓	Transfer – future from a credit account to a savings account.
4A	CR 30	DDA 20		✓	✓	Transfer – future from a credit account to a checking account.
4A	CR 30	NOW 90		✓	✓	Transfer – future from a credit account to an interest-bearing checking account.
4A	LOCR 38	SAV 10		✓	✓	Transfer – future from a line of credit account to a savings account.
4A	LOCR 38	DDA 20		✓	✓	Transfer – future from a line of credit account to a checking account.
4A	LOCR 38	NOW 90		✓	✓	Transfer – future from a line of credit account to an interest-bearing checking account.
4A	NOW 90	SAV 10		✓	✓	Transfer – future from an interest-bearing checking account to a savings account.
4A	NOW 90	DDA 20		✓	✓	Transfer – future from an interest-bearing checking account to a checking account.
4A	NOW 90	CR 30		✓	✓	Transfer – future from an interest-bearing checking account to a credit account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	**Account 1 Type**	**Account 2 Type**				
4A	NOW 90	LOCR 38		✓	✓	Transfer – future from an interest-bearing checking account to a line of credit account.
4A	NOW 90	NOW 90		✓	✓	Transfer – future from an interest-bearing checking account to an interest-bearing checking account.
4A	NOW 90	CMRCL 9A		✓	✓	Transfer – future from an interest-bearing checking account to a commercial loan.
4A	NOW 90	INSTL 9B		✓	✓	Transfer – future from an interest-bearing checking account to an installment loan.
4A	NOW 90	MRTGL 9C		✓	✓	Transfer – future from an interest-bearing checking account to a mortgage loan.
4B	NONE 00	NONE 00		✓	✓	Transfer – recurring between unspecified account types.
4B	SAV 10	SAV 10		✓	✓	Transfer – recurring from a savings account to a savings account.
4B	SAV 10	DDA 20		✓	✓	Transfer – recurring from a savings account to a checking account.
4B	SAV 10	CR 30		✓	✓	Transfer – recurring from a savings account to a credit account.
4B	SAV 10	LOCR 38		✓	✓	Transfer – recurring from a savings account to a line of credit account.
4B	SAV 10	NOW 90		✓	✓	Transfer – recurring from a savings account to an interest-bearing checking account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
4B	SAV 10	CMRCL 9A		✓	✓	Transfer – recurring from a savings account to a commercial loan.
4B	SAV 10	INSTL 9B		✓	✓	Transfer – recurring from a savings account to an installment loan.
4B	SAV 10	MRTGL 9C		✓	✓	Transfer – recurring from a savings account to a mortgage loan.
4B	DDA 20	SAV 10		✓	✓	Transfer – recurring from a checking account to a savings account.
4B	DDA 20	DDA 20		✓	✓	Transfer – recurring from a checking account to a checking account.
4B	DDA 20	CR 30		✓	✓	Transfer – recurring from a checking account to a credit account.
4B	DDA 20	LOCR 38		✓	✓	Transfer – recurring from a checking account to a line of credit account.
4B	DDA 20	NOW 90		✓	✓	Transfer – recurring from a checking account to an interest-bearing checking account.
4B	DDA 20	CMRCL 9A		✓	✓	Transfer – recurring from a checking account to a commercial loan.
4B	DDA 20	INSTL 9B		✓	✓	Transfer – recurring from a checking account to an installment loan.
4B	DDA 20	MRTGL 9C		✓	✓	Transfer – recurring from a checking account to a mortgage loan.
4B	CR 30	SAV 10		✓	✓	Transfer – recurring from a credit account to a savings account.
4B	CR 30	DDA 20		✓	✓	Transfer – recurring from a credit account to a checking account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
4B	CR 30	NOW 90		✓	✓	Transfer – recurring from a credit account to an interest-bearing checking account.
4B	LOCR 38	SAV 10		✓	✓	Transfer – recurring from a line of credit account to a savings account.
4B	LOCR 38	DDA 20		✓	✓	Transfer – recurring from a line of credit account to a checking account.
4B	LOCR 38	NOW 90		✓	✓	Transfer – recurring from a line of credit account to an interest-bearing checking account.
4B	NOW 90	SAV 10		✓	✓	Transfer – recurring from an interest-bearing checking account to a savings account.
4B	NOW 90	DDA 20		✓	✓	Transfer – recurring from an interest-bearing checking account to a checking account.
4B	NOW 90	CR 30		✓	✓	Transfer – recurring from an interest-bearing checking account to a credit account.
4B	NOW 90	LOCR 38		✓	✓	Transfer – recurring from an interest-bearing checking account to a line of credit account.
4B	NOW 90	NOW 90		✓	✓	Transfer – recurring from an interest-bearing checking account to an interest-bearing checking account.
4B	NOW 90	CMRCL 9A		✓	✓	Transfer – recurring from an interest-bearing checking account to a commercial loan.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
4B	NOW 90	INSTL 9B		✓	✓	Transfer – recurring from an interest-bearing checking account to an installment loan.
4B	NOW 90	MRTGL 9C		✓	✓	Transfer – recurring from an interest-bearing checking account to a mortgage loan.
50	NONE 00	NONE 00			✓	Payment – immediate from an unspecified account type.
50	SAV 10	NONE 00			✓	Payment – immediate from a savings account.
50	DDA 20	NONE 00			✓	Payment – immediate from a checking account.
50	CR 30	NONE 00			✓	Payment – immediate from a credit account.
50	LOCR 38	NONE 00			✓	Payment – immediate from a line of credit account.
50	NOW 90	NONE 00			✓	Payment – immediate from an interest-bearing checking account.
5A	NONE 00	NONE 00			✓	Payment – future from an unspecified account type.
5A	SAV 10	NONE 00			✓	Payment – future from a savings account.
5A	DDA 20	NONE 00			✓	Payment – future from a checking account.
5A	CR 30	NONE 00			✓	Payment – future from a credit account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
5A	LOCR 38	NONE 00			✓	Payment – future from a line of credit account.
5A	NOW 90	NONE 00			✓	Payment – future from an interest-bearing checking account.
5B	NONE 00	NONE 00			✓	Payment – recurring from an unspecified account type.
5B	SAV 10	NONE 00			✓	Payment – recurring from a savings account.
5B	DDA 20	NONE 00			✓	Payment – recurring from a checking account.
5B	CR 30	NONE 00			✓	Payment – recurring from a credit account.
5B	LOCR 38	NONE 00			✓	Payment – recurring from a line of credit account.
5B	NOW 90	NONE 00			✓	Payment – recurring from an interest-bearing checking account.
90	NONE 00	NONE 00	✓			PIN change.
91	NONE 00	NONE 00	✓			PIN verify.
9A	NONE 00	NONE 00			✓	Schedule payment – immediate from an unspecified account type.
9A	SAV 10	NONE 00			✓	Schedule payment – immediate from a savings account.
9A	DDA 20	NONE 00			✓	Schedule payment – immediate from a checking account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
9A	CR 30	NONE 00			✓	Schedule payment – immediate from a credit account.
9A	LOCR 38	NONE 00			✓	Schedule payment – immediate from a line of credit account.
9A	NOW 90	NONE 00			✓	Schedule payment – immediate from an interest-bearing checking account.
9B	NONE 00	NONE 00			✓	Schedule payment – future from an unspecified account type.
9B	SAV 10	NONE 00			✓	Schedule payment – future from a savings account.
9B	DDA 20	NONE 00			✓	Schedule payment – future from a checking account.
9B	CR 30	NONE 00			✓	Schedule payment – future from a credit account.
9B	LOCR 38	NONE 00			✓	Schedule payment – future from a line of credit account.
9B	NOW 90	NONE 00			✓	Schedule payment – future from an interest-bearing checking account.
9C	NONE 00	NONE 00			✓	Schedule payment – recurring from an unspecified account type.
9C	SAV 10	NONE 00			✓	Schedule payment – recurring from a savings account.
9C	DDA 20	NONE 00			✓	Schedule payment – recurring from a checking account.
9C	CR 30	NONE 00			✓	Schedule payment – recurring from a credit account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
9C	LOCR 38	NONE 00			✓	Schedule payment – recurring from a line of credit account.
9C	NOW 90	NONE 00			✓	Schedule payment – recurring from an interest-bearing checking account.
9D	NONE 00	NONE 00		✓	✓	Schedule transfer – future between unspecified account types.
9D	SAV 10	SAV 10		✓	✓	Schedule transfer – future from a savings account to a savings account.
9D	SAV 10	DDA 20		✓	✓	Schedule transfer – future from a savings account to a checking account.
9D	SAV 10	CR 30		✓	✓	Schedule transfer – future from a savings account to a credit account.
9D	SAV 10	LOCR 38		✓	✓	Schedule transfer – future from a savings account to a line of credit account.
9D	SAV 10	NOW 90		✓	✓	Schedule transfer – future from a savings account to an interest-bearing checking account.
9D	SAV 10	CMRCL 9A		✓	✓	Schedule transfer – future from a savings account to a commercial loan.
9D	SAV 10	INSTL 9B		✓	✓	Schedule transfer – future from a savings account to an installment loan.
9D	SAV 10	MRTGL 9C		✓	✓	Schedule transfer – future from a savings account to a mortgage loan.
9D	DDA 20	SAV 10		✓	✓	Schedule transfer – future from a checking account to a savings account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
9D	DDA 20	DDA 20		✓	✓	Schedule transfer – future from a checking account to a checking account.
9D	DDA 20	CR 30		✓	✓	Schedule transfer – future from a checking account to a credit account.
9D	DDA 20	LOCR 38		✓	✓	Schedule transfer – future from a checking account to a line of credit account.
9D	DDA 20	NOW 90		✓	✓	Schedule transfer – future from a checking account to an interest-bearing checking account.
9D	DDA 20	CMRCL 9A		✓	✓	Schedule transfer – future from a checking account to a commercial loan.
9D	DDA 20	INSTL 9B		✓	✓	Schedule transfer – future from a checking account to an installment loan.
9D	DDA 20	MRTGL 9C		✓	✓	Schedule transfer – future from a checking account to a mortgage loan.
9D	CR 30	SAV 10		✓	✓	Schedule transfer – future from a credit account to a savings account.
9D	CR 30	DDA 20		✓	✓	Schedule transfer – future from a credit account to a checking account.
9D	CR 30	NOW 90		✓	✓	Schedule transfer – future from a credit account to an interest-bearing checking account.
9D	LOCR 38	SAV 10		✓	✓	Schedule transfer – future from a line of credit account to a savings account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
9D	LOCR 38	DDA 20		✓	✓	Schedule transfer – future from a line of credit account to a checking account.
9D	LOCR 38	NOW 90		✓	✓	Schedule transfer – future from a line of credit account to an interest-bearing checking account.
9D	NOW 90	SAV 10		✓	✓	Schedule transfer – future from an interest-bearing checking account to a savings account.
9D	NOW 90	DDA 20		✓	✓	Schedule transfer – future from an interest-bearing checking account to a checking account.
9D	NOW 90	CR 30		✓	✓	Schedule transfer – future from an interest-bearing checking account to a credit account.
9D	NOW 90	LOCR 38		✓	✓	Schedule transfer – future from an interest-bearing checking account to a line of credit account.
9D	NOW 90	NOW 90		✓	✓	Schedule transfer – future from an interest-bearing checking account to an interest-bearing checking account.
9D	NOW 90	CMRCL 9A		✓	✓	Schedule transfer – future from an interest-bearing checking account to a commercial loan.
9D	NOW 90	INSTL 9B		✓	✓	Schedule transfer – future from an interest-bearing checking account to an installment loan.
9D	NOW 90	MRTGL 9C		✓	✓	Schedule transfer – future from an interest-bearing checking account to a mortgage loan.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
9E	NONE 00	NONE 00		✓	✓	Schedule transfer – recurring between unspecified account types.
9E	SAV 10	SAV 10		✓	✓	Schedule transfer – recurring from a savings account to a savings account.
9E	SAV 10	DDA 20		✓	✓	Schedule transfer – recurring from a savings account to a checking account.
9E	SAV 10	CR 30		✓	✓	Schedule transfer – recurring from a savings account to a credit account.
9E	SAV 10	LOCR 38		✓	✓	Schedule transfer – recurring from a savings account to a line of credit account.
9E	SAV 10	NOW 90		✓	✓	Schedule transfer – recurring from a savings account to an interest-bearing checking account.
9E	SAV 10	CMRCL 9A		✓	✓	Schedule transfer – recurring from a savings account to a commercial loan.
9E	SAV 10	INSTL 9B		✓	✓	Schedule transfer – recurring from a savings account to an installment loan.
9E	SAV 10	MRTGL 9C		✓	✓	Schedule transfer – recurring from a savings account to a mortgage loan.
9E	DDA 20	SAV 10		✓	✓	Schedule transfer – recurring from a checking account to a savings account.
9E	DDA 20	DDA 20		✓	✓	Schedule transfer – recurring from a checking account to a checking account.
9E	DDA 20	CR 30		✓	✓	Schedule transfer – recurring from a checking account to a credit account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
9E	DDA 20	LOCR 38		✓	✓	Schedule transfer – recurring from a checking account to a line of credit account.
9E	DDA 20	NOW 90		✓	✓	Schedule transfer – recurring from a checking account to an interest-bearing checking account.
9E	DDA 20	CMRCL 9A		✓	✓	Schedule transfer – recurring from a checking account to a commercial loan.
9E	DDA 20	INSTL 9B		✓	✓	Schedule transfer – recurring from a checking account to an installment loan.
9E	DDA 20	MRTGL 9C		✓	✓	Schedule transfer – recurring from a checking account to a mortgage loan.
9E	CR 30	SAV 10		✓	✓	Schedule transfer – recurring from a credit account to a savings account.
9E	CR 30	DDA 20		✓	✓	Schedule transfer – recurring from a credit account to a checking account.
9E	CR 30	NOW 90		✓	✓	Schedule transfer – recurring from a credit account to an interest-bearing checking account.
9E	LOCR 38	SAV 10		✓	✓	Schedule transfer – recurring from a line of credit account to a savings account.
9E	LOCR 38	DDA 20		✓	✓	Schedule transfer – recurring from a line of credit account to a checking account.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
9E	LOCR 38	NOW 90		✓	✓	Schedule transfer – recurring from a line of credit account to an interest-bearing checking account.
9E	NOW 90	SAV 10		✓	✓	Schedule transfer – recurring from an interest-bearing checking account to a savings account.
9E	NOW 90	DDA 20		✓	✓	Schedule transfer – recurring from an interest-bearing checking account to a checking account.
9E	NOW 90	CR 30		✓	✓	Schedule transfer – recurring from an interest-bearing checking account to a credit account.
9E	NOW 90	LOCR 38		✓	✓	Schedule transfer – recurring from an interest-bearing checking account to a line of credit account.
9E	NOW 90	NOW 90		✓	✓	Schedule transfer – recurring from an interest-bearing checking account to an interest-bearing checking account.
9E	NOW 90	CMRCL 9A		✓	✓	Schedule transfer – recurring from an interest-bearing checking account to a commercial loan.
9E	NOW 90	INSTL 9B		✓	✓	Schedule transfer – recurring from an interest-bearing checking account to an installment loan.
9E	NOW 90	MRTGL 9C		✓	✓	Schedule transfer – recurring from an interest-bearing checking account to a mortgage loan.
9F	NONE 00	NONE 00			✓	Scheduled payment delete.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
9G	NONE 00	NONE 00		✓	✓	Scheduled transfer delete.
9H	NONE 00	NONE 00			✓	Scheduled payment update. Allows customers to change the date, amount, period type, number of periods, or to skip the next payment.
9J	NONE 00	NONE 00		✓	✓	Scheduled transfer update. Allows customers to change the date, amount, period type, number of periods, or to skip the next transfer.
9K	NONE 00	NONE 00		✓	✓	Customer add.
9L	NONE 00	NONE 00		✓	✓	Customer inquiry.
9M	NONE 00	NONE 00		✓	✓	Customer update.
9N	NONE 00	NONE 00			✓	Customer vendor add.
9P	NONE 00	NONE 00			✓	Customer vendor deactivate.
9Q	NONE 00	NONE 00			✓	Customer vendor update.
9R	NONE 00	NONE 00			✓	Master vendor list inquiry.
9S	NONE 00	NONE 00			✓	Master vendor add.

Processing Code			TB	Enhanced TB	BP	Transaction Description
Transaction Code	Account 1 Type	Account 2 Type				
9T	NONE 00	NONE 00			✓	Customer vendor request mailing.
9V	NONE 00	NONE 00		✓	✓	Loan application.
9W	NONE 00	NONE 00		✓	✓	Miscellaneous request.
9X	NONE 00	NONE 00		✓	✓	History inquiry – all transactions.
A0	DDA 20	NONE 00	✓			Check stop payment add for a checking account.
A0	LOCR 38	NONE 00	✓			Check stop payment add for a line of credit account.
A0	NOW 90	NONE 00	✓			Check stop payment add for an interest-bearing checking account.
A0	CMRCL 9A	NONE 00	✓			Check stop payment add for a commercial loan.
A0	INSTL 9B	NONE 00	✓			Check stop payment add for an installment loan.
A0	MRTGL 9C	NONE 00	✓			Check stop payment add for a mortgage loan.
B1	NONE 00	NONE 00		✓	✓	Bank list inquiry.
B2	NONE 00	NONE 00		✓	✓	Bank branch list inquiry.
B4	NONE 00	NONE 00		✓	✓	Miscellaneous requests inquiry.

Processing Code						
Transaction Code	**Account 1 Type**	**Account 2 Type**	**TB**	**Enhanced TB**	**BP**	**Transaction Description**
B5	NONE 00	NONE 00		✓	✓	Customer account add.
B6	NONE 00	NONE 00		✓	✓	Customer account activate.
B7	NONE 00	NONE 00		✓	✓	Customer account deactivate.
B8	NONE 00	NONE 00		✓	✓	Customer account delete.
B9	NONE 00	NONE 00		✓	✓	Customer account update.
BB	NONE 00	NONE 00		✓	✓	Customer activate.
BC	NONE 00	NONE 00		✓	✓	Customer deactivate.
BD	NONE 00	NONE 00			✓	Customer vendor activate.
BE	NONE 00	NONE 00			✓	Customer vendor mask verify inquiry.

Appendix D

Object-Oriented Design

This appendix describes object-oriented design, the processes that use object-oriented design, and the objects themselves. It also provides operator instructions for building extended memory tables, warmbooting extended memory tables and objects, and sending commands for server control.

Object-Oriented Design Processes

The following processes in the BASE24 Remote Banking transaction path have been developed using what is known as object-oriented design (OOD):

- Integrated Endpoint process
- Integrated Authorization Server process
- Interface Master process

Object-oriented design divides the processing to be done according to the data involved and the action performed on that data. Each process designed this way contains one or more objects. Only one object has access to a source of data (e.g., only the Financial Institution Access object has access to the Institution Definition File). Only one object can perform a specific action (e.g., only the Authorization Combined Positive Customer object can approve a transaction). An object appears no more than once in each process.

The key to object-oriented design is that each object asks other objects for data and asks other objects to perform actions on data. Each object needs to know only which object can provide the needed data or perform the necessary actions. One object does not need to know how the other objects obtain the data or perform the action. Refer to the message flows in section 4 for examples of how objects interact when processing BASE24 Remote Banking transactions.

ACI uses the name integrated transaction services (ITS) to identify its products that use object-oriented design and uses the name procedural to identify its products that do not use object-oriented design.

The objects used in BASE24 Remote Banking processes are divided into three groups: kernel, integrated endpoint, and core. The objects that make up each of these groups are described on the following pages.

Kernel Objects

Kernel objects perform functions needed by all object-oriented processes. Each process that is developed using object-oriented design contains the kernel objects. The following table shows the kernel objects that are used with the BASE24 Remote Banking products and the processes in which each object is used. The table includes Integrated Endpoint processes (IEP), Integrated Authorization Server processes (IAS), Interface Master processes (IMP), End-of-Period processes (EOPP), and Log Permission processes (LOGP).

Object	IEP	IAS	IMP	EOPP	LOGP
Application Network Services	✓	✓	✓	✓	✓
Class	✓	✓	✓	✓	✓
Event	✓	✓	✓	✓	✓
Log Permission					✓
Process Control	✓	✓	✓	✓	✓
Trace	✓	✓	✓	✓	✓
XPNET Interface	✓	✓	✓	✓	✓

Integrated Endpoint Objects

Integrated endpoint objects maintain transaction context and the timer associated with the transaction context for certain transaction phases. They also perform message-based routing. Each Integrated Endpoint process contains kernel objects and integrated endpoint objects. Integrated endpoint objects used with the BASE24 Remote Banking products are as follows:

- Context Manager
- Destination
- Endpoint Class Handler Services
- Message ISO

Integrated Endpoint processes used with BASE24 Remote Banking products also can contain the following core objects:

- Interface Manager
- Voice Response Unit Endpoint Class Handler

These objects are considered core objects because they are application specific. The Interface Manager object is used in the Integrated Endpoint process and the Integrated Authorization Server process. The Voice Response Unit Endpoint Class Handler object is used in the Integrated Endpoint process.

Core Objects

Core objects, also known as integrated server core objects (IS core objects), provide all actions and file access needed to support integrated server applications such as the BASE24 Remote Banking products. Integrated Endpoint processes (IEP), Integrated Authorization Server processes (IAS), Interface Master processes (IMP), and End-of-Period processes (EOPP) use core objects. Each object-oriented process contains the kernel and core objects necessary to perform the desired functions. Refer to the table provided earlier in this appendix for the kernel objects used in each BASE24 Remote Banking object-oriented process. The following table shows the core objects that are used with the BASE24 Remote Banking products and the processes in which each object is used.

Object	IEP	IAS	IMP	EOPP
Account		✓		
Account Access		✓		
Authorization Combined Positive Customer		✓		
Authorization Combined Utility		✓		
Authorization Security Access		✓		
Customer		✓		
Customer Access		✓		
Customer Service Interface		✓*		
End-of-Period Process				✓
Financial Institution		✓		
Financial Institution Access		✓		
Interface Manager	✓	✓		
Interface Master Services			✓	
Internal Message Translation Interface		✓†		
Multiple Currency		✓		

Object	IEP	IAS	IMP	EOPP
Router		✓		
Transaction		✓		
Transaction History		✓		
Transaction Security		✓	✓	
Voice Response Unit Endpoint Class Handler	✓			
Voice Response Unit Interface		✓*		
Voice Response Unit Master			✓	

* An Integrated Authorization Server process contains one or more of the following objects:

- A Customer Service Interface object is used for transactions initiated at customer service representative (CSR) terminals.
- A Voice Response Unit Interface object is used for transactions acquired from RBSI-compliant devices managed by the BASE24 system or a host.

† An Integrated Authorization Server process contains the Internal Message Translation Interface object only when used with the ISO Host Interface process or the BASE24-billpay product.

Integrated Endpoint processes, Integrated Authorization Server processes, and Interface Master processes also contain the Message ISO object, which is an integrated endpoint object.

Issuer and Acquirer Objects

Object-oriented processes like the ones developed for a BASE24 Remote Banking system use a series of objects to perform transaction processing. The following objects are identified as issuers or acquirers, depending on where the transaction originates and who is to authorize the transaction:

- The Voice Response Unit Interface object is an acquirer object for transactions initiated at an RBSI-compliant device.
- The Customer Service Interface object is an acquirer object for transactions initiated at a BASE24 Remote Banking customer service representative (CSR) terminal.
- The Authorization Combined Positive Customer object is an issuer object for transactions authorized by the BASE24-telebanking product.
- The Internal Message Translation Interface object is an issuer object for transactions authorized by the host and an acquirer object for transactions initiated at the host.

Kernel Object Descriptions

Kernel objects perform functions needed by all integrated transaction services (ITS) processes, including process control, event logging, memory acquisition, communicating with outside agencies, and logging event messages. Each of the kernel objects is described briefly below. Refer to the ***BASE24 ITS Kernel Objects Reference Manual*** for additional information about the function and operation of each kernel object.

Application Network Services object	This object contains the main procedure for an integrated transaction services process. During system startup, the Application Network Services object calls all other objects in the process, directing them to initialize. The Application Network Services object allocates memory for all objects in the process to use. The Application Network Services object functions as the manager of an integrated transaction services process.
Class object	This object determines the object ID associated with a specific source class value for routing purposes. When the transport layer object in an integrated transaction services process reads an incoming message, the Class object is called to read the Source Class Map extended memory table to find the appropriate message handler object for the message.
Event object	This object provides an interface for the integrated transaction services process to the Tandem Event Management Service facility for logging purposes. The Event object provides a buffer and interface to the Event Management Service facility.
Log Permission object	This object resides in Log Permission processes only. The sole purpose of this object is to determine whether a particular event message should be logged or suppressed. The Log Permission object manages event suppression at the network level.
Process Control object	This object receives messages from an Integrated Server Control File (ISCF) and forwards them to the appropriate object. The ISCF is the mechanism by which process control commands are delivered to the objects within an integrated transaction services process.

Trace object	This object is part of the general trace facility available to all objects. The Trace object uses trace utilities to collect information at crucial points in the processing of the message. This information can be placed in a file or displayed on a terminal. Examples of this information include the following: • Selected pieces of the Internal Transaction Data (ITD) • The entire ITD • Data read from or written to files • Contents of external messages For object-oriented processes, the trace facility replaces the audit facility provided by the XPNET process. The audit facility documents messages as they move from process to process. Since most BASE24-telebanking processing is done within a single process, the XPNET audit facility is of limited benefit.
XPNET Interface object	This object receives messages through $RECEIVE for processes running under control of the XPNET process.

Integrated Endpoint Object Descriptions

The Integrated Endpoint process resides between the XPNET process and an application. The Integrated Endpoint process is an object-oriented process developed under the ACI integrated transaction services environment. The Integrated Endpoint process maintains transaction context and the timer associated with the transaction context for certain transaction phases. The process also determines the next destination of the transaction based on a combination of the message source (source class) and the message type (message class), maintains message counts, and manages network overload. Integrated endpoint objects are issued by ACI as object code only. Each of the integrated endpoint objects used with the BASE24-telebanking product is described briefly below. Refer to the ***BASE24 IEP Objects Reference Manual*** for additional information about the function and operation of each integrated endpoint object.

Context Manager object	This object stores transaction context during the life of a transaction. The object can store a copy of certain response messages received from an Integrated Server process and sets a timer associated with the response. BASE24-telebanking messages that involve transaction context are dynamic key management messages and transactions that include multiple account selection. Both the copy of the message and the timer associated with the message are considered to be context. The context is maintained until an acknowledgment is received for the response.
Destination object	This object maintains destination data for messages routed through the Integrated Endpoint process. The Destination object can be called by any object within the Integrated Endpoint process that needs to know the destination for a particular message. When called, the Destination object retrieves the destination for the source class and message class associated with the message from the Destination File extended memory table. More than one destination can exist for a particular source class and message class combination if broadcast routing is being used or if any of the message class field components contain wildcard values.

Endpoint Class Handler Services object	This object provides all the common services required by integrated endpoint objects as well as external application processes. The Endpoint Class Handler Services object provides the following services: • External message formatting • Measure counter logic • Timer management • Message destination determination by the Destination object • Retrieval and storage of context by the Context Manager object
Message ISO object	This object expands and compresses bit-mapped ISO messages to or from any specified fixed structure. The ISO message format used is defined by the object that calls the Message ISO object. Message expansion is the conversion of the bit-mapped ISO message into a fixed message structure that application objects can interpret. Within the Integrated Endpoint process, any acquirer or issuer Endpoint Class Handler object can call the Message ISO object to expand a message. Message compression is the conversion of the fixed message structure of a transaction response into a bit-mapped ISO message structure. Only Message ISO objects residing within application processes can perform message compression. It is never performed by Message ISO objects residing in the Integrated Endpoint process. The Message ISO object is typically called by the Endpoint Class Handler object. **Note:** The BASE24-telebanking Integrated Authorization Server process and Interface Master process also contain the Message ISO object. The Message ISO object in these processes performs message compression and expansion.

Core Object Descriptions

Core objects provide all actions and file or table access needed to support an integrated server application such as the BASE24-telebanking product. The BASE24-telebanking product uses the following core objects.

Account object	This object maintains account balance information, maintains account usage totals, and determines whether funds are available or account usage limits have been exceeded. This object checks and clears authorization holds that may exist for an account, as well as considering an account's overdraft limit or backup account. This object impacts account balances on response, advice, and reversal transactions. The Account object is called by the Authorization Combined Positive Customer object and calls the Account Access object to read or update records in the Positive Balance File (PBF).
Account Access object	This object reads and updates the Positive Balance File (PBF) for the Account object. This object allows database details such as file name and record format to be invisible to the Account object. The Account object calls the Account Access object to read or update records in the PBF.
Authorization Combined Positive Customer object	This object authorizes transactions using the BASE24 database. It also preauthorizes transactions and updates the BASE24 database when transactions are authorized on the host. This object can call several other objects in order to obtain the information or processing necessary to authorize a transaction. The Authorization Combined Positive Customer object can be called by the following objects: • Any acquirer object • Any issuer object • Router object
Authorization Combined Utility object	This object performs denial processing when the transaction cannot be sent to the authorizer because of a configuration error or because the destination is unavailable. The Router object can call the Authorization Combined Utility object.

Authorization Security Access object	This object reads and updates the Key Authorization File extended memory table for the Transaction Security object. This object allows database details such as extended memory table format to be invisible to the Transaction Security object. The Transaction Security object calls the Authorization Security Access object to read or update the Key Authorization File extended memory table.
Customer object	This object maintains customer information during transaction processing by calling the Customer Access object. The following objects can call the Customer object: • Any acquirer object • Authorization Combined Positive Customer object
Customer Access object	This object reads and updates information stored in the Customer Table (CSTT). It reads, but does not update, information stored in the Customer/Account Relation Table (CACT) and customer extended memory table. The customer extended memory table contains information from the Customer Allowed Transaction Table (CATT). This object allows details such as SQL table and extended memory table formats to be invisible to the Customer object. The Customer object calls the Customer Access object to read and update the customer information.
Customer Service Interface object	This object maps data between the message from a BASE24 Remote Banking Customer Service Representative (CSR) terminal and the Internal Transaction Data (ITD). The XPNET object calls the Customer Service Interface object when a transaction is initiated at a CSR terminal.

End-of-Period Process object	This object performs financial institution cutover and logical network cutover. It also creates and purges ITS Transaction Log Files (ITLFs), Exception Log Files (ELFs), and report obey files. The End-of-Period Process object resides in a separate process (the End-of-Period process) that does not perform transaction processing. **Note:** The BASE24-telebanking End-of-Period process has been developed using object-oriented design. However, this process is not part of the transaction path and is not discussed in this manual. Refer to the ***BASE24 Remote Banking End-of-Period Processing and Reporting Manual*** for information about the End-of-Period process and its use.
Financial Institution object	This object maintains financial institution data such as business dates, usage period dates, the route profile value used to read the institution routing extended memory table, and the group used to select PIN verification information from the Key Authorization File extended memory table. The Financial Institution object calls the Financial Institution Access object to read the Institution Definition File extended memory table. The Financial Institution object can be called by any core object that needs financial institution data.
Financial Institution Access object	This object reads the Institution Definition File extended memory table for the Financial Institution object. This object allows details such as the extended memory table format to be invisible to the Financial Institution object. The Financial Institution object calls the Financial Institution Access object to read the Institution Definition File extended memory table.
Interface Manager object	This object provides functions that are common across interface objects (e.g., the Customer Service Interface object and Voice Response Unit Interface object). These functions include writing records to the Exception Log File (ELF) and communicating with the Interface Master process. The interface objects call the Interface Manager object.

Interface Master Services object	This object provides services to the Voice Response Unit Master object. Services include station management, managing counters and thresholds associated with dynamic key management, and managing the Exception Log File (ELF).
Internal Message Translation Interface object	This object maps data between the Internal Transaction Data (ITD) and the Telebanking Standard Internal Message (BSTM). Data passed to and from the BASE24 ISO Host Interface process must be in the BSTM. The following objects can call the Internal Message Translation Interface object: • The Router object calls this object when a message is destined for the host. • The XPNET Interface object calls this object when a request is received from the host to be authorized by BASE24 or a response is returned from the host. • The Customer Service Interface object and Voice Response Unit Interface object can call this object to process advice and reversal messages.
Multiple Currency object	This object contains a shell for multiple currency conversion processing. Multiple currency currently is not supported.
Router object	This object determines authorization destinations using information contained in the institution routing extended memory table. The institution routing extended memory table contains information from the Processing Code Definition File (PCDF) and the Institution Routing Configuration File (IRCF). The Voice Response Unit Interface object calls the Router object to route the transaction to the primary issuer object. The Internal Message Translation Interface object calls the Router object to route a host-acquired transaction to the primary issuer object or to reroute the transaction when the issuer destination is unavailable.
Transaction object	This object maintains the Internal Transaction Data (ITD) and logs transactions to the ITS Transaction Log File (ITLF). The Transaction object is called by all objects that need access to the ITD.

Transaction History object	This object reads the Transaction History File (THF) when a transaction history inquiry is authorized on the BASE24 system. The Authorization Combined Positive Customer object calls the Transaction History object.
Transaction Security object	This object translates PINs, verifies PINs, and generates new PIN keys. This object performs these functions itself or interfaces with hardware security modules that perform these functions. The following objects can call the Transaction Security object: • Authorization Combined Positive Customer object • Customer Service Interface object • Voice Response Unit Master object • Voice Response Unit Interface object
Voice Response Unit Endpoint Class Handler object	This object determines the message type and message class of transactions originating at an RBSI-compliant device. This object also performs network overload management processing. The Voice Response Unit Endpoint Class Handler object calls the Endpoint Class Handler Services object.
Voice Response Unit Interface object	This object maps data between the BASE24 system and the RBSI-compliant device. The Voice Response Unit Interface object acquires transactions that use the ISO 8583-1993 standard.
Voice Response Unit Master object	This object performs dynamic key management and station management for all Voice Response Unit Interface objects.

Extended Memory Table Build Utility

The Extended Memory Table Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom-designed tables. Refer to the ***BASE24 ITS Kernel Configuration Manual*** for additional information about the utility**.**

The Extended Memory Table Build utility can be executed from the Configuration Control Server (CCS) screen or from a TACL prompt. Refer to the ***BASE24 ITS Kernel Files Maintenance Manual*** for a description of the CCS screens.

You can run the utility at a high or low process identification number (PIN) when you start it from a TACL prompt. The utility always runs at a low PIN when started from the CCS screens.

Supported Extended Memory Tables

This utility supports the following extended memory tables used by BASE24 Remote Banking products:

- Customer extended memory table
- Destination File extended memory table
- Event suppression extended memory table
- Institution Definition File extended memory table
- Institution routing extended memory table
- Interface extended memory table
- Interface Security File extended memory table
- Internal Message Translation File extended memory table
- Key Authorization File extended memory table
- Source Class Map File extended memory table
- Source symbolic map extended memory table
- Transaction history file extended memory table
- Transaction log file extended memory table

Refer to section 1 for a description of each extended memory table used by BASE24 Remote Banking products.

TACL Startup Command

The startup message used to start the Extended Memory Table Build utility from a TACL prompt defines the extended memory tables which are to be built. The format of the Extended Memory Table Build startup message sent to the Extended Memory Table Build program is derived from the RUN command shown below.

Syntax

[RUN] [$*location*] EMTBLO/ [HIGHPIN { ON | OFF }] [OUT $*printer*/]
LCONF *file-name*, *emt-name* [*,emt-name*]

RUN — An optional keyword for the RUN command. If this keyword is included in the command, either the Extended Memory Table Build utility must be located on the current volume and subvolume or the location variable must be included in the command. If this keyword is not included in the command, the Extended Memory Table Build utility must be located on a volume and subvolume included in the PMSEARCHLIST TACL variable.

$location — Specifies the location of the Extended Memory Table Build utility. This variable, which can include the volume, subvolume, or both, is required if the Extended Memory Table Build utility is not on the current volume or subvolume and the location of the Extended Memory Table Build utility is not included in the PMSEARCHLIST TACL variable. If this variable is included in the command, the keyword RUN must also be included in the command. The Extended Memory Table Build utility is typically located on the OC13OBJ subvolume.

HIGHPIN — An optional variable for the RUN command. If this variable is set to the value ON, the parent TACL process runs at a high PIN, and other HIGHPIN options are set appropriately, the utility will run at a high PIN. If this variable is not included or is set to the value OFF, and other HIGHPIN options are set appropriately, the utility will run at a low PIN. For more information on how different HIGHPIN options affect whether a new process runs at a high or low PIN, refer to the ***TACL Reference Manual***.

printer — Specifies the spooler location to which reports are sent. If you do not specify a spooler location, it defaults to $S.#HOLD.

file-name — Specifies the fully qualified name of the Logical Network Configuration File (LCONF) containing the EMT Build utility assigns and params.

emt-name — Specifies the name of the extended memory table to be built. The RUN statement can contain any number of extended memory table names, from a single extended memory table name to all of the valid extended memory table names. Each name must be separated by a comma.

The valid names for extended memory tables currently used by BASE24 Remote Banking products are as follows:

CUSTEMT = Customer extended memory table
DESTEMT = Destination File extended memory table
ESEMT = Event suppression extended memory table
IDFEMT = Institution Definition File extended memory table
IMTEMT = Internal Message Translation File extended memory table
INTFEMT = Interface extended memory table
IRTEEMT = Institution routing extended memory table
ISECEMT = Interface Security File extended memory table
KEYAEMT = Key Authorization File extended memory table
SCMEMT = Source Class Map File extended memory table
SSMEMT = Source symbolic map extended memory table
THFEMT = Transaction history file extended memory table
TLFEMT = Transaction log file extended memory table

Example

```
RUN \b24.$data.oc13obj.EMTBLO/HIGHPIN ON, OUT $s.#emtb/LCONF
     \b24.$data.pro1exec.l1conf, CUSTEMT, IDFEMT, IRTEEMT
```

This example command builds the customer extended memory table, the Institution Definition File extended memory table, and the institution routing extended memory table. The Extended Memory Table Build utility runs at a high PIN and reports are sent to the $S. #EMTB spooler location.

Extended Memory Table Utility Reports

The Extended Memory Table Build utility prints a Hash Table Summary report page for each extended memory table it builds, with the following exceptions:

- Interface extended memory table
- Transaction history file extended memory table

- Transaction log file extended memory table

The formats for each of these reports are shown on the following pages. Refer to the ***BASE24 ITS Kernel Configuration Manual*** for samples of the other pages generated by the Extended Memory Table Build utility.

Interface Extended Memory Table Summary Page

The Interface Extended Memory Table Summary report page is printed in summary of an attempt to build the interface extended memory table. This report page lists the source files used to build the table, the table size, the total number of data records, and totals of various interface entities included in the table.

```
BASE24            EXTENDED MEMORY TABLE BUILD REPORT            PAGE: 1


     Table Type:                 INTFEMT
     Source File:                \SYS.$REL5D.COM1DATA.ADRT
     Source File:                \SYS.$REL5D.COM1DATA.EMF
     Source File:                \SYS.$REL5D.COM1DATA.IFCF
     Source File:                \SYS.$REL5D.COM1DATA.IST
     Source File:                \SYS.$REL5D.COM1DATA.OBJ
     EMT Swap File:              \SYS.$REL5D.COM1DATA.INTFEMT

     Total Size:                 158354 bytes

     Total Data Records:         20

     Total Interface Objects:     2
     Total Destination Groups:    5
     Total Interface Instances:  21
     Total IMPs:                  3
     Total Stations:             20
```

Table Type — The acronym for the extended memory table being built.

Source File — The fully qualified name of a source file used to build this extended memory table.

EMT Swap File — The fully qualified location of the swap file used to build this extended memory table. This is the name of the actual extended memory table file on disk. The swap file is used to swap out records from disk as they are added to the memory table.

Total Size — The total size required for the interface extended memory table.

Total Data Records — The total number of data records included in the Station Map table within the interface extended memory table.

Total Interface Objects — The total number of interface objects included in the Object table within the interface extended memory table.

Total Destination Groups — The total number of destination groups in the Destination Group table within the interface extended memory table.

Total Interface Instances — The total number of interface instances in the Interface Instance table within the interface extended memory table.

Total IMPs — The total number of Interface Master processes (IMPs) in the Interface Master Process table within the interface extended memory table.

Total Stations — The total number of stations in the Station table within the interface extended memory table.

Transaction History File Extended Memory Table Summary Page

The Transaction History File Extended Memory Table Summary report page is printed in summary of an attempt to build the transaction history file extended memory table. This report page lists the source files used to build the table, the table size, the total number of Refresh and FIID table entries, the total number of Transaction History Configuration File (THCF) entries, and the total number of unique Transaction History File (THF) assigns used to build the memory table.

```
BASE24            EXTENDED MEMORY TABLE BUILD REPORT            PAGE: 1


     Table Type:                THFEMT
     Source File:               \SYS.$REL5D.COM1DATA.THCF
     EMT Swap File:             \SYS.$REL5D.COM1DATA.THFEMT


     Table Size:                2026 bytes


     Total REFR/FIID Table Records:  10
     Total THCF      Table Records:  37
     Unique THF Assigns:             6
```

Table Type — The acronym for the extended memory table being built.

Source File — The fully qualified name of a source file used to build this extended memory table.

EMT Swap File — The fully qualified location of the swap file used to build this extended memory table. This is the actual extended memory table file name. The swap file is used to swap out records from disk as they are added to the memory table.

Table Size — The total size required for the transaction history file extended memory table.

Total REFR/FIID Table Records — The total number of entries in the REFR/FIID table in the extended memory table.

Total THCF Table Records — The total number of entries in the Transaction History Configuration File (THCF) table in the extended memory table.

Unique THF Assigns — The total number of Transaction History File (THF) assigns used in the extended memory table.

Transaction Log File Extended Memory Table Summary Page

The Transaction Log File Extended Memory Table Summary report page is printed in summary of an attempt to build the transaction log file extended memory table. This report page lists the source files used to build the table, the table size, the total number of entries in the object and Log Configuration File (LCF) tables, the total number of unique interface TLF assigns, and the total number of unique terminal TLF assigns used to build the table.

```
BASE24           EXTENDED MEMORY TABLE BUILD REPORT            PAGE: 1


     Table Type:                TLFEMT
     Source File:               \SYS.$REL5D.COM1DATA.LCF
     EMT Swap File:             \SYS.$REL5D.COM1DATA.TLFEMT


     Table Size:                102 bytes


     Total Object Table Records:    1
     Total LCF Table Records:       1
     Unique Interface TLF Assigns:  1
     Unique Terminal TLF Assigns:   0
```

Table Type — The acronym for the extended memory table being built.

Source File — The fully qualified name of a source file used to build this extended memory table.

EMT Swap File — The fully qualified location of the swap file used to build this extended memory table. This is the actual extended memory table file name. The swap file is used to swap out records from disk as they are added to the memory table.

Table Size — The total size required for the memory table.

Total Object Table Records — The total number of object table entries used in the extended memory table.

Total LCF Table Records — The total number of Log Configuration File (LCF) table entries used in the extended memory table.

Unique Interface TLF Assigns — The total number of unique interface Transaction Log File (TLF) assigns used in this table.

Unique Terminal TLF Assigns — The total number of unique terminal Transaction Log File (TLF) assigns used in this table.

Warmbooting Extended Memory Tables

Once an extended memory table has been built with the Extended Memory Table Build utility, processes that use the extended memory table must be warmbooted. If multiple processes use an extended memory table, all processes must be warmbooted before any of the processes will begin using the new table. Extended Memory Table Warmboot (EWB) screens are used to issue the EMT Warmboot command, which causes objects to reread specified extended memory tables. EWB screen 1 displays all extended memory tables that can warmbooted with the EMT Warmboot command. Use EWB screen 1 to select the extended memory table to be warmbooted and display a list of all objects that use the extended memory table, then use EWB screen 2 to select the objects to be warmbooted. All processes that contain the selected objects receive EMT Warmboot commands.

Not all extended memory tables can be warmbooted using the EMT Warmboot command. If an extended memory table does not appear on Extended Memory Table (EMT) screen 1 with the EMT WARMBOOT ALLOWED field set to a value of Y, you must use the LCONF Warmboot command to reinitialize the object that uses the extended memory table. LCONF Warmboot (LWB) screen 1 is used to issue this command. Refer to the ***BASE24 ITS Kernel Files Maintenance Manual*** for a description of the EWB and LWB screens.

Warmbooting Objects

When changes are made to assigns or params in the Logical Network Configuration File (LCONF), objects that use the assigns and params must be warmbooted before the changes take effect. The following commands can be used to warmboot objects:

- Assign Warmboot
- LCONF Warmboot
- Param Warmboot

Each of these commands is performed on a specific warmboot screen. Refer to the ***BASE24 ITS Kernel Files Maintenance Manual*** for descriptions of these screens.

Assign Warmboot

The Assign Warmboot command, which is issued from the Assign Warmboot (AWB) screens, causes objects to reread an assign. Only assigns configured to the Assign File (ASGN) can be reread with the Assign Warmboot command. If a change is made to an assign that is not configured in the ASGN, only the LCONF Warmboot command causes objects to reread the assign.

AWB screen 1 displays all assigns that can be warmbooted with the Assign Warmboot command. Use AWB screen 1 to select the assign to be warmbooted and display a list of all objects that use the assign, then use AWB screen 2 to select the objects to be warmbooted. All processes that contain the selected objects receive Assign Warmboot commands.

LCONF Warmboot

The LCONF Warmboot command, which is issued from the LCONF Warmboot (LWB) screen, causes an object to reread all LCONF assigns and params that it uses. An LCONF warmboot should be done in any of the following situations:

- You are changing multiple assigns or params for a particular object
- You do not know exactly which specific assigns or params have changed
- You need to reinitialize an object
- You cannot use the Assign Warmboot, Extended Memory Table Warmboot, or Param Warmboot command

Param Warmboot

The Param Warmboot command, which is issued from the Param Warmboot (PWB) screen, causes objects to reread a param. Use PWB screen 1 to select the objects to be warmbooted. All objects that can be warmbooted using the PWB screen are shown on PWB screen 1; however, not all params used by these objects can be warmbooted with the Param Warmboot command.

The params that can be warmbooted with the Param Warmboot command are shown in the following table. The second column contains the objects that use the param. If a change is made to a param that is not shown in the following table, only the LCONF Warmboot command causes objects to reread the param.

Param	Object
CACT-PRESENT	Customer Access
CPIT-PRESENT	Customer Access
ELF-FILE-RETENTION	End-of-Period Process
IMP-TIME-LIMIT	Interface Manager
LGNT-COUNTRY-ABBR	Customer Service Interface, Internal Message Translation, Voice Response Unit Interface
LGNT-CUTOVER	Customer, Customer Service Interface, End-of-Period Process, Internal Message Translation, Transaction, Voice Response Unit Interface
LGNT-CUTOVER-LEAD-TIME	End-of-Period Process
LOGICAL-NET	Customer, Customer Service Interface, Voice Response Unit Interface
PIT-NAM-ALGO	Customer Access
PIT-PRESENT	Customer Access
ROUTE-OBJECT-ID	Internal Message Translation
RSM-TERM-MASTER-KEY	Transaction Security

Param	Object
SECURE-DEV-CONF-CMD100	Transaction Security
SECURE-DEV-CONF-TXT	Transaction Security
SECURE-DEV-TIM-LMT	Transaction Security
TLF-EXTENTS	End-of-Period Process
TLF-FILE-RETENTION	End-of-Period Process
TSEC-OBJ-MASTER	Transaction Security

Core Server Control Screen Functions

The following table lists the command syntax for currently supported function commands, the objects on which the functions can be run, and what function data, if any, is required to be entered for the command. These command functions are entered in the FUNCTION NAME field on the Server Control screen. Refer to the ***BASE24 ITS Kernel Files Maintenance Manual*** for a description of the Server Control screen, refer to the ***BASE24 Core Files Maintenance Manual*** for interface instance naming conventions, and refer to the ***NET24-XPNET Network Control Operations Guide*** for station naming conventions.

FUNCTION NAME	Object Affected	FUNCTION DATA
ELF CREATE	End-of-Period Process	MMDD (the month and day of the ELF to be created)
KPT NEW	Interface Master	I *interface-instance*[*]
MIDNIGHT	End-of-Period Process	Not applicable
RESETIDF DATES	End-of-Period Process	YYMMDD (the new year, month, and date to be used)
RPT CREATE	End-of-Period Process	MMDD (the date of the report obey file to be created)
RPT CREATE TB	End-of-Period Process	MMDD (the date of the report obey file to be created)
STATUS	Voice Response Unit Master	*interface-instance*[*] or *station-name*[*]
STATUS INTF	Voice Response Unit Master	*interface-instance*[*]

FUNCTION NAME	Object Affected	FUNCTION DATA
TLF CREATE	End-of-Period Process	MMDD (the month and day of the ITLF to be created)
RESET ECT	Log Permission	Not applicable

* This variable is optional.

Appendix E

BASE24-billpay Configuration Guidelines

This appendix discusses configuration considerations for BASE24-billpay processes. It includes recommendations for the XPNET or Pathway configuration of the following processes:

- Bank Advice process
- Pathway Billpay Server process
- Remittance Generator process
- Scheduled Transaction Initiator process
- XPNET Billpay Server process

Bank Advice Process Configuration

The Bank Advice process is configured to run under control of the XPNET process. The following items must be considered when configuring Bank Advice processes:

- Since Bank Advice processes should always be running, they should be configured with a startup logic of AUTOMATIC.
- Because each Bank Advice process services a separate History Table (HIST) partition, these processes should not be configured with the SERVICES attribute.
- One Bank Advice process must be configured for each History Table (HIST) partition that can be extracted. Therefore, the number of Bank Advice processes should equal the value of the BP-NUM-HISTORY-PART-EXT param in the LCONF.
- All Bank Advice processes should be configured with the DEFINES attribute.

Pathway Billpay Server Process Configuration

The Pathway Billpay Server process is configured as a server to the Pathway system in the Pathway Configuration File (PATHCONF). It should be configured with at least one static server and with as many additional static and dynamic servers as needed to accommodate peak demands. In addition to the standard Pathway process configuration considerations, you must also configure a startup text string for the process that adheres to the following naming convention. This text string is set using the SET SERVER command with the STARTUP parameter.

Syntax

SET SERVER STARTUP "*lconf-filename symbolic-process-name*"

lconf-filename — The fully qualified name of the Logical Network Configuration File (LCONF) that contains assigns and params for this process.

symbolic-process-name — The symbolic process name of the Pathway Billpay Server process server class.

This command provides the servers in the server class with the location of the LCONF and the symbolic process name to use to access Pathway Billpay Server process assigns and commands in the LCONF.

The LCONF filename and symbolic process name must be enclosed within double quotes (") with one or more blank spaces separating each name.

Note: The Path Server process and the matching line and station information must be configured to the XPNET system before the Pathway Billpay Server process can communicate with BASE24-telebanking and the XPNET Billpay Server process.

Example

```
SET SERVER STARTUP "$DATA.PRO1EXEC.L1CONF SV-BP"
```

In this example, the LCONF is located on the $DATA.PRO1EXEC subvolume and the server class used to access assigns and params is SV-BP.

Remittance Generator Process Configuration

The Remittance Generator process is configured to run under control of the XPNET process. The following items must be considered when configuring Remittance Generator processes:

- Since normal Remittance Generator processes should always be running, they should be configured with a startup logic of AUTOMATIC. Remittance Generator processes that run in redo mode should be configured with a startup logic of COMMAND.
- Because each Remittance Generator process services a separate History Table (HIST) partition, these processes should not be configured with the SERVICES attribute.
- One normal and redo Remittance Generator process must be configured for each History Table (HIST) partition that can be extracted and each type of remittance to be created (i.e., automated clearinghouse, single-item check, or group-item check). Therefore, the number of normal Remittance Generator processes for each type of remittance to be created should equal the value of the BP-NUM-HISTORY-PART-EXT param in the LCONF. The same is true for the number of redo Remittance Generator processes. For example, if your HIST has two partitions, you would need to configure six normal Remittance Generator processes and six redo Remittance Generator processes—one for each partition and remittance type combination.
- All Remittance Generator processes should be configured with the DEFINES attribute.

Scheduled Transaction Initiator Process Configuration

The Scheduled Transaction Initiator process is configured to run under control of the XPNET process. The following items must be considered when configuring Scheduled Transaction Initiator processes:

- Since Scheduled Transaction Initiator processes should always be running, they should be configured with a startup logic of AUTOMATIC.
- Because each Scheduled Transaction Initiator process services a separate Future Table (FUTR) partition, these processes should not be configured with the SERVICES attribute.
- One Scheduled Transaction Initiator process must be configured for each Future Table (FUTR) partition that can be extracted. Therefore, the number of Scheduled Transaction Initiator processes should equal the value of the BP-NUM-FUTURE-PART-EXT param in the LCONF.
- All Scheduled Transaction Initiator processes should be configured with the DEFINES attribute.

XPNET Billpay Server Process Configuration

The XPNET Billpay Server process is configured to run under control of the XPNET process. As many processes should be configured as needed to handle peak demand. All XPNET Billpay Server processes should be configured with a startup logic of DEMAND and must include the DEFINES attribute since these processes dynamically create defines. XPNET Billpay Server processes can be configured with the SERVICE attribute, if desired, for service-based routing.

Appendix F

BASE24-billpay Processing Details

This appendix provides in-depth processing details for several BASE24-billpay processes. This information is provided for those who need or seek a more detailed analysis of BASE24-billpay processing than that provided elsewhere in this manual. This appendix provides processing details for the following processes:

- Bank Advice process
- Pathway Billpay Server process
- Remittance Generator process
- Scheduled Transaction Initiator process
- XPNET Billpay Server process

Bank Advice Process

The Bank Advice process generates advice messages for certain transactions logged to the History Table (HIST) and sends these messages to the Integrated Authorization Server process so they can be logged to the ITS Transaction Log File (ITLF). Transactions are selected from the HIST if they meet the criteria configured in the Bank Transaction Advice Table (BKAV). The criteria includes the FIID, transaction source, transaction type, and whether the transaction was approved or denied.

Typically, only nonfinancial transactions initiated by a customer service representative are configured in the BKAV. All nonfinancial transactions initiated from a remote banking endpoint and all financial transactions initiated from any source are automatically logged to the ITLF. Therefore, if these transactions were also configured in the BKAV, they would be logged to the ITLF twice.

Initialization

During initialization, each Bank Advice process builds an image of the BKAV in memory. Therefore, each process should be stopped and restarted anytime a change is made to the BKAV. During initialization, the process also reads its own control row from the Bank Advice Control Table (BKAC), based on the HIST partition number it processes. If the process finds a BKAC row, it starts processing HIST rows with a timestamp that is greater than the timestamp of the last HIST row processed, which is also recorded in this BKAC row. If the process does not find a BKAC row, it inserts a new row using the value of the BP-HISTORY-PARTITION-NUM param assigned to the process in the LCONF, and a HIST timestamp set to January 2, 0001.

Processing

Once the process is initialized, it selects HIST rows matching the BKAV criteria with timestamps greater than the timestamp of the last HIST row processed and that have existed for at least the duration specified by the BP-LAG-TIME param in the LCONF. HIST rows are processed in TMF batches. Whenever the process finds a row that matches the BKAV criteria, it updates the BKAC row with the timestamp of the transaction and commits the TMF batch. It then formats and sends an advice message to the Integrated Authorization Server process specified by the BP-TELEBANKING-LOG param in the LCONF. This Integrated

Authorization Server process can be one already used for authorization processing or you can configure a separate Integrated Authorization Server process to only log BASE24-billpay transactions to the ITLF.

After all candidate transactions have been processed from the HIST, the Bank Advice process is suspended for the duration specified by the BP-SLEEP-INTERVAL param in the LCONF. After this duration has elapsed, the process begins processing the HIST again.

TMF Batch Processing

The process initiates a batch with the start of a TMF transaction. The process opens a cursor and retrieves HIST rows until one of the following occurs:

- The end-of-file is reached in the HIST.
- The number of HIST rows retrieved matches the value of the BP-TMF-BATCH-SIZE param in the LCONF.
- The elapsed time of the TMF transaction exceeds the value of the BP-TMF-BATCH-DURATION-TIME param in the LCONF.
- The last row retrieved from the HIST matches the last row processed from the HIST in the BKAC. In this case, the timestamp of the last row processed from the HIST in the BKAC is updated with the timestamp from the HIST row.

After one of the above events occurs, the process commits the TMF batch. The process builds an 1120 (authorization transaction advice) message and sends it to the specified Integrated Authorization Server process. No response is expected from the Integrated Authorization Server process. The Bank Advice process starts the next TMF batch as soon as it receives a reply from the XPNET process.

If any error occurs while the TMF batch is processed, the process aborts the batch and retries immediately. If the error occurs while the process is retrieving HIST rows, the size of the next batch is set to the number of successfully processed HIST rows or 1, whichever is greater. Once a batch has been successfully processed, the size of the next batch is set to the value of the BP-TMF-BATCH-SIZE param. All file input and output operations are timed to allow for timeouts, thereby eliminating the possibility of deadlock.

Advice Formatting

The Bank Advice process uses the Telebanking Standard Internal Message (BSTM) and Internal Transaction Data (ITD) to format the advice. Columns from selected HIST rows, the Bank Table (BANK), Customer Vendor Table (CVND), and the Vendor Table (VNDR); LCONF assign and param values; and hard-coded values are mapped to ITD fields as shown in the following table. Only HIST columns with nonzero or nonblank values are mapped to the advice message.

ITD Field	Mapped Value
Account 1	HIST.FROM_ACCT
Account 1 Qualifier	Not filled in
Account 2	HIST.TO_ACCT (for transfers only)
Account 2 Qualifier	Not filled in
Acquirer	
FIID	BP-ACQ-FIID param
INST-CNTRY-CDE	BP-ACQ-CNTRY-CODE param
ID-CDE	BANK.INST_ID
LGNT	LOGICAL-NET param
External Acquirer Dependent Data	
CSR User Group	HIST.LAST_FM_GRP_NUM
CSR User Number	HIST.LAST_FM_USER_NUM
Terminal Name	HIST.LAST_FM_TERM_NUM
Action Date	HIST.PMT_DATE
Additional Data	HIST.VEND_SPECIFIC_DATA
Transaction Amount	HIST.AMT (if transaction was approved); otherwise use 0
Approval Code Length	Not filled in
Authorization Time Limit	Not filled in
Local Transaction	HIST.TXN_SRC_LCT
Message Type Identifier	1120

ITD Field	Mapped Value
Primary Account Number	HIST.CUST_ID
Payee	CUSTVEND.CUSTVEND_NUM
Processing Code	HIST.TXN_TYP HIST.FROM_ACCT_TYP HIST.TO_ACCT_TYP
Product Identifier	14
Point of Service	66013066510S
Recurring Transaction Data	Not filled in
Reason Code—Message	Not filled in
Sequence Number	HIST.TXN_SRC_NUM
Source Code	HIST.TXN_SRC
Terminal Identifier	HIST.LAST_FM_TERM_NAM
Entry Timestamp	HIST.BP_GMT
Transaction Originator	7
Action Code	HIST.RSP_CODE
Transaction Amount—Cardholder Billing	Use value mapped to Transaction Amount field.
Transaction Amount—Cardholder Billing 2	If transaction is a transfer, use value mapped to Transaction Amount field; otherwise, not filled in.
Acquirer Dependent	Not filled in
Transaction Amount—Original	If HIST.AMT is not equal to 0 and the transaction was denied, use HIST. AMT; otherwise, use 0.
Customer Reference Number	HIST.REF_NUM

ITD Field	Mapped Value
Payee Description[*]	
Customer Account with Vender Vendor Name	HIST.ACCT_WITH_VEND VEND.VEND_NAME
Payment Date	HIST.PMT_DATE
Transaction Responder	7

[*] This field is filled in only if the account with vendor and vendor name are not blank.

Billpay Server Processes

The Pathway Billpay Server process and XPNET Billpay Server process authorize and log BASE24-billpay transactions. The Pathway Billpay Server process authorizes nonfinancial transactions entered from a CSR terminal, except for the scheduling of future or recurring payments or transfers. The XPNET Billpay Server process authorizes all BASE24-billpay transactions received from an RBSI-compliant device or host as well as the following BASE24-billpay transactions received from CSR terminals:

- Schedule payment – immediate
- Schedule payment – future
- Schedule payment – recurring
- Schedule transfer – future
- Schedule transfer – recurring

Program Structure

Both the Pathway Billpay Server process and XPNET Billpay Server process share a common program structure. In general, each program has a shell that accommodates its particular operating environment—Pathway or XPNET. This shell contains a format layer, transaction layer, and utility layer.

The Pathway Billpay Server process is built using a Pathmaker shell that runs in the Pathway environment. Within this shell, a Pathmaker Services layer is responsible for recognizing the request types received from the CSR Billpay Requester process and determining the processing required for each. For example, a request for updating customer information is different from a request for reading customer information.

After determining the type of service requested, the Pathmaker Services layer calls the format layer to translate the message into standard internal format. The format layer also performs a number of validations on the message. If the message has a valid format, it is passed to the transaction layer.

Error Processing

Billpay Server processes can generate two different types of errors—authorization errors and system errors. An authorization error is a transaction response returned from the Integrated Authorization Server process that indicates that the transaction failed. A system error includes any other type of error (e.g., timeout, TMF error, file lock, etc.). Transactions that fail, regardless of whether the failure was caused by an authorization or system error, are not retried. This differs from the retry processing supported by the Scheduled Transaction Initiator process, because the transactions authorized by the Billpay Server process require immediate feedback to a customer or customer service representative.

Throughout this subsection, the term *timeout* refers to the situation where the authorizer does not respond to a request within the interval defined by the BP-TXN-TIMEOUT param in the LCONF. This condition is considered to be a system error. If a response from the Integrated Authorization Server process is received before this interval expires but indicates that a timeout occurred during authorization processing, the timeout is considered to be an authorization error.

Authorization Error Processing

When the Billpay Server process receives a response from the Integrated Authorization Server process that indicates that the transaction failed, it does not perform retry processing or generate an EMS event message. The process simply logs the transaction to the History Table (HIST) and generates a response indicating that the transaction failed.

System Error Processing

If the Billpay Server process encounters an I/O error while attempting to access a database table, it inserts a row into the HIST indicating the outcome of the transaction, responds to the request, and generates an EMS event message for the I/O error. If the process cannot insert a row into the HIST, it responds to the original request and generates an EMS event message for the HIST I/O error. Depending on the severity of the error, the Billpay Server process may abend at this time. Most transactions always require a row to be inserted into the HIST.

When the Billpay Server process sends a request to the Integrated Authorization Server process, it maintains the original request information in a linked list and sets a timer based on the value of the BP-TXN-TIMEOUT param. This param should be set to an interval greater than any timeout value set for the Integrated Authorization Server process. If this timer expires before the Integrated

Authorization Server process responds, the Billpay Server process considers the transaction to have failed and inserts a row into the HIST indicating that the transaction failed. It does not generate a transaction response message. The process removes the original request message from its linked list, deallocates the memory for the message, and drops the message. A reversal is not generated for the message originator in this case.

If the Billpay Server process receives an authorization response from the Integrated Authorization Server process and cannot find the corresponding request in its linked list, it sends a reversal back to the Integrated Authorization Server process and does not update the database or generate a transaction response message. This situation should only occur if the Billpay Server process is stopped and restarted during normal processing or if the original request to the Integrated Authorization Server process timed out.

If the Billpay Server process receives a message failure for an authorization request, the process considers a system error to have occurred. In this case, the process denies the original transaction, inserts a row into the HIST, and sends a transaction response to the Integrated Authorization Server process.

If the Billpay Server process receives a message failure for a reply (i.e., a message that is not an authorization request), the process again considers a system error to have occurred. In this case, the process reverses the original transaction authorization. If this reversal in unsuccessful, the process updates the HIST row for the transaction with an appropriate condition and generates an EMS event message.

Remittance Generator Process

Remittance Generator processes create and recreate remittance data for successfully completed payment transactions. Normal Remittance Generator processes create remittance data for successfully completed payment transactions in the History Table (HIST) after the duration specified by the BP-LAG-TIME param in the LCONF has elapsed. The remittance data created by the normal Remittance Generator processes is written to remit files, with a separate file for each type of remittance and date the remittance data was created. Redo Remittance Generator processes allow remittance data for a previous day's payments to be recreated in a new remit file. The processing performed by each type of Remittance Generator process is described below.

Normal Remittance Generation Processing

Each normal Remittance Generator process periodically searches the HIST for successfully completed payments of a particular type that have existed in the HIST for at least the duration of the BP-LAG-TIME param. After all candidate payment transactions have been processed, the Remittance Generator process suspends itself for the duration of the BP-SLEEP-INTERVAL param before it repeats its search of the HIST.

HIST payment transactions are processed in batches and by the remit date. When the lag time after logical network cutover expires, the normal Remittance Generator processes close the remit files for the business day just ended, write final totals to the Payment Summary Table (PST), and open new remit files. If the new remit files do not exist, they are created. The BASE24-billpay remittance day is offset from the BASE24-billpay business day by the duration of the lag time. The lag time allows time for an immediate scheduled payment made at a remote banking device or customer service representative (CSR) terminal to be canceled before remittance data is generated for the payment.

Remit file names are specified by taking the first character of the BP-REMIT-TYPE param assigned to the process and appending the current remittance date, in YYMMDD format. Single-item and group-item check remittance data is placed in the same remit file. All remit files are created in the location specified by the BP-REMIT assign in the LCONF.

Each normal Remittance Generator process initiates a batch by starting a TMF transaction. The process opens a cursor and retrieves HIST rows until one of the following events occurs:

- The end-of-file is reached on the HIST.
- The number of rows processed from all tables reaches the value of the BP-TMF-BATCH-SIZE param in the LCONF.
- The elapsed time of the TMF transaction exceeds the value of the BP-TMF-BATCH-DURATION-TIME param in the LCONF.
- The current time exceeds the logical network cutover time plus the value of the BP-LAG-TIME param. The remittance day is cut over at this time.

As each HIST row is processed, the process updates the HIST row as follows:

- Sets the HIST.REMIT_BUS_DATE column to the current remittance date.
- Sets the HIST.REMIT_STAT column to indicate that the payment has been sent out.
- Sets the HIST.LAST_FM_OLTP_TS column to the current system time.

For each HIST row processed, the process adds a remittance data record to its remit file. The process accesses the following tables to verify HIST information and to retrieve additional information that is included in the remittance data record:

- Bank Table (BANK)
- Customer Table (CSTT)
- Customer Vendor Table (CVND)
- Personal Information Table (PIT)
- Vendor Table (VNDR)
- Vendor Branch Table (VNDB)

The process also adds entries in memory that are later used to update the Customer Vendor Summary Table (CVS) and Payment Summary Table (PST). The CVS entry consists of the customer ID, customer vendor number, and payment amount. The PST entry consists of the business date, FIID, vendor number, and payment amount.

After all HIST rows have been successfully processed for a batch, the process updates the CVS and PST. To prevent different Remittance Generator processes from colliding with one another, the entries in memory are sorted. The CVS entries are sorted by customer ID and customer vendor number. The PST entries are sorted by business date, FIID, and vendor number. Before updating the CVS and PST with these entries, the process sums together any entries with the same key column values. This eliminates the overhead of having to update the same row multiple times.

To update the PST, the Remittance Generator process attempts to read each row for which it has entries in memory. If a row exists, the process updates the PST.TOT_AMT and PST.TOT_NUM columns appropriately. If a row does not exist, the process inserts a new row into the table.

To update the CVS, the Remittance Generator process attempts to read each row for which it has entries in memory. If a row exists, the process updates the CVS.TOT_AMT and CVS.TOT_NUM columns appropriately. If a row does not exist, the process inserts a new row into the table.

After the PST and CVS are updated, the TMF transaction is committed. If any error occurs while processing the batch, the batch is aborted and retried immediately. If the error occurred while processing the HIST rows, the size of the next batch is set to the number of successfully processed HIST rows or 1, whichever is greater. If the error occurred while processing the summary tables, the size of the next batch is set to the number of successfully processed rows or 1, whichever is greater. Once a batch has been successfully processed, the size of the next batch is set to the value of the BP-TMF-BATCH-SIZE param. All file input and output operations are timed to allow for timeouts, thereby eliminating the possibility of deadlock.

Redo Remittance Generation Processing

Redo remittance generation allows you to regenerate remittance data for a previous date. Redo remittance generation may be required if a remit file is inadvertently purged or if its data becomes corrupted. Before a redo Remittance Generator process can be started, the remit file to be regenerated must either be purged or all data within the file must be purged. If a temporary Payment Summary Table already exists in the remit file subvolume, its data must also be purged prior to running the redo Remittance Generator process.

Processing performed by a redo Remittance Generator process is similar to that performed by a normal Remittance Generator process, with the following exceptions:

- The name of the remit file used is based on the date specified by the BP-REDO-DATE param rather than the current remittance date. If the remit file does not already exist, the process creates it.
- Updates are made to a temporary Payment Summary Table rather than to the permanent PST.
- Neither the permanent PST or CVS are written to or updated in any way during redo processing.
- After the temporary PST has been updated, the process compares the totals in the temporary table with those in the permanent PST. The total amount and total number are compared for each row with a match FIID and vendor number. If the totals do not match, an entry is made in a *to be logged* memory table and the difference in the total amount and count is added to a summary total.
- After all of the summary rows have been compared, one or more event messages are generated indicating the processing outcome and the process stops. If needed, these event messages can be used to manually research and correct any discrepancies. Refer to the TPBPRMTG online log message documentation file for detailed information on these event messages.

Remit File Records

The following table describes the contents of each remittance data record written to a remit file. The first column lists each field in a remittance data record. The second column identifies the table and column from which the information recorded in each field is obtained. All data written to a remit file is provided in ASCII format. If need be, data is converted to ASCII format before it is written to a remit file.

Remit File Field	Data Source
VEND-NUM	HIST.VEND_NUM
REMIT-TYP	HIST.REMIT_TYP
HIST-GMT	HIST.HIST_GMT
VEND-NAME	VNDB.VEND_BR_NAME
VEND-ADDR-1	VNDB.ADDR_1
VEND-ADDR-2	VNDB.ADDR_2

Remit File Field	Data Source
VEND-CITY	VNDB.CITY
VEND-ST	VNDB.ST
VEND-CNTRY	VNDB.CNTRY
VEND-POSTAL-CODE	VNDB.POSTAL_CODE
VEND-TO-INST-ID	VNDR.INST_ID
VEND-TO-ACCT-NUM	VNDR.ACCT_NUM
VEND-TO-ACCT-TYP	VNDR.ACCT_TYP
VEND-SPECIFIC-DATA	HIST.VEND_SPECIFIC_DATA
CUST-ID	HIST.CUST_ID
CUST-FIRST-NAME	PIT.NAM_GIVEN
CUST-LAST-NAME	PIT.NAM_FMLY
CUST-MIDDLE-INITIAL	PIT.NAM_M_I
CUST-ADDR-1	PIT.STR_ADDR_1
CUST-ADDR-2	PIT.STR_ADDR_2
CUST-CITY	PIT.CITY
CUST-ST	PIT.ST_CDE
CUST-POSTAL-CODE	PIT.POSTAL_CDE
CUST-FROM-INST-ID	BANK.INST_ID
CUST-FROM-ACCT-NUM	HIST.FROM_ACCT_NUM
CUST-FROM-ACCT-TYP	HIST.FROM_ACCT_TYP
CUST-VEND-NUM	HIST.CUST_VEND_NUM
CUST-ACCT-WITH-VEND	HIST.ACCT_WITH_VEND
AMT	HIST.AMT

Remit File Field	Data Source
REF-NUM	HIST.REF_NUM
BUS-DATE	HIST.BUS_DATE

Scheduled Transaction Initiator Process

The Scheduled Transaction Initiator process initiates payments and transfers that have been previously scheduled.

Processing Window

The Scheduled Transaction Initiator process uses a processing window to determine when future and recurring transactions can be processed. This is required so that the process does not interfere with logical network cutover and end-of-day processing.

The processing window is delimited by the values of the BP-STI-WINDOW-OPEN-TIME and BP-STI-WINDOW-CLOSE-TIME params. The processing window must be opened after logical network cutover occurs, at which time the entire logical network moves to a new business day. When the window opens, the Scheduled Transaction Initiator process begins processing any future or recurring transactions that are scheduled to be performed for the current business day. As a general rule of thumb, you should allow at least 15 minutes to elapse after the logical network cutover time, before opening the processing window. This allows time for cutover processing to complete before scheduled transaction processing begins.

Any payment transaction to be processed must also meet the aging criteria enforced by the value of the BP-STI-LAG-TIME param. This param indicates how long a payment transaction must reside in the Future Table (FUTR) before it can be processed.

The Scheduled Transaction Initiator process remains idle until the processing window opens. When the window opens, the process searches the FUTR for transactions to process until the amount of time specified by the BP-STI-LAG-TIME and BP-STI-FINAL-PROCESSING params have elapsed after the processing window closes. The process will not accept any future or recurring transactions for the current business day after the processing window has closed. The BP-STI-LAG-TIME and BP-STI-FINAL-PROCESSING params allow the Scheduled Transaction Initiator process to finish processing all transactions for the current business day prior to logical network cutover.

The processing window should be closed prior to logical network cutover such that scheduled transaction processing does not interfere with logical network cutover and end-of-day processing. As a general rule of thumb, you should allow enough time between the close of the processing window and logical network cutover, for the values of the BP-STI-LAG-TIME and BP-STI-FINAL-PROCESSING params plus an additional 15 minutes to elapse.

For example, if logical network cutover occurs at 1600 hours, the BP-STI-LAG-TIME param is set to15 minutes, and the BP-STI-FINAL-PROCESSING param is set to 3 minutes, the earliest the processing window could be closed would be 1527 hours (1600 - (15 + 15 + 3)).

For the initial payment of any recurring transaction to be made on the current business day, it must be scheduled prior to the close of the processing window. Once the window has been closed, even though the system is still on the current business day, any attempt to schedule a recurring payment transaction with the current business day as the initial payment date will be denied. Once the processing window has closed, any schedule payment – recurring transactions must specify tomorrow’s business date as the initial payment date for the transactions to be approved. The actual processing of the initial payments would then take place on the following business day when the processing window is reopened.

Normal Processing

During normal processing, the Scheduled Transaction Initiator process reads the FUTR to find all transactions in its configured partition that are scheduled to be paid before or on the current business date that have met the aging requirement of the BP-STI-LAG-TIME param. For each transaction found, the process updates the FUTR row to indicate that it is being processed, generates the corresponding financial transaction request and sends it to the Integrated Authorization Server process, where it is authorized, logged to the ITLF, and returned to the Scheduled Transaction Initiator process. When the Scheduled Transaction Initiator process receives a response indicating that the transaction was successful, it deletes the FUTR row. If an associated row in the RCUR indicates that another future transaction should be scheduled, the process inserts a new row in the FUTR for the transaction and updates the RCUR row accordingly. If the RCUR row has expired, it is deleted. Finally, the process inserts a row in the HIST to indicate that the transaction was performed.

Payment Validation Processing

Before a payment transaction is sent to the Integrated Authorization Server process to be authorized, the Scheduled Transaction Initiator process validates the payment as follows:

1. The transaction must have been in the Future Table (FUTR) for at least the amount of time specified by the BP-STI-LAG-TIME param.
2. The Verify: Status field on Vendor Table (VNDR) screen 1 must be set to the value V (Verified).
3. The payment date must be greater than or equal to the value in the Effective Date Range: Begin Date field on VNDR screen 1and less than or equal to the value in the Effective Date Range: End Date field on VNDR screen 1.
4. The Account Verify: Status field on Customer Vendor Table (CVND) screen 1 must be set to the value V (Verified).
5. The payment date must be greater than or equal to the value in the Effective Date Range: Account Begin Date field on CVND screen 1and less than or equal to the value in the Effective Date Range: Account End Date field on CVND screen 1.
6. The amount of the payment transaction must be less than or equal to the following:
    - The nonzero amount specified in the PAYMENT HIGH LIMIT field on Customer Table (CSTT) screen 1 for the customer, regardless of the financial institution's payment high limit specified in the Bank Table (BANK).
    - The amount specified in the PAYMENT HIGH LIMIT field on Bank Table (BANK) screen 1 for the customer's financial institution, if the customer's payment high limit is set to zero.

    This check is not performed if the payment high limit for both the customer and the financial institution are set to zero.
7. If the Vendor Specific Data: Desc field on Vendor Table (VNDR) screen 2 displays data, then the Vendor Specific Data field on History Table (HIST) screen 2 cannot be blank for the transaction. If the Vendor Specific Data: Desc field on Vendor Table (VNDR) screen 2 is blank, then the Vendor Specific Data field on History Table (HIST) screen 2 must also be blank for the transaction.

If a scheduled payment transaction does not pass all of the above validation criteria, it is denied and is not sent to the Integrated Authorization process for authorization.

Scheduled transfer transactions are always sent to the Integrated Authorization Server process for authorization.

Future Table (FUTR) Partitioning

The Future Table (FUTR) can be partitioned both physically and logically. The FUTR can be physically partitioned using Tandem's NonStop SQL/MP product. The FUTR is logically partitioned by configuring separate Scheduled Transaction Initiator processes to service each partition. Although it is not required that the physical and logical partitioning match, it is recommended. The first column in the primary index to the FUTR, STI_PARTITION_NUM, is a key that allows the FUTR to be partitioned both physically and logically.

When a Billpay Server process initially inserts a row into the FUTR, it uses the value of the BP-NUM-FUTURE-PART-INS param from the LCONF to calculate a value for the STI_PARTITION_NUM column. Specifically, it generates a random number between 1 and the value of the BP-NUM-FUTURE-PART-INS param. When a Scheduled Transaction Initiator process inserts a new row into the FUTR due to a recurring transaction, it uses the same method to calculate a value for the STI_PARTITION_NUM column.

A separate Scheduled Transaction Initiator process should be configured for each partition value. When a Scheduled Transaction Initiator process selects rows from the FUTR, it only selects rows from its assigned partition, which is determined by the value of the BP-FUTURE-PARTITION-NUM param configured for the process.

This method of partitioning allows you to alter both the physical and logical partitions as needed to meet performance requirements. Physical partitioning of the FUTR can be altered using NonStop SQL/MP. To alter the logical partitioning of the FUTR, Scheduled Transaction Initiator processes should be configured to the XPNET system and the BP-NUM-FUTURE-PART-INS and BP-NUM-FUTURE-PART-EXT params should be updated in the LCONF. For the change to take effect, all Billpay Server and Scheduled Transaction Initiator processes must be stopped and restarted. If you decrease the number of FUTR partitions, you should leave the BP-NUM-FUTURE-PART-EXT param at its original value until all the extra FUTR partitions are completely processed.

Error Processing

The Scheduled Transaction Initiator process categorizes errors that cause a transaction to fail into two categories—authorization errors and system errors. Authorization errors are responses returned from the Integrated Authorization Server process that indicate that the transaction failed. System errors include all other errors (e.g., timeouts, TMF errors, file locks, etc.) that cause the transaction to fail.

The Scheduled Transaction Initiator process can retry transactions that failed due to authorization errors, based on the configuration of the Retry Table (RTRY). If the action code returned from the Integrated Authorization Server process is configured in this table, it can be retried. The RTRY specifies the number of times the transaction can be retried and the length of time to delay between each retry.

You cannot configure retries for system errors. The internal logic of the Scheduled Transaction Initiator process determines whether a transaction can be retried due to a system error.

If an authorization response is not received from the Integrated Authorization Server process within the time interval specified by the BP-TXN-TIMEOUT param in the LCONF, the transaction is considered to have timed out. This is considered to be a system error. For this particular timeout error, the Scheduled Transaction Initiator process inserts a row into the History Table (HIST) with an action code of 911. The process does not generate an event message or use the RTRY for this error. If an authorization response is received from the Integrated Authorization Server process within this time interval and the action code indicates that a timeout occurred during authorization processing, the error is considered to be an authorization error.

Whenever the Scheduled Transaction Initiator process inserts a row into the HIST for a system error or a denied transaction (including the timeout condition described above), the process checks the Bank Notice Table (BKNT) to determine whether a customer notice needs to be generated for the action code. If so, the process inserts a row into the Report Action Table (RPTA). The text of the customer notice is configured in the Response Text Table (RTXT) and the customer notices are generated by the Customer Notice Report program.

Authorization Error Processing

When the Scheduled Transaction Initiator process receives an authorization response indicating that the transaction failed, it searches the RTRY for the action code returned. If the process does not find the action code in the RTRY, the transaction is not retried. The process then deletes the FUTR row, inserts a new FUTR row if indicated by the RCUR, and inserts a row into the HIST indicating the outcome of the transaction.

If the authorization response contains an action code configured in the RTRY, the process updates the FUTR row to indicate when the transaction is to be retried. The process continues to retry the transaction until it is either successful or the maximum number of retries configured in the RTRY for the last error encountered is met or exceeded.

The process maintains two retry counts in the FUTR: a transient count and a nontransient count. A flag in the RTRY indicates whether an error is considered to be transient or nontransient. A transaction can experience both transient and nontransient errors over its lifetime. The last error that occurred determines the interval to wait before the next retry and the retry limit. When the retry count for the last error meets or exceeds the retry limit configured for the error in the RTRY, the transaction is denied.

System Error Processing

If the Scheduled Transaction Initiator process encounters a retryable I/O error while attempting to retrieve a row from the FUTR, it generates an event message, delays for a set time, and then retries the transaction. If the process encounters an error that cannot be retried while attempting to retrieve a row from the FUTR, it generates an event message and attempts to retrieve the next row from the FUTR. If several system errors that cannot be retried are encountered in succession, the Scheduled Transaction Initiator process starts delaying between transactions. If the process encounters a system error that it cannot recover from, the process generates an event message and stops.

If the response to an authorization request is not received from the Integrated Authorization Server process before the interval specified by the BP-TXN-TIMEOUT param expires, the transaction is considered to have failed and is not retried. If the response is received after this timeout has occurred, the Scheduled Transaction Initiator process generates a reversal.

If the Scheduled Transaction Initiator process encounters retryable I/O error while processing an authorization response, it generates an event message, delays for a set time, and then retries the transaction. If the process encounters an I/O error that cannot be retried and the authorization is approved, the process generates a reversal in addition to its nonretryable system error processing.

If the Scheduled Transaction Initiator process cannot retrieve the FUTR row that corresponds to an authorization response, the process generates a reversal if the authorization was approved. Otherwise, the response is dropped. In either case, an event message is generated. This typically occurs when a transaction times out.

When a Scheduled Transaction Initiator process is stopped and restarted while an authorization request is outstanding, two different cases must be handled. In the first case, the authorization response is received by the restarted process before it has reread the associated FUTR row. In this case, the response is processed normally because the process can still retrieve the FUTR row. In the second case, the process rereads the FUTR row and detects that it has already sent an authorization request. If the authorization timeout interval has already expired, the transaction is handled as a timeout. If the authorization timeout interval has not yet expired, the process waits for the authorization response for the remainder of the timeout interval. If the response is received in time, it is processed normally. If not, it is treated as a timeout.

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